Quantum Scientific Computing

Lecture Notes

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Part I

Foundations

Chapter 1

Quantum Computation Basics

Learning objectives

- State the postulates of quantum mechanics in the language of qubits and registers.
- Manipulate multi-qubit states via tensor products and Dirac notation.
- Read and build simple quantum circuits from a universal gate set.
- Build a reversible binary adder from CNOT and Toffoli gates.
- Prepare states by gate synthesis and describe projective and generalized (POVM) measurements.
- Trace the quantum teleportation and superdense-coding protocols amplitude by amplitude.
- Explain measurement, the no-cloning theorem, no-signaling, and density operators.
- Quantify nonlocality through the CHSH inequality and its quantum value $2\sqrt{2}$.
- Define the basic complexity measures for quantum algorithms.

Covered this term (Fall 2026–2027). §1.1, §1.2 (excluding §1.2.4), §1.3, §1.4, §1.5.1, and §1.6. Not covered this term: §1.2.4 (nonlocality and the CHSH inequality), §1.5.2 (superdense coding), §1.7 (density operators and mixed states), and §1.8 (complexity of quantum algorithms).

This lecture develops the mathematical stage on which all of quantum computing takes place. The single postulate we need is deceptively simple: *the state of a quantum system is a unit vector in a complex inner-product space.* From this one idea we recover Dirac's bra–ket notation, the geometry of overlaps and orthogonality, the description of a single qubit and the Bloch sphere, and—via the tensor product—the exponentially large state space of many qubits, where entanglement first appears. We then turn to the operational layer: gates as unitary operators, the standard single- and multi-qubit gates, quantum circuits, universal gate sets, and the no-cloning theorem. We close with two ideas that recur throughout the course: the density-operator formalism, which lets us describe subsystems and mixtures, and the complexity measures (query complexity, gate count, circuit depth) by which we judge quantum algorithms. Everything in later lectures is built on top of this vocabulary.

1.1 Qubits, superposition, and measurement

We begin with the linear-algebraic setting.

Definition 1.1 (Complex inner-product space)**.** Let $\mathcal{H}$ be a complex vector space. An *inner product* on $\mathcal{H}$ is a map $\langle \cdot, \cdot \rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$ satisfying, for all $u, v, w \in \mathcal{H}$ and $a, b \in \mathbb{C}$:

(i) *Conjugate symmetry:* $\langle u, v\rangle = \overline{\langle v, u\rangle}$.

(ii) *Linearity in the second argument:* $\langle u,\, av + bw\rangle = a\langle u, v\rangle + b\langle u, w\rangle$. (Combined with (i), this makes the inner product *conjugate-linear* in the first argument.)

(iii) *Positive-definiteness:* $\langle u, u\rangle \geq 0$, with equality if and only if $u = 0$.

We adopt the physicists' convention that the inner product is linear in the *second* slot and conjugate-linear in the first; this is the convention that makes Dirac notation come out cleanly. Concretely, on $\mathbb{C}^N$ with $x = (x_0, \ldots, x_{N-1})^T$ and $y = (y_0, \ldots, y_{N-1})^T$,

$$\langle x, y\rangle \;=\; x^\dagger y \;=\; \sum_{i=0}^{N-1} \overline{x_i}\, y_i. \tag{1.1}$$

Definition 1.2 (Norm and unit vector)**.** The *norm* induced by the inner product is $\|u\| = \sqrt{\langle u, u\rangle}$. A vector u is a *unit vector* (or *normalized*) if $\|u\| = 1$. Two vectors u, v are *orthogonal* if $\langle u, v\rangle = 0$.

Definition 1.3 (Finite-dimensional Hilbert space)**.** A *finite-dimensional Hilbert space* is a finite-dimensional complex inner-product space $\mathcal{H}$ (in finite dimension completeness is automatic, so no further condition is needed). If $\dim \mathcal{H} = N$, then choosing an orthonormal basis gives an isomorphism $\mathcal{H} \cong \mathbb{C}^N$ that preserves the inner product. For a system of n qubits we will find $N = 2^n$.

The state-space postulate can now be stated precisely.

Remark 1.4 (State-space postulate). The state of an isolated quantum system is described by a *unit* vector in a Hilbert space $\mathcal{H}$. The space is closed under linear combination: if $u, v \in \mathcal{H}$ and $a, b \in \mathbb{C}$, then $au + bv \in \mathcal{H}$. This closure is the *superposition principle*—any linear combination of legal states is again a legal (unnormalized) state.

1.1.1 The inner product controls all geometry

Proposition 1.5 (Elementary consequences of the axioms)**.** *For all $u, v \in \mathcal{H}$ and $a \in \mathbb{C}$:*

(a) $\langle u, 0\rangle = \langle 0, u\rangle = 0$*;*

(b) $\|au\| = |a|\, \|u\|$*;*

(c) (Pythagoras) *if* $\langle u, v\rangle = 0$ *then* $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Proof. (a) By linearity in the second argument, $\langle u, 0\rangle = \langle u, 0 \cdot 0\rangle = 0\,\langle u, 0\rangle = 0$; the other equality follows from conjugate symmetry. (b) $\|au\|^2 = \langle au, au\rangle = \overline{a}\, a\, \langle u, u\rangle = |a|^2\, \|u\|^2$, and take square roots. (c) Expand $\|u + v\|^2 = \langle u + v, u + v\rangle = \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2$; the two cross terms vanish because $\langle u, v\rangle = 0$ forces $\langle v, u\rangle = \overline{\langle u, v\rangle} = 0$. □

The single most useful inequality in the whole subject is the following.

Theorem 1.6 (Cauchy–Schwarz inequality)**.** *For all $u, v \in \mathcal{H}$,*

$$|\langle u, v\rangle| \;\leq\; \|u\|\, \|v\|\,, \tag{1.2}$$

with equality if and only if u and v are linearly dependent.

Proof. If $v = 0$ both sides are 0 and u, v are trivially dependent, so assume $v \neq 0$. For any $\lambda \in \mathbb{C}$, positive-definiteness gives

$$0 \;\le\; \|u - \lambda v\|^2 \;=\; \|u\|^2 - \overline{\lambda}\langle v, u\rangle - \lambda\langle u, v\rangle + |\lambda|^2 \|v\|^2 .$$

Choose $\lambda = \dfrac{\langle v, u\rangle}{\|v\|^2}$ (the value that minimizes the right-hand side). Then $\lambda\langle u, v\rangle = \dfrac{\langle v, u\rangle\langle u, v\rangle}{\|v\|^2} = \dfrac{|\langle u, v\rangle|^2}{\|v\|^2}$, and likewise $\overline{\lambda}\langle v, u\rangle = \dfrac{|\langle u, v\rangle|^2}{\|v\|^2}$, so

$$0 \;\le\; \|u\|^2 - \frac{|\langle u, v\rangle|^2}{\|v\|^2}.$$

Rearranging yields $|\langle u, v\rangle|^2 \le \|u\|^2 \|v\|^2$, i.e. the claim. Equality holds iff $\|u - \lambda v\| = 0$, i.e. $u = \lambda v$, which is exactly linear dependence. □

Corollary 1.7 (Triangle inequality)**.** $\|u + v\| \le \|u\| + \|v\|$ *for all* $u, v \in \mathcal{H}$.

Proof. $\|u + v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\,\langle u, v\rangle + \|v\|^2 \le \|u\|^2 + 2\,|\langle u, v\rangle| + \|v\|^2 \le \|u\|^2 + 2\,\|u\|\,\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$, using $\mathrm{Re}\, z \le |z|$ and Theorem 1.6. Take square roots. □

For unit vectors the Cauchy–Schwarz inequality reads $|\langle u, v\rangle| \le 1$, so we may write

$$\langle u, v\rangle \;=\; \underbrace{\|u\|\,\|v\|}_{\text{length}}\;\underbrace{\cos\theta}_{\text{alignment}}\;\underbrace{\mathrm{e}^{\mathrm{i}\varphi}}_{\text{relative phase}}, \qquad \cos\theta = \frac{|\langle u, v\rangle|}{\|u\|\,\|v\|} \in [0, 1]. \tag{1.3}$$

The modulus measures how much the two directions coincide ($\cos\theta = 1$ means parallel, $\cos\theta = 0$ means orthogonal), while the argument φ is a *relative phase* with no analogue for real vectors. Orthogonality, $\langle u, v\rangle = 0$, is exactly the statement that the two states are perfectly distinguishable.

1.1.2 Dirac notation

Definition 1.8 (Ket and bra)**.** A vector of $\mathcal{H}$ is written as a *ket* $|x\rangle$. To each ket there corresponds a *bra* $\langle x|$, the element of the dual space $\mathcal{H}^*$ acting by $\langle x| : |y\rangle \mapsto \langle x, y\rangle$. In coordinates, if $|x\rangle$ is the column vector with entries x_i, then $\langle x|$ is the row vector $x^\dagger = (\overline{x_0}, \dots, \overline{x_{N-1}})$, the conjugate transpose.

Definition 1.9 (Inner and outer products)**.** The *inner product* of $|x\rangle$ and $|y\rangle$ is the scalar $\langle x|y\rangle = \langle x, y\rangle \in \mathbb{C}$. The *outer product* $|x\rangle\langle y|$ is the linear operator acting as

$$\big(|x\rangle\langle y|\big)\,|z\rangle = |x\rangle\;\langle y|z\rangle = \langle y|z\rangle\;|x\rangle\,.$$

In coordinates it is the matrix with entries $\big(|x\rangle\langle y|\big)_{ij} = x_i\,\overline{y_j}$.

Fix an orthonormal basis $\{|i\rangle\}_{i=0}^{N-1}$, meaning $\langle i|j\rangle = \delta_{ij}$. Any ket expands as $|x\rangle = \sum_i x_i\,|i\rangle$, and applying $\langle j|$ picks out a single coefficient:

$$\langle j|x\rangle = \sum_i x_i\,\langle j|i\rangle = \sum_i x_i\,\delta_{ji} = x_j. \tag{1.4}$$

So the j-th amplitude is literally the overlap $\langle j|x\rangle$.

Theorem 1.10 (Completeness / resolution of the identity)**.** *If $\{|i\rangle\}_{i=0}^{N-1}$ is an orthonormal basis of $\mathcal{H}$, then*

$$\sum_{i=0}^{N-1} |i\rangle\langle i| \;=\; \mathbb{I}. \tag{1.5}$$

Consequently every operator A satisfies $A = \sum_{i,j} A_{ij}\, |i\rangle\langle j|$ with $A_{ij} = \langle i|A|j\rangle$.

Proof. Let $P = \sum_i |i\rangle\langle i|$. For an arbitrary $|x\rangle = \sum_j x_j\, |j\rangle$,

$$P\,|x\rangle = \sum_i |i\rangle\,\langle i|x\rangle = \sum_i x_i\,|i\rangle = |x\rangle\,,$$

using (1.4). Since $P\,|x\rangle = |x\rangle$ for every $|x\rangle$, we have $P = \mathbb{I}$. For the second statement, insert the identity on both sides of A: $A = \mathbb{I}\,A\,\mathbb{I} = \sum_{i,j} |i\rangle\,\langle i|A|j\rangle\,\langle j| = \sum_{i,j} A_{ij}\; |i\rangle\langle j|$. □

Remark 1.11 (Projectors). For a unit vector $|u\rangle$ the operator $P_u = |u\rangle\langle u|$ is the orthogonal *projector* onto the line spanned by $|u\rangle$: it is Hermitian ($P_u^\dagger = P_u$) and idempotent, since $P_u^2 = |u\rangle\,\langle u|u\rangle\,\langle u| = |u\rangle\langle u| = P_u$ because $\langle u|u\rangle = 1$. The completeness relation of Theorem 1.10 decomposes $\mathbb{I}$ into a sum of orthogonal rank-one projectors, one per basis vector.

1.1.3 The single qubit

Definition 1.12 (Qubit and computational basis)**.** A *qubit* is a quantum system whose state space is $\mathcal{H} = \mathbb{C}^2$. The distinguished orthonormal basis

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

is called the *computational basis*. A general pure state is

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle\,, \qquad a, b \in \mathbb{C}, \qquad |a|^2 + |b|^2 = 1. \tag{1.6}$$

The coefficients a, b are called *amplitudes*; $|a|^2$ and $|b|^2$ are the probabilities of obtaining 0 or 1 in a computational-basis measurement.

The recipe just stated—outcome i occurs with probability $|\langle i|\psi\rangle|^2$, and the post-measurement state collapses to $|i\rangle$—is the *Born rule*, the measurement postulate of quantum mechanics. More generally, a projective measurement with orthogonal projectors $\{P_m\}$, $\sum_m P_m = \mathbb{I}$, returns outcome m with probability $p_m = \langle\psi|\,P_m\,|\psi\rangle$ and leaves the state $P_m\,|\psi\rangle\,/\sqrt{p_m}$; the probabilities sum to one by completeness.

Remark 1.13 (Global phase is unobservable). Measurement probabilities depend on the amplitudes only through moduli of the form $|\langle i|\psi\rangle|^2$. Replacing $|\psi\rangle$ by $\mathrm{e}^{\mathrm{i}\gamma}\,|\psi\rangle$ for a real *global phase* γ leaves every such probability unchanged, and more generally leaves $\langle\phi|A|\psi\rangle$-type quantities invariant when the same phase appears on both sides. Hence $\mathrm{e}^{\mathrm{i}\gamma}\,|\psi\rangle$ and $|\psi\rangle$ describe the same physical state: genuine states are points of the *projective space* $\mathbb{P}(\mathcal{H})$, i.e. rays rather than vectors. *Relative* phases between the terms of a superposition, by contrast, are physical and drive interference.

Proposition 1.14 (Bloch parametrization)**.** *Every single-qubit pure state can be written, up to an unobservable global phase, as*

$$|\psi\rangle = \cos\tfrac{\theta}{2}\;|0\rangle + \mathrm{e}^{\mathrm{i}\varphi}\sin\tfrac{\theta}{2}\;|1\rangle\,, \qquad \theta \in [0,\pi], \quad \varphi \in [0, 2\pi). \tag{1.7}$$

The pair (θ, φ) is the pair of spherical angles of a point on the unit sphere $S^2 \subset \mathbb{R}^3$, the Bloch sphere.

Proof. Start from (1.6) with $|a|^2 + |b|^2 = 1$. Write the amplitudes in polar form $a = r_a e^{i\alpha}$ and $b = r_b e^{i\beta}$ with $r_a, r_b \geq 0$. Since $r_a^2 + r_b^2 = 1$ we may set $r_a = \cos\frac{\theta}{2}$ and $r_b = \sin\frac{\theta}{2}$ for a unique $\theta \in [0, \pi]$ (both cosine and sine of $\theta/2$ are then nonnegative). Thus

$$|\psi\rangle = e^{i\alpha}\cos\tfrac{\theta}{2}\,|0\rangle + e^{i\beta}\sin\tfrac{\theta}{2}\,|1\rangle = e^{i\alpha}\Big(\cos\tfrac{\theta}{2}\,|0\rangle + e^{i(\beta-\alpha)}\sin\tfrac{\theta}{2}\,|1\rangle\Big).$$

The overall factor $e^{i\alpha}$ is a global phase, which by Theorem 1.13 we may discard. Setting $\varphi = \beta - \alpha \pmod{2\pi}$ gives (1.7). When $\theta \in \{0, \pi\}$ the state is $|0\rangle$ or $|1\rangle$ and φ is irrelevant (the poles of the sphere), consistent with the identification of states with points of S^2. □

Remark 1.15 (Reading the Bloch sphere). In this picture $|0\rangle$ and $|1\rangle$ sit at the north and south poles ($\theta = 0, \pi$), and *orthogonal* qubit states are *antipodal* points, not perpendicular ones—a state and its orthogonal complement are separated by angle $\theta = \pi$ on the sphere even though they are separated by 90° in $\mathbb{C}^2$. The equatorial states $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$ have $\theta = \pi/2$ and differ only in their relative phase φ.

Example 1.16 (Amplitudes and normalization)**.** The state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ has $|a|^2 = |b|^2 = \frac{1}{2}$, so a computational-basis measurement yields 0 or 1 with equal probability $\frac{1}{2}$; its Bloch angles are $\theta = \pi/2, \varphi = 0$. The state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ has the *same* measurement statistics in the computational basis but differs by the relative phase $\varphi = \pi$; indeed $\langle +|-\rangle = \frac{1}{2}(1 - 1) = 0$, so $|+\rangle$ and $|-\rangle$ are orthogonal and perfectly distinguishable by a suitable measurement.

1.2 Multi-qubit systems and tensor products

Definition 1.17 (Tensor-product state space)**.** If systems A and B have state spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, the composite system has state space the *tensor product* $\mathcal{H}_A \otimes \mathcal{H}_B$. If $\{|i\rangle_A\}$ and $\{|j\rangle_B\}$ are orthonormal bases, then $\{|i\rangle_A \otimes |j\rangle_B\}$ is an orthonormal basis of the product, with inner product

$$(\langle i| \otimes \langle j|)(|k\rangle \otimes |l\rangle) = \langle i|k\rangle\ \langle j|l\rangle\,.$$

Hence $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = (\dim \mathcal{H}_A)(\dim \mathcal{H}_B)$. For n qubits,

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}, \qquad \dim \mathcal{H} = 2^n.$$

We abbreviate $|i\rangle \otimes |j\rangle$ as $|i\rangle\,|j\rangle$ or $|ij\rangle$. The computational basis of n qubits is $\{|x\rangle : x \in \{0,1\}^n\}$, indexed by the 2^n bit strings. The exponential growth of the dimension—50 qubits already give a space of dimension $2^{50} > 10^{15}$—is one root of quantum computing's potential power: a classical description of a general state requires exponentially many amplitudes.

Remark 1.18 (The $|k\rangle$ notation for computational-basis states). Writing an integer k in a ket denotes the "one-hot" vector with a single 1 at index k and 0 elsewhere. Thus for two qubits

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} = |0\rangle_4\,, \quad |01\rangle = |1\rangle_4\,, \quad |10\rangle = |2\rangle_4\,, \quad |11\rangle = |3\rangle_4\,,$$

the four one-hot 4-vectors. Note $|00\rangle = |0\rangle_4$ is *not* the single-qubit $|0\rangle_2$; when the dimension is not clear from context we tag it, $|k\rangle_d$. In general $|x_1 \cdots x_n\rangle = |k\rangle$ with $k \in \{0, \ldots, 2^n - 1\}$ the integer whose binary representation is $x_1 \cdots x_n$; these one-hot states are exactly the computational basis, orthonormal by $\langle k|m\rangle = \delta_{km}$.

1.2.1 Product and entangled states

Definition 1.19 (Product and entangled states)**.** A state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is a *product state* (or *separable*) if it can be written as $|\psi\rangle = |\alpha\rangle_A \otimes |\beta\rangle_B$ for some $|\alpha\rangle_A \in \mathcal{H}_A$, $|\beta\rangle_B \in \mathcal{H}_B$. Otherwise $|\psi\rangle$ is *entangled*.

To see how a product state is assembled, it pays to compute a tensor product by hand, since the result is less obvious than stacking two vectors.

Example 1.20 (The tensor product of two qubits, componentwise)**.** Let $|\psi\rangle = \psi_0 |0\rangle + \psi_1 |1\rangle = \binom{\psi_0}{\psi_1}$ and $|\varphi\rangle = \binom{\varphi_0}{\varphi_1}$. Pair *each* entry of $|\psi\rangle$ with *all* of $|\varphi\rangle$:

$$|\psi\rangle \otimes |\varphi\rangle = \begin{pmatrix} \psi_0 \binom{\varphi_0}{\varphi_1} \\ \psi_1 \binom{\varphi_0}{\varphi_1} \end{pmatrix} = \begin{pmatrix} \psi_0\varphi_0 \\ \psi_0\varphi_1 \\ \psi_1\varphi_0 \\ \psi_1\varphi_1 \end{pmatrix}. \tag{1.8}$$

The four amplitudes are the four *products* of one amplitude from each qubit, in order; the outcome is a single 4-component vector describing both qubits at once. Its norm is preserved, $\sum_{i,j} |\psi_i\varphi_j|^2 = (|\psi_0|^2 + |\psi_1|^2)(|\varphi_0|^2 + |\varphi_1|^2) = 1$, so it is again a valid state. **Warning:** $|\psi\rangle \otimes |\varphi\rangle$ is *not* $|\psi\rangle$'s two entries stacked on top of $|\varphi\rangle$'s—it is the componentwise products of (1.8).

Example 1.21 (Product states built by gates)**.** Start both qubits in $|0\rangle$ and act with single-qubit gates (top qubit written on the left; "top-to-bottom" in a circuit is "left-to-right" in the ket):

$$\begin{aligned} |0\rangle \otimes |0\rangle &= |00\rangle = |0\rangle_4, \\ (X|0\rangle) \otimes |0\rangle &= |1\rangle \otimes |0\rangle = |10\rangle = |2\rangle_4, \\ |0\rangle \otimes (H|0\rangle) &= |0\rangle \otimes |+\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle + |01\rangle), \\ (H|0\rangle) \otimes (H|0\rangle) &= |+\rangle \otimes |+\rangle = \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = \tfrac{1}{2}\sum_{k=0}^{3} |k\rangle. \end{aligned}$$

The last line—all four amplitudes equal to $\frac{1}{2}$—is the uniform superposition, whose n-qubit version $H^{\otimes n}|0\cdots0\rangle$ is the standard opening move of many quantum algorithms (Theorem 1.51). Every state here is a product state; the interesting question is when a two-qubit state *cannot* be built this way—the subject of the next subsections.

Example 1.22 (A generic two-qubit state)**.** Two qubits live in $\mathbb{C}^4$ with basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. A general state is

$$|\psi\rangle = \sum_{i,j\in\{0,1\}} x_{ij} |ij\rangle, \qquad \sum_{i,j} |x_{ij}|^2 = 1.$$

For instance $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ is normalized ($4 \cdot \frac{1}{4} = 1$) and, as the reader can check, factors as $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, so it is a product state.

1.2.2 Detecting entanglement: the separability test

We can decide separability concretely, without guessing a factorization, by viewing the amplitudes as a matrix.

Proposition 1.23 (Separability test for two qubits)**.** *Write a two-qubit state as* $|\psi\rangle = \sum_{i,j\in\{0,1\}} x_{ij}\,|ij\rangle$ *and collect its amplitudes into the* 2×2 coefficient matrix $X = \left(\begin{smallmatrix} x_{00} & x_{01} \\ x_{10} & x_{11}\end{smallmatrix}\right)$*. Then* $|\psi\rangle$ *is a product state if and only if*

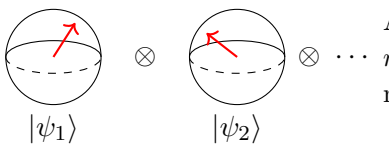
$$\det X \;=\; x_{00}x_{11} - x_{01}x_{10} \;=\; 0 \qquad (\textit{equivalently, } \operatorname{rank} X = 1).$$

Proof. If $|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle$ with $|\alpha\rangle = a_0\,|0\rangle + a_1\,|1\rangle$ and $|\beta\rangle = b_0\,|0\rangle + b_1\,|1\rangle$, then $x_{ij} = a_i b_j$, i.e. $X = \mathbf{a}\,\mathbf{b}^{\mathsf{T}}$ is an outer product; hence $\operatorname{rank} X \le 1$ and $\det X = a_0b_0\,a_1b_1 - a_0b_1\,a_1b_0 = 0$. Conversely, a nonzero 2×2 matrix with $\det X = 0$ has rank 1, so it factors as $X = \mathbf{a}\,\mathbf{b}^{\mathsf{T}}$ for some vectors $\mathbf{a}, \mathbf{b}$; reading off $x_{ij} = a_i b_j$ gives $|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle$ (the overall normalization may be split between the two factors). More generally, for a bipartite state the number of nonzero singular values of X is the *Schmidt rank*: rank 1 means product, rank > 1 means entangled. □

Example 1.24 (The test in action)**.**

- $|+\rangle\,|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$ has $X = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0\end{smallmatrix}\right)$, so $\det X = 0$: a product state (as its name already says).
- $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ has every $x_{ij} = \frac{1}{2}$, so $\det X = \frac{1}{4} - \frac{1}{4} = 0$: the product state $|+\rangle\,|+\rangle$ above.
- The Bell state $|\Phi^+\rangle$ has $X = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1\end{smallmatrix}\right)$, so $\det X = \frac{1}{2} \neq 0$: entangled—a one-line alternative to Theorem 1.27.

Remark 1.25 (Product states are the exception, not the rule)*.* A product state factors into independent single-qubit states, so it carries no correlation between the qubits: an n-qubit product state $|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$ is fixed by n points on the Bloch sphere, just $2n$ real parameters (see the figure). A general normalized n-qubit state has 2^n complex amplitudes—$2\cdot 2^n - 2$ real parameters up to global phase, exponentially many. For $n = 2$ that is a 4-parameter product family sitting inside a 6-parameter state space: the product states form a lower-dimensional (measure-zero) surface, so *almost every* multi-qubit state is entangled. Entanglement is the generic case; product states are the special, "classically describable" corner.



A product state:
n independent Bloch arrows,
no shared correlation.

1.2.3 The Bell state and entanglement

Definition 1.26 (Bell state)**.** The *Bell state* (or EPR pair) is the two-qubit state

$$|\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle). \tag{1.9}$$

Theorem 1.27 (The Bell state is entangled)**.** *There is no pair of single-qubit states* $|\alpha\rangle, |\beta\rangle \in \mathbb{C}^2$ *with* $|\Phi^+\rangle = |\alpha\rangle \otimes |\beta\rangle$*.*

Proof. Suppose for contradiction that such a factorization exists. Write $|\alpha\rangle = a_0\,|0\rangle + a_1\,|1\rangle$ and $|\beta\rangle = b_0\,|0\rangle + b_1\,|1\rangle$. Then

$$|\alpha\rangle \otimes |\beta\rangle = a_0b_0\,|00\rangle + a_0b_1\,|01\rangle + a_1b_0\,|10\rangle + a_1b_1\,|11\rangle\,.$$

Matching coefficients with $|\Phi^+\rangle = \frac{1}{\sqrt{2}}\,|00\rangle + 0\cdot|01\rangle + 0\cdot|10\rangle + \frac{1}{\sqrt{2}}\,|11\rangle$ requires

$$a_0b_0 = \tfrac{1}{\sqrt{2}}, \qquad a_0b_1 = 0, \qquad a_1b_0 = 0, \qquad a_1b_1 = \tfrac{1}{\sqrt{2}}.$$

From $a_0 b_0 = \frac{1}{\sqrt{2}} \neq 0$ we get $a_0 \neq 0$ and $b_0 \neq 0$. Then $a_0 b_1 = 0$ forces $b_1 = 0$, while $a_1 b_0 = 0$ forces $a_1 = 0$; hence $a_1 b_1 = 0 \neq \frac{1}{\sqrt{2}}$, a contradiction. Therefore no factorization exists and $|\Phi^+\rangle$ is entangled. □

Remark 1.28 (Physical meaning of entanglement). The correlations of $|\Phi^+\rangle$ have no separable description. Measuring the first qubit in the computational basis yields 0 or 1 with probability $\frac{1}{2}$ each; whatever the outcome, the post-measurement state of the second qubit is forced to the *same* value ($0 \mapsto |0\rangle$, $1 \mapsto |1\rangle$). The two outcomes are perfectly correlated even if the qubits are spatially far apart. This correlation, inexpressible as any classical mixture of independent single-qubit states, is the resource behind teleportation, superdense coding, and much of quantum advantage.

1.2.4 Nonlocality and the CHSH inequality

The correlations of a Bell pair are not merely strong; they are *stronger than any classical mechanism obeying local realism can produce*. This is the content of Bell's theorem, which we develop in the sharp form due to Clauser, Horne, Shimony, and Holt (CHSH). The remarkable point is that a philosophical assumption—that measurement outcomes are predetermined and unaffected by distant choices—makes a *quantitative*, testable prediction that quantum mechanics violates.

Two separated parties, Penelope and Odysseus, each receive one qubit of a shared pair. Each holds a measuring device with two settings. Penelope's two possible measurements return values $A_1, A_2 \in \{+1, -1\}$; Odysseus's return $B_1, B_2 \in \{+1, -1\}$. In each run each party freely picks one setting, so only one of A_1, A_2 and one of B_1, B_2 is actually observed. Consider the combination

$$S \;=\; A_1 B_1 - A_1 B_2 + A_2 B_1 + A_2 B_2 \;=\; A_1(B_1 - B_2) + A_2(B_1 + B_2). \tag{1.10}$$

Theorem 1.29 (CHSH inequality)**.** *Suppose the four quantities $A_1, A_2, B_1, B_2 \in \{+1, -1\}$ all possess definite values in every run (*realism*) and that Penelope's values do not depend on Odysseus's setting or vice versa (*locality*). Then the expectation of S obeys*

$$\big|\,\mathbb{E}[S]\,\big| \;=\; \big|\,\mathbb{E}[A_1 B_1] - \mathbb{E}[A_1 B_2] + \mathbb{E}[A_2 B_1] + \mathbb{E}[A_2 B_2]\,\big| \;\leq\; 2.$$

Proof. Under realism all four values coexist in each run, so (1.10) applies term by term. Since $B_1, B_2 \in \{+1, -1\}$, exactly one of $B_1 - B_2$ and $B_1 + B_2$ is 0 and the other is ± 2. Hence in every run one of the two products $A_1(B_1 - B_2)$, $A_2(B_1 + B_2)$ vanishes and the other equals ± 2, so $S \in \{+2, -2\}$ deterministically. The average of a quantity that is always ± 2 lies in $[-2, 2]$, and linearity of expectation splits $\mathbb{E}[S]$ into the four correlators. □

Quantum mechanics predicts a larger value. Measuring a qubit "along angle θ" in the xz-plane of the Bloch sphere means measuring the observable with eigenstates $|\theta\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$ (outcome $+1$) and its orthogonal partner (outcome -1); operationally one applies the rotation $R_y(-\theta)$ that carries $|\theta\rangle \mapsto |0\rangle$ and then measures in the computational basis.

Proposition 1.30 (Quantum correlator of a Bell pair)**.** *If Penelope measures her half of $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ along angle α and Odysseus measures his half along angle β (with the ± 1 labeling above), the correlation of their outcomes is*

$$\mathbb{E}[AB] \;=\; \cos(\alpha - \beta). \tag{1.11}$$

Proof. Applying $R_y(-\alpha)\otimes R_y(-\beta)$ to $|\Phi^+\rangle$ and using $R_y(-\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle - \sin\frac{\theta}{2}\,|1\rangle$, $R_y(-\theta)\,|1\rangle = \sin\frac{\theta}{2}\,|0\rangle + \cos\frac{\theta}{2}\,|1\rangle$, one expands and collects terms. The half-angle addition identities $\cos\frac{\alpha}{2}\cos\frac{\beta}{2} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2} = \cos\frac{\alpha-\beta}{2}$ and $\sin\frac{\alpha}{2}\cos\frac{\beta}{2} - \cos\frac{\alpha}{2}\sin\frac{\beta}{2} = \sin\frac{\alpha-\beta}{2}$ collapse the state to

$$\tfrac{1}{\sqrt{2}}\Big(\cos\tfrac{\alpha-\beta}{2}\;|00\rangle + \sin\tfrac{\alpha-\beta}{2}\;|01\rangle - \sin\tfrac{\alpha-\beta}{2}\;|10\rangle + \cos\tfrac{\alpha-\beta}{2}\;|11\rangle\Big).$$

The outcomes $00, 11$ (product $+1$) each occur with probability $\frac{1}{2}\cos^2\frac{\alpha-\beta}{2}$, and $01, 10$ (product -1) each with probability $\frac{1}{2}\sin^2\frac{\alpha-\beta}{2}$. Hence $\mathbb{E}[AB] = \cos^2\frac{\alpha-\beta}{2} - \sin^2\frac{\alpha-\beta}{2} = \cos(\alpha-\beta)$ by the double-angle identity. □

Corollary 1.31 (Tsirelson-point violation)**.** *With settings $\alpha_1 = 0$, $\alpha_2 = \frac{\pi}{2}$ for Penelope and $\beta_1 = \frac{\pi}{4}$, $\beta_2 = \frac{3\pi}{4}$ for Odysseus, the quantum prediction is*

$$\mathbb{E}[S] = \cos(\alpha_1-\beta_1) - \cos(\alpha_1-\beta_2) + \cos(\alpha_2-\beta_1) + \cos(\alpha_2-\beta_2) = 4\cos\tfrac{\pi}{4} = 2\sqrt{2} \approx 2.828,$$

which exceeds the classical bound 2 of Theorem 1.29.

Proof. Each of the four angle differences is $\mp\frac{\pi}{4}$: explicitly $\alpha_1-\beta_1 = -\frac{\pi}{4}$, $\alpha_1-\beta_2 = -\frac{3\pi}{4}$, $\alpha_2-\beta_1 = \frac{\pi}{4}$, $\alpha_2-\beta_2 = -\frac{\pi}{4}$. Since $\cos(\pm\frac{\pi}{4}) = \frac{1}{\sqrt{2}}$ and $\cos(\frac{3\pi}{4}) = -\frac{1}{\sqrt{2}}$,

$$\mathbb{E}[S] = \tfrac{1}{\sqrt{2}} - (-\tfrac{1}{\sqrt{2}}) + \tfrac{1}{\sqrt{2}} + \tfrac{1}{\sqrt{2}} = \tfrac{4}{\sqrt{2}} = 2\sqrt{2}.$$ □

Remark 1.32 (What the violation forces us to give up). The value $2\sqrt{2}$ is the maximum attainable by any quantum state and measurements (Tsirelson's bound), and it has been confirmed experimentally—most famously in the loophole-free tests recognized by the 2022 Nobel Prize in Physics. Because Theorem 1.29 follows from realism plus locality alone, its experimental violation means at least one of those assumptions fails: the pre-measurement values A_1, A_2, B_1, B_2 cannot all simultaneously exist as local "hidden variables." The correlations of $|\Phi^+\rangle$ are genuinely nonclassical, yet—as we show next in Section 1.5—they still cannot be used to signal, so no conflict with relativity arises.

1.3 Gates, universal gate sets, and circuits

The postulates of quantum mechanics say that the state of an isolated system evolves in time by a linear, inner-product-preserving map. On a finite-dimensional Hilbert space such a map is exactly a unitary operator, and in quantum computing we call it a gate.

Definition 1.33 (Unitary operator / quantum gate)**.** Let $\mathcal{H} \cong \mathbb{C}^d$ be a finite-dimensional Hilbert space. A linear operator $U : \mathcal{H} \to \mathcal{H}$ is *unitary* if

$$U^\dagger U = U U^\dagger = \mathbb{I},$$

where $U^\dagger$ is the conjugate transpose (adjoint) of U. A *quantum gate* on n qubits is a unitary operator on $\mathcal{H} = (\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^{2^n}$.

Equivalently, U is unitary iff its columns (or rows) form an orthonormal basis of $\mathbb{C}^d$, iff $U^{-1} = U^\dagger$. In particular every gate is invertible, so quantum computation is reversible.

Theorem 1.34 (Gates preserve inner products and norm)**.** *Let U be a linear operator on $\mathcal{H}$. The following are equivalent.*

(i) *U is unitary, i.e. $U^\dagger U = \mathbb{I}$.*

(ii) *U preserves inner products: $\langle U\psi|U\phi\rangle = \langle\psi|\phi\rangle$ for all $|\psi\rangle, |\phi\rangle \in \mathcal{H}$.*

(iii) *U preserves norm: $\|U\psi\| = \|\psi\|$ for all $|\psi\rangle \in \mathcal{H}$.*

Proof. (i) $\Rightarrow$ (ii). Using $\langle U\psi|U\phi\rangle = \langle\psi| U^\dagger U |\phi\rangle$ and $U^\dagger U = \mathbb{I}$, we get $\langle U\psi|U\phi\rangle = \langle\psi| \mathbb{I} |\phi\rangle = \langle\psi|\phi\rangle$.

(ii) $\Rightarrow$ (iii). Take $|\phi\rangle = |\psi\rangle$: then $\|U\psi\|^2 = \langle U\psi|U\psi\rangle = \langle\psi|\psi\rangle = \|\psi\|^2$, and taking nonnegative square roots gives $\|U\psi\| = \|\psi\|$.

(iii) $\Rightarrow$ (i). Norm preservation gives $\langle\psi| U^\dagger U |\psi\rangle = \langle\psi| \mathbb{I} |\psi\rangle$ for all $|\psi\rangle$, so the self-adjoint operator $M := U^\dagger U - \mathbb{I}$ satisfies $\langle\psi| M |\psi\rangle = 0$ for every $|\psi\rangle$. For a self-adjoint operator the quadratic form determines the operator (via the polarization identity, or by choosing $|\psi\rangle$ to be its eigenvectors), hence $M = 0$, i.e. $U^\dagger U = \mathbb{I}$. On a finite-dimensional space a left inverse is a two-sided inverse, so also $UU^\dagger = \mathbb{I}$ and U is unitary. □

Physically, Theorem 1.34 says gates map normalized states to normalized states and never change the "angle" between two states; this inner-product rigidity is exactly what will forbid cloning in Section 1.6.

1.3.1 Single-qubit gates

Definition 1.35 (Pauli, Hadamard, and phase gates)**.** The *Pauli gates* and identity on $\mathbb{C}^2$ are

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The *Hadamard gate* and the *phase gates* S and T are

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{e}^{\mathrm{i}\pi/4} \end{pmatrix}.$$

Each of these is unitary: one checks directly that $X^\dagger X = Y^\dagger Y = Z^\dagger Z = I$, that $H^\dagger H = I$, and that S, T are diagonal with unit-modulus entries. In the computational basis,

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \qquad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle,$$

so X is a *bit flip* (the quantum NOT gate) and Z is a *phase flip*: it fixes $|0\rangle$ and multiplies $|1\rangle$ by -1. The Hadamard gate creates superpositions,

$$H|0\rangle = |+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = |-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

and is self-inverse, $H^2 = I$. The phase gates leave $|0\rangle$ alone and rotate the phase of $|1\rangle$: S multiplies it by i and T by $\mathrm{e}^{\mathrm{i}\pi/4}$.

The four operators $\{I, X, Y, Z\}$ form a basis of the (complex) vector space of 2×2 matrices, so every single-qubit gate is a complex linear combination of Pauli matrices. Geometrically, a single-qubit pure state corresponds to a point on the Bloch sphere; the Pauli gates act as rotations by π about the x, y, z axes (up to a global phase). These half-turns are the special cases of a one-parameter family, which we now work out from the Pauli algebra. For a unit axis $\hat{n} \in \mathbb{R}^3$ write $\hat{n} \cdot \vec{\sigma} := n_x X + n_y Y + n_z Z$.

Proposition 1.36 (Rotation gates and their Pauli generators)**.** *The Pauli product identity* $(\vec{a}\cdot\vec{\sigma})(\vec{b}\cdot\vec{\sigma}) = (\vec{a}\cdot\vec{b})\,I + \mathrm{i}(\vec{a}\times\vec{b})\cdot\vec{\sigma}$ *gives* $(\hat{n}\cdot\vec{\sigma})^2 = I$*, so the* rotation gate *about* $\hat{n}$ *by angle* θ *has the closed form*

$$R_{\hat{n}}(\theta) := \exp\Big(-\tfrac{\mathrm{i}\theta}{2}\,\hat{n}\cdot\vec{\sigma}\Big) = \cos\tfrac{\theta}{2}\,I - \mathrm{i}\sin\tfrac{\theta}{2}\,(\hat{n}\cdot\vec{\sigma}).$$

With $\hat{n} = \hat{x}, \hat{y}, \hat{z}$ *this yields*

$$R_x(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\mathrm{i}\sin\frac{\theta}{2} \\ -\mathrm{i}\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \quad R_z(\theta) = \begin{pmatrix} \mathrm{e}^{-\mathrm{i}\theta/2} & 0 \\ 0 & \mathrm{e}^{\mathrm{i}\theta/2} \end{pmatrix},$$

the gates used for state preparation in (1.13)*. The Pauli gates are the half-turns:* $X = \mathrm{i}R_x(\pi)$*,* $Y = \mathrm{i}R_y(\pi)$*,* $Z = \mathrm{i}R_z(\pi)$*.*

Proof. Because $(\hat{n}\cdot\vec{\sigma})^2 = I$, in the series $\exp(-\mathrm{i}\alpha\,\hat{n}\cdot\vec{\sigma}) = \sum_{k\ge 0}\frac{(-\mathrm{i}\alpha)^k}{k!}(\hat{n}\cdot\vec{\sigma})^k$ (with $\alpha = \theta/2$) the even powers reduce to I and the odd powers to $\hat{n}\cdot\vec{\sigma}$, so they resum to $\cos\alpha\,I - \mathrm{i}\sin\alpha\,(\hat{n}\cdot\vec{\sigma})$. Substituting X, Y, Z gives the three matrices; e.g. $R_z(\theta) = \cos\frac{\theta}{2}I - \mathrm{i}\sin\frac{\theta}{2}Z = \mathrm{diag}(\mathrm{e}^{-\mathrm{i}\theta/2}, \mathrm{e}^{\mathrm{i}\theta/2})$. Finally $R_z(\pi) = \mathrm{diag}(\mathrm{e}^{-\mathrm{i}\pi/2}, \mathrm{e}^{\mathrm{i}\pi/2}) = -\mathrm{i}Z$, so $Z = \mathrm{i}R_z(\pi)$, and likewise for X, Y. □

Proposition 1.37 ($R_{\hat{n}}(\theta)$ rotates the Bloch vector)**.** *Write a single-qubit state as* $\rho = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$*. Conjugation by* $R_{\hat{n}}(\theta)$ *sends the Bloch vector to its rotation by angle* θ *about* $\hat{n}$*:*

$$R_{\hat{n}}(\theta)\,\rho\,R_{\hat{n}}(\theta)^\dagger = \tfrac{1}{2}(I + \vec{r}\,'\cdot\vec{\sigma}), \quad \vec{r}\,' = \cos\theta\,\vec{r} + \sin\theta\,(\hat{n}\times\vec{r}) + (1-\cos\theta)(\hat{n}\cdot\vec{r})\,\hat{n}.$$

Proof. As $R\,I\,R^\dagger = I$, only the $\vec{r}\cdot\vec{\sigma}$ part moves, so it suffices to conjugate $M := \vec{m}\cdot\vec{\sigma}$ for a generic $\vec{m}$. Put $c = \cos\frac{\theta}{2}$, $s = \sin\frac{\theta}{2}$, $N := \hat{n}\cdot\vec{\sigma}$, so $R = cI - \mathrm{i}sN$ and $R^\dagger = cI + \mathrm{i}sN$; then

$$RMR^\dagger = c^2 M + \mathrm{i}cs\,[M, N] + s^2\,NMN.$$

The product identity gives $[M, N] = 2\mathrm{i}(\vec{m}\times\hat{n})\cdot\vec{\sigma}$ and, after a short computation, $NMN = \big(2(\hat{n}\cdot\vec{m})\,\hat{n} - \vec{m}\big)\cdot\vec{\sigma}$. Hence $RMR^\dagger = \vec{m}\,'\cdot\vec{\sigma}$ with

$$\vec{m}\,' = c^2\vec{m} + 2cs\,(\hat{n}\times\vec{m}) + s^2\big(2(\hat{n}\cdot\vec{m})\,\hat{n} - \vec{m}\big),$$

and the double-angle identities $c^2 - s^2 = \cos\theta$, $2cs = \sin\theta$, $2s^2 = 1 - \cos\theta$ collapse this to Rodrigues' rotation formula. Setting $\vec{m} = \vec{r}$ proves the claim. □

Remark 1.38 (Half-angle and the $SU(2) \to SO(3)$ double cover). The state carries the half-angle $\theta/2$ while the Bloch vector turns by the full θ. Thus $R_{\hat{n}}(2\pi) = \cos\pi\,I - \mathrm{i}\sin\pi\,(\hat{n}\cdot\vec{\sigma}) = -I$: a 2π Bloch rotation returns the vector but leaves the state with a global phase -1, and only a 4π turn restores I. Equivalently, $R_{\hat{n}}(\theta) \mapsto \mathcal{R}(\hat{n}, \theta)$ is a two-to-one homomorphism $SU(2) \to SO(3)$, with $\pm R_{\hat{n}}(\theta)$ inducing the same (observable) rotation—which is why the Bloch picture faithfully represents single-qubit gates as ordinary 3D rotations.

Lemma 1.39 (Hadamard conjugation)**.** $HXH = Z$ *and* $HZH = X$*.*

Proof. Recall $H = H^\dagger$ and $H^2 = I$. Compute HXH by matrix multiplication. First

$$XH = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

and then

$$HXH = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z.$$

For the second identity, conjugate the first by H on both sides and use $H^2 = I$: from $HXH = Z$ we get $H(HXH)H = HZH$, i.e. $(H^2)X(H^2) = HZH$, so $X = HZH$. □

Theorem 1.39 is the precise statement that H is a "bridge" between the bit-flip and phase-flip pictures: a phase flip in the computational basis looks like a bit flip in the Hadamard $(|+\rangle, |-\rangle)$ basis, and vice versa. Two further identities are recorded for reference.

Proposition 1.40. *$Y = \mathrm{i}XZ$ and $S = T^2$.*

Proof. Direct computation:

$$XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \mathrm{i}XZ = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} = Y.$$

For the phase gates, T is diagonal so $T^2 = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}\pi/4})^2 = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}\pi/2}) = \mathrm{diag}(1, \mathrm{i}) = S$. □

Remark 1.41 (T is non-Clifford). The *Clifford group* is generated by $\{H, S, \mathrm{CNOT}\}$ (equivalently, the unitaries that map the Pauli group to itself under conjugation). The identities above show X, Y, Z and $S = T^2$ lie in this group. The T gate does *not*: conjugating a Pauli by T produces a superposition of Paulis rather than another Pauli. This matters because the Gottesman–Knill theorem says Clifford circuits are efficiently classically simulable; the T gate is precisely the ingredient that lifts a gate set to universality for quantum computation.

1.3.2 Multi-qubit gates

To act on more than one qubit we take tensor products. A gate A on qubit 0 and B on qubit 1 acts on the joint state as $A \otimes B$, with $(A \otimes B)(|u\rangle \otimes |v\rangle) = A|u\rangle \otimes B|v\rangle$. The most important genuinely two-qubit gate is the controlled-NOT.

Definition 1.42 (CNOT and Toffoli)**.** The *controlled-NOT* gate CNOT acts on two qubits (control a, target b) by

$$\mathrm{CNOT}\,|a, b\rangle = |a,\, a \oplus b\rangle, \qquad a, b \in \{0, 1\},$$

where $\oplus$ is addition modulo 2. In the ordered computational basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ its matrix is the 4×4 permutation unitary

$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The *Toffoli* (or CCNOT) gate acts on three qubits by $\mathrm{CCNOT}\,|a, b, c\rangle = |a,\, b,\, (a \wedge b) \oplus c\rangle$, flipping the target c exactly when both controls a, b equal 1.

Since CNOT permutes the four basis vectors it is a permutation matrix, hence unitary; the same holds for the Toffoli gate, which swaps $|110\rangle \leftrightarrow |111\rangle$. Both are their own inverse. The Toffoli gate is universal for classical reversible computation: with ancillas it can implement AND, NOT, and fan-out, so any Boolean circuit can be made reversible.

Remark 1.43 (Why XOR keeps oracles reversible). A classical oracle for $f : \{0,1\}^n \to \{0,1\}$ is realized quantumly as $U_f\,|x\rangle\,|b\rangle = |x\rangle\,|b \oplus f(x)\rangle$. Writing the answer with $\oplus$ rather than overwriting b with $f(x)$ is what makes U_f a permutation of basis states, hence unitary and reversible: applying U_f twice returns $|b \oplus f(x) \oplus f(x)\rangle = |b\rangle$, so $U_f^2 = \mathbb{I}$. A plain overwrite $b \mapsto f(x)$ is generally not injective and cannot be unitary. The CNOT rule $b \mapsto a \oplus b$ is the single-bit instance of this construction.

1.3.3 Reading quantum circuits

A *quantum circuit* is a diagram in which each horizontal wire is a qubit and gates are drawn on the wires. Three conventions govern how to read one.

(1) **Time flows left to right.** The input state sits at the far left; a gate drawn further left is applied earlier.

(2) **Controls.** On a controlled gate, a filled dot • marks a control that activates when the qubit is $|1\rangle$; an open dot ∘ marks a 0-control, which activates when the qubit is $|0\rangle$.

(3) **Matrix order is reversed.** If the circuit applies U_1 first, then U_2, the overall unitary is the *reverse-order* product $U = U_2\,U_1$, because operators act on the ket to their right. For a length-m sequence, $U = U_m U_{m-1} \cdots U_1$.

A controlled-U gate with control a can be written compactly as $c\text{-}U\ |a\rangle\,|b\rangle = |a\rangle\ U^a\,|b\rangle$, so U is applied to the target raised to the power $a \in \{0,1\}$ (nothing happens when $a = 0$). The SWAP gate exchanges two qubits, $\mathrm{SWAP}\,|a,b\rangle = |b,a\rangle$; it can be built from three CNOTs.

Definition 1.44 (Circuit depth)**.** Group the gates of a circuit into *layers*, where a layer is a maximal set of gates acting on disjoint qubits that can be executed simultaneously. The *depth* of the circuit is the number of layers, equivalently the largest number of gates on any single input-to-output path through the diagram.

Gates acting on disjoint sets of qubits commute as tensor factors and run in parallel, so depth (not total gate count) measures wall-clock runtime, and keeping depth low is central to running on noisy hardware.

Example 1.45 (A small circuit's unitary as a reverse-order product)**.** Consider one qubit with, from left to right, an H gate, then an S gate, then another H gate. The overall unitary is the reverse-order product

$$U = H\,S\,H = \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & \mathrm{i}\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1+\mathrm{i} & 1-\mathrm{i}\\ 1-\mathrm{i} & 1+\mathrm{i}\end{pmatrix}.$$

One checks $U^\dagger U = I$, confirming the composite is again a gate.

1.3.4 Building a Bell state

We now assemble single- and two-qubit gates into a circuit that creates entanglement. Prepare two qubits in $|00\rangle$, apply H to qubit 0, then apply CNOT with qubit 0 as control and qubit 1 as target.

Theorem 1.46 (Bell-state preparation)**.** *With qubit* 0 *the control and qubit* 1 *the target,*

$$\mathrm{CNOT}\,(H \otimes I)\ |00\rangle = \tfrac{1}{\sqrt{2}}(\,|00\rangle + |11\rangle\,) = |\Phi^+\rangle\,.$$

Proof. First apply $H \otimes I$. Since $H\,|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $I\,|0\rangle = |0\rangle$,

$$(H \otimes I)\,|00\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \tfrac{1}{\sqrt{2}}(\,|00\rangle + |10\rangle\,).$$

Now apply CNOT term by term using $\mathrm{CNOT}\,|a,b\rangle = |a, a \oplus b\rangle$. On $|00\rangle$ the control is 0, so the state is unchanged. On $|10\rangle$ the control is 1, so the target flips: $|10\rangle \mapsto |11\rangle$. Hence

$$\mathrm{CNOT}\ \tfrac{1}{\sqrt{2}}(\,|00\rangle + |10\rangle\,) = \tfrac{1}{\sqrt{2}}(\,|00\rangle + |11\rangle\,) = |\Phi^+\rangle\,. \qquad \square$$

By Theorem 1.27 the state $|\Phi^+\rangle$ is entangled, so this two-gate circuit turns a product input into a maximally entangled Bell pair.

1.3.5 Universal gate sets and compilation

Definition 1.47 (Universal gate set)**.** A finite set $\mathcal{G}$ of gates is *universal* if the unitaries generated by applying gates from $\mathcal{G}$ to arbitrary subsets of qubits are dense in the group of all n-qubit unitaries (up to global phase): for every target unitary U and every $\varepsilon > 0$ there is a finite circuit V over $\mathcal{G}$ with $\|U - V\| \le \varepsilon$.

Two facts underlie this definition. First, any n-qubit unitary can be written *exactly* as a product of single-qubit and two-qubit gates, $U = U_m U_{m-1} \cdots U_1$, though m may be exponential in n. Second, a fixed *finite* set already suffices to approximate every gate: the set $\mathcal{G} = \{H, T, \mathrm{CNOT}\}$ is universal. We settle for approximation because a discrete set cannot hit a continuum of unitaries exactly, but it can come within any ε.

Remark 1.48 (Solovay–Kitaev cost)*.* The Solovay–Kitaev theorem quantifies the price of this discretization: any single-qubit gate can be approximated to error ε by a circuit of $\mathcal{O}(\operatorname{polylog}(1/\varepsilon))$ gates from a universal set, and an m-gate circuit can be recompiled into the finite set using only $\mathcal{O}(m \operatorname{polylog}(m/\varepsilon))$ gates, found by an efficient classical algorithm. The overhead is thus only polylogarithmic in the inverse precision, which is why compiling ideal circuits to a fixed hardware gate set is practical.

Remark 1.49 (Why Solovay–Kitaev works (sketch))*.* The polylogarithmic bound comes from a recursion that sharpens an approximation by representing its *error* with ever-smaller gates. Assume $\mathcal{G}$ is closed under inverses. *(i) Base net.* Density of $\langle \mathcal{G} \rangle$ together with compactness of $SU(2)$ gives a fixed accuracy ε_0 reachable by *every* gate using words of bounded length. *(ii) Commutator lemma.* Any Δ near the identity, $\|\Delta - \mathbb{I}\| \le \delta$, factors as a group commutator $\Delta = VWV^\dagger W^\dagger$ up to $\mathcal{O}(\delta^{3/2})$, where V, W are only $\mathcal{O}(\sqrt{\delta})$ from the identity—writing $V = \mathrm{e}^{-\mathrm{i}A}$, $W = \mathrm{e}^{-\mathrm{i}B}$ with $A, B = \mathcal{O}(\sqrt{\delta})$, one has $VWV^\dagger W^\dagger = \mathrm{e}^{-[A,B]+\mathcal{O}(\delta^{3/2})}$ and $[A, B] = \mathcal{O}(\delta)$. *(iii) Recursion.* Given a level-$(n-1)$ approximation U_{n-1} of accuracy ε_{n-1}, decompose the residual $\Delta = U\,U_{n-1}^\dagger$ by the lemma, approximate V, W at level $n-1$, and glue the commutator onto U_{n-1}:

$$\varepsilon_n = \mathcal{O}(\varepsilon_{n-1}^{3/2}), \qquad \ell_n = 5\,\ell_{n-1} + \mathcal{O}(1).$$

The exponent $3/2$ makes the error fall doubly exponentially, so $n = \mathcal{O}(\log\log(1/\varepsilon))$ levels suffice; unrolling $\ell_n = 5^n \ell_0$ then gives $\ell(\varepsilon) = \mathcal{O}\big((\log(1/\varepsilon))^{\log 5/\log(3/2)}\big)$, with exponent ≈ 3.97. A single-qubit rotation is just the generic target $U \in SU(2)$, so the bound applies to it directly (Dawson & Nielsen, *The Solovay–Kitaev algorithm*, Quantum Inf. Comput. **6** (2006) 81).

Running an abstract circuit on a real device requires *compilation*: (i) rewriting logical gates such as H, T, CNOT into the machine's native gate set; and (ii) inserting SWAP gates to route qubits, because hardware qubits are not all pairwise connected. Throughout one minimizes gate count and depth to suppress noise and maximize fidelity.

1.3.6 Worked example: a reversible binary adder

The computational basis of n qubits is labeled by the integers $0, 1, \ldots, 2^n - 1$: we identify the string $|k_1 \cdots k_n\rangle$ with its base-two value $k = \sum_{l=1}^{n} k_l\, 2^{\,n-l}$, so that $|10\rangle = |2\rangle$, $|11\rangle = |3\rangle$, and a three-qubit register ranges over $|0\rangle, \ldots, |7\rangle$. A circuit that permutes these basis states therefore *computes an integer-valued function*, and the final computational-basis measurement returns the answer as an ordinary number. We make this concrete by building a circuit that adds two-bit numbers, which at the same time demonstrates the claim following Theorem 1.42 that the Toffoli gate is universal for classical reversible computation. Here $\oplus$ is addition modulo 2 (the XOR, computed by CNOT) and $\wedge$ is the logical AND (computed by a Toffoli onto a $|0\rangle$ ancilla).

To add A_1A_0 and B_1B_0 we work column by column, exactly as in grade-school addition but in base two. The rightmost column produces a sum bit S_0 and a carry C_1; the next column adds A_1, B_1, and the incoming carry C_1 to produce S_1 and a final carry C_2; and the result is the three-bit number $C_2S_1S_0$ (for example, $2+3$ is $10+11=101=5$). In the familiar columnar layout, with each carry written above the column it feeds:

$$\begin{array}{cccc} & C_2 & C_1 & \\ & & A_1 & A_0 \\ + & & B_1 & B_0 \\ \hline & C_2 & S_1 & S_0 \end{array} \qquad \text{e.g.} \qquad \begin{array}{cccc} & 1 & 0 & \\ & & 1 & 0 \\ + & & 1 & 1 \\ \hline & 1 & 0 & 1 \end{array}$$

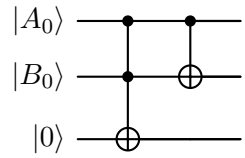
Half adder. The rightmost column has no incoming carry, so

$$S_0 = A_0 \oplus B_0, \qquad C_1 = A_0 \wedge B_0,$$

as the truth table confirms:

A_0	B_0	C_1	S_0
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

The sum bit is a single CNOT (control A_0, target B_0), and the carry is a Toffoli (controls A_0, B_0) writing $A_0 \wedge B_0$ onto a fresh $|0\rangle$ ancilla. Because the CNOT overwrites B_0, we compute the carry *first*:



Reading top to bottom, the output wires hold A_0, the sum $S_0 = A_0 \oplus B_0$, and the carry $C_1 = A_0 \wedge B_0$.

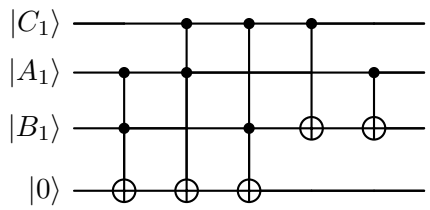
Full adder. The next column sums three bits, C_1, A_1, B_1:

$$S_1 = C_1 \oplus A_1 \oplus B_1, \qquad C_2 = (A_1 \wedge B_1) \oplus (C_1 \wedge A_1) \oplus (C_1 \wedge B_1).$$

These rules are confirmed by the truth table over the three input bits C_1, A_1, B_1:

C_1	A_1	B_1	C_2	S_1
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

The carry C_2 is accumulated onto a $|0\rangle$ ancilla by three Toffolis, one for each product; the sum S_1 is then two CNOTs targeting the B_1 wire (controls C_1 and A_1). As before we compute the carry first, since forming S_1 overwrites B_1:



The output wires now hold C_1, A_1, the sum S_1, and the carry C_2 on the ancilla.

The two-bit adder. Cascading the two blocks—the half adder's carry output C_1 feeds the full adder—gives the complete circuit for $A_1A_0 + B_1B_0$: the half adder produces S_0 and C_1 from A_0, B_0, and the full adder consumes C_1, A_1, B_1 to produce S_1 and C_2. Measuring the three wires that hold S_0, S_1, C_2 returns the binary digits of the sum. For instance, to compute $3 + 3$ one sets $A_1A_0 = B_1B_0 = 11$ by applying X to the four input qubits (initialized to $|0\rangle$) and reads the output $110_2 = 6$. The complete circuit is drawn in Figure 1.1.

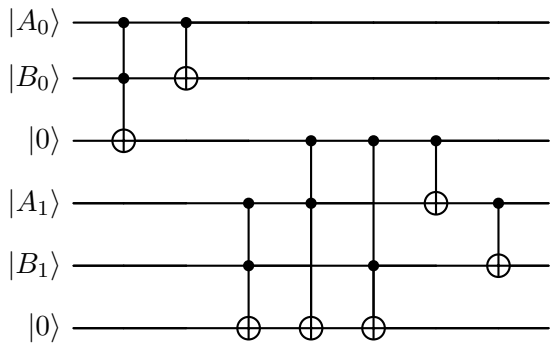


Figure 1.1: The two-bit ripple adder $A_1A_0 + B_1B_0 \mapsto C_2S_1S_0$: the half adder (first two gates) computes S_0, C_1, and the cascaded full adder computes S_1, C_2 (three Toffolis for the carry, then two CNOTs for the sum). Output wires, top to bottom: A_0, S_0, C_1, $C_1 \oplus A_1$, S_1, C_2; the sum is $C_2S_1S_0$. After Brody, Fig. 7.12.

Running the adder. On hardware every qubit starts in $|0\rangle$, so an input bit is set to 1 by a leading X (NOT) gate placed before the adder. Figure 1.2 shows three cases: $0 + 0$ (no X gates), $1 + 1 = 01 + 01$ (X on A_0 and B_0), and $1 + 2 = 01 + 10$ (X on A_0 and B_1); measuring S_0, S_1, C_2 then reads $000_2 = 0$, $010_2 = 2$, and $011_2 = 3$ respectively.

Remark 1.50 (Reversibility, ancillas, and garbage). Every gate above is a permutation of basis states, so the adder is one large permutation unitary—reversible, exactly as Theorem 1.34 requires of any gate. Reversibility is bought with bookkeeping: results are written with $\oplus$ onto fresh $|0\rangle$ ancillas rather than overwriting inputs, so the circuit also carries along the original inputs and some intermediate "garbage." Bennett's trick—copy the final answer out with CNOTs, then run the circuit backward—uncomputes the garbage and resets the ancillas to $|0\rangle$, which is what lets an adder serve as a clean subroutine inside a larger coherent algorithm. Finally, the many Toffoli gates make the adder demanding on real hardware, where gate errors accumulate with circuit size; this is an early hint of why controlling noise becomes central as circuits grow.

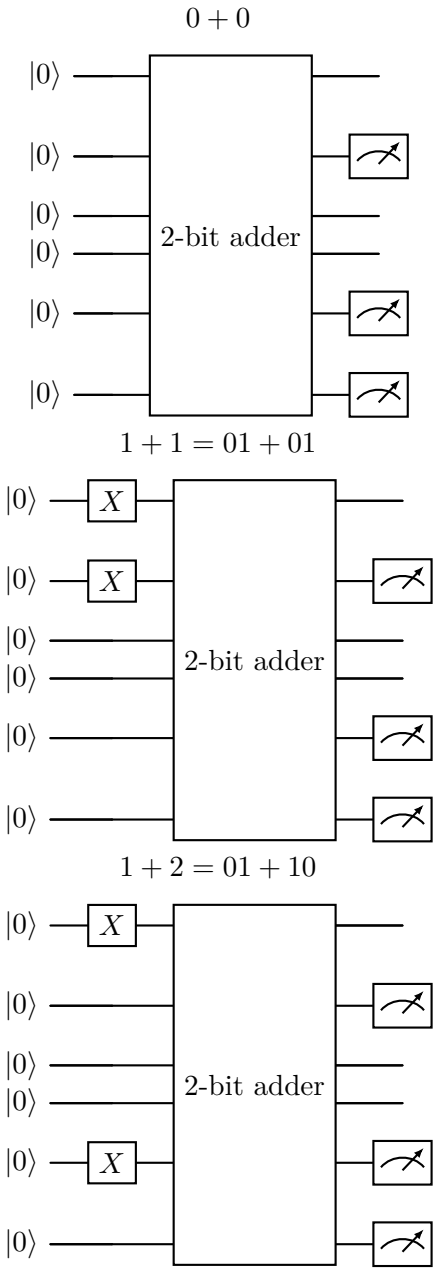


Figure 1.2: The adder of Figure 1.1 configured for three inputs (the box hides its internal gates): leading X gates load the operands and the three meters read S_0, S_1, C_2. Top to bottom the wires are $A_0, B_0, |0\rangle, A_1, B_1, |0\rangle$. After Brody, Figs. 7.13, 7.15, and 7.17.

1.4 State preparation and measurement

Every quantum algorithm is bracketed by two operations that are not gates: it must begin by *preparing* an input state and end by *measuring* an output. The gate model of Section 1.3 describes the unitary evolution in between; this section treats the two endpoints systematically, building on the computational-basis Born rule already stated in Section 1.1. Both endpoints have a cost, and that cost often dominates: a beautiful poly(n)-gate circuit is useless if loading its input takes $\mathcal{O}(2^n)$ gates, and an expectation value is only as good as the number of measurement repetitions one can afford.

1.4.1 Preparing states from $|0\cdots0\rangle$

By convention every register starts in the all-zeros state $|0\rangle^{\otimes n} = |0\cdots0\rangle$. Preparing a target state means finding a circuit that maps $|0\cdots0\rangle$ to it.

Proposition 1.51 (Basis states and the uniform superposition)**.** *Let $x = x_1\cdots x_n \in \{0,1\}^n$.*

(a) *The computational basis state $|x\rangle$ is prepared from $|0\cdots0\rangle$ by applying X to each qubit j with $x_j = 1$; that is, $\left(\bigotimes_j X^{x_j}\right)|0\cdots0\rangle = |x\rangle$, using at most n gates.*

(b) *The uniform superposition over all 2^n strings is prepared by one Hadamard per qubit,*

$$H^{\otimes n}|0\cdots0\rangle = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle\,, \tag{1.12}$$

using exactly n gates.

Proof. (a) $X^{x_j}|0\rangle = |x_j\rangle$ acts qubitwise, and the tensor product of the single-qubit results is $|x_1\rangle\cdots|x_n\rangle = |x\rangle$. (b) On one qubit $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$. Taking the tensor power and expanding the product of n sums,

$$H^{\otimes n}|0\cdots0\rangle = \bigotimes_{j=1}^{n}\tfrac{1}{\sqrt{2}}(|0\rangle+|1\rangle) = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}|x\rangle\,,$$

since choosing $|0\rangle$ or $|1\rangle$ from each factor enumerates all bit strings x exactly once. □

The uniform superposition (1.12) is the workhorse input of the query algorithms of Part I: a single depth-one layer of Hadamards places the register in an equal-weight combination of all 2^n inputs, on which an oracle can then act "in parallel."

1.4.2 Single-qubit state synthesis

A general single-qubit target is, up to global phase, the Bloch state $|\psi(\theta,\varphi)\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$ of Theorem 1.14. It is reached from $|0\rangle$ by a rotation about the y axis followed by one about the z axis, where the rotation gates are

$$R_y(\theta) = e^{-i\theta Y/2} = \begin{pmatrix}\cos\frac{\theta}{2} & -\sin\frac{\theta}{2}\\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2}\end{pmatrix}, \qquad R_z(\varphi) = e^{-i\varphi Z/2} = \begin{pmatrix}e^{-i\varphi/2} & 0\\ 0 & e^{i\varphi/2}\end{pmatrix}. \tag{1.13}$$

Proposition 1.52 (Single-qubit preparation)**.** *For any $\theta\in[0,\pi]$ and $\varphi\in[0,2\pi)$,*

$$R_z(\varphi)\,R_y(\theta)\,|0\rangle = e^{-i\varphi/2}\Big(\cos\tfrac{\theta}{2}|0\rangle + e^{i\varphi}\sin\tfrac{\theta}{2}|1\rangle\Big),$$

so up to the global phase $e^{-i\varphi/2}$ the circuit prepares $|\psi(\theta,\varphi)\rangle$ with two gates.

Proof. From (1.13), $R_y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle$. Applying the diagonal $R_z(\varphi)$ multiplies the $|0\rangle$ amplitude by $\mathrm{e}^{-\mathrm{i}\varphi/2}$ and the $|1\rangle$ amplitude by $\mathrm{e}^{\mathrm{i}\varphi/2}$:

$$R_z(\varphi)R_y(\theta)\,|0\rangle = \mathrm{e}^{-\mathrm{i}\varphi/2}\cos\tfrac{\theta}{2}\,|0\rangle + \mathrm{e}^{\mathrm{i}\varphi/2}\sin\tfrac{\theta}{2}\,|1\rangle = \mathrm{e}^{-\mathrm{i}\varphi/2}\Big(\cos\tfrac{\theta}{2}\,|0\rangle + \mathrm{e}^{\mathrm{i}\varphi}\sin\tfrac{\theta}{2}\,|1\rangle\Big).$$

The overall factor $\mathrm{e}^{-\mathrm{i}\varphi/2}$ is an unobservable global phase (Theorem 1.13). □

Example 1.53 (Preparing $|+\mathrm{i}\rangle$)**.** The state $|+\mathrm{i}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + \mathrm{i}\,|1\rangle)$ has Bloch angles $\theta = \pi/2$ (equal moduli) and $\varphi = \pi/2$ (relative phase $\mathrm{i} = \mathrm{e}^{\mathrm{i}\pi/2}$). By Theorem 1.52 it is prepared by $R_z(\pi/2)R_y(\pi/2)\,|0\rangle$: the $R_y(\pi/2)$ produces $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and $R_z(\pi/2)$ then multiplies the $|1\rangle$ amplitude by $\mathrm{e}^{\mathrm{i}\pi/2} = \mathrm{i}$ relative to the $|0\rangle$ amplitude, giving $\frac{1}{\sqrt{2}}(|0\rangle + \mathrm{i}\,|1\rangle)$ up to global phase. Note $S\,|+\rangle = |+\mathrm{i}\rangle$ as well, so a single S gate achieves the same result here.

1.4.3 General amplitude encoding

Preparing an arbitrary n-qubit state $|\psi\rangle = \sum_{x\in\{0,1\}^n} a_x\,|x\rangle$ from a classical list of the 2^n amplitudes $\{a_x\}$ is far more expensive than the special cases above.

Theorem 1.54 (Cost of exact amplitude encoding)**.** *Every n-qubit pure state $|\psi\rangle = \sum_x a_x\,|x\rangle$ can be prepared exactly from $|0\cdots0\rangle$ by a circuit of single-qubit rotations and* CNOT *gates, and $\mathcal{O}(2^n)$ gates suffice. Conversely, a counting argument shows that $\Omega(2^n)$ gates are necessary for generic states, so this scaling is optimal up to constants.*

We state the result at this level and do not reproduce the construction; the standard recipe of Möttönen, Vartiainen, Bergholm, and Salomaa, and the equivalent Shende–Bullock–Markov decomposition, prepare $|\psi\rangle$ by a cascade of uniformly controlled R_y and R_z rotations, one layer per qubit, encoding the amplitudes' moduli and phases in the rotation angles. The upper bound matches an information-theoretic lower bound: specifying a generic unit vector in $\mathbb{C}^{2^n}$ requires $\sim 2^n$ real parameters, and each gate supplies only $\mathcal{O}(1)$, so no subexponential circuit can reach all states.

Remark 1.55 (Efficient special cases)*.* The exponential cost of Theorem 1.54 is a worst case over *all* states; the states that arise in practice are usually structured, and structure buys efficiency.

- *Product states* $|\psi_1\rangle \otimes \cdots \otimes |\psi_n\rangle$ cost $\mathcal{O}(n)$ gates: prepare each qubit independently by Theorem 1.52.
- *Efficiently computable amplitudes.* If the a_x are given by a formula one can evaluate reversibly (for example a discretized smooth density, via the Grover–Rudolph procedure that recursively rotates on conditional probabilities), the state can be prepared in $\mathrm{poly}(n)$ gates.
- *Sparse states* with only s nonzero amplitudes can be prepared in $\mathcal{O}(s\,n)$ gates, independent of the full dimension 2^n.

When the amplitudes come from external data rather than a formula, one instead posits a black-box *state-preparation oracle*; this input model, and its role in quantum numerical linear algebra, is developed in Chapter 6.

1.4.4 Projective measurement

The computational-basis Born rule of Section 1.1 is the special case of a more general measurement postulate, which we now state in full.

Definition 1.56 (Projective (von Neumann) measurement)**.** An *observable* is a Hermitian operator $O = O^\dagger$ on $\mathcal{H}$. Writing its spectral decomposition as $O = \sum_m m\, P_m$, where the distinct real eigenvalues m index mutually orthogonal projectors P_m (so $P_m P_{m'} = \delta_{mm'} P_m$, $P_m^\dagger = P_m$) that resolve the identity, $\sum_m P_m = \mathbb{I}$, the associated *projective measurement* of a state $|\psi\rangle$ returns outcome m with probability

$$p_m = \langle\psi|\, P_m\, |\psi\rangle = \|P_m\,|\psi\rangle\|^2\,, \tag{1.14}$$

and leaves the post-measurement state

$$\frac{P_m\,|\psi\rangle}{\sqrt{p_m}} \qquad \text{(defined whenever } p_m > 0\text{)}. \tag{1.15}$$

The probabilities are nonnegative and sum to one: $\sum_m p_m = \langle\psi|\,(\sum_m P_m)\,|\psi\rangle = \langle\psi|\psi\rangle = 1$ by completeness. Choosing O diagonal in the computational basis, so that $P_m = |m\rangle\langle m|$ is rank one, recovers exactly the Born rule of Theorem 1.12: $p_m = \langle\psi|\,|m\rangle\langle m|\,|\psi\rangle = |\langle m|\psi\rangle|^2$ and the state collapses to $|m\rangle$. Theorem 1.56 generalizes this in two ways—the projectors may have rank > 1 (a *degenerate* outcome, as when we measure only some of the qubits), and the eigenbasis need not be the computational one.

Proposition 1.57 (Expectation value)**.** *The expected value of the observable $O = \sum_m m\, P_m$ in the state $|\psi\rangle$ is*

$$\langle O\rangle \;=\; \sum_m m\, p_m \;=\; \sum_m m\;\langle\psi|\, P_m\,|\psi\rangle \;=\; \langle\psi|\, O\,|\psi\rangle\,. \tag{1.16}$$

Proof. By definition of the mean of the outcome random variable, $\langle O\rangle = \sum_m m\, p_m$. Substituting (1.14) and pulling the sum inside the inner product, $\sum_m m\;\langle\psi|\, P_m\,|\psi\rangle = \langle\psi|\,(\sum_m m\, P_m)\,|\psi\rangle = \langle\psi|\, O\,|\psi\rangle$. □

Example 1.58 (Measuring Z and X on $|0\rangle$ and $|+\rangle$)**.** Take $O = Z = (+1)\,|0\rangle\langle 0| + (-1)\,|1\rangle\langle 1|$, with outcomes ± 1 and projectors $P_{+1} = |0\rangle\langle 0|$, $P_{-1} = |1\rangle\langle 1|$. On $|0\rangle$ the outcome is $+1$ with probability $|\langle 0|0\rangle|^2 = 1$, so $\langle Z\rangle = 1$. On $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ each outcome has probability $\frac{1}{2}$, so $\langle Z\rangle = \frac{1}{2}(+1) + \frac{1}{2}(-1) = 0$; equivalently $\langle +|\, Z\,|+\rangle = 0$. Now take $O = X = (+1)\,|+\rangle\langle +| + (-1)\,|-\rangle\langle -|$. On $|0\rangle$, $|\langle +|0\rangle|^2 = |\langle -|0\rangle|^2 = \frac{1}{2}$, so $\langle X\rangle = 0$; but on $|+\rangle$ the outcome is $+1$ with certainty and $\langle X\rangle = 1$. The two observables have complementary sharpness: $|+\rangle$ is an eigenstate of X (definite outcome) but maximally uncertain for Z, and vice versa for $|0\rangle$.

1.4.5 Measuring in an arbitrary basis

Real hardware measures only in the computational basis. To measure in some other orthonormal basis $\{|\phi_m\rangle\}$ one first rotates that basis onto the computational one with a unitary and then measures the qubits.

Proposition 1.59 (Basis change before measurement)**.** *Let $\{|\phi_m\rangle\}_m$ be an orthonormal basis and define the unitary $V = \sum_m |m\rangle\langle\phi_m|$, which satisfies $V\,|\phi_m\rangle = |m\rangle$. Measuring $|\psi\rangle$ in the basis $\{|\phi_m\rangle\}$—i.e. with projectors $|\phi_m\rangle\langle\phi_m|$—produces outcome m with the same probability as first applying V and then measuring $V\,|\psi\rangle$ in the computational basis:*

$$|\langle\phi_m|\psi\rangle|^2 = |\langle m|V|\psi\rangle|^2\,.$$

Proof. That V is unitary follows from $V^\dagger V = \sum_{m,m'} |\phi_m\rangle\,\langle m|m'\rangle\,\langle\phi_{m'}| = \sum_m |\phi_m\rangle\langle\phi_m| = \mathbb{I}$, the last equality being completeness of the orthonormal basis (Theorem 1.10). From the definition $\langle m|V|\psi\rangle = \sum_{m'} \langle m|m'\rangle\,\langle\phi_{m'}|\psi\rangle = \langle\phi_m|\psi\rangle$, and squaring the modulus gives the claim. □

Example 1.60 (Measuring in the X basis via H)**.** The Hadamard gate satisfies $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$, so $V = H = |0\rangle\langle 0|\,H + \dots$ is exactly the basis-change unitary $|0\rangle\langle +| + |1\rangle\langle -|$ for the X basis $\{|+\rangle, |-\rangle\}$. Hence a measurement of X is implemented by applying H and measuring in the computational basis, reading $|+\rangle \mapsto 0$ and $|-\rangle \mapsto 1$. For instance, on $|\psi\rangle = \cos\frac{\pi}{8}|0\rangle + \sin\frac{\pi}{8}|1\rangle$ the X-basis outcome $+$ occurs with probability $|\langle +|\psi\rangle|^2 = \frac{1}{2}\left(\cos\frac{\pi}{8} + \sin\frac{\pi}{8}\right)^2 = \cos^2\frac{\pi}{8} = \frac{2+\sqrt{2}}{4} \approx 0.854$, so $\langle X\rangle = 2\cos^2\frac{\pi}{8} - 1 = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$, in agreement with $\langle\psi|\,X\,|\psi\rangle = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$.

1.4.6 Generalized measurements and POVMs

Projective measurement is not the most general thing one can do: coupling the system to an ancilla, evolving unitarily, and then projectively measuring the ancilla implements a broader class of measurements on the system alone. These are captured by positive operator-valued measures.

Definition 1.61 (POVM)**.** A *positive operator-valued measure* (POVM) is a finite collection of operators $\{E_m\}$ on $\mathcal{H}$, called *effects*, satisfying

$$E_m \succeq 0 \quad \text{for all } m, \qquad \sum_m E_m = \mathbb{I}. \tag{1.17}$$

Applied to a state $|\psi\rangle$, it returns outcome m with probability $p_m = \langle\psi|\,E_m\,|\psi\rangle$ (more generally $p_m = \mathrm{Tr}[E_m\rho]$ for a density operator ρ).

Positivity of each E_m makes $p_m \geq 0$, and completeness $\sum_m E_m = \mathbb{I}$ makes $\sum_m p_m = 1$, so the p_m are a genuine probability distribution. A projective measurement is the special case $E_m = P_m$ where the effects are orthogonal projectors; a general POVM drops both orthogonality and idempotence, allowing more outcomes than the dimension of $\mathcal{H}$ and specifying no post-measurement state on its own.

Theorem 1.62 (Naimark dilation, statement)**.** *Every POVM $\{E_m\}$ on $\mathcal{H}$ arises from a projective measurement on a larger space: there exist an ancilla space, a fixed ancilla state, an isometry (or unitary coupling) into $\mathcal{H}\otimes\mathcal{H}_{\mathrm{anc}}$, and orthogonal projectors $\{\tilde{P}_m\}$ on the enlarged space such that $\langle\psi|\,E_m\,|\psi\rangle = \langle\Psi|\,\tilde{P}_m\,|\Psi\rangle$ for the dilated state $|\Psi\rangle$, for every $|\psi\rangle$ and every m.*

We record this result at the statement level; a constructive proof (writing $E_m = M_m^\dagger M_m$ and building the isometry from the M_m) is a standard exercise once quantum channels are in hand. The picture it justifies is the useful one: *measurement is itself a quantum channel*, an irreversible map on density operators, and every generalized measurement is "really" a projective measurement that has been given room to breathe on an ancilla. The channel viewpoint, and the distance measures that quantify how well two states can be told apart by any POVM, are developed in Chapter 2.

1.4.7 From expectations to estimates

Measurement yields a random outcome, so a single shot cannot reveal an expectation value; one repeats the prepare-measure cycle and averages. If each shot returns ± 1 with $\langle O\rangle$ the mean, the empirical average of M independent shots has standard deviation $\mathcal{O}(1/\sqrt{M})$, so estimating $\langle O\rangle$ to additive error ε requires $M = \mathcal{O}(1/\varepsilon^2)$ repetitions. This quadratic dependence on precision—the *shot-noise* or *standard quantum limit*—is a recurring cost throughout the course; we defer its precise statistical treatment (variance, Chernoff/Hoeffding bounds, and the quadratic amplitude-estimation speedup to $\mathcal{O}(1/\varepsilon)$) to Chapter 2, and the circuit that reads off $\langle\psi|U|\psi\rangle$ as such an expectation—the Hadamard test—to Chapter 4.

1.5 Quantum teleportation and superdense coding

We now assemble the pieces of the previous sections—a shared Bell pair, the H and CNOT gates, and measurement—into two protocols that show what entanglement is *for*. Teleportation transmits one unknown qubit using one shared Bell pair and two classical bits; superdense coding does the reverse, sending two classical bits with one qubit and one shared Bell pair.

1.5.1 The teleportation protocol

Penelope holds an unknown qubit $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$ that she wishes to deliver to Odysseus, who is far away. She may not know a, b and (by no-cloning, Section 1.6) cannot copy the qubit. The one resource they share is a Bell pair $|\Phi^+\rangle$ prepared earlier, when they were together: Penelope keeps one half, Odysseus the other.

Label the three qubits 1 (Penelope's message), 2 (Penelope's half of the pair), and 3 (Odysseus's half). The joint state is

$$|\psi\rangle_1 \otimes |\Phi^+\rangle_{23} = \big(a\,|0\rangle + b\,|1\rangle\big)_1 \otimes \tfrac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big)_{23}. \tag{1.18}$$

Theorem 1.63 (Teleportation)**.** *Penelope applies* CNOT *from qubit* 1 *(control) to qubit* 2 *(target), then H to qubit* 1*, then measures qubits* 1 *and* 2 *in the computational basis, obtaining two bits (m_1, m_2) each equally likely. Conditioned on (m_1, m_2), Odysseus's qubit* 3 *is*

$$X^{m_2} Z^{m_1} |\psi\rangle \quad \textit{up to nothing else,}$$

so after Odysseus applies the correction $Z^{m_1} X^{m_2}$ his qubit is exactly $|\psi\rangle$.

Proof. Expand (1.18) over qubits $2, 3$:

$$\tfrac{1}{\sqrt{2}}\big(a\,|0\rangle_1\,(|00\rangle + |11\rangle)_{23} + b\,|1\rangle_1\,(|00\rangle + |11\rangle)_{23}\big).$$

Apply $\mathrm{CNOT}_{1\to 2}$, which flips qubit 2 exactly when qubit 1 is $|1\rangle$:

$$\tfrac{1}{\sqrt{2}}\big(a\,|0\rangle_1\,(|00\rangle + |11\rangle)_{23} + b\,|1\rangle_1\,(|10\rangle + |01\rangle)_{23}\big).$$

Now apply H to qubit 1, using $H\,|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $H\,|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$\tfrac{1}{2}\Big(a(|0\rangle + |1\rangle)_1(|00\rangle + |11\rangle)_{23} + b(|0\rangle - |1\rangle)_1(|10\rangle + |01\rangle)_{23}\Big).$$

Regroup by the four values of qubits $1, 2$ (Penelope's registers), factoring out qubit 3:

$$\begin{aligned} \tfrac{1}{2}\Big(\ & |00\rangle_{12}\ (a\,|0\rangle + b\,|1\rangle)_3 + |01\rangle_{12}\ (a\,|1\rangle + b\,|0\rangle)_3 \\ & + |10\rangle_{12}\ (a\,|0\rangle - b\,|1\rangle)_3 + |11\rangle_{12}\ (a\,|1\rangle - b\,|0\rangle)_3\ \Big). \end{aligned} \tag{1.19}$$

The four terms have equal weight $\frac{1}{2}$, so each outcome (m_1, m_2) occurs with probability $\frac{1}{4}$. Reading off qubit 3 in each branch:

$$\begin{aligned} (0,0):&\ \ |\psi\rangle, \qquad (0,1):\ \ X\,|\psi\rangle, \\ (1,0):&\ \ Z\,|\psi\rangle, \quad (1,1):\ \ XZ\,|\psi\rangle \ \text{(up to a global phase)}, \end{aligned}$$

where we used $X\,|\psi\rangle = a\,|1\rangle + b\,|0\rangle$ and $Z\,|\psi\rangle = a\,|0\rangle - b\,|1\rangle$. In every branch Odysseus's state is $X^{m_2} Z^{m_1} |\psi\rangle$; since X, Z are their own inverses, applying $Z^{m_1} X^{m_2}$ (i.e. X if $m_2 = 1$, then Z if $m_1 = 1$) returns qubit 3 to $|\psi\rangle$. □

Remark 1.64 (Why teleportation needs, but does not beat, classical communication). Three features of the protocol are worth dwelling on.

- *Classical bits are indispensable.* Before Odysseus learns (m_1, m_2) his qubit is, on average over Penelope's random outcomes, the maximally mixed state $\frac{1}{2}\mathbb{I}$ (average of the four correction images of $|\psi\rangle$), carrying no information about $|\psi\rangle$. Only the two classical bits, sent over an ordinary channel, tell Odysseus which correction to apply.
- *No faster-than-light signaling.* Because Odysseus's marginal is $\frac{1}{2}\mathbb{I}$ regardless of what Penelope does, nothing Penelope performs on qubits $1, 2$ changes the statistics Odysseus can observe locally. The state is completed only once the classical message arrives, which travels no faster than light. This is the *no-signaling* principle, and it is exactly why the nonlocal correlations of Section 1.2.4 coexist peacefully with relativity.
- *The source is destroyed.* Penelope's measurement collapses qubit 1 to a computational-basis state, erasing a, b from the original. Teleportation *moves* the state rather than copying it—consistent with no-cloning (Theorem 1.67), which would otherwise be violated by an end-to-end map $|\psi\rangle \mapsto |\psi\rangle \otimes |\psi\rangle$.
- *Information, not matter.* What crosses the distance is quantum *information*—the amplitudes a, b—reconstructed on Odysseus's *pre-existing* qubit from two classical bits together with the shared entanglement; no physical qubit or mass is transported. (Brody, *The Joy of Quantum Computing*, Ch. 4, tells this as Penelope teleporting a qubit across the Aegean to the voyaging Odysseus, using an entangled pair they created before parting.)

1.5.2 Superdense coding

Running the same Bell resource "in reverse" lets Penelope send *two* classical bits by physically transmitting a *single* qubit.

Proposition 1.65 (Superdense coding)**.** *Penelope and Odysseus share* $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ *(qubit* 1 *Penelope's, qubit* 2 *Odysseus's). To send two bits* $(x, y) \in \{0, 1\}^2$*, Penelope applies* $Z^x X^y$ *to her qubit and sends it to Odysseus. Odysseus then applies* $\mathrm{CNOT}_{1\to 2}$ *followed by* H *on qubit* 1 *and measures both qubits, recovering* (x, y) *with certainty.*

Proof. The four local operations map $|\Phi^+\rangle$ to the four orthonormal Bell states:

$$(0,0)\colon\ \mathbb{I}\,|\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \qquad (0,1)\colon\ X_1\,|\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(|10\rangle + |01\rangle),$$
$$(1,0)\colon\ Z_1\,|\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \quad (1,1)\colon\ Z_1X_1\,|\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(|10\rangle - |01\rangle).$$

These four states are mutually orthogonal, hence perfectly distinguishable. The map $H_1\,\mathrm{CNOT}_{1\to 2}$ is exactly the inverse of the Bell-preparation circuit of Theorem 1.46, so it sends these four states to $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ respectively (one verifies each case as in that proof). Measuring in the computational basis therefore returns (x, y) deterministically. □

Remark 1.66 (The resource accounting). Teleportation and superdense coding are duals: one qubit of communication plus one shared entangled pair is worth two classical bits, and vice versa. Neither protocol lets entanglement alone transmit information—each requires an actual channel (two classical bits, or one qubit)—which is again the no-signaling principle at work.

1.6 The no-cloning theorem and reversibility

Classically, copying a bit is trivial. Quantum mechanically it is impossible to copy an *unknown* state, a fact discovered by Wootters, Zurek, and Dieks in 1982. A copier would be a gate U that,

given the state to copy and a fixed blank register $|s\rangle$, outputs two copies.

Theorem 1.67 (No-Cloning Theorem)**.** *There is no unitary U on $\mathcal{H}\otimes\mathcal{H}$ and blank state $|s\rangle\in\mathcal{H}$ such that*

$$U(\,|\psi\rangle\otimes|s\rangle\,)=|\psi\rangle\otimes|\psi\rangle \qquad \text{for every normalized } |\psi\rangle\in\mathcal{H},$$

unless $\dim\mathcal{H}=1$. *More precisely, any such U can clone two states $|x_1\rangle, |x_2\rangle$ simultaneously only if they are equal or orthogonal.*

Proof. Suppose such U and $|s\rangle$ exist. Apply the cloning equation to two normalized states $|x_1\rangle$ and $|x_2\rangle$:

$$U\,|x_1\rangle\,|s\rangle=|x_1\rangle\,|x_1\rangle\,, \qquad U\,|x_2\rangle\,|s\rangle=|x_2\rangle\,|x_2\rangle\,.$$

Take the inner product of these two equations. On the left, because U is unitary it preserves inner products (Theorem 1.34), and $\langle s|s\rangle=1$, so

$$(\,\langle x_1|\,\langle s|\,)U^\dagger U(\,|x_2\rangle\,|s\rangle\,)=\langle x_1|x_2\rangle\;\langle s|s\rangle=\langle x_1|x_2\rangle\,.$$

On the right, the inner product of $|x_1\rangle\,|x_1\rangle$ with $|x_2\rangle\,|x_2\rangle$ factors across the tensor product,

$$(\,\langle x_1|\,\langle x_1|\,)(\,|x_2\rangle\,|x_2\rangle\,)=\langle x_1|x_2\rangle\;\langle x_1|x_2\rangle=\langle x_1|x_2\rangle^2\,.$$

Equating the two sides, $\langle x_1|x_2\rangle=\langle x_1|x_2\rangle^2$. Writing $z=\langle x_1|x_2\rangle$, this reads $z=z^2$, i.e. $z(z-1)=0$, so $z\in\{0,1\}$. Thus $|x_1\rangle$ and $|x_2\rangle$ are either identical ($z=1$) or orthogonal ($z=0$). A universal cloner would need this for *all* pairs, but a space of dimension ≥ 2 contains states with inner product strictly between 0 and 1 (e.g. $|0\rangle$ and $|+\rangle$, with $\langle 0|+\rangle=\frac{1}{\sqrt{2}}$). This contradiction proves no universal cloner exists; only mutually orthogonal states can be cloned. □

The essence of the proof is that cloning $|\psi\rangle\mapsto|\psi\rangle\otimes|\psi\rangle$ is a nonlinear (quadratic) map of $|\psi\rangle$, whereas every quantum gate is linear; linearity forces the inner-product constraint $z=z^2$, which no continuum of states can satisfy.

Remark 1.68 (Consequences)*.*

- *No broadcasting.* Since an unknown state cannot be duplicated, it cannot be broadcast to several parties.
- CNOT *does not clone superpositions.* For $|x\rangle=a\,|0\rangle+b\,|1\rangle$,

$$\mathrm{CNOT}\,|x\rangle\,|0\rangle=a\,|00\rangle+b\,|11\rangle\neq|x\rangle\,|x\rangle=|x\rangle\otimes(a\,|0\rangle+b\,|1\rangle),$$

  because the right side contains cross terms $ab\,|01\rangle\,, ab\,|10\rangle$ absent on the left. CNOT copies only computational-basis information.
- *BB84 security.* Because an eavesdropper can neither copy the transmitted qubits nor measure them without disturbing non-orthogonal states, interception in the BB84 protocol necessarily introduces detectable errors, giving quantum key distribution its security.

Remark 1.69 (No-cloning protects no-signaling)*.* Cloning is not merely inconvenient—if it were possible it would let entanglement signal faster than light, in flat contradiction with relativity. Suppose Penelope and Odysseus share $|\Phi^+\rangle$ and Penelope encodes one bit into her *choice of measurement basis*: to send 0 she measures in the computational (z) basis, to send 1 in the Hadamard (x) basis. Her measurement leaves Odysseus's qubit in $\{|0\rangle\,,|1\rangle\}$ or in $\{|+\rangle\,,|-\rangle\}$ accordingly, but as Theorem 1.75 shows Odysseus's reduced state is $\frac{1}{2}\mathbb{I}$ either way, so a single measurement cannot reveal Penelope's basis. *If* Odysseus could clone his qubit into many copies, he could estimate its state statistically: an ensemble that is 50/50 in *both* the z and x bases (the $\frac{1}{2}\mathbb{I}$ case) versus one

sharply peaked in whichever basis Penelope used would be distinguishable, revealing Penelope's bit instantaneously over any distance. The no-cloning theorem forbids exactly this state estimation, and thereby closes the would-be superluminal channel—the same no-signaling conclusion reached constructively by teleportation in Section 1.5.

The same inner-product argument, run in reverse, forbids the deletion of an unknown state.

Theorem 1.70 (No-Deleting Theorem)**.** *There is no unitary U and fixed target $|\Sigma\rangle$ with*

$$U\big(\,|\psi\rangle \otimes |\psi\rangle\,\big) = |\psi\rangle \otimes |\Sigma\rangle \qquad \text{for all normalized } |\psi\rangle\,,$$

on a Hilbert space of dimension ≥ 2.

Proof. Assume such U exists and apply it to two states $|x_1\rangle\,, |x_2\rangle$. Taking inner products, using unitarity on the left and factoring on the right,

$$\langle x_1|x_2\rangle^2 = \langle x_1|x_2\rangle\ \langle \Sigma|\Sigma\rangle = \langle x_1|x_2\rangle\,,$$

since $\langle \Sigma|\Sigma\rangle = 1$. Again $z = \langle x_1|x_2\rangle$ satisfies $z^2 = z$, so $z \in \{0, 1\}$, which fails for non-orthogonal states. Hence no universal deleting operation exists. □

Reversibility. Because every gate U is unitary, it has an inverse $U^{-1} = U^\dagger$ that is itself a gate: any computation implemented by a sequence $U_m \cdots U_1$ can be undone by applying $U_1^\dagger \cdots U_m^\dagger$. Quantum evolution is therefore *logically reversible*, in sharp contrast to classical irreversible gates such as AND (which discards a bit and dissipates energy, by Landauer's principle). This reversibility is why classical subroutines must be embedded in the XOR-oracle form $|x\rangle\,|b\rangle \mapsto |x\rangle\,|b \oplus f(x)\rangle$ of the previous section, and why *uncomputation*—running ancilla-producing gates backward to reset scratch registers to $|0\rangle$—is a routine part of quantum algorithm design. The one intrinsically irreversible step is measurement, which collapses the state and destroys the pre-measurement superposition.

1.7 Density operators and mixed states

So far every state has been a single unit vector $|\psi\rangle$, called a *pure state*. Often, though, we know only that a system is in one of several states $|\psi_i\rangle$ with classical probabilities p_i (an *ensemble* $\{p_i, |\psi_i\rangle\}$), or we hold only part of a larger entangled system. Both situations are captured by the density-operator formalism.

Definition 1.71 (Density operator)**.** The *density operator* of the ensemble $\{p_i, |\psi_i\rangle\}$ (with $p_i \geq 0$, $\sum_i p_i = 1$) is

$$\rho \;=\; \sum_i p_i\ |\psi_i\rangle\langle\psi_i|\,. \tag{1.20}$$

A pure state $|\psi\rangle$ corresponds to the rank-one $\rho = |\psi\rangle\langle\psi|$.

For any observable measurement with projectors $\{P_m\}$ the outcome probabilities and the expectation of $M = \sum_m m\, P_m$ are

$$p(m) = \mathrm{Tr}[P_m \rho], \qquad \mathbb{E}[M] = \mathrm{Tr}[M\rho], \tag{1.21}$$

which for a pure state reduce to the Born rule of Section 1.1 since $\mathrm{Tr}[P_m\,|\psi\rangle\langle\psi|] = \langle\psi|\,P_m\,|\psi\rangle$.

Proposition 1.72 (Characterization of density operators)**.** *An operator ρ is the density operator of some ensemble if and only if*

(i) $\operatorname{Tr}\rho = 1$, *and*

(ii) $\rho \succeq 0$ *(Hermitian and positive semidefinite).*

Proof. ($\Rightarrow$) From (1.20), $\rho^\dagger = \sum_i p_i (|\psi_i\rangle\langle\psi_i|)^\dagger = \rho$ is Hermitian; for any $|\phi\rangle$, $\langle\phi|\,\rho\,|\phi\rangle = \sum_i p_i\,|\langle\phi|\psi_i\rangle|^2 \geq 0$, so $\rho \succeq 0$; and $\operatorname{Tr}\rho = \sum_i p_i \operatorname{Tr}[|\psi_i\rangle\langle\psi_i|] = \sum_i p_i = 1$. ($\Leftarrow$) A positive semidefinite Hermitian operator has a spectral decomposition $\rho = \sum_i \lambda_i\,|\lambda_i\rangle\langle\lambda_i|$ with $\lambda_i \geq 0$ and, by $\operatorname{Tr}\rho = 1$, $\sum_i \lambda_i = 1$; this exhibits ρ as the density operator of the ensemble $\{\lambda_i, |\lambda_i\rangle\}$. □

Definition 1.73 (Purity; pure vs. mixed)**.** The *purity* of ρ is $\operatorname{Tr}[\rho^2] \in (0, 1]$. A state is *pure* iff $\rho^2 = \rho$ (equivalently $\operatorname{Tr}[\rho^2] = 1$, equivalently ρ is a rank-one projector); otherwise $\operatorname{Tr}[\rho^2] < 1$ and ρ is called *mixed*.

For example, the maximally mixed single-qubit state $\rho = \frac{1}{2}\mathbb{I}$ has $\operatorname{Tr}[\rho^2] = \frac{1}{2}$: it describes a qubit that is $|0\rangle$ or $|1\rangle$ with probability $\frac{1}{2}$ each, and no measurement can distinguish it from the even mixture of $|+\rangle$ and $|-\rangle$—the ensemble decomposition is not unique, only ρ is physical.

1.7.1 The partial trace and reduced states

To describe one part of a composite system we "trace out" the rest.

Definition 1.74 (Partial trace)**.** Let $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ with orthonormal basis $\{|j\rangle_B\}$ of $\mathcal{H}_B$. The *partial trace over* B is the linear map sending an operator M on $\mathcal{H}$ to the operator on $\mathcal{H}_A$

$$\operatorname{Tr}_B[M] \;=\; \sum_j (\mathbb{I}_A \otimes \langle j|_B)\, M\, (\mathbb{I}_A \otimes |j\rangle_B). \tag{1.22}$$

On product operators it acts as $\operatorname{Tr}_B[A \otimes B] = A\,\operatorname{Tr}[B]$. The *reduced density operator* of subsystem A is $\rho_A = \operatorname{Tr}_B[\rho]$; it yields the correct statistics (1.21) for every measurement performed on A alone.

Example 1.75 (Reduced state of a Bell pair)**.** Take the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ with density operator

$$\rho = |\Phi^+\rangle\langle\Phi^+| = \tfrac{1}{2}\big(\,|00\rangle + |11\rangle\,\big)\big(\,\langle 00| + \langle 11|\,\big) = \tfrac{1}{2}\big(\,|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|\,\big).$$

Trace out the second qubit. Using $\operatorname{Tr}_B[|ab\rangle\langle cd|] = |a\rangle\langle c|\;\langle d|b\rangle$ (so only terms with $b = d$ survive),

$$\rho_A = \operatorname{Tr}_B[\rho] = \tfrac{1}{2}\big(\,|0\rangle\langle 0|\,\underbrace{\langle 0|0\rangle}_{1} + |0\rangle\langle 1|\,\underbrace{\langle 1|0\rangle}_{0} + |1\rangle\langle 0|\,\underbrace{\langle 0|1\rangle}_{0} + |1\rangle\langle 1|\,\underbrace{\langle 1|1\rangle}_{1}\,\big) = \tfrac{1}{2}\big(\,|0\rangle\langle 0| + |1\rangle\langle 1|\,\big) = \tfrac{1}{2}\mathbb{I}.$$

So the reduced state of one half of a Bell pair is *maximally mixed*, with purity $\operatorname{Tr}[\rho_A^2] = \frac{1}{2} < 1$, even though the global state is pure ($\operatorname{Tr}[\rho^2] = 1$). The missing information lives entirely in the correlations between the two qubits: this loss of purity upon restriction to a subsystem is a hallmark of entanglement.

Unitary evolution acts on density operators by conjugation, $\rho \mapsto U\rho U^\dagger$, which for a pure state reproduces $|\psi\rangle \mapsto U\,|\psi\rangle$. Not every physical operation is unitary, however; the general (completely positive, trace-preserving) quantum channels—and the distance measures that compare density operators—are the subject of the next lecture. For most of this course we work with pure states and unitary operations, but the density operator remains the right tool whenever we discuss subsystems, noise, or statistical mixtures.

1.8 Complexity of quantum algorithms

Finally we fix the yardsticks by which quantum algorithms are judged. Let n be the number of qubits needed to represent the input.

Gate complexity. The *gate complexity* of a circuit is the total number of elementary (one- and two-qubit) gates it contains. A quantum algorithm is *efficient* if its gate complexity is $\mathcal{O}(\mathrm{poly}(n))$. Because measurement outcomes are probabilistic, we are satisfied if the algorithm produces the correct answer with high enough probability p: for a decision problem we require $p > 2/3$ (or merely $p > 1/2 + 1/\mathrm{poly}(n)$), and repeating the circuit $\mathcal{O}(\log(1/\delta))$ times with a majority vote drives the error below any $\delta > 0$ by the Chernoff bound.

Query complexity. Often the input is supplied by a black-box unitary—a *quantum oracle*—and we count only the number of calls to it, ignoring the cost of the surrounding gates. For a Boolean function $f : \{0,1\}^n \to \{0,1\}$ two standard oracles are the XOR (or bit) oracle $U_f |x\rangle |b\rangle = |x\rangle |b \oplus f(x)\rangle$ and the phase oracle

$$U_f |x\rangle = (-1)^{f(x)} |x\rangle , \tag{1.23}$$

the latter returning the answer as a *phase kickback*. Query complexity is the central cost measure for the algorithms of Part I (Deutsch–Jozsa, Grover, phase estimation): quantum speedups are most cleanly stated as reductions in the number of oracle calls. Note that having query access to f does not mean we know f globally—finding an x with $f(x) = 0$ can still be hard.

Circuit depth. Group the gates into layers of mutually commuting, disjoint-qubit gates (Theorem 1.44); the *depth* is the number of layers, i.e. the longest input-to-output path. Since gates in a layer run in parallel, depth is the quantum analogue of wall-clock time. It is also the binding physical constraint on near-term hardware, whose qubits stay coherent only for a limited *coherence time*; one often trades a larger gate count or more repetitions for a shallower circuit that fits within the coherence budget.

Space and ancillas. The qubits of an algorithm split into *system* registers (holding the data of interest) and *ancilla* registers (scratch space needed to make operations unitary). Whenever possible one *uncomputes* ancillas back to $|0\rangle$ so they can be reused; a register that can be freed this way is a *working register*, and its cost is often not charged in asymptotic analyses.

Asymptotic notation. We measure all of these costs asymptotically in n (and in accuracy parameters such as $1/\varepsilon$). Recall that $g(n) = \mathcal{O}(h(n))$ means $|g(n)| \le C\, h(n)$ for some constant $C > 0$ and all large n; we write $\mathrm{poly}(n)$ for $\mathcal{O}(n^c)$ and $\mathrm{polylog}(n)$ for $\mathrm{poly}(\log n)$. An *exponential* speedup replaces a classical $\mathcal{O}(2^n)$ cost with a quantum $\mathrm{poly}(n)$ cost, while Grover's algorithm gives a *quadratic* speedup, $\mathcal{O}(2^n) \to \mathcal{O}(2^{n/2})$. These are the comparisons we will make throughout.

Exercises

Exercise 1.1. Let $|\psi\rangle = 2\mathrm{i}\,|0\rangle - |1\rangle \in \mathbb{C}^2$ (unnormalized). Find $\||\psi\rangle\|$ and the normalized state $|\hat{\psi}\rangle$, and give the probabilities of measuring 0 and 1 in the computational basis.

Exercise 1.2. Show that $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ form an orthonormal basis of $\mathbb{C}^2$, and express $|0\rangle$ and $|1\rangle$ in this basis.

Exercise 1.3. Using the equality case of the Cauchy–Schwarz inequality (Theorem 1.6), prove that two *unit* vectors $|u\rangle, |v\rangle$ satisfy $|\langle u|v\rangle| = 1$ if and only if $|v\rangle = \mathrm{e}^{\mathrm{i}\gamma}|u\rangle$ for some real γ—i.e. they are the same physical state.

Exercise 1.4. Let $|u\rangle$ be a unit vector and $P = |u\rangle\langle u|$. Prove that $P^2 = P$ and $P^\dagger = P$, and compute $\mathrm{Tr}(P)$.

Exercise 1.5. Find the Bloch angles (θ, φ) of the state $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}\mathrm{e}^{\mathrm{i}\pi/3}|1\rangle$.

Exercise 1.6. Decide whether $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$ is a product state. If so, exhibit an explicit factorization; if not, prove it is entangled.

Exercise 1.7. Verify by matrix multiplication that H is unitary and self-inverse ($H^\dagger = H$ and $H^2 = I$). Then, using Theorem 1.39 and Theorem 1.40, compute HYH.

Exercise 1.8. Compute $\mathrm{CNOT}|+\rangle|+\rangle$ and decide whether the result is entangled. Then exhibit a unitary U and blank state $|s\rangle$ with $U|0\rangle|s\rangle = |0\rangle|0\rangle$ and $U|1\rangle|s\rangle = |1\rangle|1\rangle$, and explain why this does not contradict the no-cloning theorem (Theorem 1.67).

Exercise 1.9. Consider the two-qubit state $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ but now measure only the *first* qubit. Using the density operator, (a) compute the reduced state $\rho_A = \mathrm{Tr}_B[|\psi\rangle\langle\psi|]$ and its purity $\mathrm{Tr}[\rho_A^2]$; (b) give the probability of outcome 0 on the first qubit and the resulting (normalized) state of the second qubit. Compare with the reduced state of the product state $|+\rangle|+\rangle$.

Exercise 1.10. In the teleportation protocol (Theorem 1.63), suppose Penelope's measurement yields $(m_1, m_2) = (1, 0)$. (a) Write down Odysseus's pre-correction state and verify it equals $Z|\psi\rangle$. (b) State the correction Odysseus applies and confirm it restores $|\psi\rangle$. (c) Averaging over all four equally likely outcomes *before* the classical bits arrive, show that Odysseus's reduced state is $\frac{1}{2}\mathbb{I}$, and explain in one sentence why this is what forbids superluminal signaling.

Exercise 1.11. Verify one case of superdense coding (Theorem 1.65). Starting from $|\Phi^+\rangle$, apply $Z^1X^1 = ZX$ to qubit 1 to encode $(x, y) = (1, 1)$, then apply $\mathrm{CNOT}_{1\to 2}$ followed by H on qubit 1. Show that a computational-basis measurement returns 11 with certainty. (Fix the global-phase convention $ZX|0\rangle = |1\rangle$, $ZX|1\rangle = -|0\rangle$.)

Exercise 1.12. Using the quantum correlator $\mathbb{E}[AB] = \cos(\alpha - \beta)$ of Theorem 1.30, consider the CHSH settings $\alpha_1 = 0$, $\alpha_2 = \frac{\pi}{2}$, $\beta_1 = \frac{\pi}{4}$, $\beta_2 = \frac{3\pi}{4}$. (a) Evaluate the four correlators and confirm $\mathbb{E}[S] = 2\sqrt{2}$. (b) Show that no choice of angles makes $|\mathbb{E}[S]|$ exceed $2\sqrt{2}$ for this state, by maximizing $f(\alpha_1, \alpha_2, \beta_1, \beta_2) = \cos(\alpha_1 - \beta_1) - \cos(\alpha_1 - \beta_2) + \cos(\alpha_2 - \beta_1) + \cos(\alpha_2 - \beta_2)$. *Hint:* fix β_1, β_2 and optimize over α_1, α_2 first, writing each of the two α-dependent pairs as a single cosine of the form $R\cos(\alpha - \phi)$.

Exercise 1.13. Give explicit rotation angles that prepare the state $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}\mathrm{e}^{\mathrm{i}\pi/2}|1\rangle$ from $|0\rangle$ as $R_z(\varphi)R_y(\theta)|0\rangle$, using Theorem 1.52. State the values of θ and φ, and verify that the resulting state agrees with $|\psi\rangle$ up to a global phase.

Exercise 1.14. Let $|\psi\rangle = \cos\frac{\pi}{8}|0\rangle + \sin\frac{\pi}{8}|1\rangle$ and let O be the observable measured "along angle $\vartheta = \frac{\pi}{4}$", i.e. the Hermitian operator with eigenstates $|\vartheta\rangle = \cos\frac{\vartheta}{2}|0\rangle + \sin\frac{\vartheta}{2}|1\rangle$ (outcome $+1$) and its orthogonal partner (outcome -1). (a) Write the two projectors $P_{\pm 1}$. (b) Compute the outcome probabilities $p_{\pm 1}$ for $|\psi\rangle$ and the expectation $\langle O\rangle$. (c) Identify the basis-change unitary V of Theorem 1.59 that reduces this to a computational-basis measurement.

Exercise 1.15. Consider the three operators $E_0 = \frac{1}{2}|0\rangle\langle 0|$, $E_1 = \frac{1}{2}|1\rangle\langle 1|$, and $E_2 = \frac{1}{2}\mathbb{I}$ on $\mathbb{C}^2$. (a) Verify that $\{E_0, E_1, E_2\}$ is a valid POVM (Theorem 1.61): check each $E_m \succeq 0$ and $\sum_m E_m = \mathbb{I}$. (b) Compute the three outcome probabilities on $|0\rangle$ and on $|+\rangle$. (c) Explain why this cannot be a projective measurement, and why an outcome of 2 tells you nothing about the input qubit.

Exercise 1.16. Using the half- and full-adder circuits of Section 1.3.6: (a) trace the two-bit adder on the input $1 + 2$ (that is, $A_1A_0 = 01$ and $B_1B_0 = 10$) and confirm the measured output is $011_2 = 3$. (b) Count the Toffoli and CNOT gates in the complete two-bit adder. (c) The full adder leaves a scratch ancilla holding C_2 alongside the intact inputs; explain, using Bennett's trick, how one would uncompute the intermediate carries to reset any scratch ancillas to $|0\rangle$ once the sum bits have been copied out, and why this matters when the adder is a subroutine in a longer computation.

Further reading

- Lin Lin, *Lecture Notes on Quantum Algorithms for Scientific Computation*, Ch. 1 (postulates, density operator, quantum circuits, no-cloning, complexity) — `qasc_notes.pdf`.
- Lin Lin, *live notes*, Ch. 2 (states, tensor products, entanglement, mixed states) — `live_notes_0429.pdf`.
- A. Childs, *Lecture Notes on Quantum Algorithms*, Ch. 1–2 (quantum circuit model and universality) — `material/`.
- J. Brody, *The Joy of Quantum Computing*, Princeton University Press, 2025, Ch. 4–7 (teleportation, no-cloning and no-signaling, the CHSH/Bell inequality, reversible arithmetic) — `material/`.
- An intuitive, step-by-step treatment of the tensor product and the computational-basis $|k\rangle$ notation (product states) — `material/productstate_zhihu.pdf`.
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Ch. 1–2, for a complete textbook treatment.

Chapter 2

Quantum Channels, Distances, and Statistical Estimates

Learning objectives

- Describe quantum operations as completely positive trace-preserving maps.
- Recognize decoherence as a dephasing channel that decays density-matrix coherences.
- Compare states and channels using trace distance, fidelity, and the diamond norm.
- Apply concentration inequalities to bound measurement error.
- Use statistical amplification and the Cramér–Rao bound to reason about shot budgets.

The previous lecture treated states as pure vectors and evolutions as unitaries. Real devices are noisier and richer than that: a qubit couples to its environment, a measurement discards information, and every number we read off a quantum computer arrives as a *sample* from a probability distribution. This lecture builds the three tools that let us reason quantitatively about these facts. First, the language of *quantum channels*—completely positive trace-preserving (CPTP) maps—which is the most general physically admissible evolution of a density operator and subsumes unitaries, noise, and measurement. Second, the *distance measures* (trace distance, fidelity, diamond norm) that tell us how well one state or channel approximates another, and how those errors accumulate. Third, the *statistical machinery*—concentration inequalities, amplification, and the Cramér–Rao bound—that converts a target accuracy ε and confidence $1-\delta$ into a concrete number of measurement repetitions (a "shot budget"). Together these underlie the error analysis of every algorithm in the rest of the course.

2.1 Quantum channels (CPTP maps)

A quantum channel generalizes both the unitary evolution of an isolated system and the stochastic evolution of a classical probability distribution. It is the right object whenever a system interacts with an environment, undergoes noise, or is measured. We work with density operators (Section 1.7); recall that a density operator on $\mathbb{C}^N$ is a positive semidefinite operator with unit trace. Write $L(\mathbb{C}^N)$ for the space of linear operators on $\mathbb{C}^N$ and $\mathcal{D}(\mathbb{C}^N)$ for the set of density operators.

Definition 2.1 (Superoperator)**.** A *superoperator* is a linear map $\mathcal{Q} : L(\mathbb{C}^N) \to L(\mathbb{C}^M)$, i.e. a linear map that sends operators to operators. We write its action on an operator A as $\mathcal{Q}[A]$. The *identity channel* $\mathcal{I} : L(\mathbb{C}^N) \to L(\mathbb{C}^N)$ is $\mathcal{I}[A] = A$.

To be a legal evolution of density operators, a superoperator must preserve the trace and positivity—but positivity alone is not enough, because the system may be entangled with an untouched ancilla.

Definition 2.2 (Positive, trace preserving, completely positive)**.** A linear map $\mathcal{Q} : L(\mathbb{C}^N) \to L(\mathbb{C}^M)$ is

(i) *trace preserving* (TP) if $\mathrm{Tr}(\mathcal{Q}[A]) = \mathrm{Tr}(A)$ for all A;

(ii) *positive* if $\mathcal{Q}[A] \succeq 0$ whenever $A \succeq 0$;

(iii) *completely positive* (CP) if for every integer $K \geq 1$ the extended map $\mathcal{I}_K \otimes \mathcal{Q}$ on $L(\mathbb{C}^K \otimes \mathbb{C}^N)$ is positive, where $\mathcal{I}_K$ is the identity channel on $L(\mathbb{C}^K)$.

Complete positivity is strictly stronger than positivity, and it is exactly the condition that makes local action on one half of an entangled state produce a valid state. The canonical witness is the transpose.

Example 2.3 (A positive map that is not CP)**.** Let $T : L(\mathbb{C}^2) \to L(\mathbb{C}^2)$ be the transpose in the computational basis, $T[A] = A^\top$. Since A and $A^\top$ have the same spectrum, T is positive. But apply $\mathcal{I} \otimes T$ (the partial transpose on the second qubit) to the Bell state $|\Phi^+\rangle\langle\Phi^+|$ with $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$(\mathcal{I} \otimes T)\big[|\Phi^+\rangle\langle\Phi^+|\big] = \tfrac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

whose eigenvalues are $\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\}$. A negative eigenvalue means the output is not a density operator, so T is *not* completely positive. This is why complete positivity, not mere positivity, is the correct requirement.

Definition 2.4 (Quantum channel)**.** A *quantum channel* is a linear map $\mathcal{Q} : L(\mathbb{C}^N) \to L(\mathbb{C}^M)$ that is completely positive and trace preserving (CPTP). Such a map sends every density operator to a density operator, even when acting on part of a larger entangled system.

2.1.1 The Kraus representation

The most useful concrete description of a channel is the operator-sum, or Kraus, form.

Theorem 2.5 (Kraus representation)**.** *A linear map $\mathcal{Q} : L(\mathbb{C}^N) \to L(\mathbb{C}^M)$ is a quantum channel if and only if there exist operators $\{K_j\}_{j=1}^{R} \subset \mathbb{C}^{M \times N}$ (with $R \leq NM$) satisfying the* completeness relation $\sum_j K_j^\dagger K_j = \mathbb{I}_N$ *such that*

$$\mathcal{Q}[\rho] \;=\; \sum_{j=1}^{R} K_j \, \rho \, K_j^\dagger. \tag{2.1}$$

The K_j are called Kraus operators.

Proof of the "if" direction. Suppose $\mathcal{Q}$ has the form (2.1) with $\sum_j K_j^\dagger K_j = \mathbb{I}_N$. For complete positivity, take any $K \geq 1$ and any $X \succeq 0$ on $\mathbb{C}^K \otimes \mathbb{C}^N$. Then

$$(\mathcal{I}_K \otimes \mathcal{Q})[X] = \sum_j (\mathbb{I}_K \otimes K_j)\, X\, (\mathbb{I}_K \otimes K_j)^\dagger,$$

and each summand has the form $AXA^\dagger \succeq 0$ because $X \succeq 0$; a sum of positive operators is positive, so $\mathcal{I}_K \otimes \mathcal{Q}$ is positive for every K. For trace preservation, the cyclic property of the trace gives

$$\mathrm{Tr}(\mathcal{Q}[\rho]) = \sum_j \mathrm{Tr}(K_j \rho K_j^\dagger) = \mathrm{Tr}\Big(\rho \sum_j K_j^\dagger K_j\Big) = \mathrm{Tr}(\rho\, \mathbb{I}_N) = \mathrm{Tr}(\rho). \qquad \square$$

The converse—every channel admits a Kraus form—is the Choi–Kraus theorem; it follows from the Choi–Jamiołkowski isomorphism, which associates to $\mathcal{Q}$ the positive *Choi operator* $\sigma = (\mathcal{I}_N \otimes \mathcal{Q})[|\gamma\rangle\langle\gamma|]$ built from the unnormalized maximally entangled vector $|\gamma\rangle = \sum_i |i\rangle\,|i\rangle$; the eigenvectors of σ, suitably reshaped, are the Kraus operators. We use the forward direction constantly and cite the converse. A useful corollary is Choi's criterion: $\mathcal{Q}$ is CP if and only if its Choi operator is positive semidefinite, so complete positivity need only be checked for an ancilla of the input dimension.

2.1.2 Examples of channels

Example 2.6 (Unitary channel)**.** For a unitary $U \in \mathbb{C}^{N\times N}$, the map $\mathcal{U}[\rho] = U\rho U^\dagger$ is a channel with the single Kraus operator $K_0 = U$; the completeness relation is $U^\dagger U = \mathbb{I}_N$. This recovers the pure-state rule $|\psi\rangle \mapsto U\,|\psi\rangle$ of Section 1.3. The identity channel is the special case $U = \mathbb{I}$.

Example 2.7 (Bit-flip and phase-flip (dephasing) channels)**.** The *bit-flip channel* applies X with probability $1-p$ and does nothing with probability p:

$$\mathcal{Q}_{\mathrm{bf}}[\rho] = p\,\rho + (1-p)\,X\rho X, \qquad 0 \le p \le 1,$$

with Kraus operators $K_0 = \sqrt{p}\,\mathbb{I}$ and $K_1 = \sqrt{1-p}\,X$ (indeed $K_0^\dagger K_0 + K_1^\dagger K_1 = p\,\mathbb{I} + (1-p)\mathbb{I} = \mathbb{I}$). The *phase-flip* or *dephasing channel* replaces X by Z,

$$\mathcal{Q}_{\mathrm{pf}}[\rho] = p\,\rho + (1-p)\,Z\rho Z,$$

and suppresses off-diagonal coherences while leaving the diagonal populations untouched.

Example 2.8 (Depolarizing channel)**.** The *depolarizing channel* on $\mathbb{C}^N$ leaves ρ intact with probability p and replaces it by the maximally mixed state with probability $1-p$:

$$\mathcal{Q}_{\mathrm{dp}}[\rho] = p\,\rho + (1-p)\,\frac{\mathbb{I}}{N}, \qquad 0 \le p \le 1.$$

It is the standard isotropic noise model; a Kraus form for one qubit uses the Paulis $\{\mathbb{I}, X, Y, Z\}$ with appropriate weights.

Example 2.9 (Amplitude-damping channel)**.** Energy dissipation, in which the excited state $|1\rangle$ decays to $|0\rangle$ with probability γ, is modeled by the Kraus operators

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}, \qquad 0 \le \gamma \le 1.$$

One checks $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{I}$. Unlike dephasing, amplitude damping is not unital ($\mathcal{Q}[\mathbb{I}] \neq \mathbb{I}$): it drives every state toward $|0\rangle$.

Example 2.10 (Measurement channel)**.** A projective measurement in an orthonormal basis $\{|m\rangle\}$, when the outcome is recorded but then forgotten, is the channel

$$\mathcal{Q}_{\mathrm{meas}}[\rho] = \sum_m |m\rangle\langle m|\ \rho\ |m\rangle\langle m| = \sum_m \langle m|\rho|m\rangle\ |m\rangle\langle m|\,,$$

with Kraus operators $K_m = |m\rangle\langle m|$; completeness is the resolution of the identity $\sum_m |m\rangle\langle m| = \mathbb{I}$ (Theorem 1.10). Its output is diagonal in the measurement basis—a *classical state* encoding the outcome distribution. This shows that classical stochastic processes are a special case of quantum channels.

2.1.3 Decoherence as a channel: dephasing and the "cat" state

The phase-flip channel of Theorem 2.7 is the discrete face of a physical process—*decoherence*, the loss of quantum coherence to an environment—and tracking its effect on a density matrix gives concrete intuition for why noisy qubits behave classically. Write a single-qubit state in the computational basis as

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix},$$

so the diagonal entries ρ_{00}, ρ_{11} are the *populations* (measurement probabilities in the Z basis) and the off-diagonal $\rho_{01} = \overline{\rho_{10}}$ is the *coherence* that encodes the relative phase of a superposition.

Example 2.11 (Dephasing decays the off-diagonal element)**.** Apply the phase-flip channel $\mathcal{Q}_{\mathrm{pf}}[\rho] = p\,\rho + (1-p)\,Z\rho Z$ of Theorem 2.7. Since $Z\rho Z$ flips the sign of the off-diagonal entries ($Z\,|0\rangle\langle 1|\,Z = -\,|0\rangle\langle 1|$) while fixing the diagonal, one qubit passing through the channel gives

$$\mathcal{Q}_{\mathrm{pf}}[\rho] = \begin{pmatrix} \rho_{00} & (2p-1)\rho_{01} \\ (2p-1)\rho_{10} & \rho_{11} \end{pmatrix}.$$

The populations are untouched, but the coherence is multiplied by $2p-1 \in [-1,1]$. Interpreting the channel as one time step of duration Δt and setting $2p-1 = \mathrm{e}^{-\Delta t/T_2}$ for a *dephasing time* $T_2 > 0$, repeated application for a total time t multiplies ρ_{01} by $(2p-1)^{t/\Delta t} = \mathrm{e}^{-t/T_2}$: the off-diagonal element decays exponentially,

$$\rho_{01}(t) = \mathrm{e}^{-t/T_2}\,\rho_{01}(0), \tag{2.2}$$

while the diagonal is preserved. In the limit $t \to \infty$ (equivalently $p \to \frac{1}{2}$) the state becomes $\mathrm{diag}(\rho_{00}, \rho_{11})$, exactly the output of the measurement channel $\mathcal{Q}_{\mathrm{meas}}$ of Theorem 2.10. Decoherence thus turns a coherent superposition into a classical probability distribution over the basis states *without ever performing a measurement*: it is measurement enacted by an inattentive environment.

Stinespring dilation (below) makes the mechanism precise: the qubit becomes entangled with an environment, and discarding the environment is exactly the partial trace that suppresses the coherence. A pure superposition $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ has density matrix $|\psi\rangle\langle\psi| = \frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ with full coherence and purity $\mathrm{Tr}(|\psi\rangle\langle\psi|^2) = 1$; the fully dephased state $\frac{1}{2}\mathbb{I}$ has zero coherence and purity $\frac{1}{2}$. These are physically distinct states—and the trace distance of Section 2.2 quantifies exactly how distinguishable they are: $D(|\psi\rangle\langle\psi|\,, \frac{1}{2}\mathbb{I}) = \frac{1}{2}$, so a single optimal measurement identifies which one was prepared only three times in four. This is the formal content of the "Schrödinger's cat" slogan: a genuine superposition $\frac{1}{\sqrt{2}}(|\mathrm{alive}\rangle + |\mathrm{dead}\rangle)$ (nonzero coherence) is a different object from the classical mixture $\frac{1}{2}(|\mathrm{alive}\rangle\langle\mathrm{alive}| + |\mathrm{dead}\rangle\langle\mathrm{dead}|)$ (zero coherence), and decoherence is the channel that maps the former to the latter.

2.1.4 Stinespring dilation

The examples above are irreversible, yet each can be seen as a unitary on a larger system followed by discarding an ancilla. This is the content of the Stinespring dilation theorem, the structural counterpart of the Kraus form.

Theorem 2.12 (Stinespring dilation)**.** *For any quantum channel $\mathcal{Q} : L(\mathbb{C}^N) \to L(\mathbb{C}^M)$ there exist an ancilla space $\mathbb{C}^R$ (with $R \le NM$) and an isometry $V : \mathbb{C}^N \to \mathbb{C}^M \otimes \mathbb{C}^R$ (so $V^\dagger V = \mathbb{I}_N$) with*

$$\mathcal{Q}[\rho] = \mathrm{Tr}_R(V\rho V^\dagger). \tag{2.3}$$

Equivalently, there is a unitary U on a joint system and a fixed ancilla state $|0\rangle$ with $\mathcal{Q}[\rho] = \mathrm{Tr}_R\big(U(\rho \otimes |0\rangle\langle 0|)U^\dagger\big)$.

Proof. Take a Kraus form $\mathcal{Q}[\rho] = \sum_{j=1}^{R} K_j \rho K_j^\dagger$ (Theorem 2.5) and fix an orthonormal basis $\{|j\rangle\}$ of the ancilla $\mathbb{C}^R$. Define $V = \sum_j K_j \otimes |j\rangle$, i.e. $V|\psi\rangle = \sum_j (K_j|\psi\rangle) \otimes |j\rangle$. Then

$$\langle\psi|V^\dagger V|\psi\rangle = \sum_j \|K_j|\psi\rangle\|^2 = \sum_j \langle\psi|K_j^\dagger K_j|\psi\rangle = \langle\psi|\psi\rangle\,,$$

using completeness, so $V^\dagger V = \mathbb{I}_N$ and V is an isometry. Moreover

$$V\rho V^\dagger = \Big(\sum_i K_i \otimes |i\rangle\Big)\rho\Big(\sum_j K_j^\dagger \otimes \langle j|\Big) = \sum_{i,j} (K_i \rho K_j^\dagger) \otimes |i\rangle\langle j|\,.$$

Tracing out the ancilla, $\mathrm{Tr}(|i\rangle\langle j|) = \langle j|i\rangle = \delta_{ij}$, leaves $\sum_j K_j \rho K_j^\dagger = \mathcal{Q}[\rho]$, which is (2.3). To obtain the unitary form, extend the isometry V (defined on the subspace $|\psi\rangle \otimes |0\rangle$) to a full unitary U on the joint space; then $U(\rho \otimes |0\rangle\langle 0|)U^\dagger = V\rho V^\dagger$ and tracing out the ancilla again gives $\mathcal{Q}[\rho]$. □

Stinespring's theorem says that *every* quantum channel, however noisy, is a unitary interaction with an environment followed by discarding the environment. All quantum evolution is unitary once the system is taken large enough.

2.2 Distances between states and channels

To say that an algorithm "approximately" prepares a state or implements a unitary, we need metrics. We first fix the operator norms and then define the two central state distances—trace distance and fidelity—and the channel distance, the diamond norm.

Definition 2.13 (Schatten norms)**.** For $A \in \mathbb{C}^{M\times N}$ with singular values $\sigma_i(A)$, the *trace norm* is $\|A\|_1 = \sum_i \sigma_i(A) = \mathrm{Tr}\,|A|$ with $|A| = \sqrt{A^\dagger A}$, the *Hilbert–Schmidt norm* is $\|A\|_2 = (\sum_i \sigma_i(A)^2)^{1/2}$, and the *operator norm* is $\|A\|_\infty = \max_i \sigma_i(A)$. If $A \succeq 0$ then $\|A\|_1 = \mathrm{Tr}(A)$.

These obey the triangle inequality and Hölder's inequality $\|AB\|_1 \le \|A\|_\infty \|B\|_1$; we use the latter freely.

2.2.1 Trace distance

Definition 2.14 (Trace distance)**.** The *trace distance* between $\rho, \sigma \in \mathcal{D}(\mathbb{C}^N)$ is

$$D(\rho,\sigma) = \tfrac{1}{2}\,\|\rho - \sigma\|_1\,. \tag{2.4}$$

The factor $\frac{1}{2}$ puts $D \in [0,1]$, since $\|\rho-\sigma\|_1 \le \|\rho\|_1 + \|\sigma\|_1 = 2$.

Trace distance is the quantum generalization of the classical total variation distance $\frac{1}{2}\sum_i |p_i - q_i|$: for diagonal (classical) states the two coincide. Its importance is operational.

Theorem 2.15 (Operational meaning of trace distance)**.** *For any $\rho, \sigma \in \mathcal{D}(\mathbb{C}^N)$,*

$$D(\rho,\sigma) = \max_{0 \preceq M \preceq \mathbb{I}} \mathrm{Tr}\big[M(\rho-\sigma)\big], \tag{2.5}$$

the maximum ranging over all measurement operators $0 \preceq M \preceq \mathbb{I}$ and achieved by the projector onto the positive part of $\rho - \sigma$. Hence $D(\rho,\sigma)$ is the largest gap between the probabilities of any single measurement outcome on ρ versus σ: it is the best chance of correctly distinguishing the two states with one measurement.

Proof. Write $\Delta = \rho - \sigma$; it is Hermitian with $\operatorname{Tr}\Delta = 0$. Decompose $\Delta = \Delta_+ - \Delta_-$ into its positive and negative parts (orthogonal support, both $\succeq 0$). Since $\operatorname{Tr}\Delta = 0$ we have $\operatorname{Tr}\Delta_+ = \operatorname{Tr}\Delta_-$, so $\|\Delta\|_1 = \operatorname{Tr}\Delta_+ + \operatorname{Tr}\Delta_- = 2\operatorname{Tr}\Delta_+$ and $D(\rho,\sigma) = \operatorname{Tr}\Delta_+$. For any $0 \preceq M \preceq \mathbb{I}$,

$$\operatorname{Tr}[M\Delta] = \operatorname{Tr}[M\Delta_+] - \operatorname{Tr}[M\Delta_-] \le \operatorname{Tr}[M\Delta_+] \le \operatorname{Tr}[\Delta_+],$$

using $\operatorname{Tr}[M\Delta_-] \ge 0$ (product of positive operators) and $M \preceq \mathbb{I}$. Equality holds for $M = P$, the projector onto the support of Δ_+: then $\operatorname{Tr}[P\Delta] = \operatorname{Tr}\Delta_+ = D(\rho,\sigma)$. □

A fundamental consequence: channels can only make states *less* distinguishable.

Proposition 2.16 (Contractivity under channels)**.** *For any quantum channel $\mathcal{Q}$ and any ρ, σ, $D(\mathcal{Q}[\rho], \mathcal{Q}[\sigma]) \le D(\rho,\sigma)$.*

Proof. Let P be the projector attaining $D(\mathcal{Q}[\rho], \mathcal{Q}[\sigma])$ in (2.5), and decompose $\rho - \sigma = \Delta_+ - \Delta_-$ as above with $\operatorname{Tr}\Delta_+ = D(\rho,\sigma)$. Since $\mathcal{Q}$ is positive, $\mathcal{Q}[\Delta_-] \succeq 0$, so $\operatorname{Tr}[P\,\mathcal{Q}[\Delta_-]] \ge 0$ and

$$D(\mathcal{Q}[\rho], \mathcal{Q}[\sigma]) = \operatorname{Tr}[P\,\mathcal{Q}[\Delta_+]] - \operatorname{Tr}[P\,\mathcal{Q}[\Delta_-]] \le \operatorname{Tr}[P\,\mathcal{Q}[\Delta_+]] \le \operatorname{Tr}[\mathcal{Q}[\Delta_+]] = \operatorname{Tr}[\Delta_+],$$

where the last equality is trace preservation. Thus $D(\mathcal{Q}[\rho], \mathcal{Q}[\sigma]) \le \operatorname{Tr}\Delta_+ = D(\rho,\sigma)$. □

2.2.2 Fidelity

Definition 2.17 (Fidelity)**.** The *fidelity* between $\rho, \sigma \in \mathcal{D}(\mathbb{C}^N)$ is

$$F(\rho,\sigma) = \operatorname{Tr}\sqrt{\rho^{1/2}\sigma\rho^{1/2}} = \left\|\rho^{1/2}\sigma^{1/2}\right\|_1. \tag{2.6}$$

It is symmetric, satisfies $0 \le F \le 1$, and $F(\rho,\sigma) = 1$ iff $\rho = \sigma$.

Fidelity measures overlap rather than distinguishability. For pure states it reduces to the modulus of the inner product.

Example 2.18 (Fidelity of pure states)**.** If $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\varphi\rangle\langle\varphi|$ then $\rho^{1/2}\sigma\rho^{1/2} = |\langle\psi|\varphi\rangle|^2\,|\psi\rangle\langle\psi|$ is rank one, so $F(\rho,\sigma) = |\langle\psi|\varphi\rangle|$. If only $\rho = |\psi\rangle\langle\psi|$ is pure, $F(|\psi\rangle\langle\psi|, \sigma) = \sqrt{\langle\psi|\sigma|\psi\rangle}$.

Fidelity is not itself a metric (it fails the triangle inequality and equals 1, not 0, for identical states), but $\arccos F$—the Bures angle—is a metric, and fidelity is monotone *up* under channels: $F(\mathcal{Q}[\rho], \mathcal{Q}[\sigma]) \ge F(\rho,\sigma)$. Trace distance and fidelity are two views of the same closeness, quantitatively tied by the Fuchs–van de Graaf inequalities.

Theorem 2.19 (Fuchs–van de Graaf)**.** *For any $\rho, \sigma \in \mathcal{D}(\mathbb{C}^N)$,*

$$1 - F(\rho,\sigma) \;\le\; D(\rho,\sigma) \;\le\; \sqrt{1 - F(\rho,\sigma)^2}. \tag{2.7}$$

Proof for pure states. The general bounds follow from Uhlmann's theorem; we prove the sharp pure-state case, which also shows the upper bound is attained. Let $\rho = |\psi\rangle\langle\psi|$, $\sigma = |\varphi\rangle\langle\varphi|$ and set $F = |\langle\psi|\varphi\rangle| = \cos\theta$ with $\theta \in [0, \frac{\pi}{2}]$. Choose a basis with $|\psi\rangle = |0\rangle$ and $|\varphi\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle$. In the $\{|0\rangle, |1\rangle\}$ block,

$$\rho - \sigma = \begin{pmatrix} 1-\cos^2\theta & -\cos\theta\sin\theta \\ -\cos\theta\sin\theta & -\sin^2\theta \end{pmatrix} = \begin{pmatrix} \sin^2\theta & -\cos\theta\sin\theta \\ -\cos\theta\sin\theta & -\sin^2\theta \end{pmatrix},$$

whose eigenvalues are $\pm\sin\theta$. Hence $\|\rho - \sigma\|_1 = 2\sin\theta$ and $D(\rho,\sigma) = \sin\theta = \sqrt{1-F^2}$, attaining the upper bound. Since $\sin\theta \ge 1 - \cos\theta = 1 - F$ for $\theta \in [0, \frac{\pi}{2}]$, the lower bound holds too. □

Remark 2.20 (Why both bounds matter). When states are close, write $F = \cos\theta$ with θ small; then (2.7) reads $\frac{1}{2}\theta^2 \lesssim D \lesssim \theta$, a quadratic gap. Trace distance directly bounds measurement statistics ($|\mathrm{Tr}[M\rho] - \mathrm{Tr}[M\sigma]| \le D(\rho,\sigma)$ for every M), while fidelity is often the easier quantity to estimate experimentally (e.g. via the swap test). An infidelity $1-F = \varepsilon$ guarantees only $D \le \sqrt{2\varepsilon}$ in general, which is the conversion one uses when propagating error bounds.

2.2.3 The diamond norm

For channels, the naive induced trace norm $\|\mathcal{Q}\|_{1\to1} = \sup_{\|X\|_1\le1} \|\mathcal{Q}[X]\|_1$ is not stable under tensoring with an untouched ancilla, exactly as positivity was not stable in Theorem 2.3. The correct channel distance stabilizes it.

Definition 2.21 (Diamond norm)**.** The *diamond norm* of a linear map $\mathcal{Q} : L(\mathbb{C}^N) \to L(\mathbb{C}^M)$ is

$$\|\mathcal{Q}\|_\diamond = \sup_{k\ge1} \|\mathcal{Q}\otimes\mathcal{I}_k\|_{1\to1} = \sup_{k\ge1} \sup_{\|X\|_1\le1} \|(\mathcal{Q}\otimes\mathcal{I}_k)[X]\|_1 , \tag{2.8}$$

where $\mathcal{I}_k$ is the identity on $L(\mathbb{C}^k)$. The supremum is achieved already at $k = N$. The *diamond distance* between channels $\mathcal{Q}, \mathcal{Q}'$ is $\|\mathcal{Q} - \mathcal{Q}'\|_\diamond$.

For a quantum channel $\|\mathcal{Q}\|_\diamond = 1$; the interesting quantity is the distance between two channels, which upper-bounds the largest difference in *any* measurement statistic even when the channels act on part of an entangled input. Two facts make it usable in practice: it is submultiplicative and subadditive, so errors of composed channels add linearly. The following worked example is the workhorse bound for gate error.

Example 2.22 (Diamond distance of unitary channels)**.** Let $\mathcal{U}[\rho] = U\rho U^\dagger$ and $\mathcal{V}[\rho] = V\rho V^\dagger$. Using the identity $A\rho A^\dagger - B\rho B^\dagger = A\rho(A^\dagger - B^\dagger) + (A-B)\rho B^\dagger$ and Hölder's inequality $\|Y_1 X Y_2\|_1 \le \|Y_1\|_\infty \|X\|_1 \|Y_2\|_\infty$ (with $\|U\|_\infty = \|V\|_\infty = 1$) gives, after stabilizing,

$$\|\mathcal{U} - \mathcal{V}\|_\diamond \;\le\; 2\,\|U - V\|_\infty . \tag{2.9}$$

Thus if each of the m gates of a circuit is implemented to operator-norm error $\|U_i - V_i\|_\infty \le \eta$, subadditivity of the diamond norm yields a total error $\le 2m\eta$: circuit error grows only *linearly* in depth. This is the bound that justifies gate-synthesis accuracy targets.

2.3 Elementary concentration inequalities

We now turn to the statistics of reading out a quantum computer. A measurement produces a sample; estimating a quantity means averaging many samples. Three classical inequalities control how fast an empirical average concentrates around its mean.

Theorem 2.23 (Markov's inequality)**.** *If $X \ge 0$ is a random variable and $t > 0$, then $\mathbb{P}(X \ge t) \le \mathbb{E}[X]/t$.*

Proof. Since $X \ge 0$, we have $X \ge t\,\mathbf{1}_{X\ge t}$ pointwise, where $\mathbf{1}_{X\ge t}$ is the indicator of $\{X \ge t\}$. Taking expectations, $\mathbb{E}[X] \ge t\,\mathbb{E}[\mathbf{1}_{X\ge t}] = t\,\mathbb{P}(X \ge t)$; divide by t. □

Theorem 2.24 (Chebyshev's inequality)**.** *If X has finite mean $\mathbb{E}[X]$ and variance $\mathrm{Var}(X)$, then for $t > 0$, $\mathbb{P}(|X - \mathbb{E}[X]| \ge t) \le \mathrm{Var}(X)/t^2$.*

Proof. The event $|X - \mathbb{E}[X]| \geq t$ equals $(X - \mathbb{E}[X])^2 \geq t^2$. Apply Markov's inequality (Theorem 2.23) to the nonnegative variable $(X - \mathbb{E}[X])^2$: $\mathbb{P}((X - \mathbb{E}[X])^2 \geq t^2) \leq \mathbb{E}[(X - \mathbb{E}[X])^2]/t^2 = \mathrm{Var}(X)/t^2$. □

Chebyshev converts a variance into a probabilistic error bar, but its tail decays only polynomially. Hoeffding's inequality gives exponential tails for bounded variables.

Theorem 2.25 (Hoeffding's inequality)**.** *Let $X_1, \ldots, X_n$ be independent with $a_i \leq X_i \leq b_i$, and let $\bar{X} = \frac{1}{n}\sum_i X_i$. Then for any $t > 0$,*

$$\mathbb{P}\big(\bar{X} - \mathbb{E}[\bar{X}] \geq t\big) \leq \exp\Big(-\frac{2n^2t^2}{\sum_i (b_i - a_i)^2}\Big), \tag{2.10}$$

and the same bound holds for $\mathbb{P}(\bar{X} - \mathbb{E}[\bar{X}] \leq -t)$.

Proof. Set $Y_i = X_i - \mathbb{E}[X_i]$, so $Y_i \in [a_i - \mathbb{E}X_i,\, b_i - \mathbb{E}X_i]$ is centered, and let $S = \sum_i Y_i$, so $\bar{X} - \mathbb{E}[\bar{X}] = S/n$. For $\lambda > 0$, the exponential Markov inequality gives $\mathbb{P}(S \geq nt) = \mathbb{P}(\mathrm{e}^{\lambda S} \geq \mathrm{e}^{\lambda nt}) \leq \mathrm{e}^{-\lambda nt}\mathbb{E}[\mathrm{e}^{\lambda S}]$, and independence factorizes $\mathbb{E}[\mathrm{e}^{\lambda S}] = \prod_i \mathbb{E}[\mathrm{e}^{\lambda Y_i}]$. Hoeffding's lemma bounds each factor: for a zero-mean $Y \in [a, b]$, $\mathbb{E}[\mathrm{e}^{\lambda Y}] \leq \exp(\lambda^2 (b-a)^2/8)$. Hence

$$\mathbb{P}(S \geq nt) \leq \exp\Big(-\lambda nt + \tfrac{\lambda^2}{8}\sum_i (b_i - a_i)^2\Big).$$

Minimizing over λ (optimal $\lambda = 4nt/\sum_i (b_i - a_i)^2$) yields (2.10); the lower tail is symmetric. □

Example 2.26 (Bernoulli / Chernoff form)**.** If each $X_i \in \{0, 1\}$ is Bernoulli with mean p (a single-shot measurement outcome), then $b_i - a_i = 1$ and (2.10) gives, for the sample mean $\bar{X}_n$,

$$\mathbb{P}\Big(\big|\bar{X}_n - p\big| \geq \varepsilon\Big) \leq 2\exp(-2n\varepsilon^2). \tag{2.11}$$

This is the bound we use to size shot counts below. A variance-aware refinement, Bernstein's inequality, replaces the exponent by $-n\varepsilon^2/(2p(1-p) + 2\varepsilon/3)$ and is sharper when p is near 0 or 1.

2.4 Statistical amplification and shot budgets

Two distinct tasks recur: (i) *boosting* the success probability of a correct/incorrect decision, and (ii) *estimating* a continuous quantity to additive accuracy ε. Both reduce to the concentration bounds above.

2.4.1 Majority vote and median amplification

Suppose one run of an algorithm returns the correct bit with probability $p > \frac{1}{2}$. Run it n times (say n even, ties counted as failures) and take the majority; the answer is wrong only if the count $S = \sum_i X_i$ of correct runs satisfies $S \leq n/2$. Since $\mathbb{E}[S] = np$, the Hoeffding bound (2.11) gives

$$\mathbb{P}(S \leq n/2) = \mathbb{P}\big(S - np \leq -(p - \tfrac{1}{2})n\big) \leq \exp\big(-2(p - \tfrac{1}{2})^2 n\big). \tag{2.12}$$

To push the failure probability below δ it therefore suffices to take

$$n \;\geq\; \frac{\log(1/\delta)}{2(p - \frac{1}{2})^2} \;=\; \mathcal{O}\Big(\frac{\log(1/\delta)}{(p - \frac{1}{2})^2}\Big)\,: \tag{2.13}$$

the number of repetitions grows only *logarithmically* in the inverse failure probability. The same idea, applied to a real-valued estimator via the *median* of n independent runs, exponentially suppresses the chance that the estimate leaves a target interval: if each run lands outside an interval I with probability $\epsilon < \frac{1}{2}$, the median lands outside I with probability at most $\frac{1}{2}\big(2\sqrt{\epsilon(1-\epsilon)}\big)^n$, which decays exponentially in n. This "median-of-means" trick is the standard way to convert a constant-confidence estimator into a high-confidence one.

2.4.2 Shot count for additive error

Now suppose we want to estimate a probability p (for instance $p = \langle\psi| M |\psi\rangle$ obtained by measuring the observable M) to additive accuracy ε with confidence $1-\delta$. Setting the right-hand side of (2.11) equal to δ and solving for n gives the shot budget

$$n \;\geq\; \frac{\log(2/\delta)}{2\varepsilon^2} \;=\; \mathcal{O}\Big(\frac{\log(1/\delta)}{\varepsilon^2}\Big). \tag{2.14}$$

The characteristic $n = \Theta(\varepsilon^{-2})$ scaling is the *standard quantum limit* (or shot-noise limit): halving the error quadruples the cost. It is ubiquitous—every expectation-value estimate on near-term hardware pays it.

Example 2.27 (Chebyshev refinement near the extremes)**.** The Hoeffding budget (2.14) is independent of p, which is pessimistic when $p \approx 0$ or 1. Since a Bernoulli variable has variance $p(1-p)$, the sample mean has variance $p(1-p)/n$, and Chebyshev's inequality (Theorem 2.24) gives the alternative budget

$$n \;\geq\; \frac{p(1-p)}{\varepsilon^2\delta}.$$

This vanishes as $p \to 0$ or $p \to 1$—matching the intuition that a near-certain outcome is cheap to estimate—at the cost of a worse $1/\delta$ (rather than $\log(1/\delta)$) dependence on the confidence. Bernstein's inequality interpolates, combining the variance factor $p(1-p)$ with the logarithmic confidence dependence.

The lesson for shot budgeting: use Hoeffding (2.14) for a worst-case, confidence-robust estimate; use the variance (Chebyshev/Bernstein) when p is known to be near an extreme; and use majority/median amplification (2.13) whenever the task is a decision rather than a continuous estimate.

2.5 Parameter estimation and the Cramér–Rao bound

The shot bounds above are upper bounds on cost. The Cramér–Rao bound is a matching *lower* bound: no unbiased estimator can do better than the inverse Fisher information. This is what tells us when an estimator is optimal, and it sets up the quantum metrology of Chapter 4.

Definition 2.28 (Score and Fisher information)**.** Let $\{P(\nu \mid \theta) : \nu \in \Sigma\}$ be a statistical model, smooth in the scalar parameter θ with $P(\nu \mid \theta) > 0$. The *score* is $S(\theta \mid \nu) = \partial_\theta \log P(\nu \mid \theta)$, and the *Fisher information* of one sample is its variance,

$$I_F(\theta) = \mathbb{E}_\nu\big[S(\theta \mid \nu)^2\big] = \sum_{\nu\in\Sigma} P(\nu \mid \theta)\Big(\frac{\partial_\theta P(\nu \mid \theta)}{P(\nu \mid \theta)}\Big)^2. \tag{2.15}$$

For N independent samples the Fisher information is additive, $I_F^{(N)}(\theta) = N\, I_F(\theta)$.

The score has mean zero: $\mathbb{E}_\nu[S] = \sum_\nu \partial_\theta P(\nu \mid \theta) = \partial_\theta \sum_\nu P(\nu \mid \theta) = \partial_\theta 1 = 0$. This is the key identity behind the bound.

Theorem 2.29 (Cramér–Rao bound)**.** *Any unbiased estimator $\hat{\theta}$ of θ built from N independent samples satisfies*

$$\mathrm{Var}(\hat{\theta}) \;\geq\; \frac{1}{I_F^{(N)}(\theta)} \;=\; \frac{1}{N\, I_F(\theta)}. \tag{2.16}$$

Proof. Let $S = S(\theta \mid D)$ be the score of the joint data D, with $\mathbb{E}[S] = 0$ and $\mathrm{Var}(S) = I_F^{(N)}(\theta)$. Unbiasedness means $\mathbb{E}_D[\hat{\theta}] = \theta$, so

$$\mathrm{Cov}(\hat{\theta}, S) = \mathbb{E}_D[\hat{\theta}\, S] = \sum_D \hat{\theta}(D)\, \partial_\theta P(D \mid \theta) = \partial_\theta \sum_D \hat{\theta}(D) P(D \mid \theta) = \partial_\theta\, \theta = 1,$$

using $S = \partial_\theta P / P$ and $\mathbb{E}[\hat{\theta} S] = \mathrm{Cov}(\hat{\theta}, S)$ (as $\mathbb{E}[S] = 0$). The Cauchy–Schwarz inequality for covariance, $\mathrm{Cov}(\hat{\theta}, S)^2 \leq \mathrm{Var}(\hat{\theta})\, \mathrm{Var}(S)$, then gives $1 \leq \mathrm{Var}(\hat{\theta})\, I_F^{(N)}(\theta)$, which rearranges to (2.16). □

An estimator meeting (2.16) with equality is called *efficient*.

Example 2.30 (Fisher information of a biased coin / single-qubit measurement)**.** Estimating the success probability p of a Bernoulli outcome (a qubit measured in the computational basis, $P(1 \mid p) = p$, $P(0 \mid p) = 1 - p$) has scores $\partial_p \log P(1 \mid p) = 1/p$ and $\partial_p \log P(0 \mid p) = -1/(1-p)$, so

$$I_F(p) = p \cdot \frac{1}{p^2} + (1-p) \cdot \frac{1}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

The Cramér–Rao bound then reads $\mathrm{Var}(\hat{p}) \geq p(1-p)/N$, and the sample mean (the observed frequency of 1s) is unbiased with exactly this variance—so it is efficient. This recovers, and shows the optimality of, the Chebyshev shot budget of Theorem 2.27.

2.5.1 The quantum Cramér–Rao bound and metrological scaling

When the data come from measuring a parameterized state $\rho(\theta)$, the *measurement itself* is a design choice. Optimizing the classical Fisher information over all POVMs $\{E_\nu\}$ defines the *quantum Fisher information* $F_Q(\theta) = \sup_{\{E_\nu\}} I_F(\theta; \{E_\nu\})$, and the quantum Cramér–Rao bound states that for N copies of $\rho(\theta)$,

$$\mathrm{Var}(\hat{\theta}) \;\geq\; \frac{1}{N\, F_Q(\theta)}. \tag{2.17}$$

For a pure family $\rho(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|$ one has the closed form $F_Q(\theta) = 4\big(\langle\partial_\theta\psi|\partial_\theta\psi\rangle - |\langle\psi|\partial_\theta\psi\rangle|^2\big)$.

Example 2.31 (Single-qubit phase, and the two scaling regimes)**.** For $|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + \mathrm{e}^{\mathrm{i}\theta}|1\rangle)$ we have $|\partial_\theta\psi\rangle = \frac{\mathrm{i}}{\sqrt{2}}\mathrm{e}^{\mathrm{i}\theta}|1\rangle$, giving $\langle\partial_\theta\psi|\partial_\theta\psi\rangle = \frac{1}{2}$ and $\langle\psi|\partial_\theta\psi\rangle = \frac{\mathrm{i}}{2}$, hence $F_Q(\theta) = 4(\frac{1}{2} - \frac{1}{4}) = 1$. So (2.17) gives $\mathrm{Var}(\hat{\theta}) \geq 1/N$, and an X-basis measurement (whose outcome probabilities are $\frac{1}{2}(1 \pm \cos\theta)$) achieves classical Fisher information 1, saturating the bound. Using N *independent* probes each acquiring phase θ yields $\mathrm{Var}(\hat{\theta}) = \mathcal{O}(1/N)$—standard-quantum-limit scaling, the metrological face of the ε^{-2} shot count. By instead letting a single probe accumulate phase for a time proportional to N (equivalently, using an N-fold-entangled probe), the effective generator scales up and one reaches $\mathrm{Var}(\hat{\theta}) = \mathcal{O}(1/N^2)$, the *Heisenberg limit*. Quantum phase estimation, developed in Chapter 4, is precisely the algorithm that realizes this quadratic improvement.

Exercises

Exercise 2.1. Show that the bit-flip channel $\mathcal{Q}_{\mathrm{bf}}[\rho] = p\,\rho + (1-p)X\rho X$ (with $0 \le p \le 1$) is CPTP by exhibiting Kraus operators and verifying the completeness relation $\sum_j K_j^\dagger K_j = \mathbb{I}$. Then compute $\mathcal{Q}_{\mathrm{bf}}[|0\rangle\langle 0|]$ and interpret the result.

Exercise 2.2. Consider the amplitude-damping channel of Theorem 2.9 with Kraus operators $K_0 = \mathrm{diag}(1, \sqrt{1-\gamma})$ and $K_1 = \sqrt{\gamma}\,|0\rangle\langle 1|$. Verify that it is trace preserving, and show that it is *not* unital by computing $\mathcal{Q}[\mathbb{I}]$.

Exercise 2.3. Compute the trace distance $D(\rho,\sigma)$ between the single-qubit states $\rho = |0\rangle\langle 0|$ and $\sigma = |+\rangle\langle +|$ with $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Then compute the fidelity $F(\rho,\sigma)$ and verify that the pair satisfies the Fuchs–van de Graaf inequalities (2.7).

Exercise 2.4. You wish to estimate a measurement probability p to additive accuracy $\varepsilon = 10^{-2}$ with confidence $1-\delta = 0.99$. Using the Hoeffding bound (2.14), give a sufficient number of shots n. Then, assuming you know $p \le 0.05$, use the Chebyshev budget of Theorem 2.27 to give an alternative n and compare.

Exercise 2.5. A decision algorithm returns the correct bit with probability $p = 0.6$ on each run. Using (2.13), how many independent runs n suffice so that a majority vote is correct with probability at least $1-\delta$ for $\delta = 10^{-3}$? Explain why the dependence on δ is logarithmic rather than polynomial.

Exercise 2.6. Consider a Hadamard-test-style estimator in which a qubit is measured in the X-basis and yields outcome $+$ with probability $P(+\mid\theta) = \frac{1}{2}(1+\cos\theta)$ and $-$ with probability $\frac{1}{2}(1-\cos\theta)$. Compute the classical Fisher information $I_F(\theta)$ for the parameter θ, and state the Cramér–Rao lower bound on $\mathrm{Var}(\hat\theta)$ for N independent measurements. For which θ is the information largest, and for which is it zero?

Exercise 2.7. Let $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ pass through the dephasing channel of Theorem 2.11 for a time t, so its density matrix is $\rho(t) = \frac{1}{2}\left(\begin{smallmatrix} 1 & \mathrm{e}^{-t/T_2} \\ \mathrm{e}^{-t/T_2} & 1 \end{smallmatrix}\right)$. Compute the purity $\mathrm{Tr}(\rho(t)^2)$ and the trace distance $D(\rho(t), \frac{1}{2}\mathbb{I})$ to the completely mixed state as functions of t. Verify that at $t=0$ the state is pure with $D = \frac{1}{2}$, and that both quantities decay monotonically to their fully mixed values as $t \to \infty$.

Further reading

- Lin Lin, *live notes*, Ch. 3 (probability, quantum channels and CPTP maps, Kraus and Stinespring representations, trace distance, fidelity, diamond norm) — `live_notes_0429.pdf`.
- Lin Lin, *live notes*, Ch. 8 (concentration inequalities, statistical amplification, parameter estimation, classical and quantum Cramér–Rao bounds) — `live_notes_0429.pdf`.
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Ch. 8–9 (quantum operations, trace distance and fidelity).
- J. Watrous, *The Theory of Quantum Information*, Ch. 2–3 (channels, Choi representation, diamond norm), for a rigorous treatment.

- D. C. Brody, *The Joy of Quantum Computing*, Princeton University Press, 2025, Ch. 13–14 (the density-matrix picture; decoherence as entanglement with the environment and the Schrödinger's-cat superposition versus classical mixture) — accessible motivation for Section 2.1.3.

Chapter 3

Grover's Algorithm and Amplitude Amplification

Learning objectives

- Trace the historical arc from Deutsch's algorithm to unstructured search.
- Derive the Grover iteration geometrically.
- Construct the phase oracle and diffusion operator as explicit circuits.
- Generalize to amplitude amplification and state its quadratic speedup.
- Sketch the query lower bound for unstructured search.

Covered this term (Fall 2026–2027). the entire chapter (§3.1–§3.5).

The previous lectures fixed the language of states, gates, and measurement. This lecture uses that language to build the two archetypal query algorithms of the subject. We begin with the oracle model, in which the only access to an unknown Boolean function is a black-box unitary, and cost is measured in *queries* to that box rather than in elementary gates. Within this model, Deutsch's algorithm and its descendants (Deutsch–Jozsa, Bernstein–Vazirani) demonstrate a clean and deterministic quantum advantage; the tools they introduce—phase kickback and the Hadamard transform—recur throughout the course. We then turn to the flagship application, Grover's algorithm for *unstructured search*, which finds a marked item among N candidates using only $\mathcal{O}(\sqrt{N})$ queries, a quadratic speedup over the classical $\mathcal{O}(N)$. The mechanism reduces to a strikingly simple geometric picture: the search lives inside a two-dimensional plane, and each iteration is a rotation toward the solution. We prove the rotation picture, compute the optimal number of iterations, generalize to amplitude amplification, and close with the matching $\Omega(\sqrt{N})$ lower bound that shows the algorithm is optimal.

3.1 Deutsch's algorithm

We are given a Boolean function $f : \{0,1\}^n \to \{0,1\}$ whose internal structure is hidden from us. We do not have the code of f; we may only feed inputs and receive outputs through a black box. To use f inside a quantum circuit we need its action to be reversible, since every quantum gate is unitary. The standard device for this is the following.

Definition 3.1 (Oracle / query model)**.** Let $f : \{0,1\}^n \to \{0,1\}$. The *oracle* (or *quantum oracle*) for f is the operator U_f acting on $n+1$ qubits—an n-qubit *input register* and a single-qubit *target*

register—defined on the computational basis by

$$U_f\,|x\rangle\,|y\rangle = |x\rangle\,|y \oplus f(x)\rangle\,, \qquad x \in \{0,1\}^n,\ y \in \{0,1\},$$

where $\oplus$ denotes addition modulo 2. The *query complexity* of a task is the minimum number of applications of U_f (over all algorithms that succeed) needed to solve it.

The exclusive-or in the target register—rather than simply overwriting y with $f(x)$—is precisely what makes the oracle reversible.

Lemma 3.2 (The oracle is unitary)**.** *For every $f : \{0,1\}^n \to \{0,1\}$, the operator U_f of Theorem 3.1 is unitary and is its own inverse: $U_f^\dagger U_f = U_f^2 = \mathbb{I}$.*

Proof. The set $\{|x\rangle\,|y\rangle\}$ over $x \in \{0,1\}^n$, $y \in \{0,1\}$ is an orthonormal basis of the 2^{n+1}-dimensional state space. The map $(x,y) \mapsto (x,\, y \oplus f(x))$ is a bijection of the index set to itself, because it is inverted by itself: applying it twice gives $(x,\, y \oplus f(x) \oplus f(x)) = (x,y)$ since $f(x) \oplus f(x) = 0$. Hence U_f merely permutes an orthonormal basis, so it is unitary, and $U_f^2 = \mathbb{I}$. Equivalently, in matrix form $\langle x'y'|U_f^\dagger U_f|xy\rangle = \delta_{xx'}\delta_{yy'}$, i.e. $U_f^\dagger U_f = \mathbb{I}$. □

Classical versus quantum access. A classical algorithm can only submit one input x at a time and read back the single bit $f(x)$; setting the target to $y = 0$ gives $U_f\,|x\rangle\,|0\rangle = |x\rangle\,|f(x)\rangle$. To learn a *global* property of f this way may force us to examine almost all 2^n inputs. A quantum algorithm can instead prepare a uniform superposition of all inputs and apply U_f once, acting on every input in parallel:

$$U_f\Big(\tfrac{1}{\sqrt{N}} \sum_{x\in\{0,1\}^n} |x\rangle\Big)\,|0\rangle = \tfrac{1}{\sqrt{N}} \sum_x |x\rangle\,|f(x)\rangle\,, \qquad N = 2^n.$$

Parallelism alone is not enough—measuring this state returns one random pair $(x, f(x))$, no better than a classical query. The extra ingredient that converts parallelism into advantage is *interference*, engineered through phase kickback and the Hadamard transform.

3.1.1 Phase kickback

The oracle writes $f(x)$ into the target register as a bit. We would prefer to have $f(x)$ appear as a *phase* on the input register, since phases are what interfere. The following observation—perhaps the single most important trick in the lecture—accomplishes exactly this by choosing the target register cleverly.

Lemma 3.3 (Phase kickback)**.** *Let $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Then for every $x \in \{0,1\}^n$,*

$$U_f\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle\,.$$

In particular the target qubit is left in $|-\rangle$ and may be discarded, so on the input register the oracle acts as the effective diagonal unitary $\tilde{U}_f\,|x\rangle = (-1)^{f(x)}\,|x\rangle$.

Proof. Expand the target register and apply Theorem 3.1 to each branch:

$$U_f\,|x\rangle\,|-\rangle = \tfrac{1}{\sqrt{2}}(U_f\,|x\rangle\,|0\rangle - U_f\,|x\rangle\,|1\rangle\,) = \tfrac{1}{\sqrt{2}}(\,|x\rangle\,|f(x)\rangle - |x\rangle\,|1 \oplus f(x)\rangle\,).$$

Consider the two cases for the bit $f(x)$. If $f(x) = 0$ the target factor is $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle = (+1)\,|-\rangle$. If $f(x) = 1$ it is $\frac{1}{\sqrt{2}}(|1\rangle - |0\rangle) = -\,|-\rangle = (-1)\,|-\rangle$. Both cases are captured by the single formula $(-1)^{f(x)}\,|-\rangle$, whence $U_f\,|x\rangle\,|-\rangle = (-1)^{f(x)}\,|x\rangle\,|-\rangle$. Since the target factor is unchanged and unentangled from the input register, it factors out and can be dropped. □

Remark 3.4. The name *kickback* reflects that the eigenvalue $(-1)^{f(x)}$ of the target qubit's controlled action is "kicked back" onto the control (input) register. The target qubit acts purely as a passive signal state. This is the same mechanism that, with a target eigenstate of a general unitary, powers quantum phase estimation later in the course (Chapter 4).

Applying U_f to a uniform superposition with the target in $|-\rangle$ therefore produces, on the input register,

$$\tfrac{1}{\sqrt{N}}\sum_x |x\rangle \;\xmapsto{U_f}\; \tfrac{1}{\sqrt{N}}\sum_x (-1)^{f(x)}\,|x\rangle\,.$$

Every value of f is now recorded in a relative sign. It remains to read out these signs; the tool for that is the Hadamard transform.

3.1.2 The Hadamard transform

We will use repeatedly the action of $H^{\otimes n}$ on a basis state. Recall for a single qubit $H|a\rangle = \frac{1}{\sqrt{2}}\sum_{b\in\{0,1\}}(-1)^{ab}|b\rangle$.

Lemma 3.5 (Hadamard identity)**.** *For $x \in \{0,1\}^n$, writing $x\cdot z = \bigoplus_{i=1}^n x_i z_i$ for the bitwise inner product modulo 2,*

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}}\sum_{z\in\{0,1\}^n}(-1)^{x\cdot z}\,|z\rangle\,.$$

Proof. Take the tensor product of the single-qubit rule over the n qubits:

$$H^{\otimes n}|x\rangle = \bigotimes_{i=1}^{n}\Big(\tfrac{1}{\sqrt{2}}\sum_{z_i\in\{0,1\}}(-1)^{x_i z_i}|z_i\rangle\Big) = \frac{1}{\sqrt{2^n}}\sum_{z\in\{0,1\}^n}(-1)^{\sum_i x_i z_i}|z\rangle\,.$$

Because the phase is ± 1, the exponent $\sum_i x_i z_i$ may be reduced modulo 2, giving $(-1)^{x\cdot z}$. □

We also record the orthogonality relation that makes the interference exact:

$$\frac{1}{2^n}\sum_{x\in\{0,1\}^n}(-1)^{x\cdot w} = \delta_{w,0^n}, \qquad w\in\{0,1\}^n, \tag{3.1}$$

which holds because the sum factors as $\prod_{i=1}^n \frac{1}{2}\sum_{x_i}(-1)^{x_i w_i}$, and each factor equals 1 if $w_i = 0$ and 0 if $w_i = 1$.

3.1.3 The Deutsch algorithm

Consider the simplest nontrivial case, $n = 1$, so $f:\{0,1\}\to\{0,1\}$. There are four such functions. We wish to decide a global property of f using as few queries as possible.

Definition 3.6 (Constant and balanced functions)**.** A function $f:\{0,1\}^n\to\{0,1\}$ is *constant* if $f(x)$ takes the same value on all 2^n inputs. It is *balanced* if $f(x) = 0$ on exactly half of the inputs and $f(x) = 1$ on the other half. For $n = 1$, f is constant iff $f(0) = f(1)$ and balanced iff $f(0)\neq f(1)$.

Classically, deciding whether f is constant or balanced for $n = 1$ requires *two* queries: knowing $f(0)$ alone tells us nothing about $f(1)$. Deutsch's[1] algorithm decides it with a single query.

[1] David Deutsch (b. 1953) is a physicist at the Centre for Quantum Computation, Clarendon Laboratory, University of Oxford. His 1985 paper *Quantum theory, the Church–Turing principle and the universal quantum computer* introduced the universal quantum Turing machine and the first quantum algorithm; he also pioneered quantum logic gates and quantum error correction, and is a leading advocate of the Everett ("many-worlds") interpretation and the founder of constructor theory.

Theorem 3.7 (Correctness of the Deutsch algorithm)**.** *Prepare the two-qubit state* $|0\rangle|1\rangle$*, apply* H *to each qubit, apply* U_f*, apply* H *to the first qubit, and measure the first qubit. The outcome is* 0 *if* f *is constant and* 1 *if* f *is balanced. Exactly one query to* U_f *is used.*

Proof. After the first Hadamards, $H|0\rangle \otimes H|1\rangle = |+\rangle|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|-\rangle$. Applying U_f and using phase kickback (Theorem 3.3) on each branch,

$$U_f \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|-\rangle = \tfrac{1}{\sqrt{2}}((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle)|-\rangle .$$

Now apply H to the first qubit, using $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$\tfrac{1}{2}\Big[\big((-1)^{f(0)} + (-1)^{f(1)}\big)|0\rangle + \big((-1)^{f(0)} - (-1)^{f(1)}\big)|1\rangle\Big]|-\rangle .$$

If f is constant, $f(0) = f(1)$, the $|1\rangle$ amplitude vanishes and the $|0\rangle$ amplitude is ± 1; the first qubit is $\pm|0\rangle$ and measurement yields 0 with certainty. If f is balanced, $f(0) \neq f(1)$, the $|0\rangle$ amplitude vanishes and the first qubit is $\pm|1\rangle$; measurement yields 1 with certainty. The oracle is called once. □

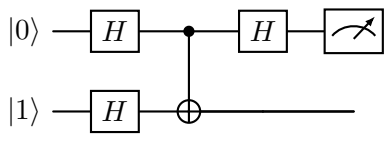
Notice that the answer is read from the *first* qubit, even though classically one would expect the "information" to live in the target qubit that received $f(x)$. This inversion is the hallmark of interference.

Example 3.8 (Deutsch on $f(a) = a \oplus 1$)**.** Take $f(0) = 1$, $f(1) = 0$, which is balanced. After the query the first register is $\frac{1}{\sqrt{2}}((-1)^1|0\rangle + (-1)^0|1\rangle) = \frac{1}{\sqrt{2}}(-|0\rangle + |1\rangle) = -|-\rangle$. Applying H gives $H(-|-\rangle) = -|1\rangle$, so the measurement returns 1, correctly reporting "balanced." By contrast, for the constant function $f \equiv 1$ the first register after the query is $\frac{1}{\sqrt{2}}(-|0\rangle - |1\rangle) = -|+\rangle$, and $H(-|+\rangle) = -|0\rangle$ returns 0, reporting "constant."

Example 3.9 (Explicit circuits: the four one-bit oracles and Deutsch's circuit)**.** For $n = 1$ there are exactly four functions $f : \{0,1\} \to \{0,1\}$, and each oracle $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$ is a small explicit circuit on the input qubit x (top) and target qubit y (bottom):

$$\begin{array}{llll} f \equiv 0 \text{ (const)}: & U_f = \mathbb{I}, & f(x) = x \text{ (bal)}: & U_f = \mathrm{CNOT}_{x\to y}, \\ f \equiv 1 \text{ (const)}: & U_f = \mathbb{I} \otimes X, & f(x) = x \oplus 1 \text{ (bal)}: & U_f = (\mathbb{I} \otimes X)\,\mathrm{CNOT}_{x\to y}. \end{array}$$

Take the balanced $f(x) = x$, whose oracle is a single CNOT. The full Deutsch circuit—H on both qubits, the oracle, H on the input, then measure the input—is



Tracing it through: $|0\rangle|1\rangle \xrightarrow{H\otimes H} |+\rangle|-\rangle \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|-\rangle = |-\rangle|-\rangle \xrightarrow{H\otimes\mathbb{I}} |1\rangle|-\rangle$, so the input qubit is measured as 1: "balanced," in one query. For the constant $f \equiv 0$ the oracle is the identity, the input qubit stays $|+\rangle \xrightarrow{H} |0\rangle$, and the measurement returns 0: "constant."

3.1.4 The Deutsch–Jozsa algorithm

We now generalize to n qubits under the following *promise*: f is either constant or balanced (Theorem 3.6), and we are guaranteed that one of these holds. The task is to decide which. Classically, in the worst case one must query $2^{n-1}+1$ inputs: after seeing 2^{n-1} identical outputs the function could still be either constant or balanced, and one more query is needed. Deutsch–Jozsa[2] decides it with a single quantum query.

The circuit is the unifying pattern $H^{\otimes n}$–U_f–$H^{\otimes n}$ on the n-qubit input register, with the target qubit held in $|-\rangle$.

Theorem 3.10 (Correctness of Deutsch–Jozsa)**.** *Prepare $|0\rangle^{\otimes n}$ on the input register and $|-\rangle$ on the target. Apply $H^{\otimes n}$, then U_f, then $H^{\otimes n}$, and measure the input register. If f is constant the outcome is 0^n with probability 1; if f is balanced the outcome 0^n occurs with probability 0. Thus measuring all zeros certifies "constant," and any nonzero outcome certifies "balanced." One query to U_f is used.*

Proof. By Theorem 3.5, $H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}}\sum_x |x\rangle$ with $N = 2^n$. Applying U_f with the target in $|-\rangle$ triggers phase kickback (Theorem 3.3) on every branch, giving (dropping the unchanged target factor)

$$\tfrac{1}{\sqrt{N}}\sum_x (-1)^{f(x)}|x\rangle\,.$$

Apply $H^{\otimes n}$ once more and use Theorem 3.5 again:

$$H^{\otimes n}\tfrac{1}{\sqrt{N}}\sum_x (-1)^{f(x)}|x\rangle = \frac{1}{N}\sum_z \Big(\sum_x (-1)^{f(x)}(-1)^{x\cdot z}\Big)|z\rangle\,.$$

The amplitude of the outcome $z = 0^n$ is obtained by setting $z = 0^n$, so $x \cdot z = 0$:

$$\alpha_{0^n} = \frac{1}{N}\sum_{x\in\{0,1\}^n} (-1)^{f(x)}.$$

If f is constant, every term is the same sign and $\alpha_{0^n} = \pm 1$, so the outcome is 0^n with probability $|\alpha_{0^n}|^2 = 1$. If f is balanced, exactly $N/2$ terms equal $+1$ and $N/2$ equal -1, so they cancel and $\alpha_{0^n} = 0$: the outcome 0^n never occurs. In either case a single query suffices, and the two cases are distinguished with certainty. □

Remark 3.11 (Gate complexity of the oracle, and the role of the promise). The "one query" of Theorem 3.10 counts *calls* to U_f, which the query model (Theorem 3.1) treats as a unit-cost black box. The *gate* complexity—the number of elementary gates needed to *build* U_f—is a separate question, and it depends on the particular function f, *not* on whether f is constant or balanced. The promise is a global property of f's truth table; it says nothing about how hard f is to compute.

- *Constant.* $f \equiv 0$ gives $U_f = \mathbb{I}$ (no gates); $f \equiv 1$ flips the target unconditionally, $U_f = \mathbb{I}^{\otimes n} \otimes X$ (one gate). Constant oracles are trivial.
- *Balanced.* The gate count spans the whole range: a single parity $f(x) = x_i$ needs one CNOT, and $f(x) = s \cdot x$ needs at most n CNOTs (Theorem 3.12), but a *generic* balanced function is as hard as a generic Boolean function—by a counting (Shannon) argument almost all n-bit functions require $\Theta(2^n/n)$ gates.

[2] Richard Jozsa (b. 1953) is the Leigh Trapnell Professor of Quantum Physics at the University of Cambridge (DAMTP), where he heads the Centre for Quantum Information and Foundations. With Deutsch he generalized the 1985 result to the Deutsch–Jozsa algorithm (*Rapid solution of problems by quantum computation*, 1992)—the first quantum algorithm exponentially faster than any deterministic classical one—and he is one of the six co-authors of the 1993 quantum-teleportation protocol, as well as a co-inventor of the state fidelity and the swap test.

So the constant/balanced classification does *not* fix the gate cost: two balanced functions can differ by an exponential factor. The dramatic 1-vs-$(2^{n-1}+1)$ Deutsch–Jozsa separation is therefore a statement about *query* complexity only. If U_f must be compiled from gates, that cost is paid once and is incurred by *any* algorithm—classical or quantum—so it cannot be charged against the comparison; the quantum advantage is purely in the number of oracle calls. (Phase kickback reuses the same bit oracle U_f with the target held in $|-\rangle$, so reading f as a phase adds no gates beyond the one-time $|-\rangle$ preparation.)

3.1.5 The Bernstein–Vazirani algorithm

The Deutsch–Jozsa algorithm returns a single bit of information (constant vs. balanced). The Bernstein–Vazirani[3] algorithm uses the very same circuit to extract an entire hidden n-bit string in one query, provided f has a special linear structure.

Definition 3.12 (Bernstein–Vazirani function)**.** Fix a hidden string $s \in \{0,1\}^n$. The associated *linear* Boolean function is

$$f_s(x) = s \cdot x = \bigoplus_{i=1}^{n} s_i x_i \pmod 2,$$

the parity of the number of positions at which s and x are both 1. The Bernstein–Vazirani problem is: given oracle access to U_f for such an f_s, determine s.

Classically, one recovers s bit by bit. Querying $x = e_i$ (the string with a single 1 in position i) returns $f_s(e_i) = s_i$; this determines one bit per query, so n queries are needed and sufficient. The quantum algorithm needs one.

Theorem 3.13 (Correctness of Bernstein–Vazirani)**.** *For $f = f_s$ of Theorem 3.12, running the circuit of Theorem 3.10—$H^{\otimes n}$, then U_f, then $H^{\otimes n}$, with the target in $|-\rangle$—leaves the input register in the state $|s\rangle$. Measuring recovers s with certainty using a single query.*

Proof. As before, after $H^{\otimes n}$ and the single query with phase kickback the input register is

$$\tfrac{1}{\sqrt{N}} \sum_x (-1)^{f_s(x)} |x\rangle = \tfrac{1}{\sqrt{N}} \sum_x (-1)^{s\cdot x} |x\rangle .$$

Applying $H^{\otimes n}$ and using Theorem 3.5,

$$H^{\otimes n} \tfrac{1}{\sqrt{N}} \sum_x (-1)^{s\cdot x} |x\rangle = \frac{1}{N} \sum_z \Big(\sum_x (-1)^{s\cdot x} (-1)^{x\cdot z} \Big) |z\rangle = \frac{1}{N} \sum_z \Big(\sum_x (-1)^{x\cdot(s\oplus z)} \Big) |z\rangle ,$$

where $s \cdot x \oplus x \cdot z = x \cdot (s \oplus z)$ because the inner product is linear in its second argument modulo 2. By the orthogonality relation (3.1), $\frac{1}{N}\sum_x (-1)^{x\cdot(s\oplus z)} = \delta_{s\oplus z,\,0^n} = \delta_{z,s}$. Hence only the term $z = s$ survives and the register is exactly $|s\rangle$. Measuring in the computational basis returns s with probability 1. □

Equivalently, $H^{\otimes n}|s\rangle = \frac{1}{\sqrt{N}}\sum_x (-1)^{s\cdot x}|x\rangle$ by Theorem 3.5, so the post-query state is $H^{\otimes n}|s\rangle$; a second $H^{\otimes n}$ inverts it (since $H^{\otimes n}$ is its own inverse) and returns $|s\rangle$.

[3]The algorithm is due to Ethan Bernstein and Umesh Vazirani. Vazirani is the Roger A. Strauch Professor of EECS at the University of California, Berkeley, and directs the Berkeley Quantum Computation Center; Bernstein was his doctoral student there. Their 1993 paper *Quantum complexity theory* defined the quantum Turing machine and the class BQP, and their algorithm—a special case of Deutsch–Jozsa—recovers an n-bit hidden string in a single query, the seed of the first oracle separation between BPP and BQP.

Remark 3.14 (The unifying pattern and its limits). Several points deserve emphasis. *(i) Deterministic advantage.* All three algorithms succeed with probability exactly 1; the advantage is not a matter of a lucky measurement. For Deutsch–Jozsa the separation is 1 quantum query versus $2^{n-1}+1$ classical queries in the worst case, but only in the *deterministic* setting: a handful of random classical queries distinguishes constant from balanced with high confidence, so the bounded-error classical query complexity is $\mathcal{O}(1)$. For Bernstein–Vazirani the separation is 1 versus n and survives even with bounded error. *(ii) The template.* Every algorithm here is the circuit $H^{\otimes n} U_f H^{\otimes n}$ applied to $|0\rangle^{\otimes n}$ with the target in $|-\rangle$: phase kickback writes f into signs, and the two Hadamard layers read those signs out by interference. This template recurs throughout quantum algorithms, from Simon's problem to Shor's. *(iii) Queries vs. gates.* We count calls to U_f, treating the oracle as a unit-cost black box; the gate cost of *building* U_f for a given f must be analyzed separately and need not be small.

3.2 The unstructured search problem

We now turn to the flagship query problem. The word *unstructured* is essential: there is no order, index, or exploitable regularity among the items, so the trick behind binary search is unavailable.

Definition 3.15 (Unstructured search)**.** Let $N = 2^n$ and let $f\colon \{0,1\}^n \to \{0,1\}$ be a Boolean function to which we have only *oracle* access (Theorem 3.1). We are promised that there is a unique marked item $x_0 \in \{0,1\}^n$ with

$$f(x) = \begin{cases} 1, & x = x_0, \\ 0, & x \neq x_0. \end{cases}$$

The goal is to determine x_0 using as few queries to U_f as possible.

Each application of U_f counts as one query, and we measure the cost of an algorithm by its query complexity. Because the function has no structure, the oracle is the only way to learn anything about x_0: the string x_0 cannot be read off directly.

Proposition 3.16 (Classical lower bound)**.** *Any classical (possibly randomized) algorithm that identifies x_0 with constant success probability must make $\Omega(N)$ queries to f in the worst case.*

Sketch. A classical algorithm learns nothing about x_0 from a query $f(x)$ unless $x = x_0$. An adversary may answer $f(x) = 0$ for every queried x until only one unqueried candidate remains, which is consistent with x_0 being that last item. Thus to be certain, an algorithm must probe $N - 1$ items in the worst case. A randomized algorithm querying a uniformly random subset of size k finds x_0 with probability k/N, so achieving constant success probability requires $k = \Omega(N)$ queries. □

This abstract task captures many concrete situations: searching an unsorted database, deciding satisfiability of a constraint problem (treat any satisfying assignment as a marked item and let the oracle test assignments), or inverting a function. In every case the common structure is the same: *package "is this candidate a solution?" into an oracle*, and the classical exhaustive cost is $\mathcal{O}(N)$. Grover's algorithm attacks all of them at once.

The phase oracle. As in Section 3.1, it is convenient to convert the bit-flip oracle U_f into a *phase* oracle. Preparing the target qubit in $|-\rangle$ and applying phase kickback (Theorem 3.3) gives $U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle$; the target factors out, and on the n-qubit register the oracle acts as $|x\rangle \mapsto (-1)^{f(x)} |x\rangle$, simply attaching a minus sign to the marked basis state $|x_0\rangle$.

3.3 Grover's search algorithm

Two operators drive the algorithm. Both are *reflections*, and their product is a rotation. We first define them, prove they are genuine reflections, and then assemble the Grover iterate and analyze it geometrically.

Throughout, let

$$|\psi_0\rangle = H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{N}} \sum_{x\in\{0,1\}^n} |x\rangle$$

be the uniform superposition over all N basis states. This is the algorithm's starting state; every candidate carries amplitude $1/\sqrt{N}$.

Definition 3.17 (Marking reflection and diffusion operator)**.** The *marking reflection* (phase oracle) is

$$R_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|,$$

which flips the sign of the amplitude on $|x_0\rangle$ and leaves the orthogonal complement fixed. The *diffusion operator* is

$$R_{\psi_0} = 2|\psi_0\rangle\langle\psi_0| - \mathbb{I},$$

which fixes $|\psi_0\rangle$ and flips the sign of everything orthogonal to it. It can be written equivalently as

$$R_{\psi_0} = H^{\otimes n}(2|0^n\rangle\langle 0^n| - \mathbb{I})H^{\otimes n},$$

which is how it is implemented as a circuit.

Lemma 3.18 (Both operators are reflections)**.** *Each of R_{x_0} and R_{ψ_0} is Hermitian, unitary, and an involution ($R^2 = \mathbb{I}$); geometrically each is a reflection of the Hilbert space about a hyperplane.*

Proof. Let $|u\rangle$ be a unit vector and set $P = |u\rangle\langle u|$, so $P = P^\dagger$ and $P^2 = P$ (an orthogonal projector). Consider $R = \mathbb{I} - 2P$. Then

$$R^\dagger = \mathbb{I} - 2P^\dagger = \mathbb{I} - 2P = R,$$

so R is Hermitian. Moreover

$$R^2 = (\mathbb{I} - 2P)^2 = \mathbb{I} - 4P + 4P^2 = \mathbb{I} - 4P + 4P = \mathbb{I},$$

so R is an involution. A Hermitian involution is unitary, since $R^\dagger R = R^2 = \mathbb{I}$. Its action is $R|u\rangle = -|u\rangle$ and $R|w\rangle = |w\rangle$ for every $|w\rangle \perp |u\rangle$, which is exactly a reflection about the hyperplane $|u\rangle^\perp$.

Applying this with $|u\rangle = |x_0\rangle$ gives $R_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|$ directly. For the diffusion operator, note $R_{\psi_0} = 2|\psi_0\rangle\langle\psi_0| - \mathbb{I} = -(\mathbb{I} - 2|\psi_0\rangle\langle\psi_0|)$; by the computation above $\mathbb{I} - 2|\psi_0\rangle\langle\psi_0|$ is a Hermitian unitary involution, and negating it preserves all three properties. Explicitly $R_{\psi_0}|\psi_0\rangle = |\psi_0\rangle$ and $R_{\psi_0}|w\rangle = -|w\rangle$ for $|w\rangle \perp |\psi_0\rangle$, so R_{ψ_0} is the reflection through the axis $|\psi_0\rangle$. The identity $R_{\psi_0} = H^{\otimes n}(2|0^n\rangle\langle 0^n| - \mathbb{I})H^{\otimes n}$ follows because $H^{\otimes n}$ is unitary and $H^{\otimes n}|0^n\rangle = |\psi_0\rangle$. □

Intuitively R_{ψ_0} implements an "inversion about the mean": writing a state as $\sum_x a_x|x\rangle$ with mean amplitude $\bar{a} = \frac{1}{N}\sum_x a_x$, one checks that $R_{\psi_0}\sum_x a_x|x\rangle = \sum_x(2\bar{a} - a_x)|x\rangle$, reflecting each amplitude about $\bar{a}$ (Exercise 3.1).

Definition 3.19 (Grover iterate)**.** The *Grover iterate* is the composition of the two reflections,

$$G = R_{\psi_0}R_{x_0} = (2|\psi_0\rangle\langle\psi_0| - \mathbb{I})(\mathbb{I} - 2|x_0\rangle\langle x_0|).$$

One iteration first marks the solution (via R_{x_0}) and then reflects about the mean (via R_{ψ_0}).

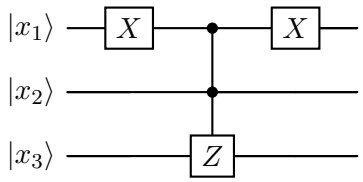
3.3.1 Building the two reflections as circuits

So far R_{x_0} and R_{ψ_0} are abstract operators. Before analyzing them it is instructive to see that both are ordinary circuits built from standard gates.

The marking reflection as a phase oracle. Recall from Section 3.1 that the bit-flip oracle U_f becomes the phase oracle $|x\rangle \mapsto (-1)^{f(x)}|x\rangle$ once the target qubit is held in $|-\rangle$: by phase kickback (Theorem 3.3) the eigenvalue $(-1)^{f(x)}$ is written onto the input register, leaving the ancilla in $|-\rangle$ to be reused on the next query. For the single-solution predicate of Theorem 3.15 this is exactly $R_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|$, the operator that flips the sign of $|x_0\rangle$ alone. It has a transparent gate-level implementation. Start from the multiply-controlled Z gate on n qubits, which multiplies $|1^n\rangle$ by -1 and fixes every other basis state, since $Z|1\rangle = -|1\rangle$ fires only when all n qubits are 1. This is precisely the phase oracle for the marked string 1^n. To mark an *arbitrary* $x_0 \in \{0,1\}^n$ instead, conjugate the multiply-controlled Z by X (NOT) gates placed on exactly those qubits where x_0 has a 0:

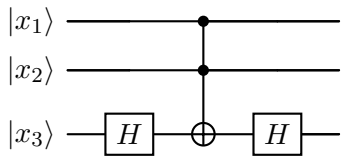
$$R_{x_0} = \left(\bigotimes_{i:\,(x_0)_i=0} X_i\right) (\text{mult. ctrl-}Z) \left(\bigotimes_{i:\,(x_0)_i=0} X_i\right).$$

The leading NOTs turn $|x_0\rangle$ into $|1^n\rangle$; the controlled-Z attaches the -1; the trailing NOTs restore $|x_0\rangle$. Every other basis state either fails to become $|1^n\rangle$ (so it is untouched) or has its two NOT layers cancel, so it emerges unchanged. For $n = 3$ and $x_0 = |011\rangle$, for instance, one places an X on the single 0-qubit before and after a doubly controlled Z, giving the circuit for R_{x_0}



Reading it through: the leading X on q_1 sends $|011\rangle \mapsto |111\rangle$, the doubly-controlled Z multiplies $|111\rangle$ by -1, and the trailing X restores $|011\rangle$; every other basis state either never reaches $|111\rangle$ or has its two X layers cancel, so only $|x_0\rangle = |011\rangle$ picks up the sign—exactly $R_{x_0} = \mathbb{I} - 2|011\rangle\langle 011|$.

Realizing the controlled-Z by kickback. The doubly (or multiply) controlled Z is itself typically compiled from a controlled-X (Toffoli, for $n = 3$) sandwiched between two Hadamards on the target, using the identity $Z = HXH$: with the target prepared by the first H in $|-\rangle$, the multi-controlled NOT kicks back a factor -1 exactly when all controls are 1, and the second H returns the target to its computational value. This is the same phase-kickback mechanism of Theorem 3.3, now used to synthesize the diagonal phase gate rather than to query an external oracle. Concretely the doubly-controlled Z above is



a Toffoli conjugated by Hadamards on the target wire ($Z = HXH$).

Remark 3.20 (We never build R_{x_0} from x_0). The circuit above hard-codes the marked string—X gates on the 0-bits of a *given* x_0—so it is illustrative only; if we already knew x_0 there would be nothing to search for. In a real search R_{x_0} is the phase oracle compiled from the *verifier* f

of Theorem 3.15, the reversible circuit that *tests* whether a candidate is a solution. Building it needs only the ability to *recognize* a solution, not to *locate* one—the "hard to find, easy to check" structure of search. The marked string x_0 is whatever input makes f fire; Grover surfaces it at the final measurement, after which one classical evaluation of f confirms it.

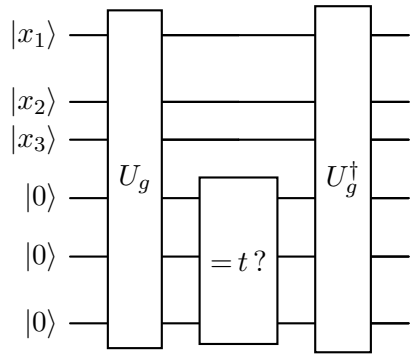
Example 3.21 (Compiling R_{x_0} from a verifier)**.** Most searches take the form "find x with $g(x) = t$" for an efficiently computable g and a known target t (satisfiability, function inversion, and database matching all fit this mold). The verifier is $f(x) = [\, g(x) = t \,]$, and R_{x_0} is assembled in three stages that use only g and t—never x_0:

(1) *Compute.* A reversible circuit U_g writes the result into a fresh ancilla register: $|x\rangle\,|0^m\rangle \mapsto |x\rangle\,|g(x)\rangle$.

(2) *Mark.* Flip the phase iff the ancilla equals t—the X-conjugated multiply-controlled Z of the previous figure, now applied to the *ancilla* register with Xs on the bits where $t_j = 0$. Only the known target t enters.

(3) *Uncompute.* Apply $U_g^\dagger$ to reset the ancilla to $|0^m\rangle$ (Bennett), disentangling it from the data.

The net action on the data register is $|x\rangle \mapsto (-1)^{[\,g(x)=t\,]}\,|x\rangle = R_{x_0}\,|x\rangle$, with $x_0 = g^{-1}(t)$ the solution we never had to find:



For instance take $g(x) = (x_1,\ x_1 \oplus x_2,\ x_1x_2 \oplus x_3)$—three CNOT/Toffoli writes into the ancilla—and $t = 101$. This oracle marks the unique $x_0 = 110$, since $g(110) = 101$, yet its construction referred only to the circuit for g and the target t. Here g is small enough to invert by hand, but the identical recipe compiles the oracle for a thousand-variable SAT instance or a cryptographic one-way g—in size proportional to the cost of evaluating f—while x_0 stays out of reach until amplitude amplification surfaces it.

Remark 3.22 (Examples of search oracles). The verifier g (equivalently $f = [\, g(x) = t \,]$) can be *any* efficiently computable function—anything for which one can write a fast "is x a solution?" check. Grover is the generic tool when checking is easy but the problem is *unstructured*. Representative oracles:

- *Symmetric-key search.* x is a secret key k, $g(k) = \mathrm{Enc}_k(P_0)$ for a known plaintext P_0, and t the observed ciphertext; x_0 is the key. The quadratic speedup halves the effective key length (e.g. $2^{128} \to 2^{64}$ for AES-128), the reason post-quantum guidance doubles symmetric key sizes.
- *Hash preimage.* $g = H$ a cryptographic hash and t a target digest; find x with $H(x) = t$.
- *Satisfiability / constraint problems.* $g(x) = \varphi(x)$ with $t = 1$: "does x satisfy every clause?"—likewise Sudoku, graph coloring, and scheduling.

- *Unstructured table.* $g(x) = L[x]$ for a fixed unsorted table L; find the index matching t. E.g. $L = [5, 2, 7, 4, 1, 6, 0, 3]$ and $t = 4$ give $x_0 = 3$: with no structure, trying entries is the only option—the purest Grover setting.
- *Optimization by thresholding.* $g(x) = [\,\mathrm{cost}(x) \le \theta\,]$; binary-searching θ reduces minimization to a sequence of searches.

If instead g has exploitable structure (periodicity for Shor, linearity for Bernstein–Vazirani, convexity for classical optimization), a tailored algorithm beats Grover; Grover is the universal fallback, giving a quadratic speedup over blind search whenever all one has is a checker.

The diffusion operator as a circuit. The diffusion reflection admits the factorization already recorded in Theorem 3.17,

$$R_{\psi_0} = H^{\otimes n}\big(2\,|0^n\rangle\langle 0^n| - \mathbb{I}\big)H^{\otimes n}.$$

The inner operator $2\,|0^n\rangle\langle 0^n| - \mathbb{I}$ is a reflection about $|0^n\rangle$: it fixes $|0^n\rangle$ and negates everything else, which (up to a global -1) is the same multiply-controlled phase gate as above, now conditioned on the all-0 string and so obtained by wrapping the controlled-Z in NOTs on *every* qubit. The flanking Hadamard layers map $|\psi_0\rangle \leftrightarrow |0^n\rangle$, so conjugation converts "reflect about $|0^n\rangle$" into "reflect about $|\psi_0\rangle$"—the inversion about the mean. Thus a full Grover iterate $G = R_{\psi_0}R_{x_0}$ is: one phase-oracle call, then a layer of H, a reflection about $|0^n\rangle$, and a final layer of H.

The catch, and why the algorithm is still useful. Building R_{x_0} as above appears to require knowing x_0, which is the very thing we are searching for. In genuine applications one does *not* hardcode x_0: the oracle is a reversible circuit for a *predicate* $f(x) = [\,x \text{ is a solution}\,]$—"does this candidate satisfy the constraints?"—that can be evaluated without knowing which inputs pass. The phase oracle then marks whatever inputs happen to satisfy f, and Grover's iteration amplifies them. Counting oracle calls (Theorem 3.1) rather than elementary gates is what makes this separation of concerns clean: the cost of compiling f into a circuit is charged separately, exactly as in Theorem 3.14.

3.3.2 The invariant plane and the rotation picture

The power of the algorithm comes from the fact that G never leaves a fixed two-dimensional plane, inside which it acts as an ordinary rotation.

Define the normalized state orthogonal to $|x_0\rangle$ that lives in the "bad" subspace:

$$|\psi_0^\perp\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq x_0} |x\rangle\,.$$

Then $\langle x_0|\psi_0^\perp\rangle = 0$, and the uniform state decomposes as

$$|\psi_0\rangle = \frac{1}{\sqrt{N}}\,|x_0\rangle + \frac{\sqrt{N-1}}{\sqrt{N}}\,|\psi_0^\perp\rangle = \sin\frac{\theta}{2}\,|x_0\rangle + \cos\frac{\theta}{2}\,|\psi_0^\perp\rangle\,, \tag{3.2}$$

where the angle $\theta \in (0, \pi)$ is defined by

$$\sin\frac{\theta}{2} = \frac{1}{\sqrt{N}}, \qquad \theta = 2\arcsin\frac{1}{\sqrt{N}} \approx \frac{2}{\sqrt{N}} \quad (N \gg 1). \tag{3.3}$$

So $|\psi_0\rangle$ makes a small angle $\theta/2$ with the axis $|\psi_0^\perp\rangle$.

Theorem 3.23 (G is a rotation in the plane $\mathcal{P}$)**.** *Let $\mathcal{P} = \operatorname{span}\{|x_0\rangle, |\psi_0^\perp\rangle\}$. Then G maps $\mathcal{P}$ into itself, and restricted to $\mathcal{P}$ (in the ordered orthonormal basis $\{|\psi_0^\perp\rangle, |x_0\rangle\}$) it is the rotation by the angle θ of (3.3):*

$$G|_{\mathcal{P}} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Proof. First note that both reflections preserve $\mathcal{P}$. The vector $|x_0\rangle$ lies in $\mathcal{P}$ and $|\psi_0\rangle \in \mathcal{P}$ by (3.2), so $|\psi_0^\perp\rangle \in \mathcal{P}$ as well. The operator R_{x_0} negates $|x_0\rangle$ and fixes every vector orthogonal to $|x_0\rangle$; in particular it fixes $|\psi_0^\perp\rangle$, so it maps $\mathcal{P}$ to itself. The operator R_{ψ_0} fixes $|\psi_0\rangle \in \mathcal{P}$ and negates its orthogonal complement; the in-plane vector orthogonal to $|\psi_0\rangle$ inside $\mathcal{P}$ stays in $\mathcal{P}$, so R_{ψ_0} also preserves $\mathcal{P}$. Hence $G = R_{\psi_0}R_{x_0}$ preserves $\mathcal{P}$.

We now use the plane-geometry fact that *the product of two reflections is a rotation by twice the angle between their axes* (Exercise 3.7). Inside $\mathcal{P}$, the mirror line of R_{x_0} is the axis $|\psi_0^\perp\rangle$ (the fixed direction of R_{x_0} within $\mathcal{P}$), and the mirror line of R_{ψ_0} is the axis $|\psi_0\rangle$. By (3.2) the angle between $|\psi_0^\perp\rangle$ and $|\psi_0\rangle$ is $\theta/2$, so G is a rotation by $2 \cdot (\theta/2) = \theta$.

For a self-contained verification, compute the matrix directly. Using $\sin\frac{\theta}{2} = 1/\sqrt{N}$, $\cos\frac{\theta}{2} = \sqrt{(N-1)/N}$ and (3.2), apply $R_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|$:

$$R_{x_0}|\psi_0^\perp\rangle = |\psi_0^\perp\rangle, \qquad R_{x_0}|x_0\rangle = -|x_0\rangle.$$

Thus in the basis $\{|\psi_0^\perp\rangle, |x_0\rangle\}$, $R_{x_0} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, a reflection about the $|\psi_0^\perp\rangle$ axis. Next $R_{\psi_0} = 2|\psi_0\rangle\langle\psi_0| - \mathbb{I}$ is the reflection about the axis $|\psi_0\rangle$, which sits at angle $\theta/2$ from $|\psi_0^\perp\rangle$; a reflection about a line at angle ϕ from the first basis vector has matrix $\left(\begin{smallmatrix} \cos 2\phi & \sin 2\phi \\ \sin 2\phi & -\cos 2\phi \end{smallmatrix}\right)$, so with $\phi = \theta/2$,

$$R_{\psi_0} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}.$$

Multiplying,

$$G = R_{\psi_0}R_{x_0} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

which is exactly the rotation by θ. □

Theorem 3.24 (State after k iterations)**.** *With θ as in (3.3),*

$$G^k|\psi_0\rangle = \sin\frac{(2k+1)\theta}{2}\,|x_0\rangle + \cos\frac{(2k+1)\theta}{2}\,|\psi_0^\perp\rangle.$$

Consequently the probability of measuring the marked item after k iterations is $p_k = \sin^2((2k+1)\theta/2)$.

Proof. By Theorem 3.23, G restricted to $\mathcal{P}$ is rotation by θ, so G^k is rotation by $k\theta$. The initial state (3.2) sits at angle $\theta/2$ measured from the $|\psi_0^\perp\rangle$ axis (its $|x_0\rangle$-component is $\sin(\theta/2)$). Rotating by $k\theta$ increases this angle to $\theta/2 + k\theta = (2k+1)\theta/2$. Hence the $|x_0\rangle$-component becomes $\sin((2k+1)\theta/2)$ and the $|\psi_0^\perp\rangle$-component becomes $\cos((2k+1)\theta/2)$, as claimed. Measuring in the computational basis yields x_0 with probability equal to the squared $|x_0\rangle$-amplitude, namely $p_k = \sin^2((2k+1)\theta/2)$. □

Example 3.25 (A fully worked run: $N = 16$, marked item $x_0 = 5$)**.** It is worth watching every amplitude move once, on a case small enough to write out yet large enough that several iterations are needed—and large enough to see what happens if one runs *too* many. Take four qubits, $N =$

16, and mark the basis state $|x_0\rangle = |5\rangle = |0101\rangle$. From (3.3), $\sin(\theta/2) = 1/\sqrt{16} = 1/4$, so $\theta = 2\arcsin(1/4) \approx 28.96°$ and $(\pi/2\theta) - \frac{1}{2} \approx 2.64$; rounding gives $k^* = 3$ iterations. Because the amplitudes stay uniform across the fifteen unmarked states throughout, it suffices to track two numbers per step: the common amplitude a on the unmarked states and the amplitude g on $|x_0\rangle$. Recall that the oracle R_{x_0} flips the sign of g and that the diffusion R_{ψ_0} sends every amplitude $a_x \mapsto 2\bar{a} - a_x$, reflecting it about the mean $\bar{a}$ (Exercise 3.1).

Start. $|\psi_0\rangle = H^{\otimes 4}|0000\rangle$ gives every amplitude equal to $1/4$, so measuring now returns $|x_0\rangle$ with probability $1/16$.

Iteration 1. The oracle sends g: $1/4 \mapsto -1/4$ while the fifteen unmarked amplitudes are unchanged. The mean is $\bar{a} = (15 \cdot \frac{1}{4} - \frac{1}{4})/16 = \frac{7}{32}$. Inversion about the mean gives, for an unmarked amplitude, $2\bar{a} - \frac{1}{4} = \frac{3}{16}$, and for the marked one $2\bar{a} + \frac{1}{4} = \frac{11}{16}$, so

$$G|\psi_0\rangle = \frac{3}{16}\sum_{x\neq 5}|x\rangle + \frac{11}{16}|5\rangle, \qquad p_1 = \left(\tfrac{11}{16}\right)^2 = \frac{121}{256} \approx 0.473.$$

Iteration 2. The oracle flips the marked amplitude to $-\frac{11}{16}$; the new mean is $\bar{a} = (15 \cdot \frac{3}{16} - \frac{11}{16})/16 = \frac{17}{128}$. Inversion yields unmarked $2\bar{a} - \frac{3}{16} = \frac{5}{64}$ and marked $2\bar{a} + \frac{11}{16} = \frac{61}{64}$, so

$$G^2|\psi_0\rangle = \frac{5}{64}\sum_{x\neq 5}|x\rangle + \frac{61}{64}|5\rangle, \qquad p_2 = \left(\tfrac{61}{64}\right)^2 = \frac{3721}{4096} \approx 0.908.$$

Iteration 3 (the optimum). The oracle flips the marked amplitude to $-\frac{61}{64}$; the mean is $\bar{a} = (15 \cdot \frac{5}{64} - \frac{61}{64})/16 = \frac{7}{512}$. Inversion yields unmarked $2\bar{a} - \frac{5}{64} = -\frac{13}{256}$ and marked $2\bar{a} + \frac{61}{64} = \frac{251}{256}$, so

$$G^3|\psi_0\rangle = -\frac{13}{256}\sum_{x\neq 5}|x\rangle + \frac{251}{256}|5\rangle, \qquad p_3 = \left(\tfrac{251}{256}\right)^2 = \frac{63001}{65536} \approx 0.961.$$

This is the maximum: $p_3 \approx 0.961$ already exceeds the guarantee $1 - 1/N = 15/16$ of Theorem 3.26, and the small *negative* amplitude on the unmarked states shows the state has just edged past $|x_0\rangle$.

Iterations 4 and 5 (over-rotation). Nothing stops us from continuing, but the geometry now works against us: each further G rotates the state by another θ, carrying it *away* from $|x_0\rangle$. Running the same recurrence twice more gives marked amplitudes $\frac{781}{1024}$ and then $\frac{1451}{4096}$,

$$p_4 = \left(\tfrac{781}{1024}\right)^2 = \frac{609961}{1048576} \approx 0.582, \qquad p_5 = \left(\tfrac{1451}{4096}\right)^2 = \frac{2105401}{16777216} \approx 0.126.$$

The success probability falls from 0.961 back down toward its starting value—a concrete instance of the over-rotation warned of in Theorem 3.27. All six stages are collected in Table 3.1; the entries agree with the closed form $p_k = \sin^2((2k+1)\theta/2)$ of Theorem 3.24.

3.3.3 Choosing the number of iterations

Theorem 3.24 shows the amplitude on $|x_0\rangle$ is maximized when the angle $(2k+1)\theta/2$ is as close as possible to $\pi/2$, i.e. when the state is (nearly) aligned with $|x_0\rangle$. This fixes the optimal iteration count.

Theorem 3.26 (Optimal number of iterations)**.** *Choose k so that $(2k+1)\theta/2$ is closest to $\pi/2$; the optimal choice is*

$$k^* = \left\lfloor \frac{\pi}{4}\sqrt{N} \right\rfloor \qquad (\textit{rounding } \tfrac{\pi}{2\theta} - \tfrac{1}{2}).$$

With this choice the success probability satisfies $p_{k^*} \geq 1 - 1/N$, so a single run succeeds with probability ≈ 1. The algorithm therefore uses $k^* = \mathcal{O}(\sqrt{N})$ oracle queries.

iteration k	angle $\frac{(2k+1)\theta}{2}$	unmarked amp.	marked amp. ($\|5\rangle$)	$p_k = \|g\|^2$
0 (start)	14.5°	1/4	1/4	$1/16 \approx 0.063$
1	43.4°	3/16	11/16	$121/256 \approx 0.473$
2	72.4°	5/64	61/64	$3721/4096 \approx 0.908$
3 (optimum)	101.3°	−13/256	251/256	$63001/65536 \approx 0.961$
4	130.3°	−171/1024	781/1024	$609961/1048576 \approx 0.582$
5	159.3°	−989/4096	1451/4096	$2105401/16777216 \approx 0.126$

Table 3.1: Amplitudes after each Grover iteration for $N = 16$, $x_0 = 5$ (Theorem 3.25). All fifteen unmarked states share a common amplitude at every stage. The success probability climbs to its peak at the optimum $k^* = 3$ and then *falls* as iterations 4 and 5 over-rotate the state past $|x_0\rangle$; each row matches the closed form $p_k = \sin^2((2k+1)\theta/2)$.

Proof. We want $(2k+1)\theta/2 \approx \pi/2$, i.e. $2k+1 \approx \pi/\theta$, i.e. $k \approx \pi/(2\theta) - 1/2$. Using the approximation $\theta \approx 2/\sqrt{N}$ from (3.3) valid for large N,

$$k \approx \frac{\pi}{2\theta} - \frac{1}{2} \approx \frac{\pi}{2} \cdot \frac{\sqrt{N}}{2} - \frac{1}{2} = \frac{\pi}{4}\sqrt{N} - \frac{1}{2},$$

and rounding to the nearest nonnegative integer gives $k^* = \lfloor (\pi/4)\sqrt{N} \rfloor$. Because k is an integer we cannot in general hit the angle $\pi/2$ exactly, but the residual angular error is at most $\theta/2$ (half a rotation step): choosing k^* so that $(2k^*+1)\theta/2$ lies within $\theta/2$ of $\pi/2$ gives

$$p_{k^*} = \sin^2 \frac{(2k^*+1)\theta}{2} = \cos^2\Big(\frac{\pi}{2} - \frac{(2k^*+1)\theta}{2}\Big) \geq \cos^2 \frac{\theta}{2} = 1 - \sin^2 \frac{\theta}{2} = 1 - \frac{1}{N}.$$

Each iterate $G = R_{\psi_0} R_{x_0}$ makes exactly one oracle call (inside R_{x_0}), so the total query cost is $k^* = \mathcal{O}(\sqrt{N})$. □

This is the headline result: $\mathcal{O}(\sqrt{N})$ queries versus the classical $\mathcal{O}(N)$, a quadratic speedup. The gain comes from operating on *amplitudes* rather than probabilities: G rotates the amplitude linearly in k, and probability is its square, so the linear growth of the angle produces the square-root scaling in k.

Remark 3.27 (Do not over-rotate). More iterations is *not* always better. Since $p_k = \sin^2((2k+1)\theta/2)$ oscillates, once $(2k+1)\theta/2$ passes $\pi/2$ the success probability decreases again—the state "rotates past" $|x_0\rangle$. One must therefore stop at k^*; running the loop too long is counterproductive. If N is unknown one can still succeed with $\mathcal{O}(\sqrt{N})$ expected queries by trying geometrically increasing guesses for the number of iterations and choosing the number at random within each guess.

3.3.4 Running Grover in practice

The state-after-k formula (Theorem 3.24) and the optimal count k^* (Theorem 3.26) are all we need to search in earnest. The two examples below put them to work and make a point easily lost in the geometry: the algorithm never uses the *value* of the marked string x_0, only the oracle that recognizes it.

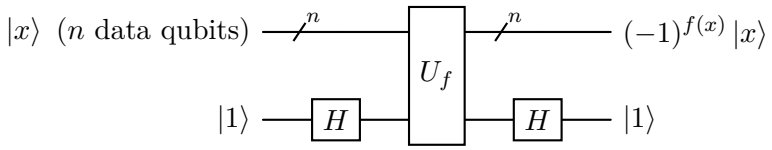
Example 3.28 (Running Grover in practice: we never use x_0)**.** Theorem 3.25 tracked the amplitude on $|x_0\rangle = |5\rangle$ for illustration, but the algorithm itself never knows x_0: the only access to f is through

the black-box oracle R_{x_0}, and the mark $|5\rangle$ was baked into that oracle by whoever posed the search, not by us. What, then, do we actually run?

Where the oracle comes from. We are not handed x_0; we are handed a way to *recognize* it—a predicate f with $f(x) = 1$ exactly when $x = x_0$, supplied as the reversible *bit* oracle $U_f\colon |x\rangle\,|y\rangle \mapsto |x\rangle\,|y \oplus f(x)\rangle$ of Theorem 3.1, which writes the *value* of f into an extra qubit. Grover, on the other hand, needs the *phase* oracle $R_{x_0} = \mathbb{I} - 2\,|x_0\rangle\langle x_0|$, which writes no bit but attaches a sign to $|x_0\rangle$. These are the same operator viewed through a single ancilla. Prepare that ancilla in $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$; since the controlled flip $|y\rangle \mapsto |y \oplus f(x)\rangle$ either leaves $|-\rangle$ alone ($f(x) = 0$) or sends it to $-|-\rangle$ ($f(x) = 1$), phase kickback (Theorem 3.3) gives

$$U_f(|x\rangle \otimes |-\rangle) = (-1)^{f(x)}\,|x\rangle \otimes |-\rangle = (R_{x_0}\,|x\rangle) \otimes |-\rangle\,, \qquad R_{x_0} = \sum_x (-1)^{f(x)}\,|x\rangle\langle x| = \mathbb{I} - 2\,|x_0\rangle\langle x_0|\,,$$

the last equality because $(-1)^{f(x)} = +1$ everywhere except at the single string $x = x_0$, where it is -1. In words: *with the target held in $|-\rangle$, the bit oracle U_f acts on the data register exactly as the phase oracle R_{x_0}*—the ancilla emerges unchanged and factors off, carrying the sign back onto $|x\rangle$. Concretely,



the ancilla's two H gates turn $|1\rangle$ into $|-\rangle$ and back, so it returns to $|1\rangle$ and may be reused or discarded; everything between them, read on the top register, *is* R_{x_0}. Note what this circuit contains: the gates of U_f and one ancilla, and *never* x_0 *itself*—the minus sign lands on whichever input satisfies f, wherever that happens to be. (If U_f allocates scratch qubits while evaluating f, uncompute them afterwards so they return to $|0\rangle$ and leave the data register unentangled.) This is the compilation of Theorem 3.21, and Theorem 3.29 below carries it out for a concrete, genuinely unknown x_0.

The one quantity we *do* know is the size of the search space, $N = 16$ (and, here, that there is a single solution, $M = 1$). That is enough, because by Theorem 3.26 the optimal iteration count depends only on N,

$$k^* = \left\lfloor \tfrac{\pi}{4}\sqrt{N} \right\rfloor = \left\lfloor \tfrac{\pi}{4} \cdot 4 \right\rfloor = \lfloor \pi \rfloor = 3,$$

not on *which* item is marked. So the procedure is:

(1) Prepare $|\psi_0\rangle = H^{\otimes 4}\,|0000\rangle$ and apply the Grover iterate $G = R_{\psi_0} R_{x_0}$ exactly $k^* = 3$ times. Here R_{x_0} is the oracle we are *handed*: we run it as a black box, and although its subscript x_0 records that it marks a single item (so $R_{x_0} = \mathbb{I} - 2\,|x_0\rangle\langle x_0|$), the value of that item is unknown to us—the circuit never reads it.

(2) Measure the four qubits, obtaining a candidate string $y \in \{0,1\}^4$.

(3) *Verify*: evaluate the predicate once. If $f(y) = 1$, output y; otherwise discard it and repeat from (1).

The verification in step (3) is a single classical evaluation of f, and it is what makes the algorithm trustworthy without ever referring to x_0: a wrong guess is detected and costs nothing but a repeat. By Theorem 3.25 the measurement in step (2) returns the true $x_0 = 5$ with probability $p_3 = 63001/65536 \approx 0.961$, so the expected number of attempts is $1/p_3 \approx 1.04$—almost always the first one. Each attempt uses $k^* = 3$ oracle calls inside the Grover iterations plus one f-evaluation to verify, for an expected total of about four queries.

Contrast this with brute force: testing items one at a time finds the marked string in 8 queries on average and 15 in the worst case. Even at $N = 16$ Grover is already ahead, and the gap widens as $\sqrt{N}$ against N.

One caveat: the recipe used $M = 1$ to fix k^*. If the number of solutions M is *unknown*, one cannot pin down k^* in advance—too few iterations and the amplitude is still small, too many and it over-rotates (Theorem 3.27). The standard fix is to try exponentially growing guesses $k = 1, 2, 4, 8, \ldots$, drawing the actual count at random below each guess and verifying after every run; this still finds a solution in $\mathcal{O}(\sqrt{N})$ expected queries. The multi-solution case is treated below.

Example 3.29 (A search whose solution we genuinely do not know: inverting a function)**.** Theorem 3.28 still *named* the answer, $x_0 = 5$, so that we could watch the amplitudes move; in a real search we would not know it. Here is the same algorithm applied to a problem whose solution is *not* available in advance.

Let $g : \{0,1\}^4 \to \{0,1\}^4$ be a function we can evaluate but not easily invert—think of a toy block cipher $g(x) = E_x(m_0)$ that encrypts a fixed, known plaintext m_0 under the 4-bit key x. Somebody chose a secret key x_0 and published the ciphertext $t = g(x_0)$. Our task is to recover the key: find x with $g(x) = t$. Assume the key is unique for this t, so there is a single marked item—but *we have no idea which of the sixteen strings it is*, and no shortcut to find out short of search.

Building the oracle without knowing x_0. The predicate $f(x) = \mathbf{1}[g(x) = t]$ is something we *can* compute even though we cannot invert g: run the circuit for g on x, compare the output to the published t, and set a flag bit. Turning this verifier into the phase oracle $R_{x_0} = \mathbb{I} - 2\,|x_0\rangle\langle x_0|$ is exactly the construction of Theorem 3.21—compute f into an ancilla held in $|-\rangle$, which stamps a minus sign on precisely the satisfying input, then uncompute the scratch. At no point does this circuit contain x_0; it contains g and t, and the mark *emerges* at whatever input happens to satisfy $g(x) = t$.

Running it. The quantum circuit is now *identical* to that of Theorem 3.25: prepare $|\psi_0\rangle = H^{\otimes 4}\,|0000\rangle$, apply $G = R_{\psi_0} R_{x_0}$ for $k^* = \lfloor (\pi/4)\sqrt{16} \rfloor = 3$ iterations, and measure. Because the amplitude analysis of Theorem 3.25 never used the *value* of x_0—only that there was one marked item among $N = 16$—the state traces out the very same trajectory, and the measurement returns the hidden key with probability $p_3 = 63001/65536 \approx 0.961$, *whatever that key turns out to be.* We then evaluate g once on the measured string to confirm $g(y) = t$; on the rare miss we simply repeat. Only at this verification step do we learn the answer—the quantum part discovered it without ever being told.

The contrast with classical key search is the whole point. Not knowing x_0, a classical attacker can only try keys one at a time, expecting 8 and worst-case 15 evaluations of g; Grover needs $k^* = 3$ queries to the g-based oracle plus one verification. This quadratic speedup for brute-force inversion—in effect halving the key length an adversary must contend with—is the standard reason symmetric-key sizes are doubled to stay secure against quantum attack.

3.4 Multiple solutions and amplitude amplification

The single-solution analysis extends in two directions: to several marked items at once, and—more broadly—to amplifying the success amplitude of *any* state preparation, not just the uniform superposition. Both reuse the two-reflection rotation of Theorem 3.23 verbatim, changing only what plays the role of the "good" subspace.

3.4.1 Multiple solutions

Suppose now that there are $M \geq 1$ marked items, $S = \{x : f(x) = 1\}$ with $|S| = M$. Everything goes through with the "good" subspace replaced by the span of the marked basis states. Let

$$|\text{good}\rangle = \frac{1}{\sqrt{M}} \sum_{x \in S} |x\rangle, \qquad |\text{bad}\rangle = \frac{1}{\sqrt{N-M}} \sum_{x \notin S} |x\rangle,$$

so that $|\psi_0\rangle = \sqrt{M/N}\,|\text{good}\rangle + \sqrt{(N-M)/N}\,|\text{bad}\rangle$. Define $R_S = \mathbb{I} - 2\sum_{x \in S} |x\rangle\langle x|$ (the phase oracle now flips all marked items) and keep the same diffusion R_{ψ_0}. The identical argument shows $G = R_{\psi_0} R_S$ is a rotation by θ in $\text{span}\{|\text{good}\rangle, |\text{bad}\rangle\}$ with

$$\sin\frac{\theta}{2} = \sqrt{\frac{M}{N}}, \qquad \text{equivalently} \qquad \sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2\sqrt{\frac{M}{N}}\sqrt{\frac{N-M}{N}} \approx 2\sqrt{\frac{M}{N}}.$$

The optimal iteration count becomes

$$k^* = \left\lfloor \frac{\pi}{4}\sqrt{\frac{N}{M}} \right\rfloor = \mathcal{O}\left(\sqrt{\frac{N}{M}}\right),$$

so more solutions make the search faster.

Example 3.30 (The exact case $N = 4$, $M = 1$)**.** Here $\sin(\theta/2) = 1/\sqrt{4} = 1/2$, so $\theta/2 = \pi/6$ and $\theta = \pi/3$. After a single iteration ($k = 1$) the angle is

$$\frac{(2 \cdot 1 + 1)\theta}{2} = \frac{3\theta}{2} = \frac{3}{2} \cdot \frac{\pi}{3} = \frac{\pi}{2},$$

so by Theorem 3.24 the state is $G|\psi_0\rangle = \sin(\pi/2)\,|x_0\rangle + \cos(\pi/2)\,|\psi_0^\perp\rangle = |x_0\rangle$. Thus for $N = 4$ a *single* Grover iteration finds the marked item with certainty—a rare instance where the algorithm is exactly deterministic.

Example 3.31 (A multi-solution estimate)**.** Suppose a SAT instance on $n = 30$ variables has $M = 2^{10}$ satisfying assignments among $N = 2^{30}$ candidates. Classical exhaustive search needs on the order of $N/M = 2^{20} \approx 10^6$ trials to hit a solution. Grover instead needs about

$$\frac{\pi}{4}\sqrt{\frac{N}{M}} = \frac{\pi}{4}\sqrt{2^{20}} = \frac{\pi}{4} \cdot 2^{10} \approx 804$$

oracle queries—the square-root improvement in action.

3.4.2 Amplitude amplification

Grover's construction generalizes far beyond search.

Theorem 3.32 (Amplitude amplification)**.** *Let $\mathcal{A}$ be any quantum procedure preparing a state*

$$\mathcal{A}|0\rangle = \sqrt{p_0}\,|\psi_{\text{good}}\rangle + \sqrt{1-p_0}\,|\psi_{\text{bad}}\rangle,$$

where a projector P recognizes the good part ($P|\psi_{\text{good}}\rangle = |\psi_{\text{good}}\rangle$, $P|\psi_{\text{bad}}\rangle = 0$) and $p_0 = \langle 0|\mathcal{A}^\dagger P \mathcal{A}|0\rangle$ is the initial success probability. Define the two reflections $S_P = \mathbb{I} - 2P$ and $S_0 =$

$\mathcal{A}(2\,|0\rangle\langle 0| - \mathbb{I})\mathcal{A}^\dagger$, *and set* $Q = S_0 S_P$. *Then* Q *rotates by an angle* θ *with* $\sin^2(\theta/2) = p_0$ *inside* $\operatorname{span}\{|\psi_{\text{good}}\rangle, |\psi_{\text{bad}}\rangle\}$, *and after*

$$k^* = \left\lfloor \frac{\pi}{4 \arcsin \sqrt{p_0}} \right\rfloor = \mathcal{O}(1/\sqrt{p_0})$$

iterations the good state is obtained with probability $\Omega(1)$.

Sketch. The state $\mathcal{A}\,|0\rangle$ decomposes in the two-dimensional space $\operatorname{span}\{|\psi_{\text{good}}\rangle, |\psi_{\text{bad}}\rangle\}$ exactly as $|\psi_0\rangle$ did in (3.2), with $\sin(\theta/2) = \sqrt{p_0}$. The reflection S_P negates the good component and fixes the bad one; the reflection $S_0 = \mathcal{A}(2\,|0\rangle\langle 0| - \mathbb{I})\mathcal{A}^\dagger$ is, by conjugating the elementary reflection $2\,|0\rangle\langle 0| - \mathbb{I}$ by the unitary $\mathcal{A}$, the reflection about the axis $\mathcal{A}\,|0\rangle$. The proof of Theorem 3.23 applies verbatim with $|x_0\rangle$ replaced by $|\psi_{\text{good}}\rangle$ and $|\psi_0\rangle$ by $\mathcal{A}\,|0\rangle$: the product of the two reflections is a rotation by θ. After k iterations the good amplitude is $\sin((2k+1)\theta/2)$, maximized at $k^* \approx \pi/(2\theta) - 1/2$ with $\theta = 2 \arcsin \sqrt{p_0}$, giving $k^* = \mathcal{O}(1/\sqrt{p_0})$ since $\theta \geq 2\sqrt{p_0}$ for small p_0. □

This is a quadratic improvement over the $\mathcal{O}(1/p_0)$ repetitions that a classical prepare-and-check strategy would require to see the good outcome. Grover search is the special case $\mathcal{A} = H^{\otimes n}$, $p_0 = M/N$.

3.5 Query complexity lower bounds

Grover's algorithm is not just good; it is asymptotically the best possible.

Theorem 3.33 (Query lower bound)**.** *Any quantum algorithm that finds the marked item* x_0 *(or even merely decides whether a marked item exists) with constant success probability must make* $\Omega(\sqrt{N})$ *oracle queries.*

Idea. Run the algorithm both on the "empty" oracle $f \equiv 0$ and on the oracle marking a single item x_0. Let $|\phi_k\rangle$ be the state after k queries in the empty case and $|\phi_k^{x_0}\rangle$ the state in the marked case, and define the total deviation

$$D_k = \sum_{x_0} \left\| \, |\phi_k^{x_0}\rangle - |\phi_k\rangle \, \right\|^2 .$$

A hybrid argument shows each additional query can increase D_k by only a bounded amount: $D_k \leq 4k^2$. On the other hand, to reliably distinguish which item is marked, the marked-case states for different x_0 must be nearly orthogonal to the empty-case state, which forces $D_k = \Omega(N)$. Combining $\Omega(N) \leq D_k \leq 4k^2$ gives $k = \Omega(\sqrt{N})$. □

Remark 3.34 (Grover is optimal)*.* The upper bound of Theorem 3.26 and the lower bound above match up to constant factors: both are $\Theta(\sqrt{N})$. Hence Grover's algorithm is optimal for unstructured search—no quantum algorithm can do asymptotically better. The quadratic speedup is the best achievable when the only access to the problem is through an unstructured oracle; a super-polynomial speedup would require exploitable structure, as in Shor's algorithm or the phase-estimation-based methods of Chapter 4.

Exercises

Exercise 3.1. Show that the diffusion operator implements "inversion about the mean": for any state $\sum_x a_x\,|x\rangle$ with mean amplitude $\bar{a} = \frac{1}{N}\sum_x a_x$,

$$R_{\psi_0} \sum_x a_x\,|x\rangle = \sum_x (2\bar{a} - a_x)\,|x\rangle .$$

Exercise 3.2. List all four functions $f : \{0,1\} \to \{0,1\}$ and classify each as constant or balanced. For each, compute by hand the first-qubit measurement outcome produced by the Deutsch algorithm of Theorem 3.7.

Exercise 3.3. Prove that U_f of Theorem 3.1 is Hermitian, and use this together with $U_f^2 = \mathbb{I}$ to conclude that its eigenvalues are ± 1. Then show directly (without invoking Theorem 3.3) that if the target qubit is initialized in $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ instead of $|-\rangle$, then $U_f |x\rangle |+\rangle = |x\rangle |+\rangle$ for all x, so no phase is kicked back; explain why $|-\rangle$ is the useful choice.

Exercise 3.4. For $n = 2$, let $f(x_1x_2) = x_1 \oplus x_2$. Verify that f is balanced and trace the Deutsch–Jozsa state through $H^{\otimes 2}$, U_f, $H^{\otimes 2}$, confirming that the amplitude of $|00\rangle$ is 0. In the Bernstein–Vazirani setting with hidden string s, suppose instead the oracle is for the affine function $g(x) = s \cdot x \oplus c$ with an unknown constant bit c; show that the algorithm of Theorem 3.13 still outputs s with certainty, and identify where the constant c goes.

Exercise 3.5. Take $N = 2$ (one qubit, $n = 1$) with the single marked item x_0. Compute θ from (3.3), apply one Grover iterate to $|\psi_0\rangle$, and interpret the result. Why is the case $N = 2$ slightly special? Then, for general N with $k^* = \lfloor (\pi/4)\sqrt{N} \rfloor$, show that the success probability satisfies $p_{k^*} \geq 1 - 1/N$ (you may use that the residual angle from $\pi/2$ is at most $\theta/2$).

Exercise 3.6. Consider $N = 2^{20}$. Approximately how many iterations k^* maximize the success probability? What happens qualitatively to the success probability if one instead runs $2k^*$ iterations? Explain using $p_k = \sin^2((2k+1)\theta/2)$. Separately, for $M = 4$ marked items among $N = 1024$, find $\sin(\theta/2)$, the optimal iteration count k^*, and compare with the single-solution count for the same N.

Exercise 3.7. Prove the plane-geometry fact used in Theorem 3.23: in a real two-dimensional plane, the composition of a reflection about a line ℓ_1 and a reflection about a line ℓ_2 (with ℓ_2 at angle α from ℓ_1) is a rotation by 2α.

Exercise 3.8. Building the phase oracle. On three qubits, let $x_0 = |101\rangle$ be the marked state. Following Section 3.3.1, write the marking reflection $R_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|$ as a doubly controlled Z conjugated by X gates, specifying which qubits carry the X gates. Verify by hand that your circuit multiplies $|101\rangle$ by -1 and leaves the other seven basis states fixed (it suffices to check $|101\rangle$, one basis state that becomes $|111\rangle$ under the leading NOTs, and one that does not). Then run one Grover iterate on the uniform state $|\psi_0\rangle = H^{\otimes 3}|000\rangle$ with this x_0 and report the resulting amplitude on $|x_0\rangle$ and the success probability p_1; confirm it matches Theorem 3.25.

Further reading

- Lin Lin, *qasc notes*, Ch. 2 (query model, Deutsch–Jozsa and Bernstein–Vazirani, Grover's algorithm) — `qasc_notes.pdf`.
- Lin Lin, *live notes*, Ch. 11 (amplitude amplification and its applications) — `live_notes_0429.pdf`.
- A. M. Childs, *Lecture Notes on Quantum Algorithms* (the oracle model, Grover search, amplitude amplification, and the hybrid-argument $\Omega(\sqrt{N})$ lower bound) — `material/`.
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Ch. 6 (quantum search algorithms and the search lower bound).

- D. Brody, *The Joy of Quantum Computing*, Princeton Univ. Press, 2025, Ch. 2 (Deutsch's algorithm and the phase oracle) and Ch. 8 (a concrete walkthrough of Grover's search: oracle construction from a multiply-controlled Z, the inversion-about-the-mean geometry, and the worked $N = 8$ example) — `material/Brody_The Joy of Quantum Computing_Princeton_2025.pdf`.

Chapter 4

Quantum Fourier Transform and Phase Estimation

Learning objectives

- Implement the quantum Fourier transform and analyze its gate cost.
- Realize modular integer addition as an application of the QFT (Draper's adder).
- Build quantum phase estimation on top of the QFT.
- Use the Hadamard test to estimate expectation values.
- Analyze the accuracy and success probability of QPE.

Covered this term (Fall 2026–2027). §4.1–§4.4. Not covered this term: §4.5 (Heisenberg-limited and iterative variants).

Quantum phase estimation (QPE) is one of the central primitives of quantum computing: it extracts an eigenvalue phase of a unitary operator and, in doing so, serves as the common engine behind Shor's factoring algorithm, the quantum linear-system solver of Chapter 10, amplitude estimation, and quantum counting. Its beating heart is the *quantum Fourier transform* (QFT), the quantum analogue of the discrete Fourier transform, which can be realized with only $\mathcal{O}(n^2)$ gates on n qubits. This lecture develops both. We begin with the Hadamard test, a one-ancilla diagnostic that reads out the expectation value $\langle\psi|U|\psi\rangle$ and already furnishes a rudimentary phase estimator. We then construct the QFT, prove it is unitary, derive its product representation and the efficient circuit that follows, and count its gates. With the QFT in hand we assemble standard QPE around controlled powers of U and an inverse QFT, analyze its precision and the number of ancilla qubits it needs, contrast it with Kitaev's bit-by-bit method, and close with the caveats of eigenstate preparation and a note on Heisenberg-limited iterative variants.

Throughout we work with n qubits, write $N = 2^n$ for the dimension of the state space $(\mathbb{C}^2)^{\otimes n} \cong \mathbb{C}^N$, and abbreviate the primitive N-th root of unity by $\omega := \mathrm{e}^{\mathrm{i}2\pi/N}$, so that $\omega^N = 1$. Computational basis states are labeled by integers $j \in \{0, \ldots, N-1\}$, and we freely identify j with its n-bit binary string $j_1 j_2 \cdots j_n$, where j_1 is the most significant bit:

$$j = \sum_{l=1}^{n} j_l \, 2^{n-l}, \qquad |j\rangle = |j_1\rangle \otimes \cdots \otimes |j_n\rangle \,.$$

4.1 The Hadamard test

Before estimating an eigenphase directly, we build a tool that reads out the expectation value $\langle\psi|U|\psi\rangle$ of a unitary U on a known state $|\psi\rangle$ using a single ancilla qubit. The phase-estimation problem it serves is the following.

Definition 4.1 (Phase-estimation problem)**.** Let U be a unitary operator on $\mathbb{C}^N$ and let $|\psi\rangle$ be one of its eigenvectors, so that $U|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi}|\psi\rangle$ with $\varphi \in [0,1)$. The *phase-estimation problem* is: given the ability to apply controlled powers of U and to prepare $|\psi\rangle$, produce an estimate $\tilde{\varphi}$ of the eigenphase φ to a prescribed additive precision ϵ, $|\tilde{\varphi} - \varphi| \le \epsilon$, with high probability.

Because U is unitary, every eigenvalue has modulus one and can be written as $\mathrm{e}^{\mathrm{i}2\pi\varphi}$ with $\varphi \in [0,1)$; the phase φ is exactly the information we wish to recover. One might hope to extract it classically: for any basis state $|j\rangle$ with $\langle j|\psi\rangle \neq 0$, $\langle j|U|\psi\rangle \,/\, \langle j|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi}$. This element-wise division requires reading amplitudes of an unknown quantum state and cannot be carried out efficiently, since measurement collapses the state. The Hadamard test provides an efficient quantum alternative for the expectation value.

Definition 4.2 (Hadamard test)**.** The *Hadamard test* for a unitary U and state $|\psi\rangle$ uses one ancilla qubit initialized to $|0\rangle$ and the system register in $|\psi\rangle$. The circuit applies a Hadamard gate H to the ancilla, a controlled-U (with the ancilla as control), and a second Hadamard to the ancilla, and then measures the ancilla in the computational basis. In the *imaginary* variant, a phase gate $S^\dagger = \mathrm{diag}(1, -\mathrm{i})$ is applied to the ancilla between the controlled-U and the final Hadamard. The circuit is drawn in Figure 4.1.

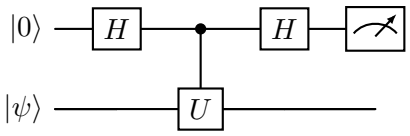


Figure 4.1: The (real) Hadamard test: one ancilla, H–controlled-U–H, then measure, giving $\mathbb{P}(0) = \frac{1}{2}(1 + \Re\langle\psi|U|\psi\rangle)$. Inserting $S^\dagger$ on the ancilla just before the final H (the imaginary variant) instead yields $\frac{1}{2}(1 + \Im\langle\psi|U|\psi\rangle)$.

Theorem 4.3 (Hadamard test)**.** *In the real Hadamard test the probability of measuring the ancilla in $|0\rangle$ is*

$$p(0) = \tfrac{1}{2}\big(1 + \Re\langle\psi|U|\psi\rangle\big).$$

In the imaginary variant (with the $S^\dagger$ gate inserted) it is

$$p(0) = \tfrac{1}{2}\big(1 + \Im\langle\psi|U|\psi\rangle\big).$$

Proof. Start from $|0\rangle \otimes |\psi\rangle$. The first Hadamard produces $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\psi\rangle$. The controlled-$U$ applies U to the system exactly when the ancilla is $|1\rangle$, giving $\frac{1}{\sqrt{2}}(|0\rangle \otimes |\psi\rangle + |1\rangle \otimes U|\psi\rangle)$. The second Hadamard sends $|0\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|1\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, so the $|0\rangle$-ancilla branch of the output is $\frac{1}{2}(|\psi\rangle + U|\psi\rangle)$. Hence

$$p(0) = \left\| \tfrac{1}{2}(|\psi\rangle + U|\psi\rangle) \right\|^2 = \tfrac{1}{4}\big(2 + \langle\psi|U|\psi\rangle + \langle\psi|U^\dagger|\psi\rangle\big) = \tfrac{1}{2}\big(1 + \Re\langle\psi|U|\psi\rangle\big),$$

using $\langle\psi|U^\dagger|\psi\rangle = \overline{\langle\psi|U|\psi\rangle}$, so the two cross terms sum to $2\Re\langle\psi|U|\psi\rangle$. For the imaginary variant, inserting $S^\dagger$ multiplies the $|1\rangle$ branch by $-\mathrm{i}$ before the final Hadamard; the $|0\rangle$-ancilla branch becomes $\frac{1}{2}(|\psi\rangle - \mathrm{i}\,U|\psi\rangle)$, and

$$p(0) = \tfrac{1}{4}(2 + \mathrm{i}\langle\psi|U|\psi\rangle - \mathrm{i}\langle\psi|U^\dagger|\psi\rangle) = \tfrac{1}{2}(1 + \Im\langle\psi|U|\psi\rangle),$$

since $\mathrm{i}(\langle\psi|U|\psi\rangle - \overline{\langle\psi|U|\psi\rangle}) = \mathrm{i}\cdot 2\mathrm{i}\,\Im\langle\psi|U|\psi\rangle = -2\,\Im\langle\psi|U|\psi\rangle$ and tracking the overall sign of the cross terms gives the stated $+\Im$ result. □

Running both circuits and combining the two estimates recovers the full complex number $\langle\psi|U|\psi\rangle = \Re\langle\psi|U|\psi\rangle + \mathrm{i}\,\Im\langle\psi|U|\psi\rangle$. Each probability lies in $[0,1]$ because the real and imaginary parts of $\langle\psi|U|\psi\rangle$ lie in $[-1,1]$.

Remark 4.4 (Swap test as a special case). Take the "system" to consist of two registers holding $|\phi\rangle$ and $|\psi\rangle$, and let U be the swap operator that exchanges them. Then $\langle\phi,\psi|\mathrm{SWAP}|\phi,\psi\rangle = \langle\phi|\psi\rangle\langle\psi|\phi\rangle = |\langle\phi|\psi\rangle|^2$, which is real, so the real Hadamard test gives $p(0) = \frac{1}{2}(1 + |\langle\phi|\psi\rangle|^2)$. This is exactly the well-known *swap test* for estimating the overlap of two states.

A first phase estimator. The Hadamard test already yields a rudimentary phase estimator. Applying Theorem 4.3 with the eigenrelation $\langle\psi|U|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi}$, the real test gives (writing $p(1)$ for the outcome $|1\rangle$) $p(1) = \frac{1}{2}(1 - \cos 2\pi\varphi)$, from which $\varphi = \pm\arccos(1-2p(1))/(2\pi)$ (mod 1). Because cos is even, the real test alone cannot distinguish φ from $1-\varphi$; the imaginary variant gives $p(1) = \frac{1}{2}(1-\sin 2\pi\varphi)$, and the sign of $\sin 2\pi\varphi$ resolves which half of $[0,1)$ contains φ. This estimator learns φ only through the probability $p(1)$, which must be estimated by repeated sampling. To resolve φ to additive error ϵ one needs the empirical frequency accurate to $\mathcal{O}(\epsilon)$, and by the variance of a Bernoulli mean this requires $\mathcal{O}(1/\epsilon^2)$ repetitions. This quadratic cost is the price of using only U itself; the QFT-based method below reduces it to $\mathcal{O}(1/\epsilon)$ by accessing higher powers of U.

4.2 The quantum Fourier transform

The QFT is the quantum analogue of the classical discrete Fourier transform (DFT). It sits at the heart of QPE and of Shor's factoring algorithm.

Definition 4.5 (Quantum Fourier transform)**.** Let $N = 2^n$ and $\omega = \mathrm{e}^{\mathrm{i}2\pi/N}$. The *quantum Fourier transform* is the linear operator QFT on $\mathbb{C}^N$ acting on computational basis states by

$$\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1}\mathrm{e}^{\mathrm{i}2\pi jk/N}\,|k\rangle = \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1}\omega^{jk}\,|k\rangle\,, \qquad j = 0,1,\ldots,N-1.$$

Equivalently, in the computational basis QFT is the $N\times N$ matrix with entries $(\mathrm{QFT})_{kj} = N^{-1/2}\omega^{jk}$, a normalized Vandermonde matrix in the powers of ω.

By linearity, if $|\psi\rangle = \sum_j x_j\,|j\rangle$ then $\mathrm{QFT}\,|\psi\rangle = \sum_k y_k\,|k\rangle$ with $y_k = N^{-1/2}\sum_j \omega^{jk}x_j$, exactly the classical DFT of the amplitude vector. The essential point is that the input is a single basis state $|j\rangle$ while the output is an *equal-magnitude* superposition of all N basis states; the integer j is encoded entirely in the relative phases ω^{jk}. For $j=0$ every phase is 1, so $\mathrm{QFT}\,|0\rangle = N^{-1/2}\sum_k|k\rangle = H^{\otimes n}\,|0\rangle^{\otimes n}$: on the all-zeros input the QFT is just a layer of Hadamards.

We first compute the transform explicitly in the two smallest cases; its general properties—unitarity, a product (tensor) form, and an efficient circuit—then follow and are seen at work in these examples.

Example 4.6 (The $n = 2$ QFT, worked in full)**.** Take $n = 2$, so $N = 4$ and $\omega = e^{i2\pi/4} = i$. The definition $\mathrm{QFT}\,|j\rangle = \frac{1}{2}\sum_{k=0}^{3} i^{jk}\,|k\rangle$ is the matrix

$$\mathrm{QFT}_4 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix},$$

whose columns are $\mathrm{QFT}\,|0\rangle, \ldots, \mathrm{QFT}\,|3\rangle$. Writing the labels in binary ($|0\rangle = |00\rangle$, $|1\rangle = |01\rangle$, $|2\rangle = |10\rangle$, $|3\rangle = |11\rangle$) and factoring each column into (left qubit) $\otimes$ (right qubit):

$$\begin{aligned}
\mathrm{QFT}\,|00\rangle &= \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) &&= \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \\
\mathrm{QFT}\,|01\rangle &= \tfrac{1}{2}(|00\rangle + i\,|01\rangle - |10\rangle - i\,|11\rangle) &&= \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \tfrac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle), \\
\mathrm{QFT}\,|10\rangle &= \tfrac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) &&= \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \\
\mathrm{QFT}\,|11\rangle &= \tfrac{1}{2}(|00\rangle - i\,|01\rangle - |10\rangle + i\,|11\rangle) &&= \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \tfrac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle).
\end{aligned}$$

Each factorization is read off *by inspection*: a minus sign sits wherever the *left* qubit is $|1\rangle$ (so the left output factor is $\frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$), and a factor of i wherever the *right* qubit is $|1\rangle$ (fixing the right factor).

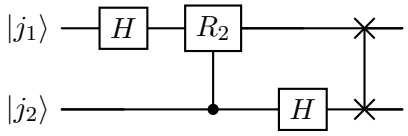
The circuit. Every output factor is a Hadamard carrying an extra phase, so three gates build the QFT:

(i) H on qubit 1 makes $\frac{1}{\sqrt{2}}(|0\rangle + e^{i2\pi\,0.j_1}\,|1\rangle)$;

(ii) a controlled-R_2 (control qubit 2) folds in the next bit, upgrading the phase to $0.j_1j_2$—here $R_2 = P(\pi/2) = \mathrm{diag}(1, i)$ with $P(\theta) = \mathrm{diag}(1, e^{i\theta})$;

(iii) H on qubit 2 makes $\frac{1}{\sqrt{2}}(|0\rangle + e^{i2\pi\,0.j_2}\,|1\rangle)$.

This yields $\frac{1}{\sqrt{2}}(|0\rangle + e^{i2\pi\,0.j_1j_2}\,|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{i2\pi\,0.j_2}\,|1\rangle)$—the *mirror image* of the QFT product form (Theorem 4.11), i.e. the circuit's output *before* the final bit-reversal SWAP:



The swap. For $|j_1j_2\rangle = |01\rangle$ the three gates give $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, whereas the table above has $\mathrm{QFT}\,|01\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ —the *same two factors in the opposite order.* A final SWAP of the two qubits reconciles them, and likewise for every input: the circuit outputs the factors bit-reversed, and the SWAP undoes it. (Brody, *The Joy of Quantum Computing*, Ch. 9, tabulates these four cases; the full bit-reversal bookkeeping is worked out for $n = 3$ next.)

Example 4.7 (The $n = 3$ QFT and the bit-reversal step, in full)**.** Take $n = 3$, $N = 8$, $\omega = e^{i2\pi/8} = e^{i\pi/4}$, and compute $\mathrm{QFT}\,|5\rangle$ two ways. Here $j = 5 = (101)_2$, so $j_1 = 1$, $j_2 = 0$, $j_3 = 1$.

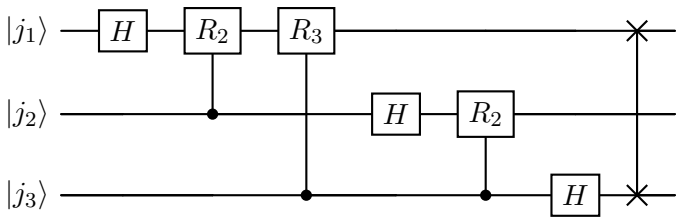
From the definition. Theorem 4.5 gives the eight amplitudes $N^{-1/2}\omega^{5k}$ directly. Since $\omega^5 = e^{i5\pi/4} = (-1 - i)/\sqrt{2}$, the successive powers ω^{5k} for $k = 0, \ldots, 7$ are 1, ω^5, $\omega^{10} = \omega^2 = i$, $\omega^{15} = \omega^7$, $\omega^{20} = \omega^4 = -1$, $\omega^{25} = \omega$, $\omega^{30} = \omega^6 = -i$, $\omega^{35} = \omega^3$, so

$$\mathrm{QFT}\,|5\rangle = \frac{1}{2\sqrt{2}}\Big(|0\rangle - \tfrac{1+i}{\sqrt{2}}\,|1\rangle + i\,|2\rangle + \tfrac{1-i}{\sqrt{2}}\,|3\rangle - |4\rangle + \tfrac{1+i}{\sqrt{2}}\,|5\rangle - i\,|6\rangle - \tfrac{1-i}{\sqrt{2}}\,|7\rangle\Big). \tag{4.1}$$

From the circuit, before its final SWAP. Built qubit by qubit, the Hadamard–controlled-phase network writes on qubits $1, 2, 3$ the phases $0.j_1j_2j_3 = 0.101_2 = \frac{5}{8}$, $0.j_2j_3 = 0.01_2 = \frac{1}{4}$, and $0.j_3 = 0.1_2 = \frac{1}{2}$—the *mirror image* of the QFT product form Theorem 4.11. Its pre-SWAP output is therefore

$$|\psi_{\text{circ}}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi\cdot 5/8}\,|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + \mathrm{i}\,|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \tag{4.2}$$

where $\mathrm{e}^{\mathrm{i}2\pi\cdot 5/8} = \mathrm{e}^{\mathrm{i}5\pi/4} = -(1+\mathrm{i})/\sqrt{2}$. The circuit realizes Equation (4.2) qubit by qubit: an H on qubit 1 writes the leading phase $0.j_1$, a controlled-R_2 from qubit 2 and a controlled-R_3 from qubit 3 fold in the bits j_2, j_3 to complete $0.j_1j_2j_3$; an H on qubit 2 and a controlled-R_2 from qubit 3 build $0.j_2j_3$; and an H on qubit 3 builds $0.j_3$:



The bit-reversal reconciliation. Expanding Equation (4.2) as a sum over basis states $|k_1k_2k_3\rangle$—keeping the tensor factors in their written order—gives amplitudes indexed by $k_1k_2k_3$ that do *not* match Equation (4.1) term by term; for instance the product assigns the factor $-(1+\mathrm{i})/\sqrt{2}$ to $|100\rangle = |4\rangle$, whereas the definition assigns it to $|001\rangle = |1\rangle$. The two agree precisely after *bit reversal* $k_1k_2k_3 \mapsto k_3k_2k_1$, implemented by the final SWAP of qubits 1 and 3: this exchanges $|1\rangle \leftrightarrow |4\rangle$ and $|3\rangle \leftrightarrow |6\rangle$ (the strings that are not palindromes), carrying Equation (4.2) exactly onto Equation (4.1). In short, the product representation outputs the amplitudes in bit-reversed order, and the SWAP layer is exactly the permutation that undoes it.

Factoring by inspection (Brody's route). The product form can also be read off Equation (4.1) directly, bypassing Theorem 4.11: the amplitude of each single-1 string fixes one output qubit's relative phase. The coefficients of $|100\rangle, |010\rangle, |001\rangle$ are -1, i, $\mathrm{e}^{\mathrm{i}5\pi/4}$, so

$$\mathrm{QFT}\,|5\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes \tfrac{1}{\sqrt{2}}(|0\rangle + \mathrm{i}\,|1\rangle) \otimes \tfrac{1}{\sqrt{2}}(|0\rangle + \mathrm{e}^{\mathrm{i}5\pi/4}\,|1\rangle),$$

which is Equation (4.2) with its tensor factors reversed—exactly the bit reversal the SWAP performs. A multi-1 amplitude is then the product of the relevant single-1 phases: e.g. the coefficient of $|011\rangle$ is $\mathrm{e}^{\mathrm{i}\pi/2} \cdot \mathrm{e}^{\mathrm{i}5\pi/4} = \mathrm{e}^{\mathrm{i}7\pi/4}$, matching Equation (4.1)—a quick consistency check (Brody, *The Joy of Quantum Computing*, Ch. 9).

Remark 4.8 (Same kernel as the DFT, but no free lunch). The kernel $N^{-1/2}\mathrm{e}^{\mathrm{i}2\pi jk/N}$ is identical to that of the symmetrically normalized classical DFT; the QFT applies it to the 2^n amplitudes of an n-qubit state *in superposition*. It is tempting to conclude an exponential speedup for Fourier analysis, but after the transform we hold the coefficients y_k only as amplitudes, and a measurement returns a single basis state $|k\rangle$ with probability $|y_k|^2$. We cannot read out all N amplitudes; estimating them to good precision would need exponentially many repetitions. The power of the QFT is *structural*: embedded inside a larger algorithm (phase estimation, period finding) it converts information hidden in phases into measurement statistics, so that few measurements suffice. The circuit below uses $\mathcal{O}(n^2) = \mathcal{O}((\log N)^2)$ gates, versus $\mathcal{O}(N \log N)$ operations for the classical FFT and $\mathcal{O}(N^2)$ for the naive DFT.

Unitarity

Because the QFT is realized as a quantum circuit, it must be unitary. The engine of the proof is the following elementary identity.

Lemma 4.9 (Geometric-series identity)**.** *Let $N \geq 1$ and $m \in \mathbb{Z}$. Then*

$$\sum_{k=0}^{N-1} \mathrm{e}^{\mathrm{i}2\pi mk/N} = N\,[\,m \equiv 0 \bmod N\,],$$

where the bracket is 1 if $m \equiv 0 \pmod N$ and 0 otherwise. In particular, for $m = j - j'$ with $j, j' \in \{0, \ldots, N-1\}$ we have $\sum_{k=0}^{N-1} \mathrm{e}^{\mathrm{i}2\pi(j-j')k/N} = N\,\delta_{j,j'}$.

Proof. Write $r = \mathrm{e}^{\mathrm{i}2\pi m/N}$, so the sum is the geometric series $\sum_{k=0}^{N-1} r^k$. If $m \equiv 0 \pmod N$ then $r = 1$ and every term is 1, giving N. Otherwise $r \neq 1$ and $\sum_{k=0}^{N-1} r^k = (r^N - 1)/(r-1) = (\mathrm{e}^{\mathrm{i}2\pi m} - 1)/(r-1) = 0$, since $r^N = \mathrm{e}^{\mathrm{i}2\pi m} = 1$ for integer m while the denominator is nonzero. For the final statement, if $|j - j'| < N$ then $j - j' \equiv 0 \pmod N$ forces $j = j'$. □

Theorem 4.10 (The QFT is unitary)**.** *The operator of Theorem 4.5 satisfies* $\mathrm{QFT}^\dagger \mathrm{QFT} = \mathbb{I}$*, so* QFT *is unitary.*

Proof. The adjoint has entries $(\mathrm{QFT}^\dagger)_{jk} = \overline{(\mathrm{QFT})_{kj}} = N^{-1/2}\omega^{-jk}$. Inserting a resolution of the identity over $\{|k\rangle\}$,

$$\langle j'|\mathrm{QFT}^\dagger \mathrm{QFT}|j\rangle = \sum_{k=0}^{N-1} \overline{(\mathrm{QFT})_{kj'}}\,(\mathrm{QFT})_{kj} = \frac{1}{N}\sum_{k=0}^{N-1} \omega^{(j-j')k} = \frac{1}{N}\,N\,\delta_{j,j'} = \delta_{j,j'},$$

by Theorem 4.9. As j, j' range over the basis this says $\mathrm{QFT}^\dagger \mathrm{QFT} = \mathbb{I}$, so QFT is unitary. □

Since QFT is unitary, $\mathrm{QFT}^{-1} = \mathrm{QFT}^\dagger$ acts by $\mathrm{QFT}^\dagger\,|j\rangle = N^{-1/2}\sum_k \omega^{-jk}\,|k\rangle$, differing from QFT only by conjugating every phase; the inverse circuit is obtained by reversing the gate order and negating each phase angle.

Product representation

To build a circuit we express $\mathrm{QFT}\,|j\rangle$ qubit by qubit. We use the binary fraction $0.j_l j_{l+1}\cdots j_n := \sum_{s=l}^{n} j_s 2^{-(s-l+1)}$; note $0.j_n = j_n/2$ and $0.j_1\cdots j_n = j/N$.

Theorem 4.11 (Product representation)**.** *For every $j = j_1 j_2 \cdots j_n$ (binary),*

$$\mathrm{QFT}\,|j\rangle = \bigotimes_{l=1}^{n} \frac{1}{\sqrt{2}}\Big(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi\,0.j_{n-l+1}\cdots j_n}\,|1\rangle\Big) = \frac{1}{2^{n/2}}\big(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi\,0.j_n}\,|1\rangle\big) \otimes \cdots \otimes \big(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi\,0.j_1\cdots j_n}\,|1\rangle\big).$$

That is, output qubit l carries the binary-fraction phase $0.j_{n-l+1}\cdots j_n$: the first *output qubit carries the shortest fraction $0.j_n$ and the* last *carries the full $0.j_1\cdots j_n$. In particular the n output qubits are unentangled. (The Hadamard–controlled-phase circuit of Theorem 4.12 naturally produces these factors in the* opposite *qubit order—qubit l carrying $0.j_l\cdots j_n$—so a final layer of SWAP gates,* bit reversal*, reverses the register to realize* QFT*.)*

Proof. Expand the summation index $k = k_1 \cdots k_n$ in binary, $k = \sum_l k_l 2^{n-l}$. The exponent factorizes because the exponential of a sum is a product:

$$\mathrm{e}^{\mathrm{i}2\pi j(\sum_l k_l 2^{n-l})/2^n} = \prod_{l=1}^{n} \mathrm{e}^{\mathrm{i}2\pi j k_l 2^{n-l}/2^n} = \prod_{l=1}^{n} \mathrm{e}^{\mathrm{i}2\pi j k_l/2^l}.$$

Substitute this into the definition of the QFT and write the basis ket in binary, $|k\rangle = |k_1\rangle \otimes \cdots \otimes |k_n\rangle$. Each summand becomes a product of n factors whose l-th factor depends only on the single bit k_l; the one sum over all 2^n strings k therefore unfolds into n independent bit-sums and distributes across the tensor product ("a sum of products is a product of sums"):

$$\begin{aligned}
\mathrm{QFT}\,|j\rangle &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \mathrm{e}^{\mathrm{i}2\pi jk/N} |k\rangle = \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \Big(\prod_{l=1}^{n} \mathrm{e}^{\mathrm{i}2\pi j k_l/2^l} \Big) |k_1\rangle \otimes \cdots \otimes |k_n\rangle \\
&= \frac{1}{\sqrt{N}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \bigotimes_{l=1}^{n} \mathrm{e}^{\mathrm{i}2\pi j k_l/2^l} |k_l\rangle = \frac{1}{\sqrt{N}} \bigotimes_{l=1}^{n} \Big(\sum_{k_l=0}^{1} \mathrm{e}^{\mathrm{i}2\pi j k_l/2^l} |k_l\rangle \Big) \\
&= \frac{1}{\sqrt{N}} \bigotimes_{l=1}^{n} \big(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi j/2^l} |1\rangle \big).
\end{aligned}$$

The first line is the factorization above regrouped so that factor l carries the ket $|k_l\rangle$; the second line is the distributive law, pulling each independent bit-sum $\sum_{k_l}$ inside its own tensor slot; and the last line evaluates each two-term sum over $k_l \in \{0,1\}$ (the $k_l = 0$ term is $|0\rangle$, the $k_l = 1$ term is $\mathrm{e}^{\mathrm{i}2\pi j/2^l} |1\rangle$). To simplify $\mathrm{e}^{\mathrm{i}2\pi j/2^l}$, write $j/2^l = \sum_s j_s 2^{n-s-l}$ and split into the terms with $n - s \geq l$ (a nonnegative integer, contributing $\mathrm{e}^{\mathrm{i}2\pi(\text{integer})} = 1$) and those with $n - s < l$, i.e. $s = n-l+1, \ldots, n$, which give $0.j_{n-l+1} \cdots j_n$. Hence $\mathrm{e}^{\mathrm{i}2\pi j/2^l} = \mathrm{e}^{\mathrm{i}2\pi\, 0.j_{n-l+1}\cdots j_n}$, and distributing $1/\sqrt{N} = \prod_l 2^{-1/2}$ gives $\mathrm{QFT}\,|j\rangle = \bigotimes_{l=1}^{n} \frac{1}{\sqrt{2}}(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi\, 0.j_{n-l+1}\cdots j_n} |1\rangle)$—exactly the stated form, with output qubit l carrying $0.j_{n-l+1} \cdots j_n$ (so qubit 1 carries $0.j_n$ and qubit n carries the full $0.j_1 \cdots j_n$). The output is a tensor product, hence unentangled. □

The circuit and its gate count

The product representation suggests synthesizing the QFT one qubit at a time, using two gate families: the Hadamard H and the controlled-phase gates $R_k = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}2\pi/2^k})$ (so $R_1 = Z$, $R_2 = S$, $R_3 = T$). The construction rests on the single-qubit identity, valid for $j_m \in \{0,1\}$,

$$H\,|j_m\rangle = \frac{1}{\sqrt{2}}\big(|0\rangle + \mathrm{e}^{\mathrm{i}\pi j_m} |1\rangle\big) = \frac{1}{\sqrt{2}}\big(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi\, 0.j_m} |1\rangle\big),$$

because $\mathrm{e}^{\mathrm{i}\pi j_m} = (-1)^{j_m}$ and $0.j_m = j_m/2$. A Hadamard therefore produces the leading term of the required phase; the lower-order bits are folded in by controlled-R_k gates, since a controlled-R_k applies exactly the phase $\mathrm{e}^{\mathrm{i}2\pi j_{m+k-1}/2^k}$ when its control holds the bit j_{m+k-1}. The full circuit processes qubits $1, 2, \ldots, n$: on qubit m apply H, then for $k = 2, \ldots, n-m+1$ a controlled-R_k with target m and control $m+k-1$. A final layer of SWAP gates (*bit reversal*) restores the correct qubit order; the general pattern is drawn in Figure 4.2, and worked out concretely for $n = 2$ and $n = 3$ in Theorems 4.6 and 4.7 above.

Theorem 4.12 (Gate complexity of the QFT)**.** *The circuit realizes the QFT on n qubits using exactly n Hadamard gates, $\binom{n}{2} = n(n-1)/2$ controlled-phase gates, and at most $\lfloor n/2 \rfloor$ SWAP gates, for a total of $n + n(n-1)/2 + \lfloor n/2 \rfloor = \mathcal{O}(n^2)$ elementary gates. Since $n = \log_2 N$, this is $\mathcal{O}((\log N)^2)$, polylogarithmic in N.*

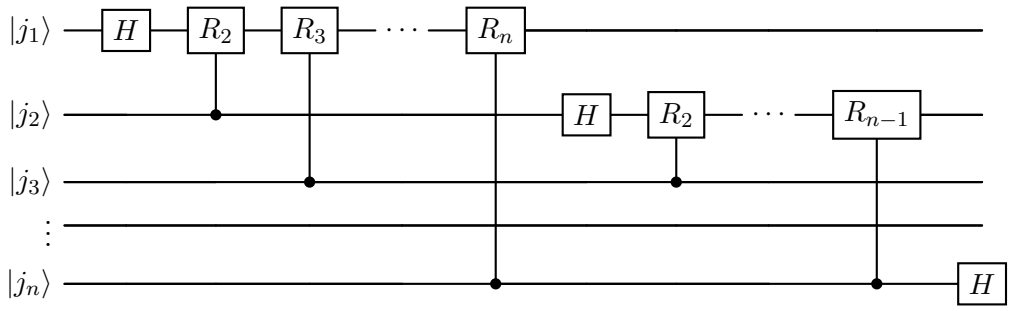


Figure 4.2: The quantum Fourier transform on n qubits. Qubit m receives a Hadamard followed by controlled phase rotations $R_k = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}2\pi/2^k})$ from the qubits below it, folding the lower-order bits into its phase. A final layer of SWAPs (bit reversal, omitted here) reverses the qubit order. The count is n Hadamards, $\binom{n}{2}$ controlled phases, and $\lfloor n/2 \rfloor$ SWAPs—$\mathcal{O}(n^2)$ gates (Theorem 4.12).

Proof. On qubit m we apply one Hadamard, giving n in total. Afterward we fold in the bits $j_{m+1}, \ldots, j_n$ with controlled-$R_2, \ldots, R_{n-m+1}$, that is $n-m$ controlled-phase gates on qubit m. Summing, $\sum_{m=1}^{n}(n-m) = n(n-1)/2 = \binom{n}{2}$. Finally bit reversal exchanges qubit m with $n-m+1$ for $m = 1, \ldots, \lfloor n/2 \rfloor$, using $\lfloor n/2 \rfloor$ SWAP gates. Adding the three contributions gives the claim. □

Remark 4.13 (Approximate QFT). The controlled-R_k gates apply a phase $\mathrm{e}^{\mathrm{i}2\pi/2^k}$ that is exponentially small in k; dropping all gates with $k > k_{\max} = \mathcal{O}(\log(n/\varepsilon))$ yields the *approximate QFT*, within error ε in operator norm while using only $\mathcal{O}(n \log(n/\varepsilon))$ gates. Either way the gate count is far below the $\mathcal{O}(N \log N)$ of the classical FFT, though (Theorem 4.8) this is not by itself an end-to-end speedup.

Application: addition in the Fourier basis (the Draper adder)

The QFT is not only the engine of phase estimation; it also turns integer *addition* into a product of single-qubit phase rotations. Let the target register be n qubits holding $j \in \{0, \ldots, N-1\}$, and fix a classical constant a. We seek the *constant adder*

$$\mathrm{ADD}_a : |j\rangle \longmapsto |(j+a) \bmod N\rangle .$$

Because the QFT diagonalizes the cyclic shift—Exercise 4.5 is the case $a = 1$, with $\mathrm{ADD}_1 |j\rangle = |(j+1) \bmod N\rangle$—the operator ADD_a is diagonal in the Fourier basis, and the following identity makes this explicit.

Proposition 4.14 (Addition as Fourier-basis phases)**.** *With $\omega = \mathrm{e}^{\mathrm{i}2\pi/N}$ and the diagonal unitary $\Phi_a := \mathrm{diag}(\omega^{ak})_{k=0}^{N-1}$,*

$$\mathrm{ADD}_a = \mathrm{QFT}^\dagger \, \Phi_a \, \mathrm{QFT}, \qquad \textit{equivalently} \qquad \mathrm{QFT}\, |(j+a) \bmod N\rangle = \Phi_a \, \mathrm{QFT}\, |j\rangle .$$

Moreover Φ_a factorizes into single-qubit phase gates: on qubit l (bit value k_l, most significant first) it acts as $P_l^{(a)} = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}2\pi a\, 2^{\,n-l}/N})$, so $\Phi_a = \bigotimes_{l=1}^{n} P_l^{(a)}$.

Proof. Using $\mathrm{QFT}\,|j\rangle = N^{-1/2} \sum_k \omega^{jk} |k\rangle$ and $\mathrm{QFT}^\dagger |k\rangle = N^{-1/2} \sum_m \omega^{-km} |m\rangle$,

$$\mathrm{QFT}^\dagger \Phi_a \mathrm{QFT}\, |j\rangle = \frac{1}{N} \sum_m \Big(\sum_k \omega^{k(j+a-m)} \Big) |m\rangle = \sum_m \delta_{m,\,(j+a) \bmod N} \; |m\rangle = |(j+a) \bmod N\rangle ,$$

since the inner geometric sum equals N exactly when $m \equiv j + a \pmod N$ and vanishes otherwise (Theorem 4.9). For the factorization, write $k = \sum_{l=1}^{n} k_l\, 2^{n-l}$; then $\omega^{ak} = \prod_l \big(\mathrm{e}^{\mathrm{i}2\pi a 2^{n-l}/N}\big)^{k_l}$, which is exactly $\bigotimes_l P_l^{(a)}$ acting on $|k_1 \cdots k_n\rangle$. □

The circuit is therefore strikingly simple: *apply the QFT, apply n single-qubit phase gates, apply the inverse QFT*. Adding a classical constant costs the two QFTs—$\mathcal{O}(n^2)$ gates, or $\mathcal{O}(n \log(n/\varepsilon))$ with the approximate QFT of Theorem 4.13—plus n single-qubit rotations, and, remarkably, *no entangling gates in between*. This is Draper's adder.

To add one *quantum* register to another, $|a\rangle\,|j\rangle \mapsto |a\rangle\,|(j+a) \bmod N\rangle$, promote each middle phase $P_l^{(a)}$ to a phase *controlled* by the qubits of the first register: writing $a = \sum_r a_r 2^{n-r}$, the phase $\mathrm{e}^{\mathrm{i}2\pi a 2^{n-l}/N}$ is built bit by bit from the a_r using controlled-R_k gates, exactly the gate family of the QFT itself. The two-register adder then uses $\mathcal{O}(n^2)$ controlled-phase gates between the QFT pair and needs *no carry ancillas*—in contrast to a ripple-carry adder, which propagates carries through Toffoli gates and requires an extra register.

Remark 4.15 (From adders to circulant matrices: a bridge to Chapter 7). The constant adders are the powers of the cyclic shift, $\mathrm{ADD}_a = V^a$ with $V = \mathrm{ADD}_1$, and Theorem 4.14 shows they are simultaneously diagonalized by the QFT with eigenvalues ω^{ak}. Any *circulant* matrix is a linear combination of these shifts, $C = \sum_a c_a V^a$, and is therefore diagonalized by the QFT, $C = \mathrm{QFT}^\dagger\, \mathrm{diag}\big(\sum_a c_a \omega^{ak}\big)_k\, \mathrm{QFT}$. Restricting the sum to a few small $|a|$ yields a *banded* operator such as a finite-difference Laplacian. This reappears later, twice over: the linear combination $\sum_a c_a(\cdot)$ is precisely what the LCU block-encoding calculus of Chapter 7 implements at the operator level—with each shift V^a supplied by the adder built here—and the QFT diagonalization is a workhorse behind the Hamiltonian simulation of such operators in Chapter 13. The quantum adder is thus the elementary unitary from which circulant and finite-difference block encodings are assembled.

Application: revealing periodicity (toward period finding)

A classical Fourier transform turns a signal into its *frequencies*; the QFT does the same for the amplitudes of a quantum state, and this is the property that powers Shor's algorithm. Suppose the amplitudes of $|\psi\rangle = \sum_x a_x\,|x\rangle$ are *periodic* with period $r \mid N$—nonzero only on an evenly spaced comb $x \in \{x_0,\, x_0 + r,\, x_0 + 2r, \dots\}$ of N/r equal spikes. Then, by the same geometric sum that proved unitarity, the QFT is supported *only* on the multiples of N/r:

$$\mathrm{QFT}\Big(\sqrt{\tfrac{r}{N}} \sum_{m=0}^{N/r-1} |x_0 + mr\rangle\Big) = \frac{1}{\sqrt{r}} \sum_{\ell=0}^{r-1} \mathrm{e}^{\mathrm{i}2\pi \ell x_0/r} \,\Big|\, \ell\, \tfrac{N}{r} \,\Big\rangle.$$

A comb of spacing r in the computational basis becomes a comb of spacing N/r after the transform; the offset x_0 hides in the phases $\mathrm{e}^{\mathrm{i}2\pi \ell x_0/r}$, which a computational-basis measurement cannot see. Measuring therefore returns a multiple of N/r, from which the period r is read off—this is the heart of period finding.

Example 4.16 (Reading a period off the QFT)**.** Take $n = 3$ ($N = 8$) and $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |4\rangle)$, whose amplitude sequence $\frac{1}{\sqrt{2}}, 0, 0, 0, \frac{1}{\sqrt{2}}, 0, 0, 0$ repeats with period $r = 4$. Since $\mathrm{QFT}\,|4\rangle = \frac{1}{\sqrt{8}} \sum_k \mathrm{e}^{\mathrm{i}\pi k}\,|k\rangle = \frac{1}{\sqrt{8}} \sum_k (-1)^k\,|k\rangle$,

$$\mathrm{QFT}\,|\psi\rangle = \tfrac{1}{\sqrt{2}}\big(\mathrm{QFT}\,|0\rangle + \mathrm{QFT}\,|4\rangle\big) = \tfrac{1}{2}\big(\,|0\rangle + |2\rangle + |4\rangle + |6\rangle\,\big).$$

The output lives on the multiples of $2 = N/r$, so a measurement yields one of $\{0, 2, 4, 6\}$ and we recover $r = N/2 = 4$. Dually, the period-2 state $\frac{1}{2}(|0\rangle + |2\rangle + |4\rangle + |6\rangle)$ transforms to a state on multiples of 4, giving $r = 2$: the QFT trades a spacing of r for a spacing of N/r.

Remark 4.17 (When the period does not divide N). If $r \nmid N$ the comb no longer lands on exact grid points: the QFT output is *peaked* near the multiples of N/r rather than exactly on them, and the period is recovered from the measured value by continued fractions. This is precisely Shor's order-finding situation, developed in Chapter 5. (Brody, *The Joy of Quantum Computing*, Ch. 9, works these period examples out by hand on a small simulator.)

4.3 Quantum phase estimation via the QFT

The goal. We are handed a unitary U together with one of its eigenstates $|\psi\rangle$, $U|\psi\rangle = \lambda|\psi\rangle$, and we want the eigenvalue λ. Every eigenvalue of a unitary has unit modulus, so $\lambda = \mathrm{e}^{\mathrm{i}\theta}$ for some angle θ. A measurement returns only 0s and 1s, so rather than chase θ directly we chase the *fraction*

$$\varphi = \frac{\theta}{2\pi} \in [0, 1), \qquad \lambda = \mathrm{e}^{\mathrm{i}\theta} = \mathrm{e}^{\mathrm{i}2\pi\varphi}, \tag{4.3}$$

and record it in binary, $\varphi = 0.\varphi_1\varphi_2\cdots = \sum_{\ell\geq 1}\varphi_\ell 2^{-\ell}$; with t ancilla qubits a measurement returns t fractional bits of φ. (This φ is Brody's j. For $U = Z$ on $|\psi\rangle = |1\rangle$, $\lambda = -1 = \mathrm{e}^{\mathrm{i}\pi}$ gives $\varphi = \frac{1}{2} = 0.1_2$—a far friendlier target than $\theta = \pi$, and one that three measured bits pin down exactly.)

Phase kickback, one control qubit. The engine of the algorithm is a single move. Put one control qubit on a controlled-U whose target carries $|\psi\rangle$. If the control is $|0\rangle$ the gate does nothing; if it is $|1\rangle$ then U acts and returns $U|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi}|\psi\rangle$. By linearity a control in the superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ gives

$$\text{c-}U\ \tfrac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)|\psi\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + \mathrm{e}^{\mathrm{i}2\pi\varphi}|1\rangle\big)|\psi\rangle\,,$$

so the eigenphase is *kicked back* onto the *control* while $|\psi\rangle$ is returned untouched (Theorem 3.3). The eigenstate acts as a passive catalyst; all the information lands in the control's relative phase.

From one control qubit to t. Give each of t ancillas its own power: ancilla k (for $k = 0, \ldots, t-1$) controls U^{2^k}. Since $U^{2^k}|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi 2^k}|\psi\rangle$, the same kickback writes the phase $\mathrm{e}^{\mathrm{i}2\pi\varphi 2^k}$ onto ancilla k—and only the t powers $U, U^2, \ldots, U^{2^{t-1}}$ are ever needed, not all 2^t. Preparing the ancillas in $H^{\otimes t}|0^t\rangle = 2^{-t/2}\sum_j |j\rangle$ and applying all t controlled powers leaves them in the *phase-kickback state*

$$\frac{1}{\sqrt{2^t}}\sum_{j=0}^{2^t-1}\mathrm{e}^{\mathrm{i}2\pi\varphi j}|j\rangle \otimes |\psi\rangle\,,$$

with $|\psi\rangle$ unchanged and the eigenphase spread across the ancilla amplitudes as a pure "frequency" φ. The final step reads that frequency off: when φ has an exact t-bit expansion this register is *precisely* $\mathrm{QFT}\,|2^t\varphi\rangle$ (Theorem 4.19), so an *inverse* QFT turns it into the basis state $|2^t\varphi\rangle$ whose bits spell out φ.

Definition 4.18 (Standard QPE)**.** *Standard quantum phase estimation* uses t ancilla qubits and proceeds in three steps:

1. **Superposition:** apply $H^{\otimes t}$ to the ancilla, with the system in $|\psi\rangle$: $|0^t\rangle|\psi\rangle \mapsto 2^{-t/2}\sum_j |j\rangle|\psi\rangle$.
2. **Controlled powers:** apply controlled-U^{2^k} from the k-th ancilla for $k = 0, \ldots, t-1$, producing $2^{-t/2}\sum_j \mathrm{e}^{\mathrm{i}2\pi\varphi j}|j\rangle|\psi\rangle$.

3. **Inverse-QFT readout:** apply QFT^{-1} to the ancilla and measure it in the computational basis, obtaining $|\tilde{\varphi}\rangle$.

We use the convention that $\mathrm{QFT}\,|k\rangle = 2^{-t/2}\sum_j \mathrm{e}^{\mathrm{i}2\pi jk/2^t}\,|j\rangle$, so $\mathrm{QFT}^{-1}\,|j\rangle = 2^{-t/2}\sum_k \mathrm{e}^{-\mathrm{i}2\pi jk/2^t}\,|k\rangle$. Figure 4.3 shows the circuit for $t = 3$ ancillas.

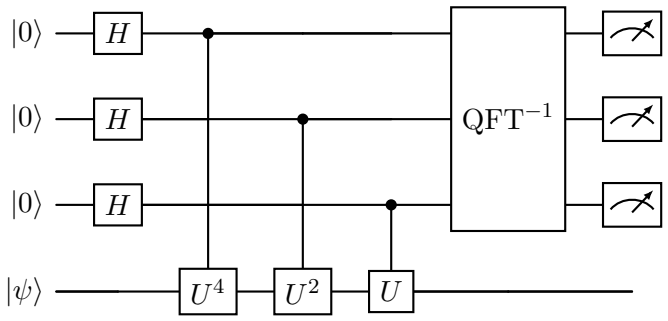


Figure 4.3: Standard quantum phase estimation with $t = 3$ ancillas on an eigenstate $U\,|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi}\,|\psi\rangle$: Hadamards prepare the ancillas, ancilla k controls U^{2^k} (kicking the phase $\mathrm{e}^{\mathrm{i}2\pi\varphi 2^k}$ onto it), an inverse QFT concentrates the amplitude, and measuring the ancillas returns the t-bit estimate $\tilde{\varphi}$ of φ. The eigenstate register is returned unchanged.

Theorem 4.19 (Exact QPE)**.** *If $\varphi = k/2^t$ for some integer $k \in \{0, \ldots, 2^t - 1\}$, then standard QPE measures the outcome k with probability 1, so $\tilde{\varphi} = k/2^t = \varphi$ exactly.*

Proof. After the controlled powers the ancilla is in $2^{-t/2}\sum_j \mathrm{e}^{\mathrm{i}2\pi\varphi j}\,|j\rangle$. Substituting $\varphi = k/2^t$ gives amplitudes $\mathrm{e}^{\mathrm{i}2\pi jk/2^t}/\sqrt{2^t}$, which is precisely $\mathrm{QFT}\,|k\rangle$. Therefore $\mathrm{QFT}^{-1}(\mathrm{QFT}\,|k\rangle) = |k\rangle$, and measuring yields k deterministically with $\tilde{\varphi} = k/2^t = \varphi$. □

Example 4.20 (Worked exact case, $t = 3$)**.** Let $\varphi = 0.101_2 = \frac{5}{8}$, so $k = 5$ and $2^t = 8$. After Step 2 the ancilla holds $\frac{1}{\sqrt{8}}\sum_{j=0}^{7} \mathrm{e}^{\mathrm{i}2\pi(5/8)j}\,|j\rangle = \mathrm{QFT}\,|5\rangle$. The inverse QFT returns $|5\rangle = |101\rangle$, and measurement reads 101, i.e. $\tilde{\varphi} = 5/8 = \varphi$ with certainty—a single run gives the exact answer.

Example 4.21 (QPE on a phase gate, traced end to end)**.** We run standard QPE on a concrete single-qubit unitary whose eigenphase is exactly dyadic, tracing every register. Let $U = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}\pi/4})$ be the phase gate (Brody's $P(\pi/4)$), take the eigenstate $|\psi\rangle = |1\rangle$, and use $t = 3$ ancillas. From $U\,|1\rangle = \mathrm{e}^{\mathrm{i}\pi/4}\,|1\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi}\,|1\rangle$ we read off $\varphi = \frac{1}{8} = 0.001_2$, so we expect the exact outcome $k = 1$.

Step 1 (superposition). With the system in $|1\rangle$, the Hadamards make $\frac{1}{\sqrt{8}}\sum_{j=0}^{7} |j\rangle \otimes |1\rangle$.

Step 2 (controlled powers). The three controlled operations are controlled-U, controlled-U^2, and controlled-U^4. Since $U^{2^m}\,|1\rangle = \mathrm{e}^{\mathrm{i}\pi 2^m/4}\,|1\rangle$, they may be replaced by the simpler gates $U = P(\pi/4)$, $U^2 = P(\pi/2)$, $U^4 = P(\pi)$ on the eigenstate. Each controlled power kicks its phase back onto its ancilla, and because $\mathrm{e}^{\mathrm{i}2\pi\varphi j} = \mathrm{e}^{\mathrm{i}\pi j/4}$ the ancilla register becomes

$$\frac{1}{\sqrt{8}}\sum_{j=0}^{7} \mathrm{e}^{\mathrm{i}2\pi(1/8)j}\,|j\rangle \otimes |1\rangle = \mathrm{QFT}\,|1\rangle \otimes |1\rangle\,,$$

using Theorem 4.5 with $N = 8$. Written out over the ancilla, the eight amplitudes are $\frac{1}{\sqrt{8}}\mathrm{e}^{\mathrm{i}\pi j/4}$ for $j = 0, \ldots, 7$, i.e. the phases $1, \omega, \mathrm{i}, \omega^3, -1, \omega^5, -\mathrm{i}, \omega^7$ with $\omega = \mathrm{e}^{\mathrm{i}\pi/4}$.

Step 3 (inverse-QFT readout). Because the ancilla is exactly $\mathrm{QFT}\,|1\rangle$, applying QFT^{-1} returns $|1\rangle = |001\rangle$ deterministically. Measuring reads 001, so $\tilde{\varphi} = 0.001_2 = \frac{1}{8}$ and the recovered eigenvalue is $\mathrm{e}^{\mathrm{i}2\pi\tilde{\varphi}} = \mathrm{e}^{\mathrm{i}\pi/4}$—exactly the phase of U on $|1\rangle$. This is the noiseless idealization of Brody's hardware run, whose histogram is dominated by the outcome 001. The companion cases $U = Z = \mathrm{diag}(1, -1)$ with $|\psi\rangle = |1\rangle$ (giving $\varphi = \frac{1}{2}$, outcome 100) and $|\psi\rangle = |0\rangle$ (giving $\varphi = 0$, outcome 000) are read off the same way, since Z^{2^m} acts trivially for $m \geq 1$.

Remark 4.22 (When the eigenphase is not dyadic). If instead we pick a unitary with $\varphi = \frac{1}{3}$ (not of the form $k/8$), Step 2 still leaves the ancilla in $\frac{1}{\sqrt{8}} \sum_j \mathrm{e}^{\mathrm{i}2\pi(1/3)j}\,|j\rangle$, but this is no longer any $\mathrm{QFT}\,|k\rangle$. The inverse QFT then spreads the measurement over the levels $k/8$, peaking at the nearest one; the exact probabilities computed from the geometric-sum amplitude of Section 4.4 place weight ≈ 0.688 on $k = 3$ ($\frac{3}{8}$, the closest three-bit approximation to $0.333\ldots$), ≈ 0.175 on $k = 2$, and the remainder on the other six outcomes. A single run therefore returns the best three-bit estimate with high—but no longer unit—probability, exactly the peaked distribution analyzed next.

Remark 4.23 (Reading the eigenvalue off the bits; superposed eigenstates). Concretely, a $t = 3$ measurement string $b_1 b_2 b_3$ is read as the binary fraction $\tilde{\varphi} = 0.b_1 b_2 b_3$, and the eigenvalue is $\lambda = \mathrm{e}^{\mathrm{i}2\pi\tilde{\varphi}}$:

$b_1b_2b_3$	$\tilde{\varphi}$	λ	$b_1b_2b_3$	$\tilde{\varphi}$	λ
000	0	1	100	$\frac{1}{2}$	-1
001	$\frac{1}{8}$	$\mathrm{e}^{\mathrm{i}\pi/4}$	101	$\frac{5}{8}$	$\mathrm{e}^{\mathrm{i}5\pi/4}$
010	$\frac{1}{4}$	i	110	$\frac{3}{4}$	$-\mathrm{i}$
011	$\frac{3}{8}$	$\mathrm{e}^{\mathrm{i}3\pi/4}$	111	$\frac{7}{8}$	$\mathrm{e}^{\mathrm{i}7\pi/4}$

So QPE turns 0/1 measurement outcomes into a point on the unit circle (Brody, *The Joy of Quantum Computing*, Ch. 10, Table 10.1). When the input is a *superposition* of eigenstates rather than a single one, QPE instead *samples* the eigenvalues by overlap—the mechanism analyzed in Section 4.5.

4.4 Analysis of phase estimation

When φ is not a dyadic rational $k/2^t$, the inverse QFT no longer produces a single basis state; the measured value concentrates near the best t-bit approximation of φ, with amplitudes forming a peaked geometric sum. Write $\tilde{\varphi}_k = k/2^t$ and $T = 2^t$. The amplitude of outcome k after the inverse QFT is

$$\gamma_k = \frac{1}{T} \sum_{j=0}^{T-1} \mathrm{e}^{\mathrm{i}2\pi(\varphi - \tilde{\varphi}_k)j} = \frac{1}{T} \frac{1 - \mathrm{e}^{\mathrm{i}2\pi T(\varphi - \tilde{\varphi}_k)}}{1 - \mathrm{e}^{\mathrm{i}2\pi(\varphi - \tilde{\varphi}_k)}}.$$

When $\varphi = \tilde{\varphi}_k$ the sum telescopes to $\gamma_k = 1$; otherwise $|\gamma_k|^2$ is sharply peaked at the k closest to $T\varphi$.

Theorem 4.24 (Precision bound and qubit count)**.** *Let $|\varphi - \tilde{\varphi}_k|_1$ denote the distance from φ to $\tilde{\varphi}_k = k/T$ measured modulo 1 (the wrap-around distance on the circle). Then*

$$\mathbb{P}\big(|\varphi - \tilde{\varphi}_k|_1 \geq \epsilon\big) \leq \frac{1}{2T\epsilon} + \frac{1}{2(T\epsilon)^2}.$$

Consequently, to obtain the leading ℓ bits of φ with success probability at least $1 - \epsilon$ it suffices to take $t = \ell + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$ ancilla qubits.

Proof sketch. Let b be the integer with b/T the best t-bit approximation of φ, and set $\delta = \varphi - b/T$ with $|\delta| \le 1/(2T)$. Summing the geometric series and using $\left|1 - \mathrm{e}^{\mathrm{i}\theta}\right| = 2\,|\sin(\theta/2)|$ gives, for outcome $k = b + m$,

$$|\gamma_{b+m}|^2 = \frac{1}{T^2}\,\frac{\sin^2(\pi T(\delta - m/T))}{\sin^2(\pi(\delta - m/T))}.$$

For $|m| \ge e$ the numerator is at most 1 and, using $|\sin \pi x| \ge 2\,|x|$ for $|x| \le \frac{1}{2}$, the denominator is bounded below so that $|\gamma_{b+m}|^2 \lesssim 1/(4(m - \frac{1}{2})^2)$. Summing the tails $\sum_{|m|\ge e} |\gamma_{b+m}|^2$ and comparing with an integral yields a bound of order $1/e$; converting the level distance e to the phase distance $\epsilon = e/T$ produces the leading $1/(2T\epsilon)$ term, with $1/(2(T\epsilon)^2)$ the next order (the standard Nielsen–Chuang estimate). For the qubit count, resolving n bits needs $\epsilon_{\mathrm{res}} = 2^{-n}$; setting $t = n + p$ gives $T\epsilon_{\mathrm{res}} = 2^p$, and demanding $\frac{1}{2\cdot 2^p} + \frac{1}{2\cdot 2^{2p}} \le \epsilon$ yields $2^p \ge 2 + 1/(2\epsilon)$, hence $t = n + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$. □

The extra $\lceil \log_2(2 + 1/(2\epsilon)) \rceil$ qubits beyond the n bits of resolution are exactly the overhead paid to suppress the geometric-sum tails.

Kitaev's bit-by-bit method

An alternative to the full ancilla register reads the binary digits of φ one at a time by amplifying the phase through powers of U.

Lemma 4.25 (Phase amplification)**.** *If $U\,|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi}\,|\psi\rangle$, then $U^{2^j}\,|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi(2^j\varphi)}\,|\psi\rangle$, so a Hadamard test with U^{2^j} in place of U reads the amplified phase $2^j\varphi \pmod 1$.*

Proof. Applying the eigenrelation 2^j times gives $U^{2^j}\,|\psi\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi 2^j}\,|\psi\rangle$; the Hadamard-test probability depends on the phase only through $\mathrm{e}^{\mathrm{i}2\pi(2^j\varphi)}$, and phases are defined modulo 1. □

Suppose φ has an exact d-bit expansion $\varphi = (.\varphi_{d-1}\cdots\varphi_0)_2$. By Theorem 4.25, using U^{2^j} shifts the binary point j places left, so $2^j\varphi = (.\varphi_{d-j-1}\cdots\varphi_0)_2 \pmod 1$. When only one fractional bit remains ($j = d - 1$) the amplified phase is 0 or $\frac{1}{2}$, and the Hadamard test outputs $p(1) \in \{0, 1\}$, so a single measurement determines φ_0 deterministically. Working from $j = d - 1$ down to $j = 0$ and estimating each amplified phase to constant precision (e.g. within $1/16$, enough to round to the correct bit given the already-recovered lower bits), classical post-processing recovers all d bits.

Remark 4.26 (Depth-versus-samples tradeoff). The dominant cost is applying U^{2^j}, which uses 2^j applications of U. Summing over the required powers, $\mathcal{O}(\sum_j 2^j) = \mathcal{O}(2^d) = \mathcal{O}(1/\epsilon)$ since $2^{-d} \sim \epsilon$. Kitaev's method thus attains $\mathcal{O}(1/\epsilon)$ total query cost by using *deeper* circuits (higher powers of U) rather than *more* samples—a full order of magnitude better than the $\mathcal{O}(1/\epsilon^2)$ single-qubit estimator of Section 4.1.

4.5 Heisenberg-limited and iterative variants

Standard QPE assumes the system register holds a genuine eigenstate $|\psi\rangle$. In practice one prepares a superposition of eigenstates. Let U have spectral decomposition $U\,|\psi_k\rangle = \mathrm{e}^{\mathrm{i}2\pi\varphi_k}\,|\psi_k\rangle$ and suppose the input is $|\Psi\rangle = \sum_k c_k\,|\psi_k\rangle$ with $p_k = |c_k|^2$, $\sum_k p_k = 1$. By linearity QPE acts on each eigencomponent independently: the ancilla becomes correlated with each $|\psi_k\rangle$ through its own phase φ_k. Measuring the ancilla therefore yields an estimate of one eigenphase φ_k with probability p_k, and the system collapses to the corresponding $|\psi_k\rangle$:

$$|\Psi\rangle \xrightarrow{\text{QPE}} (\tilde{\varphi}_k,\, |\psi_k\rangle) \quad \text{with probability } p_k. \tag{4.4}$$

Remark 4.27 (Targeting and applications). To be confident of obtaining a particular target phase φ_0, one uses prior knowledge—e.g. that $\varphi_0 \in (a, b)$ while all other $\varphi_i > b$— and either post-selects on the measured interval or ensures the overlap $|c_0|^2$ is large. This eigenstate-superposition behavior is precisely what makes QPE useful downstream: in Shor's algorithm the input is an easily prepared superposition over eigenstates of the modular-multiplication operator, and QPE returns a random eigenphase from which the order is extracted by continued fractions; in the quantum linear-system solver of Chapter 10 the input $|b\rangle$ is a superposition of eigenvectors of the matrix A, and QPE writes each eigenvalue into an ancilla so a controlled rotation can implement $\lambda \mapsto 1/\lambda$.

Example 4.28 (Order finding: an easily prepared eigenstate superposition)**.** The eigenstate-superposition mechanism is what powers the order-finding heart of Shor's algorithm. Take the modular-multiplication unitary $U|J\rangle = |4J \bmod 15\rangle$ on four qubits, so $U|1\rangle = |4\rangle$ and $U|4\rangle = |16 \bmod 15\rangle = |1\rangle$; on the two-dimensional invariant subspace $\mathrm{span}\{|1\rangle, |4\rangle\}$ the operator U swaps the two basis vectors. Its eigenvectors are $|u_s\rangle = \frac{1}{\sqrt{2}}(|1\rangle + (-1)^s |4\rangle)$ for $s \in \{0, 1\}$, with $U|u_s\rangle = (-1)^s |u_s\rangle = \mathrm{e}^{\mathrm{i}2\pi(s/2)} |u_s\rangle$, so the eigenphases are $\varphi_s = s/2 \in \{0, \frac{1}{2}\}$; here the order is $r = 2$. Preparing a true eigenstate would require knowing r in advance, but the trivially prepared computational state $|1\rangle$ already decomposes as an equal superposition,

$$|1\rangle = \tfrac{1}{\sqrt{2}} |u_0\rangle + \tfrac{1}{\sqrt{2}} |u_1\rangle, \qquad p_0 = p_1 = \tfrac{1}{2}.$$

Running QPE with $|1\rangle$ in the system register therefore returns $\tilde{\varphi}_0 = 0$ (outcome 000) or $\tilde{\varphi}_1 = \frac{1}{2}$ (outcome 100 for $t = 3$), each with probability $\frac{1}{2}$. No eigenstate preparation is needed—the phase-kickback and collapse of Equation (4.4) do the work. The full order-finding algorithm—recovering r from the measured s/r by continued fractions, and thence the factors of 15—is developed in Chapter 5.

Iterative phase estimation and the Heisenberg limit. The full t-qubit register of Theorem 4.18 can be traded for a single ancilla reused t times: in *iterative phase estimation* one measures the bits of φ from least to most significant, feeding each recovered bit back as a classical phase correction on the ancilla before the next controlled power U^{2^k}. This reproduces the QFT readout with one ancilla and classical feed-forward, at the cost of an adaptive, sequential circuit. Both standard and iterative QPE reach a precision that scales as $\epsilon \sim 1/\mathcal{T}$ with the *total* evolution time $\mathcal{T} = \sum_k 2^k = \mathcal{O}(1/\epsilon)$, rather than the $\epsilon \sim 1/\sqrt{\mathcal{T}}$ of the sampling-based estimator of Section 4.1. This $1/\mathcal{T}$ scaling is the *Heisenberg limit*, the optimal precision-versus-resource tradeoff permitted by quantum mechanics; the shot-noise $1/\sqrt{\mathcal{T}}$ scaling is what one obtains from independent repetitions. The connection is made quantitative by the Fisher-information and Cramér–Rao framework of Chapter 2: coherently applying U^{2^k} increases the Fisher information about φ quadratically in the evolution time, saturating the Cramér–Rao bound at the Heisenberg-limited rate, whereas incoherent repetition only adds Fisher information linearly.

Exercises

Exercise 4.1. For $n = 2$ ($N = 4$), compute $\mathrm{QFT}|1\rangle$ directly from Theorem 4.5 and verify the result using the product representation (Theorem 4.11).

Exercise 4.2. Prove that $\mathrm{QFT}\,\mathrm{QFT}^\dagger = \mathbb{I}$ as well, by evaluating $\langle k'|\mathrm{QFT}\,\mathrm{QFT}^\dagger|k\rangle$ with Theorem 4.9. Show also that for $n = 1$ ($N = 2$, $\omega = -1$) the QFT equals the Hadamard gate, $\mathrm{QFT} = H$.

Exercise 4.3. Using Theorem 4.9, show that QFT maps the uniform superposition $N^{-1/2}\sum_{j=0}^{N-1}|j\rangle$ to the basis state $|0\rangle$, and interpret this as the statement that a constant signal has only a zero-frequency component.

Exercise 4.4. Write down, gate by gate, the $n=3$ QFT circuit (Hadamards, controlled-R_k gates, and SWAPs), and count the total number of gates. Confirm that it matches Theorem 4.12.

Exercise 4.5. Consider the "shift by 1" operator V defined by $V|j\rangle = |(j+1) \bmod N\rangle$. Show that each $|\psi_k\rangle := \mathrm{QFT}|k\rangle$ satisfies $V|\psi_k\rangle = \mathrm{e}^{-\mathrm{i}2\pi k/N}|\psi_k\rangle$, so the QFT diagonalizes the cyclic shift.

Exercise 4.6. (Draper adder.) Take $n=3$, $N=8$, and the constant $a=3$. Write down the three single-qubit phase gates $P_l^{(3)} = \mathrm{diag}(1, \mathrm{e}^{\mathrm{i}2\pi\cdot 3\cdot 2^{3-l}/8})$ of Theorem 4.14, and verify by direct computation that $\mathrm{QFT}^\dagger(\bigotimes_l P_l^{(3)})\mathrm{QFT}$ sends $|6\rangle \mapsto |1\rangle$, consistent with $6+3=9\equiv 1 \pmod 8$. Identify where the modular wrap-around enters.

Exercise 4.7. Verify the real Hadamard-test identity of Theorem 4.3 for a diagonal single-qubit unitary $U=\mathrm{diag}(1,\mathrm{e}^{\mathrm{i}2\pi\varphi})$ acting on the eigenstate $|\psi\rangle=|1\rangle$: compute $p(0)$ and $p(1)$ directly. Then show the imaginary variant gives $p(0)=\frac{1}{2}(1+\Im\langle\psi|U|\psi\rangle)$, and explain why the real test alone cannot distinguish φ from $1-\varphi$.

Exercise 4.8. Let $\varphi = 0.011_2 = \frac{3}{8}$ and $t=3$. Write the ancilla state after the controlled powers in standard QPE, identify it as $\mathrm{QFT}|k\rangle$ for the correct k, and conclude the measurement outcome.

Exercise 4.9. For $\varphi=\frac{1}{3}$ (not dyadic) and $t=3$, compute the probabilities $|\gamma_k|^2$ for $k=2$ and $k=3$ (the nearest levels $\frac{2}{8}$ and $\frac{3}{8}$) using $|\gamma_k|^2 = \frac{1}{T^2}\sin^2(\pi T(\varphi-k/T))/\sin^2(\pi(\varphi-k/T))$ with $T=8$, and comment on which outcome is most likely.

Exercise 4.10. Using Theorem 4.24, determine how many ancilla qubits t are needed to estimate φ to $n=6$ bits of precision with success probability at least 0.9 (so $\epsilon=0.1$).

Exercise 4.11. Explain why Kitaev's bit-by-bit method achieves query complexity $\mathcal{O}(1/\epsilon)$ while the single-qubit estimator of Section 4.1 needs $\mathcal{O}(1/\epsilon^2)$ samples. Identify precisely which resource—circuit depth versus number of repetitions—each method spends, and connect this to the Heisenberg versus shot-noise scaling of Section 4.5.

Exercise 4.12. Compute $\mathrm{QFT}|5\rangle$ for $n=3$ from Theorem 4.5. Separately, write the output of the QFT *circuit before its final SWAP*—the mirror image of the product representation Theorem 4.11, with qubit 1 carrying the full fraction $0.j_1j_2j_3$. Expand both over basis states $|k_1k_2k_3\rangle$ and exhibit the explicit bit-reversal permutation (the final SWAP of qubits 1 and 3) that maps the pre-SWAP output onto $\mathrm{QFT}|5\rangle$, identifying which basis states are exchanged and which are fixed.

Exercise 4.13. Run standard QPE with $t=3$ ancillas on the phase gate $U=\mathrm{diag}(1,\mathrm{e}^{\mathrm{i}\pi/4})$ and eigenstate $|\psi\rangle=|1\rangle$. Identify φ, write the ancilla state after the controlled powers as $\mathrm{QFT}|k\rangle$, and give the measurement outcome. Then repeat for $U=Z$ with $|\psi\rangle=|1\rangle$ and with $|\psi\rangle=|0\rangle$, explaining in each case why the controlled powers U^{2^m} with $m\geq 1$ may be omitted.

Further reading

- Lin Lin, *qasc notes*, Ch. 3 (quantum Fourier transform and phase estimation).
- Lin Lin, *live notes*, Ch. 16 (phase estimation and Fourier transforms).

- J. Brody, *The Joy of Quantum Computing*, Princeton University Press, 2025, Ch. 9 (QFT and inverse QFT) and Ch. 10 (quantum phase estimation), for concrete small-qubit walkthroughs and the order-finding application.
- T. G. Draper, *Addition on a quantum computer*, arXiv:quant-ph/0008033 (2000), for the Fourier-basis (QFT) adder.

Chapter 5

Applications of Phase Estimation

Learning objectives

- Estimate ground-state energies with QPE.
- Set up amplitude estimation and its quadratic advantage.
- Formulate the eigenvalue problem and near-optimal ground-state preparation.
- Cast order finding—the quantum core of Shor's factoring algorithm—as phase estimation of a modular-multiplication unitary, and recover the order by continued fractions.

Phase estimation (Chapter 4) is not an end in itself but a workhorse subroutine. This lecture assembles its three most consequential applications in scientific computing. First, we apply QPE to the unitary $\mathrm{e}^{-\mathrm{i}Ht}$ to read off the eigenphases—and hence the energies—of a Hamiltonian H, obtaining the *ground-state energy estimation* algorithm whose cost is governed by the overlap between the input state and the true ground state. Second, we show that *amplitude estimation*—the task of learning the weight $\sin^2(\theta/2)$ of a "good" component in a prepared state—is nothing but phase estimation of the Grover iterate of Chapter 3, and that this reduction turns the classical $\mathcal{O}(1/\epsilon^2)$ Monte Carlo cost into a quadratically better $\mathcal{O}(1/\epsilon)$. Third, we move from estimating an eigenvalue to *preparing* the corresponding eigenstate, describing at the statement level how a spectral filter—built from the quantum singular value transformation (QSVT) of Chapter 9—prepares the ground state with a cost that scales as the inverse square root of the overlap and the inverse of the spectral gap. Finally, we close with the *lower bounds* that show these algorithms are essentially optimal: a metrological Heisenberg limit for phase estimation, a Grover-type $\Omega(1/\sqrt{p_0})$ and $\Omega(1/\Delta)$ bound for ground-state preparation, and a pointer to the no-fast-forwarding theorem that will recur in Part III.

Throughout, H denotes a Hermitian matrix on $\mathbb{C}^N$ with $N = 2^n$ and eigendecomposition

$$H\,|\psi_j\rangle = \lambda_j\,|\psi_j\rangle\,, \qquad \lambda_0 \le \lambda_1 \le \cdots \le \lambda_{N-1},$$

so that λ_0 is the *ground-state energy* and $|\psi_0\rangle$ the *ground state*. The *spectral gap* is $\Delta := \lambda_1 - \lambda_0$. We follow the QPE conventions of Chapter 4: eigenphases live in $[0, 1)$, a t-ancilla register resolves them to $T = 2^t$ levels, and $\tilde{\varphi}_k = k/T$ denotes a candidate estimate.

5.1 Ground-state energy estimation

The most direct application of QPE to physics is to estimate the lowest energy of a Hamiltonian. Two Hamiltonians recur as running examples.

Example 5.1 (Transverse-field Ising model)**.** The one-dimensional transverse-field Ising model on n sites with nearest-neighbor coupling is

$$H = -\sum_{i=1}^{n-1} Z_i Z_{i+1} - g \sum_{i=1}^{n} X_i,$$

where X_i, Z_i are Pauli operators acting on site i and g tunes the field. The matrix H has dimension 2^n, exponential in the number of sites, so classical diagonalization is intractable for large n.

Remark 5.2 (The Ising Hamiltonian written out for $n = 1, 2, 3$). It helps to see H as an explicit matrix. With Z_i denoting Z on qubit i and the identity elsewhere (likewise X_i), the diagonal of H is the classical Ising energy $-\sum_i Z_i Z_{i+1}$ (a number for each computational string), and every off-diagonal entry equals $-g$ and connects two strings differing in a single bit (one X_i).

$n = 1$. The coupling sum is empty, leaving only the field:

$$H = -gX = \begin{pmatrix} 0 & -g \\ -g & 0 \end{pmatrix}, \qquad \lambda = \pm g, \quad \text{ground state } |+\rangle .$$

$n = 2$. $H = -Z_1 Z_2 - g(X_1 + X_2)$, where each single-site operator is a tensor product with the identity on the other qubit,

$$Z_1 = Z \otimes I = \mathrm{diag}(1, 1, -1, -1), \qquad Z_2 = I \otimes Z = \mathrm{diag}(1, -1, 1, -1),$$

and $X_1 = X \otimes I$, $X_2 = I \otimes X$ (so $Z_i |b_1 b_2\rangle = (-1)^{b_i} |b_1 b_2\rangle$—the sign is set by the i-th bit—and hence $Z_1 Z_2 = \mathrm{diag}(1, -1, -1, 1)$). In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$H = \begin{pmatrix} -1 & -g & -g & 0 \\ -g & 1 & 0 & -g \\ -g & 0 & 1 & -g \\ 0 & -g & -g & -1 \end{pmatrix},$$

whose diagonal $(-1, 1, 1, -1)$ is $-Z_1 Z_2$ (it is -1 when the two spins agree in the Z-basis, $|00\rangle, |11\rangle$). At the critical field $g = 1$ the ground energy is $\lambda_0 = -\sqrt{5} \approx -2.236$.

$n = 3$. $H = -(Z_1 Z_2 + Z_2 Z_3) - g(X_1 + X_2 + X_3)$; in the basis $|000\rangle, \ldots, |111\rangle$,

$$H = \begin{pmatrix} -2 & -g & -g & 0 & -g & 0 & 0 & 0 \\ -g & 0 & 0 & -g & 0 & -g & 0 & 0 \\ -g & 0 & 2 & -g & 0 & 0 & -g & 0 \\ 0 & -g & -g & 0 & 0 & 0 & 0 & -g \\ -g & 0 & 0 & 0 & 0 & -g & -g & 0 \\ 0 & -g & 0 & 0 & -g & 2 & 0 & -g \\ 0 & 0 & -g & 0 & -g & 0 & 0 & -g \\ 0 & 0 & 0 & -g & 0 & -g & -g & -2 \end{pmatrix}.$$

The diagonal $(-2, 0, 2, 0, 0, 2, 0, -2)$ reads off $-(Z_1 Z_2 + Z_2 Z_3)$ bitwise—e.g. $|000\rangle$ has both neighbor pairs aligned (-2), $|010\rangle$ has both frustrated $(+2)$—and the $-g$'s are exactly the edges of the Boolean cube $\{0, 1\}^3$ (each string joined to its three single-bit-flip neighbors). At $g = 1$, $\lambda_0 \approx -3.494$.

In each case the diagonal is the cheap, local Ising part and the transverse field supplies all the off-diagonal structure. The dimension 2^n doubles with every added site; the lowest eigenvalue λ_0 is the ground-state energy that the phase-estimation and block-encoding methods of this chapter and Chapter 9 estimate without ever forming this matrix.

Example 5.3 (Fermionic Hamiltonian in second quantization)**.** The central Hamiltonian of quantum chemistry describes electrons occupying a fixed set of n single-particle *spin-orbitals* $\{\chi_1, \dots, \chi_n\}$. After the Born–Oppenheimer approximation freezes the nuclei, the electronic energy is governed by the *second-quantized* Hamiltonian

$$H = \sum_{i,j=1}^{n} T_{ij}\, \hat{a}_i^\dagger \hat{a}_j + \sum_{i,j,k,l=1}^{n} V_{ijkl}\, \hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k,$$

whose ingredients we now unpack.

Occupation-number states. Instead of tracking electron *positions*, second quantization tracks which spin-orbitals are *occupied*. A basis state is a bit string $|f_1 f_2 \cdots f_n\rangle$ with $f_i \in \{0,1\}$ recording whether orbital i is empty or filled; these span the 2^n-dimensional Fock space. The Pauli exclusion principle—at most one electron per spin-orbital—is automatic, since $f_i \le 1$.

Creation and annihilation operators. The operator $\hat{a}_i^\dagger$ *adds* an electron to orbital i and $\hat{a}_i$ *removes* one,

$$\hat{a}_i^\dagger \,|\dots f_i{=}0 \dots\rangle = (-1)^{s_i}\,|\dots f_i{=}1 \dots\rangle\,, \qquad \hat{a}_i \,|\dots f_i{=}1 \dots\rangle = (-1)^{s_i}\,|\dots f_i{=}0 \dots\rangle\,,$$

while $\hat{a}_i^\dagger$ annihilates an already-filled orbital and $\hat{a}_i$ an empty one ($\hat{a}_i^\dagger\,|f_i{=}1\rangle = 0$, $\hat{a}_i\,|f_i{=}0\rangle = 0$). Because electrons are *fermions*, the wavefunction must change sign under exchange; this is carried by the factor $(-1)^{s_i}$ with $s_i = \sum_{p<i} f_p$ counting the electrons in earlier orbitals. Equivalently, the operators obey the *canonical anticommutation relations*

$$\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}\,\mathbb{I}, \qquad \{\hat{a}_i, \hat{a}_j\} = \{\hat{a}_i^\dagger, \hat{a}_j^\dagger\} = 0, \qquad \{A, B\} := AB + BA.$$

It is the *anti*commutator—not the commutator of bosons—that is the algebraic fingerprint of fermionic statistics.

The two terms. The coefficients are molecular integrals precomputed classically from the chosen orbitals (in atomic units). The *one-electron integrals*

$$T_{ij} = \int \chi_i^*(\mathbf{r}) \Big(-\tfrac{1}{2}\nabla^2 + V_{\text{nuc}}(\mathbf{r}) \Big) \chi_j(\mathbf{r})\, d\mathbf{r}$$

hold the kinetic energy and electron–nucleus attraction, and $\hat{a}_i^\dagger \hat{a}_j$ moves one electron from orbital j to orbital i. The *two-electron integrals* $V_{ijkl} = \iint \chi_i^*(\mathbf{r}_1)\chi_j^*(\mathbf{r}_2)\, r_{12}^{-1}\, \chi_k(\mathbf{r}_1)\chi_l(\mathbf{r}_2)\, d\mathbf{r}_1 d\mathbf{r}_2$ encode the Coulomb repulsion between electrons, and $\hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k$ scatters a pair of them. The lowest eigenvalue λ_0 of H is the molecular ground-state energy sought by electronic-structure theory.

The Jordan–Wigner transform. To run on a quantum computer we identify each spin-orbital with one qubit—occupied $\leftrightarrow |1\rangle$, empty $\leftrightarrow |0\rangle$—so a Fock basis state *is* a computational basis state. The single-qubit operators

$$\tfrac{1}{2}(X + \mathrm{i}Y) = |0\rangle\langle 1| \quad \text{(lowering, “annihilate”)}, \qquad \tfrac{1}{2}(X - \mathrm{i}Y) = |1\rangle\langle 0| \quad \text{(raising, “create”)}$$

flip one qubit between filled and empty. On their own, however, operators on different sites *commute*, whereas fermionic operators must *anticommute*. The Jordan–Wigner transform fixes this by attaching a string of Z's,

$$\hat{a}_i = Z^{\otimes(i-1)} \otimes \tfrac{1}{2}(X + \mathrm{i}Y) \otimes \mathbb{I}^{\otimes(n-i)}, \qquad \hat{a}_i^\dagger = Z^{\otimes(i-1)} \otimes \tfrac{1}{2}(X - \mathrm{i}Y) \otimes \mathbb{I}^{\otimes(n-i)}.$$

The leading $Z^{\otimes(i-1)}$ evaluates the parity $(-1)^{s_i}$ of the orbitals *below* i (recall $Z\,|1\rangle = -\,|1\rangle$, $Z\,|0\rangle = |0\rangle$), supplying precisely the fermionic sign; and because Z anticommutes with $X \pm \mathrm{i}Y$ on any

shared qubit, two such operators on different modes acquire the required extra minus sign, so the anticommutation relations above are reproduced exactly. Substituting these expressions turns every $\hat{a}_i^\dagger \hat{a}_j$ and $\hat{a}_i^\dagger \hat{a}_j^\dagger \hat{a}_l \hat{a}_k$ into a sum of Pauli strings, yielding a qubit Hamiltonian $H = \sum_\alpha c_\alpha P_\alpha$ of exactly the form QPE and the simulation methods of Part III can handle.

Cost. The price of the Z string is *locality*: a single $\hat{a}_i$ acts nontrivially on up to i qubits, so Jordan–Wigner Hamiltonian terms can be $\mathcal{O}(n)$-local. Alternative encodings (Bravyi–Kitaev, parity) reduce this to $\mathcal{O}(\log n)$-local operators at the cost of a more intricate mapping. Estimating λ_0 of the resulting qubit Hamiltonian is the central task of quantum chemistry on a quantum computer.

From eigenphase to energy

To bring QPE to bear we need a unitary whose eigenphases encode the eigenvalues of H. The natural choice is the time-evolution operator. Assume, after an affine rescaling of H if necessary, that all eigenvalues lie in $(0, \frac{1}{2})$, and set

$$U = \mathrm{e}^{\mathrm{i}2\pi H}, \qquad \text{so that} \qquad U\,|\psi_j\rangle = \mathrm{e}^{\mathrm{i}2\pi\lambda_j}\,|\psi_j\rangle\,. \tag{5.1}$$

Then the eigenphase of U on $|\psi_j\rangle$ is exactly λ_j, and running QPE on U estimates λ_j. Producing U from a description of H is the *Hamiltonian simulation* problem, the subject of Chapter 11; for now we assume U is available exactly and count queries to it.

The catch is that we cannot prepare the exact ground state $|\psi_0\rangle$—if we could, the problem would be largely solved. Instead we are given an *approximate* initial state,

$$|\phi\rangle = \sum_{k=0}^{N-1} c_k\,|\psi_k\rangle\,, \qquad p_0 := |\langle\phi|\psi_0\rangle|^2 = |c_0|^2\,, \tag{5.2}$$

whose *overlap* p_0 with the ground state is bounded below but generally less than 1. By linearity, QPE acts on each eigencomponent of $|\phi\rangle$ independently (Theorem 4.27): a single run returns an estimate of the eigenphase λ_k with probability $p_k = |c_k|^2$, collapsing the system to $|\psi_k\rangle$. In particular the ground-state energy λ_0 is returned with probability p_0 per run; the full circuit is Figure 5.1.

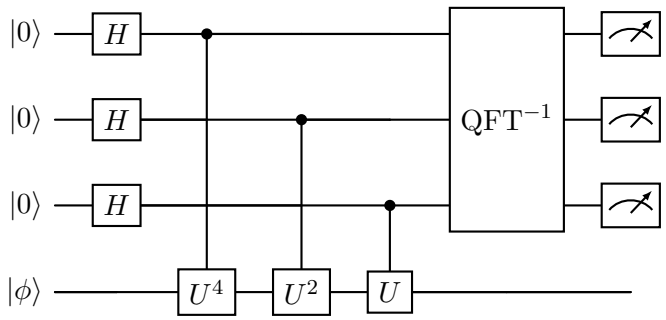


Figure 5.1: Ground-state energy estimation: phase estimation on $U = \mathrm{e}^{\mathrm{i}2\pi H}$ (Equation (5.1)), fed the trial state $|\phi\rangle$ of ground-state overlap p_0 (Equation (5.2)). The controlled powers U^{2^k} are controlled Hamiltonian simulations; by linearity a single run returns an estimate of some eigenenergy λ_k with probability $|c_k|^2$, so the ground energy λ_0 appears with probability p_0. The system register (one wire here) carries all n system qubits.

The min-over-runs estimator

Because λ_0 is the *smallest* energy, a natural strategy is:

Definition 5.4 (Ground-state energy estimation by repeated QPE)**.** Run QPE on $U = \mathrm{e}^{\mathrm{i}2\pi H}$ with input $|\phi\rangle$ a total of M times, recording the phase estimates $\tilde{\varphi}^{(1)}, \ldots, \tilde{\varphi}^{(M)}$. Return the minimum $\min_\ell \tilde{\varphi}^{(\ell)}$ as the estimate of λ_0.

If every λ_k had an exact t-bit representation, one run would return λ_0 exactly with probability p_0, and $M = \mathcal{O}(p_0^{-1} \log \delta^{-1})$ runs would suffice to see it at least once with probability $1 - \delta$. The subtlety is the *aliasing effect*: when λ_k is not a dyadic rational, the geometric-sum tails of Section 4.4 put nonzero weight on outcomes $\tilde{\varphi}_{k'} < \lambda_0$, i.e. QPE can report an energy *below* the true ground-state energy. Taking the minimum over runs makes such underestimates dangerous, and controlling them is what drives the cost.

Theorem 5.5 (Overlap-dependent cost of ground-state energy estimation)**.** *Let H have all eigenvalues in $(\epsilon, \frac{1}{2})$ and let p_0 be the initial overlap in (5.2). The estimator of Theorem 5.4 returns $\hat{\lambda}_0$ with $\left|\hat{\lambda}_0 - \lambda_0\right| \le \epsilon$ with probability at least $1 - \delta$ provided one uses*

$$t = d + \mathcal{O}(\log p_0^{-1} + \log \delta^{-1} + \log\log \delta^{-1})$$

ancilla qubits (with $\epsilon = 2^{-d}$), circuit depth $T = \mathcal{O}((\epsilon p_0 \delta)^{-1} \log \delta^{-1})$, and $M = \mathcal{O}(p_0^{-1} \log \delta^{-1})$ repetitions. The total query cost is

$$\mathcal{O}(\epsilon^{-1} \delta^{-1} p_0^{-2} (\log \delta^{-1})^2).$$

Proof sketch. Write $T = 2^t$ and let $\tilde{\varphi}_{k'} = k'/T$ be the measurement grid. By the Fejér-kernel analysis of Section 4.4, the probability of outcome k' on a single run is $\sum_k |c_k|^2 F_T(\lambda_k - \tilde{\varphi}_{k'})$ with a kernel obeying $F_T(x) \le 1/(4T^2 |x|_1^2)$, where $|\cdot|_1$ is the wrap-around distance. *Underestimation:* when $\tilde{\varphi}_{k'} \le \lambda_0 - \epsilon$ we have $|\lambda_k - \tilde{\varphi}_{k'}|_1 \ge \lambda_0 - \tilde{\varphi}_{k'} \ge \epsilon$ for every k, so summing the tail gives

$$\mathbb{P}(\tilde{\varphi}_{k'} \le \lambda_0 - \epsilon) \le \frac{1}{4T\epsilon} + \frac{1}{4(T\epsilon)^2} \le \frac{\delta'}{2},$$

on setting $T\epsilon = \delta'^{-1}$. Over M runs the minimum underestimates with probability at most $M\delta'/2$, so choosing $\delta' = M^{-1}\delta$ keeps it below $\delta/2$. *Overestimation:* conditioned on hitting $|\psi_0\rangle$ (probability p_0), the same kernel bound gives $\mathbb{P}(|\tilde{\varphi}_{k'} - \lambda_0|_1 < \epsilon) \ge p_0(1 - \delta')$, so a single run overshoots by ϵ with probability at most $1 - p_0/2$; over M runs the minimum overshoots with probability $\le (1 - p_0/2)^M \le \mathrm{e}^{-p_0 M/2}$, which is $\le \delta/2$ once $M = \lceil 2p_0^{-1} \log(2/\delta) \rceil$. Combining $\delta' = M^{-1}\delta$ with $T\epsilon = \delta'^{-1}$ yields the stated qubit count, depth, and—multiplying depth by repetitions—the total cost. □

Remark 5.6 (The overlap p_0 is the bottleneck). Theorem 5.5 exposes a delicate interplay. Suppressing overshoot forces $M = \mathcal{O}(p_0^{-1})$ repetitions; but more repetitions raise the chance that *some* run undershoots, which forces $\delta' = \mathcal{O}(M^{-1}) = \mathcal{O}(p_0)$ and hence circuit depth $T = \mathcal{O}(\delta'^{-1}) = \mathcal{O}(p_0^{-1})$. The product is $TM = \mathcal{O}(p_0^{-2})$: plain QPE has a *quadratic* dependence on the inverse overlap. When p_0 is exponentially small—as it can be for large chemical systems—this is prohibitive. Section 5.3 shows how spectral filtering improves the dependence to $\mathcal{O}(p_0^{-1/2})$.

Remark 5.7 (Energy estimation does not need a gap). Estimating λ_0 does not require a positive spectral gap Δ: any estimate λ with $|\lambda - \lambda_0| < \epsilon$ is a good energy estimate even if it happens to come from an excited state clustered near the ground state (Weyl's inequality guarantees eigenvalue stability). Preparing the ground *state*, by contrast, becomes ill-posed as $\Delta \to 0$: the ground and first-excited eigenvectors are then nearly degenerate and cannot be separated by any filter (the Davis–Kahan $\sin\theta$ theorem). This distinction organizes Section 5.3.

5.2 Amplitude estimation

Grover's algorithm (Chapter 3) rotates a state toward a marked "good" subspace; amplitude *estimation* instead *measures* the weight of that subspace, and it does so by feeding the Grover iterate into phase estimation. The result is a quadratic speedup over classical sampling for a huge range of estimation tasks.

Definition 5.8 (Amplitude estimation)**.** Let U_ϕ prepare a state $U_\phi|0^n\rangle = |\phi_0\rangle$ that decomposes as

$$|\phi_0\rangle = \sqrt{P}\,|\phi_{\text{good}}\rangle + \sqrt{1-P}\,|\phi_{\text{bad}}\rangle\,, \qquad \sqrt{P} = \sin\frac{\theta}{2}, \tag{5.3}$$

where $|\phi_{\text{good}}\rangle \perp |\phi_{\text{bad}}\rangle$. The *amplitude-estimation problem* is to estimate P (equivalently the angle θ) to additive precision ϵ.

Classically, if one can only sample—prepare $|\phi_0\rangle$, measure, and check whether the outcome is good—then P is a Bernoulli mean, and by the variance of a Bernoulli estimator one needs $\mathcal{O}(1/\epsilon^2)$ samples to resolve P to additive error ϵ. The quantum algorithm does better by working coherently.

Theorem 5.9 (Amplitude estimation as phase estimation of the Grover iterate)**.** *Let $R_{\text{good}} = \mathbb{I} - 2\,|\phi_{\text{good}}\rangle\langle\phi_{\text{good}}|$ reflect about the good subspace and $R_{\phi_0} = \mathbb{I} - 2\,|\phi_0\rangle\langle\phi_0|$ reflect about the initial state, and let $G = R_{\phi_0}R_{\text{good}}$ be the Grover iterate. On the invariant plane $\mathcal{P} = \text{span}\{|\phi_{\text{good}}\rangle\,, |\phi_{\text{bad}}\rangle\}$, G acts as rotation by θ and has eigenstates*

$$|\phi_\pm\rangle = \tfrac{1}{\sqrt{2}}\big(|\phi_{\text{good}}\rangle \pm \mathrm{i}\,|\phi_{\text{bad}}\rangle\big), \qquad G\,|\phi_\pm\rangle = \mathrm{e}^{\pm\mathrm{i}\theta}\,|\phi_\pm\rangle\,.$$

The prepared state has overlap $|\langle\phi_0|\phi_\pm\rangle|^2 = \frac{1}{2}$ with each eigenstate, so phase estimation of G on input $|\phi_0\rangle$ returns an estimate of $+\theta$ or $-\theta$, each with probability $\frac{1}{2}$, and no post-selection is needed.

Proof. By the amplitude-amplification analysis of Section 3.4, G restricted to $\mathcal{P}$ is the rotation $\left(\begin{smallmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{smallmatrix}\right)$ in the basis $\{|\phi_{\text{good}}\rangle\,, |\phi_{\text{bad}}\rangle\}$. A real rotation by θ has eigenvalues $\mathrm{e}^{\pm\mathrm{i}\theta}$ with eigenvectors $\frac{1}{\sqrt{2}}(|\phi_{\text{good}}\rangle \pm \mathrm{i}\,|\phi_{\text{bad}}\rangle) = |\phi_\pm\rangle$, as stated. Writing $|\phi_0\rangle = \sin(\theta/2)\,|\phi_{\text{good}}\rangle + \cos(\theta/2)\,|\phi_{\text{bad}}\rangle$ and projecting,

$$|\langle\phi_\pm|\phi_0\rangle|^2 = \tfrac{1}{2}\left|\sin\tfrac{\theta}{2} \mp \mathrm{i}\cos\tfrac{\theta}{2}\right|^2 = \tfrac{1}{2}(\sin^2\tfrac{\theta}{2} + \cos^2\tfrac{\theta}{2}) = \tfrac{1}{2}.$$

Thus $|\phi_0\rangle = \frac{1}{\sqrt{2}}(\mathrm{e}^{\mathrm{i}\alpha_+}\,|\phi_+\rangle + \mathrm{e}^{\mathrm{i}\alpha_-}\,|\phi_-\rangle)$ is an equal superposition of the two eigenstates; QPE (Theorem 4.18) applied to G collapses onto $|\phi_\pm\rangle$ with probability $\frac{1}{2}$ each and returns an estimate of the corresponding eigenphase $\pm\theta$. Since $P = \sin^2(\theta/2)$ is even in θ, either outcome determines P. □

The resulting circuit—prepare $|\phi_0\rangle = U_\phi|0^n\rangle$, then phase-estimate the Grover iterate G—is shown in Figure 5.2.

Query count and the quadratic advantage

Let QPE use $t = d + \lceil\log\delta^{-1}\rceil$ ancilla qubits, so it estimates θ to precision $\epsilon' = 2^{-d}$ with success probability $1-\delta$ (Theorem 4.24). The relationship $P = \sin^2(\theta/2)$ converts an error in θ into an error in P.

Theorem 5.10 (Amplitude estimation error and cost)**.** *With $\hat{P} = \sin^2(\tilde{\theta}/2)$ the estimator built from a QPE estimate $\tilde{\theta}$ of θ with $\left|\tilde{\theta} - \theta\right| \le \epsilon'$,*

$$\left|\hat{P} - P\right| \le \sqrt{P(1-P)}\,\epsilon' + \frac{\epsilon'^2}{4}. \tag{5.4}$$

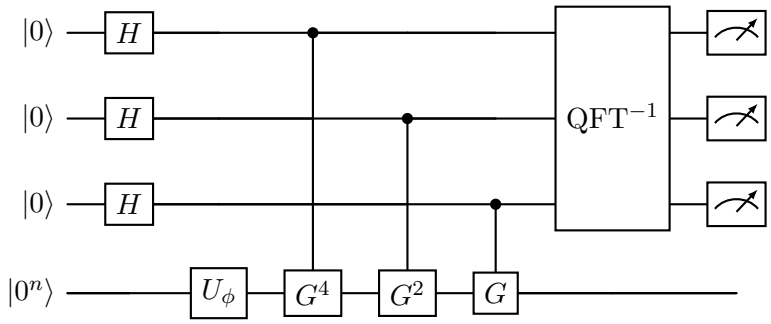


Figure 5.2: Amplitude estimation: prepare $|\phi_0\rangle = U_\phi |0^n\rangle$, then run phase estimation on the Grover iterate $G = R_{\phi_0} R_{\text{good}}$, whose eigenvalues on the invariant plane are $\mathrm{e}^{\pm i\theta}$ (Theorem 5.9). The measured t-bit estimate of θ yields $P = \sin^2(\theta/2)$ to additive error $\mathcal{O}(2^{-t})$, i.e. $\mathcal{O}(1/\epsilon)$ queries to G—quadratically fewer than the $\mathcal{O}(1/\epsilon^2)$ samples of classical Monte Carlo.

Consequently, if P is bounded away from 0 and 1 one takes $\epsilon' = \mathcal{O}(\epsilon)$ and QPE of G costs $\mathcal{O}(1/\epsilon)$ queries to G (hence to U_ϕ); if P is small one estimates it to multiplicative error ϵ by taking $\epsilon' = \sqrt{P}\,\epsilon$, at cost $\mathcal{O}(P^{-1/2}\epsilon^{-1})$. In both cases this is a quadratic improvement over the $\mathcal{O}(\epsilon^{-2})$ and $\mathcal{O}(P^{-1}\epsilon^{-2})$ sample counts of Monte Carlo estimation.

Proof. Using the product-to-sum identities,

$$\sin^2\tfrac{\tilde\theta}{2} - \sin^2\tfrac{\theta}{2} = \sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2}\sin(\tilde\theta-\theta) + (1-2\sin^2\tfrac{\theta}{2})\sin^2\tfrac{\tilde\theta-\theta}{2}.$$

The first term has magnitude $\sqrt{P(1-P)}\,\left|\sin(\tilde\theta-\theta)\right| \le \sqrt{P(1-P)}\,\epsilon'$ since $\sin(\theta/2)\cos(\theta/2) = \tfrac{1}{2}\sin\theta = \sqrt{P(1-P)}$ and $|\sin x| \le |x|$; the second is bounded by $|1-2P|\sin^2(\epsilon'/2) \le \epsilon'^2/4$. Adding gives (5.4). When $P(1-P) = \Theta(1)$ the linear term dominates and $\epsilon' = \mathcal{O}(\epsilon)$ suffices, so QPE queries G a total of $\mathcal{O}(1/\epsilon') = \mathcal{O}(1/\epsilon)$ times. When P is small, $\left|\hat P - P\right| \approx \sqrt{P}\,\epsilon'$, and demanding this be $< P\epsilon$ forces $\epsilon' = \sqrt{P}\,\epsilon$ and cost $\mathcal{O}(\epsilon'^{-1}) = \mathcal{O}(P^{-1/2}\epsilon^{-1})$. The classical Monte Carlo counts follow from the Bernoulli variance $P(1-P)$. □

Example 5.11 (Accelerating the Hadamard test)**.** Recall the Hadamard test of Section 4.1, which estimates $\Re\langle\psi|U|\psi\rangle$ through the outcome probability $p(0) = \tfrac{1}{2}(1 + \Re\langle\psi|U|\psi\rangle)$. Let $|\psi\rangle = U_\psi|0^n\rangle$. The combined circuit that applies H to an ancilla, controlled-U_ψ then controlled-U, produces

$$|\phi_0\rangle = \tfrac{1}{2}|0\rangle\left(|\psi\rangle + U|\psi\rangle\right) + \tfrac{1}{2}|1\rangle\left(|\psi\rangle - U|\psi\rangle\right) = \sqrt{P}\;|\phi_{\text{good}}\rangle + \sqrt{1-P}\;|\phi_{\text{bad}}\rangle\,,$$

with the good subspace flagged by the ancilla $|0\rangle$, so $P = p(0)$. Here $R_{\text{good}} = (\mathbb{I} - 2|0\rangle\langle 0|)\otimes\mathbb{I}^{\otimes n} = -Z\otimes\mathbb{I}^{\otimes n}$ is a single-qubit reflection and R_{ϕ_0} is built from the state-preparation unitary. Running amplitude estimation on $G = R_{\phi_0}R_{\text{good}}$ estimates $p(0)$—and hence $\Re\langle\psi|U|\psi\rangle$—to precision ϵ using $\mathcal{O}(1/\epsilon)$ queries, rather than the $\mathcal{O}(1/\epsilon^2)$ repetitions of naive sampling.

Remark 5.12 (In what sense is amplitude estimation "Heisenberg-limited"?). Amplitude estimation is often called Heisenberg-limited because it reduces to phase estimation of the Grover eigenphase and achieves error $\mathcal{O}(1/\mathcal{T})$ in the "Grover time" $\mathcal{T} = \mathcal{O}(2^t)$—the same $1/\mathcal{T}$ scaling as the metrological Heisenberg limit of Section 4.5. The interpretation differs, though: the metrological limit concerns a phase generated by physical time evolution $\mathrm{e}^{-\mathrm{i}Ht}$, whereas here G is a *constructed* unitary and $\mathcal{T}$ counts applications of the state-preparation oracle.

5.3 Eigenvalue problems and ground-state preparation

We now move from estimating λ_0 to *preparing* $|\psi_0\rangle$. As noted in Theorem 5.7, this requires a positive spectral gap Δ. The QPE-based approach of Section 5.1 already prepares $|\psi_0\rangle$ as a by-product—the system collapses to $|\psi_k\rangle$ when the ancilla reads λ_k—but at cost $\mathcal{O}(p_0^{-2})$ in the inverse overlap. The modern approach replaces the slowly-decaying Fejér kernel of QPE by a *sharp spectral filter* implemented with QSVT (Chapter 9), improving the overlap dependence to $\mathcal{O}(p_0^{-1/2})$.

The filtering idea

Suppose we know a threshold μ separating the ground energy from the rest, $\lambda_0 \le \mu - \frac{\Delta}{2} < \mu + \frac{\Delta}{2} \le \lambda_1$. Consider the (shifted) step function

$$\theta_\mu(x) = \begin{cases} 1, & x < \mu, \\ 0, & x > \mu, \end{cases}$$

so that the matrix function $\theta_\mu(H) = |\psi_0\rangle\langle\psi_0|$ is exactly the projector onto the ground state. If we could apply $\theta_\mu(H)$ to the initial state $|\phi\rangle$, we would obtain $\theta_\mu(H)\,|\phi\rangle = c_0\,|\psi_0\rangle$, which after normalization is precisely $|\psi_0\rangle$. QSVT lets us apply not the exact step function—which is discontinuous and would need infinite-degree polynomials—but a polynomial F that approximates it away from the forbidden window $(\mu - \frac{\Delta}{2}, \mu + \frac{\Delta}{2})$.

Definition 5.13 (Spectral filter polynomial)**.** A real even polynomial F is an (ϵ', μ, Δ)-*ground-state filter* if

$$|F(x) - 1| \le \epsilon' \quad \forall x \in [0, \mu - \tfrac{\Delta}{2}], \qquad |F(x)| \le \epsilon' \quad \forall x \in [\mu + \tfrac{\Delta}{2}, 1].$$

Such an F can be built from a polynomial approximation to the sign function, with degree $\deg F = \mathcal{O}(\Delta^{-1}\log(1/\epsilon'))$.

The degree bound is the crux: resolving a gap Δ costs $\mathcal{O}(1/\Delta)$ in polynomial degree, and hence in the number of block-encoding queries, while the precision ϵ' enters only logarithmically. Applying $F(H)$ to $|\phi\rangle = \sum_k c_k\,|\psi_k\rangle$,

$$F(H)\,|\phi\rangle = \sum_k c_k F(\lambda_k)\,|\psi_k\rangle = \underbrace{c_0 F(\lambda_0)}_{\approx c_0}\,|\psi_0\rangle + \sum_{k\ge 1} \underbrace{c_k F(\lambda_k)}_{\mathcal{O}(\epsilon')}\,|\psi_k\rangle\,,$$

so the excited components are suppressed to $\mathcal{O}(\epsilon')$ while the ground component is preserved. The QSVT circuit realizes $F(H)$ as a block encoding: on input $|0^m\rangle\,|\phi\rangle$ it outputs $|0^m\rangle\,F(H)\,|\phi\rangle + |\perp\rangle$, and measuring the ancilla flag $|0^m\rangle$ succeeds with probability $\approx p_0$, leaving the system in a state of fidelity $1 - \mathcal{O}(\epsilon'^2/p_0)$ with $|\psi_0\rangle$.

Near-optimal cost

Naively repeating until the flag reads $|0^m\rangle$ costs $\mathcal{O}(p_0^{-1})$ trials; amplitude amplification (Section 3.4) boosts the success amplitude and reduces this to $\mathcal{O}(p_0^{-1/2})$. This is the near-optimal ground-state preparation result.

Theorem 5.14 (Ground-state preparation with a known energy window)**.** *Let U_H be an (α, m)-block encoding of H with spectral gap $\Delta = \lambda_1 - \lambda_0 > 0$, let $|\phi\rangle = U_I\,|0^n\rangle$ have overlap $p_0 =$*

$|\langle\phi|\psi_0\rangle|^2 > 0$, and let μ satisfy $\lambda_0 \le \mu - \frac{\Delta}{2} < \mu + \frac{\Delta}{2} \le \lambda_1$. Then the ground state $|\psi_0\rangle$ can be prepared to fidelity at least $1-\epsilon$ using

$$\mathcal{O}(\alpha\,\Delta^{-1}\,p_0^{-1/2}\,\log(p_0^{-1}\epsilon^{-1}))$$

queries to U_H and its inverse, and $\mathcal{O}(p_0^{-1/2})$ queries to U_I and its inverse.

Proof idea. Take F an (ϵ', μ, Δ)-filter of degree $\mathcal{O}(\Delta^{-1}\log(1/\epsilon'))$ (Theorem 5.13) and implement $F(H)$ by QSVT, which uses $\mathcal{O}(\deg F)$ queries to $U_H, U_H^\dagger$ per application. The post-selected state has fidelity $1 - \mathcal{O}(\epsilon'^2/p_0)$ with $|\psi_0\rangle$, so $\epsilon' = \Theta(\sqrt{\epsilon p_0})$ gives fidelity $1-\epsilon$ and makes $\log(1/\epsilon') = \mathcal{O}(\log(p_0^{-1}\epsilon^{-1}))$. Amplitude amplification on the flag $|0^m\rangle$, whose success amplitude is $\Theta(\sqrt{p_0})$, needs $\mathcal{O}(p_0^{-1/2})$ rounds; multiplying the per-round query depth $\mathcal{O}(\alpha\Delta^{-1}\log(1/\epsilon'))$ by these rounds gives the stated U_H-query count, and each round uses $\mathcal{O}(1)$ calls to U_I. □

Remark 5.15 (Removing the a priori energy estimate). Theorem 5.14 assumes a known window μ. In practice μ is unknown and must itself be computed. A binary search over μ, in which each step solves a *fuzzy bisection* decision ("is λ_0 below $\mu - h$ or above $\mu + h$?") by estimating $\|F(H)\,|\phi\rangle\|$ with amplitude estimation, locates λ_0 to precision ϵ using $\widetilde{\mathcal{O}}(\alpha\,\epsilon^{-1}p_0^{-1/2}\log\delta^{-1})$ queries to U_H. The precision ϵ plays the role of an effective gap, so this achieves Heisenberg-limited $\mathcal{O}(1/\epsilon)$ scaling in the energy precision *without* assuming a positive gap; when a gap Δ is known one runs it only to precision $\Delta/2$ and then invokes Theorem 5.14.

Remark 5.16 (Why $p_0^{-1/2}$ and not p_0^{-1}). The square-root improvement over plain QPE (Theorem 5.6) is exactly the amplitude-amplification gain of Section 5.2: preparing the ground state is "finding a marked item" (the $|\psi_0\rangle$ component of $|\phi\rangle$, of weight p_0), and amplitude amplification finds it in $\mathcal{O}(p_0^{-1/2})$ rounds rather than the $\mathcal{O}(p_0^{-1})$ of incoherent sampling. The sharp QSVT filter is what makes the "marking" clean enough for amplification to apply.

5.4 Lower bounds

Are the algorithms above wasteful, or essentially optimal? Three lower bounds—one metrological, two from unstructured search—show that their key resource dependences cannot be improved.

The Heisenberg limit for phase estimation

For estimating an eigenphase φ through queries to $U(t) = \mathrm{e}^{-\mathrm{i}Ht}$, the natural resource is the total evolution time $\mathcal{T} = \sum_p |t_p|$. As discussed in Section 4.5, coherent phase accumulation attains precision $\epsilon \sim 1/\mathcal{T}$ (the *Heisenberg limit*), quadratically better than the $\epsilon \sim 1/\sqrt{\mathcal{T}}$ shot-noise scaling of incoherent repetition. This is not merely achievable but optimal.

Theorem 5.17 (Heisenberg-limit lower bound)**.** *Consider any protocol that applies controlled evolutions $U(t_1), \ldots, U(t_N)$ to a fixed eigenstate $|\psi\rangle$ with $U(t)\,|\psi\rangle = \mathrm{e}^{\mathrm{i}t\varphi}\,|\psi\rangle$, interleaved with arbitrary unitaries acting only on the ancilla register, and produces a locally unbiased estimator $\hat{\varphi}$ of φ. Then, with total evolution time $\mathcal{T} = \sum_p |t_p|$,*

$$\mathrm{Var}(\hat{\varphi}) \ge \frac{1}{4\mathcal{T}^2}.$$

Hence no protocol in this model beats the Heisenberg scaling $\epsilon = \Omega(1/\mathcal{T})$.

Proof idea. Each controlled query $U(t_p)$ imprints an effective phase rotation $V_p(\varphi) = (\mathbb{I} - \Pi_p) + e^{it_p\varphi}\Pi_p$ on the ancilla, where Π_p is the projector onto the "activated" subspace, so $\|\partial_\varphi V_p\| = |t_p|\,\|\Pi_p\| \le |t_p|$. The final state $|\Psi(\varphi)\rangle$ is a product of the V_p and ancilla unitaries; by the triangle inequality $\|\partial_\varphi \Psi\| \le \sum_p |t_p| = \mathcal{T}$. For a pure-state family the quantum Fisher information obeys $F_Q \le 4\,\|\partial_\varphi \Psi\|^2 \le 4\mathcal{T}^2$, and the quantum Cramér–Rao bound (Chapter 2) gives $\mathrm{Var}(\hat{\varphi}) \ge 1/F_Q \ge 1/(4\mathcal{T}^2)$. □

The Fisher-information viewpoint also explains why the *spacing* of evolution times matters: only exponentially growing times $t_p \propto 2^p$—as in Kitaev's method and standard QPE—accumulate the quadratic Fisher information $\sum_p t_p^2 = \Theta(\mathcal{T}^2)$ needed to saturate Theorem 5.17. A uniform grid $t_p = p\tau$ gives only $\sum_p t_p^2 = \Theta(\mathcal{T}^2/N)$ and reaches precision $\epsilon = \Theta(\mathcal{T}^{-2/3})$, between shot noise and the Heisenberg limit.

Search-based bounds for ground-state preparation

The cost $\mathcal{O}(p_0^{-1/2}\Delta^{-1})$ of Theorem 5.14 is optimal in *both* parameters, as one sees by reducing unstructured search—which needs $\Omega(\sqrt{N})$ queries (Chapter 3)—to ground-state preparation.

Theorem 5.18 (Overlap lower bound)**.** *There is a family of block-encoded Hamiltonians with constant gap $\Delta = \Omega(1)$ and overlap $p_0 \to 0$ for which preparing $|\psi_0\rangle$ to fidelity at least $\sqrt{3}/2$ requires $\Omega(p_0^{-1/2})$ queries to U_H.*

Proof idea. Take $U_H = U_t = \mathbb{I} - 2\,|t\rangle\langle t|$ for an unknown marked string t; it is Hermitian and unitary, is its own $(1,0)$-block encoding, has $|t\rangle$ as its unique ground state with gap 2, and the uniform superposition $|u\rangle = H^{\otimes n}\,|0^n\rangle$ has overlap $|\langle u|t\rangle|^2 = 1/N$, so $p_0 = 1/N$ and $\sqrt{p_0} = 1/\sqrt{N}$. Preparing the ground state to fidelity $\ge \sqrt{3}/2$ and measuring returns t with probability $\ge 3/4$. If this used $o(p_0^{-1/2}) = o(\sqrt{N})$ queries it would solve unstructured search in $o(\sqrt{N})$ queries, contradicting Grover's lower bound. □

Theorem 5.19 (Gap lower bound)**.** *There is a family with constant overlap $p_0 = \Omega(1)$ and gap $\Delta \to 0$ for which preparing $|\psi_0\rangle$ to fidelity at least $\sqrt{3}/2$ requires $\Omega(\Delta^{-1})$ queries to U_H.*

Proof idea. With $U_t = \mathbb{I} - 2\,|t\rangle\langle t|$ and $D = \mathbb{I} - 2\,|u\rangle\langle u|$ (both reflections implementable by Hadamards and one oracle call), the interpolated Hamiltonian $H(\tfrac{1}{2}) = \tfrac{1}{2}(D + U_t)$ acts nontrivially only on $\mathrm{span}\{|u\rangle, |t\rangle\}$, where it has eigenvalues $\pm 1/\sqrt{N}$: the gap is $\Delta = 2/\sqrt{N} \to 0$, while its ground state has constant overlap $\tfrac{1}{2} + \mathcal{O}(1/\sqrt{N})$ with both $|u\rangle$ and $|t\rangle$. Preparing that ground state and measuring yields t with constant probability, so $o(\Delta^{-1}) = o(\sqrt{N})$ queries would again break Grover's bound. □

No fast-forwarding, and a look ahead to Part III

Both bounds above trace back to the optimality of Grover search. A related and deeper obstruction governs the *simulation* of Hamiltonians and the solution of linear systems: the *no-fast-forwarding theorem* states that simulating a generic Hamiltonian for time t requires $\Omega(t)$ queries in the worst case— one cannot compress long-time dynamics into sublinear cost. This is why the quantum linear-system solvers previewed in Chapter 10 inherit a linear dependence on the condition number κ (the QLSP lower bound is $\Omega(\kappa)$, proved by encoding Hamiltonian simulation into a linear system), and why the Hamiltonian-simulation algorithms of Chapters 11 and 12 aim for cost *linear* in t rather than sublinear. We will meet the no-fast-forwarding theorem in detail when we build those

algorithms in Part III; for now it stands as the fundamental limit that phase estimation, amplitude estimation, and eigenstate preparation all press against.

5.5 Order finding and Shor's algorithm

The applications so far have been drawn from scientific computing. We close with the algorithm that first made phase estimation famous: *Shor's factoring algorithm*, whose sole quantum ingredient is *order finding*, and whose order-finding routine is nothing but QPE (Chapter 4) applied to a modular-multiplication unitary. Beyond its historical weight, it is the cleanest illustration of the recurring pattern of this lecture—encode a number-theoretic quantity in the eigenphase of an efficiently implementable unitary, then read that phase out with QPE—and it is why a large fault-tolerant quantum computer would break the RSA cryptosystem.

The number-theory wrapper: factoring reduces to order finding

Let $N = PQ$ be a product of two large primes, the modulus of an RSA public key. Classically the best known factoring algorithms run in time $e^{\mathcal{O}((\log N)^{1/3}(\log\log N)^{2/3})}$ (the number field sieve), superpolynomial in the bit length $\log N$. Shor's insight is that factoring reduces, in randomized polynomial classical time, to the problem of computing a *multiplicative order*.

Definition 5.20 (Multiplicative order)**.** For a coprime to N (that is, $\gcd(a, N) = 1$), the *order* of a modulo N is the least positive integer r with

$$a^r \equiv 1 \pmod{N}.$$

The classical wrapper around order finding is:

1. Pick $a \in \{2, \ldots, N-1\}$ uniformly at random. Compute $g = \gcd(a, N)$ by Euclid's algorithm. If $g > 1$ then g is already a nontrivial factor of N—return it. Otherwise $\gcd(a, N) = 1$ and we continue.
2. Use the quantum order-finding routine below to compute the order r of a mod N. *This is the only quantum step.*
3. If r is odd, or if $a^{r/2} \equiv -1 \pmod{N}$ (equivalently $a^{r/2} \bmod N = N - 1$), the run is unlucky; discard it and return to step 1 with a fresh a.
4. Otherwise compute $\gcd(a^{r/2} - 1,\, N)$ and $\gcd(a^{r/2} + 1,\, N)$. Each is a nontrivial factor of N.

Why does step 4 work? Since $a^r \equiv 1$, the difference of squares gives

$$(a^{r/2} - 1)(a^{r/2} + 1) = a^r - 1 \equiv 0 \pmod{N}, \tag{5.5}$$

so $N \mid (a^{r/2}-1)(a^{r/2}+1)$. Minimality of r forbids $a^{r/2} \equiv 1$ (that would make $r/2$ a smaller order), and step 3 has excluded $a^{r/2} \equiv -1$; hence N divides neither factor on the left of (5.5) alone, so its prime factors must split between the two factors. Consequently $\gcd(a^{r/2} \mp 1, N)$ is a nontrivial divisor of N. A standard number-theory estimate shows that a random a yields an even r with $a^{r/2} \not\equiv -1$ with probability at least $1/2$ when N has at least two distinct odd prime factors, so $\mathcal{O}(1)$ repetitions suffice on average. Every step but order finding—random choice, gcd, the parity test—is elementary classical arithmetic.

Order finding as phase estimation

Fix a with $\gcd(a, N) = 1$ and work on the state space $\mathbb{C}^{2^n}$ with $n = \lceil \log_2 N \rceil$, whose computational basis labels the residues $\{0, 1, \ldots, N-1\}$ (and unused labels $\geq N$). Define the *modular-multiplication unitary*

$$U_a |y\rangle = |ay \bmod N\rangle \quad (0 \leq y < N), \qquad U_a |y\rangle = |y\rangle \ \ (y \geq N). \tag{5.6}$$

Because multiplication by a permutes the residues coprime to N (its inverse is multiplication by $a^{-1} \bmod N$), U_a is a permutation matrix, hence unitary, and it is implemented with $\mathcal{O}((\log N)^2)$ elementary gates by modular-exponentiation circuits. The point is that U_a has an eigenbasis whose eigenphases expose the order r.

Lemma 5.21 (Eigenphases of the modular-multiplication unitary)**.** *Let r be the order of a mod N. For each integer $s \in \{0, 1, \ldots, r-1\}$ the state*

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \mathrm{e}^{-2\pi \mathrm{i}\, sk/r} \, |a^k \bmod N\rangle$$

is an eigenstate of U_a with eigenvalue $\mathrm{e}^{2\pi \mathrm{i}\, s/r}$:

$$U_a |u_s\rangle = \mathrm{e}^{2\pi \mathrm{i}\, s/r} |u_s\rangle .$$

Moreover $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$.

Proof. Applying U_a shifts $k \mapsto k+1$ in the cyclic group generated by a:

$$U_a |u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \mathrm{e}^{-2\pi \mathrm{i} sk/r} \, |a^{k+1} \bmod N\rangle = \mathrm{e}^{2\pi \mathrm{i} s/r} \, \frac{1}{\sqrt{r}} \sum_{k'=0}^{r-1} \mathrm{e}^{-2\pi \mathrm{i} sk'/r} \, |a^{k'} \bmod N\rangle ,$$

where we reindexed $k' = k+1$ and used $a^r \equiv 1$ so that the sum is periodic with period r; the reindexing multiplies each term by $\mathrm{e}^{2\pi \mathrm{i} s/r}$. The final identity follows because $\frac{1}{r} \sum_{s=0}^{r-1} \mathrm{e}^{-2\pi \mathrm{i} sk/r} = \delta_{k,0}$ picks out the $k = 0$ term $|a^0 \bmod N\rangle = |1\rangle$. □

Theorem 5.21 places us exactly in the QPE setting of Chapter 4: the eigenphase of U_a on $|u_s\rangle$ is $\varphi_s = s/r \in [0, 1)$. Running QPE on U_a with the t-qubit ancilla register of Chapter 4 therefore returns a t-bit estimate of some s/r. Two facts make this practical without knowing the eigenstates $|u_s\rangle$:

- *Cheap input state.* We cannot prepare an individual $|u_s\rangle$, but the last identity of Theorem 5.21 says the trivially-prepared state $|1\rangle$ is the uniform superposition of all the $|u_s\rangle$. By linearity (as in Equation (5.2) for energy estimation), QPE on input $|1\rangle$ returns an estimate of s/r for a uniformly random $s \in \{0, \ldots, r-1\}$.
- *Controlled powers.* QPE needs controlled $U_a^{2^j}$, and $U_a^{2^j} |y\rangle = |a^{2^j} y \bmod N\rangle$ is again a modular multiplication (by the precomputable constant $a^{2^j} \bmod N$), so the whole circuit is modular exponentiation, costing $\mathcal{O}((\log N)^2)$ gates per controlled power and $\mathcal{O}((\log N)^3)$ overall.

Assembling the two ingredients gives the *order-finding circuit* of Figure 5.3, the quantum core of Shor's algorithm.

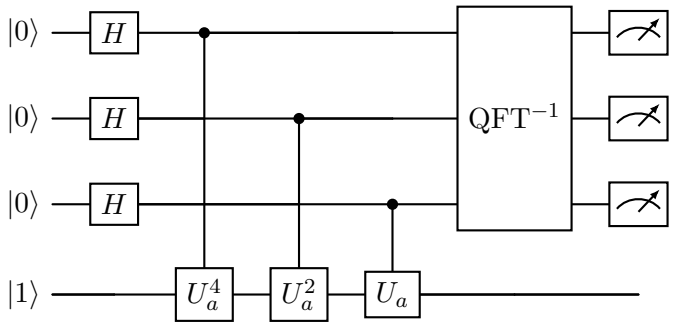


Figure 5.3: Order finding by phase estimation on the modular-multiplication unitary $U_a|y\rangle = |ay \bmod N\rangle$ (Equation (5.6)). The work register is prepared in the trivially-available $|1\rangle = \frac{1}{\sqrt{r}}\sum_s |u_s\rangle$ (Theorem 5.21), and each controlled $U_a^{2^k}$ is a modular multiplication by the precomputed constant $a^{2^k} \bmod N$ (modular exponentiation). A run returns a t-bit estimate of a random s/r; continued fractions then recover the order r.

Recovering r by continued fractions

QPE returns a dyadic rational $\tilde{\varphi} = k/2^t \approx s/r$. With $t = 2\log_2 N + \mathcal{O}(1)$ ancilla qubits, $\tilde{\varphi}$ is within $1/2N^2$ of the true s/r; since two distinct fractions with denominators below N differ by more than $1/N^2$, the continued-fraction expansion of $\tilde{\varphi}$ has a convergent s'/r' (in lowest terms) that equals s/r exactly. When $\gcd(s, r) = 1$ this convergent's denominator r' *is* r; otherwise it is a divisor of r, and testing $a^{r'} \equiv 1$ (or a few small multiples of r') recovers r. Continued fractions thus convert the approximate phase readout into an exact integer order.

Example 5.22 (Factoring $N = 15$ with $a = 7$)**.** Take $N = 15$ and, having drawn $a = 7$ with $\gcd(7, 15) = 1$, compute the order: $7^1 = 7$, $7^2 = 49 \equiv 4$, $7^3 \equiv 7 \cdot 4 = 28 \equiv 13$, $7^4 \equiv 7 \cdot 13 = 91 \equiv 1 \pmod{15}$, so $r = 4$. By Theorem 5.21 the unitary U_7 has eigenphases $s/4 \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$. Order finding uses $n = 4$ qubits for the residue register (since $15 = 1111_2$). With even a modest ancilla register these phases are dyadic, so QPE on input $|1\rangle$ returns one of the readouts

$$0.00_2 = \tfrac{0}{4}, \quad 0.01_2 = \tfrac{1}{4}, \quad 0.10_2 = \tfrac{1}{2}, \quad 0.11_2 = \tfrac{3}{4},$$

each with probability $\frac{1}{4}$. A nonzero outcome, say $\tilde{\varphi} = 0.01_2 = 1/4$, already presents $r = 4$ as its denominator; the continued-fraction step is trivial here. (The outcome 0 carries no information about r and is simply discarded.) Steps 3–4 of the wrapper then finish the job: $r = 4$ is even and $7^{r/2} = 7^2 = 49 \equiv 4 \not\equiv -1 \pmod{15}$, so

$$\gcd(7^2 - 1,\, 15) = \gcd(48, 15) = 3, \qquad \gcd(7^2 + 1,\, 15) = \gcd(50, 15) = 5,$$

recovering both prime factors, $15 = 3 \times 5$. (These match Brody's worked run of the order-finding circuit on real hardware, where the dominant outcomes for $a \in \{2, 7, 8, 13\}$ are exactly $\{00, 01, 10, 11\}$ $\frac{1}{4}$, giving $r = 4$.)

The continued-fraction machinery earns its keep only when r is not a power of two, so that s/r is a repeating binary fraction and the QPE readout is a genuine approximation. Factoring $N = 21$ with $a = 2$ (order $r = 6$) is the standard such example and is developed in Exercise 5.7.

Cost and the exponential speedup

Assembling the pieces: the residue and ancilla registers use $\mathcal{O}(\log N)$ qubits; QPE issues $\mathcal{O}(\log N)$ controlled powers of U_a, each a modular exponentiation of gate cost $\mathcal{O}((\log N)^2)$ (or $\widetilde{\mathcal{O}}(\log N)$ with fast multiplication), for a total quantum cost of $\mathcal{O}((\log N)^3)$; and the classical wrapper—gcd, continued fractions, and $\mathcal{O}(1)$ expected retries—adds only polynomially in $\log N$. The whole algorithm therefore runs in time polynomial in the bit length, an *exponential* speedup over the superpolynomial classical number field sieve. Because RSA's security rests precisely on the presumed classical hardness of factoring $N = PQ$, a scalable fault-tolerant machine running Shor's algorithm would recover the private key from the public modulus—this is the sense in which phase estimation "breaks the internet." The order-finding core is a strict generalization of the QPE reasoning we used for energies in Section 5.1 and for the Grover angle in amplitude estimation (Section 5.2): in every case a quantity of interest is written as the eigenphase of an efficiently implementable unitary and extracted by Chapter 4's phase-estimation circuit.

Exercises

Exercise 5.1. *Energy resolution versus ancilla count.* Suppose H has been rescaled so that $U = \mathrm{e}^{\mathrm{i}2\pi H}$ has all eigenphases in $[0, 1)$, and QPE is run on an exact eigenstate with t ancilla qubits. Show that the additive resolution in the energy λ is 2^{-t}, and determine how many ancilla qubits are needed to resolve λ_0 to 10^{-3} (in the rescaled units). If the physical energy range is $[E_{\min}, E_{\max}]$ with $E_{\max} - E_{\min} = W$, what is the resolution in physical units?

Exercise 5.2. *Amplitude-estimation query count.* Using Theorem 5.10, determine the number of applications of the Grover iterate G needed to estimate an amplitude P with $P(1 - P) = \Theta(1)$ to additive precision $\epsilon = 10^{-3}$, and compare with the number of independent samples a classical Monte Carlo estimator would need for the same precision. State the ratio of the two costs as a function of ϵ.

Exercise 5.3. *Overlap requirement for success probability $\geq \frac{1}{2}$.* In the QPE-based ground-state routine of Section 5.1, a single run returns (an estimate of) λ_0 with probability p_0. Show that to obtain λ_0 at least once with probability $\geq \frac{1}{2}$ in M independent runs one needs $M \geq \log 2 / \log((1 - p_0)^{-1})$, and simplify this for small p_0. Conclude that $M = \mathcal{O}(p_0^{-1})$ runs suffice.

Exercise 5.4. *Eigenphases of the Grover iterate.* Verify by direct matrix computation that the rotation $\left(\begin{smallmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{smallmatrix}\right)$ has eigenvalues $\mathrm{e}^{\pm\mathrm{i}\theta}$ with eigenvectors $\frac{1}{\sqrt{2}}(1, \mp\mathrm{i})^{\mathsf{T}}$, and confirm the overlaps $|\langle\phi_0|\phi_\pm\rangle|^2 = \frac{1}{2}$ used in Theorem 5.9.

Exercise 5.5. *Filter degree versus gap.* A ground-state filter (Theorem 5.13) has degree $\deg F = \mathcal{O}(\Delta^{-1} \log(1/\epsilon'))$. Explain why the degree must grow at least like Δ^{-1} (hint: a degree-D polynomial bounded by 1 on $[-1, 1]$ cannot change from ≈ 1 to ≈ 0 across an interval much shorter than $\mathcal{O}(1/D)$; recall the behavior of Chebyshev polynomials). Deduce how the query depth of Theorem 5.14 scales as $\Delta \to 0$ and connect this to the gap lower bound of Theorem 5.19.

Exercise 5.6. *Quadratic versus square-root overlap dependence.* Contrast the total cost of plain QPE ground-state energy estimation (Theorem 5.5, $\mathcal{O}(p_0^{-2})$ ignoring logs) with the QSVT-based preparation of Theorem 5.14 ($\mathcal{O}(p_0^{-1/2})$). Explain, in one or two sentences each, which mechanism produces each power of p_0, and why amplitude amplification is responsible for the improvement (cf. Theorem 5.16).

Exercise 5.7. *Factoring $N = 21$ with $a = 2$ by continued fractions.* Verify that the order of $2 \bmod 21$ is $r = 6$. With a 5-bit ancilla register, order finding on U_2 (Equation (5.6)) returns readouts approximating $s/6$; one such readout is $0.00101_2 = 5/32$. Compute the continued-fraction expansion of $5/32$, list its convergents, and identify the convergent whose denominator is the order $r = 6$ (check $2^6 \equiv 1 \pmod{21}$ and that no smaller candidate denominator works). Then complete Shor's algorithm: confirm r is even and $2^{r/2} \not\equiv -1$, and compute $\gcd(2^3 - 1, 21)$ and $\gcd(2^3 + 1, 21)$ to recover the factors of 21.

Exercise 5.8. *Eigenphases of the modular-multiplication unitary.* For $N = 15$ and $a = 7$, list the cycle $\{7^k \bmod 15 : k = 0, 1, 2, 3\}$ and use it to write the four eigenstates $|u_s\rangle$ ($s = 0, 1, 2, 3$) of U_7 from Theorem 5.21 explicitly. Verify directly that $U_7 |u_1\rangle = \mathrm{e}^{2\pi \mathrm{i}/4} |u_1\rangle = \mathrm{i} |u_1\rangle$, and confirm that $\frac{1}{2} \sum_{s=0}^{3} |u_s\rangle = |1\rangle$, so that preparing the easy state $|1\rangle$ is equivalent to sampling a uniformly random eigenphase $s/4$.

Further reading

- Lin Lin, *qasc notes*, Ch. 4 (ground-state energy estimation, amplitude estimation, HHL, and the Poisson-equation example).
- Lin Lin, *live notes*, Ch. 16 (amplitude estimation, Heisenberg limit, and the Cramér–Rao viewpoint).
- Lin Lin, *live notes*, Ch. 18 (eigenvalue problems, near-optimal ground-state preparation, and the matching lower bounds).
- J. Brody, *The Joy of Quantum Computing*, Princeton University Press, 2025, Ch. 10–11 (a hands-on account of order finding via phase estimation and Shor's factoring algorithm, including worked order-finding circuits for $a \bmod 15$ and the continued-fraction recovery of r when factoring 21).

Part II

Quantum Numerical Linear Algebra

Chapter 6

Input Models: State Preparation and Block Encoding

Learning objectives

- Describe the state-preparation oracle as the *vector* input model and state when it can be implemented efficiently.
- Define block encoding as the *matrix* input model, and read off the subnormalization factor α, ancilla count a, and error ϵ.
- Apply a block encoding to a prepared state to perform *matrix–vector multiplication*, and read its cost off α and the output norm.
- Combine input models by *matrix addition*, *vector addition*, and *matrix–matrix multiplication*, tracking how α and the ancilla count compose.
- Construct block encodings of 1-sparse and s-sparse matrices from entry and location oracles.
- Distinguish Hermitian from general block encodings and explain why the distinction matters for qubitization.
- Recognize the block-encoding relation in the general projector form $A = \alpha\,\Pi' U_A\,\Pi$, valid beyond the computational basis.

Part I treated quantum states and unitary circuits as the primitive objects. Part II turns to *quantum numerical linear algebra*: solving linear systems, applying matrix functions, simulating dynamics. Every such task begins with the same question—the *input model*: how does a matrix $A \in \mathbb{C}^{N\times N}$ (with $N = 2^n$), which is generally *not* unitary and may be dense, enter a quantum computer, whose only native operations are unitary? This lecture develops the two answers that underpin the entire QNLA toolbox. The first, the *state-preparation oracle*, loads a classical vector into the amplitudes of a quantum state and is the input model for the right-hand side of a linear system. The second, *block encoding*, embeds A (rescaled) as the top-left corner of a larger unitary U_A; it is the input model for the matrix itself and the single abstraction on which linear combination of unitaries (LCU, Chapter 7), qubitization and quantum signal processing (QSP, Chapter 8), the quantum singular value transformation (QSVT, Chapter 9), and quantum linear-system solvers (Chapter 10) are all built. We give the definition, quantify its cost with the *subnormalization factor* α, and construct block encodings of sparse matrices explicitly from a pair of oracles—one for entry values, one for nonzero locations. Throughout, A is an n-qubit matrix and $\|A\|$ denotes the spectral (operator) norm.

6.1 State-preparation oracles

Many algorithms are posed relative to a fixed input vector: the linear-system problem $Ax = b$ needs the right-hand side b; ground-state energy estimation (Section 5.1) needs an initial guess $|\phi\rangle$; a Monte Carlo estimand is encoded in a prepared amplitude (Section 5.2). In each case the vector must be loaded into the amplitudes of an n-qubit register. This is the vector input model.

Definition 6.1 (State-preparation oracle)**.** Let $u \in \mathbb{C}^N$ be a nonzero vector with normalized version $|u\rangle = u/\|u\|_2 = \sum_{i=0}^{N-1} u_i |i\rangle / \|u\|_2$. A *state-preparation oracle* for u is an n-qubit unitary O_u with

$$O_u |0^n\rangle = |u\rangle . \tag{6.1}$$

The oracle is *efficient* if O_u is implemented by a circuit of size $\mathrm{poly}(n) = \mathrm{polylog}(N)$.

Only the direction of u survives normalization; the norm $\|u\|_2$ is a classical scalar that must be tracked separately (it reappears, for example, as the scaling of the solution of a linear system). The essential subtlety is that (6.1) is trivial to *write down* but not always cheap to *implement*.

Proposition 6.2 (Cost of exact amplitude loading)**.** *For any $|u\rangle \in \mathbb{C}^N$ there exists a state-preparation unitary O_u satisfying (6.1) using $\mathcal{O}(N) = \mathcal{O}(2^n)$ elementary gates and no ancilla qubits. In general this cost cannot be reduced below $\Omega(2^n/n)$ gates for a worst-case $|u\rangle$.*

Proof sketch. Prepare $|u\rangle$ by a sequence of controlled single-qubit rotations, one per qubit, that build up the amplitudes conditionally on the bits already set. Concretely, the first qubit is rotated so that the probabilities of its two branches match the total weight of u on states with first bit 0 versus 1; conditioned on that bit, the second qubit is rotated to split the weight again; and so on. Level k of this tree requires 2^{k-1} controlled rotations (one per branch), so the total is $\sum_{k=1}^{n} 2^{k-1} = 2^n - 1 = \mathcal{O}(N)$ rotations, each with phase and magnitude angles read off from partial sums of $|u_i|^2$ and $\arg u_i$. For the lower bound, a general $|u\rangle$ carries $\Theta(2^n)$ independent real parameters, while a circuit of g two-qubit gates on n qubits has only $\mathcal{O}(g)$ continuous parameters; matching the parameter counts forces $g = \Omega(2^n/n)$. □

The message of Theorem 6.2 is that *arbitrary* amplitude loading is exponentially expensive and destroys any hope of a quantum speedup on its own. Efficient state preparation is possible only for *structured* vectors.

Example 6.3 (When state preparation is efficient)**.** Three broad structures admit $\mathrm{polylog}(N)$-size preparation.

- *Product (unentangled) states.* If $|u\rangle = \bigotimes_{k=1}^{n} |v_k\rangle$ with each $|v_k\rangle$ a single qubit, then $O_u = \bigotimes_k R_k$ is n single-qubit rotations. The uniform superposition $\mathsf{H}^{\otimes n} |0^n\rangle$ is the special case $|v_k\rangle = |+\rangle$.
- *Efficiently integrable distributions.* If $|u_i|^2 = p(i)$ for a probability density p whose cumulative distribution is classically computable, the Grover–Rudolph procedure loads $|u\rangle$ with $\mathcal{O}(n)$ controlled rotations, each rotation angle set from an integral of p over a dyadic subinterval.
- *Sparse vectors.* If u has only $d = \mathrm{poly}(n)$ nonzero entries whose indices and values are known, $|u\rangle$ can be prepared with $\mathcal{O}(dn)$ gates by loading the d amplitudes one at a time.

When no such structure is available one may fall back on quantum random-access memory (QRAM), which supplies O_u in $\mathcal{O}(n)$ *depth* at the price of $\mathcal{O}(N)$ hardware; whether QRAM is a realistic resource is a standing debate.

We henceforth treat O_u as a given oracle and count *queries* to it, exactly as we counted queries to $\mathrm{e}^{-\mathrm{i}Ht}$ in Chapter 5. The state-preparation oracle is the vector half of the input model; the rest of the lecture builds the matrix half.

6.2 Block encoding: definition and factor

A non-unitary A cannot be realized directly by a circuit. The idea of block encoding is to place a rescaled copy of A in one *block* of a larger unitary and to access it by projecting onto that block with ancilla qubits.

The one-ancilla picture

Start with the simplest case. Suppose we can find an $(n+1)$-qubit unitary U_A whose top-left n-qubit block equals A,

$$U_A = \begin{pmatrix} A & * \\ * & * \end{pmatrix}, \tag{6.2}$$

where $*$ denotes blocks irrelevant to the computation. For any n-qubit state $|b\rangle$, feed in the flagged state $|0\rangle|b\rangle$:

$$U_A|0\rangle|b\rangle = |0\rangle\ A|b\rangle + |1\rangle|\perp\rangle, \tag{6.3}$$

for some (unnormalized) state $|\perp\rangle$. To see why, split every $(n+1)$-qubit vector by its ancilla bit into a *top* block (ancilla $|0\rangle$) and a *bottom* block (ancilla $|1\rangle$), each an n-qubit vector. (We order the ancilla as the *leading* register, so $|0\rangle|b\rangle = |0\rangle \otimes |b\rangle = \begin{pmatrix} |b\rangle \\ 0 \end{pmatrix}$ and $|1\rangle|b\rangle = \begin{pmatrix} 0 \\ |b\rangle \end{pmatrix}$: the ancilla's $|0\rangle$ versus $|1\rangle$ amplitude selects the top versus bottom block.) In this split the flagged input has vanishing bottom block, $|0\rangle|b\rangle = \begin{pmatrix} |b\rangle \\ 0 \end{pmatrix}$, so, writing the lower-left block of U_A as C,

$$U_A|0\rangle|b\rangle = \begin{pmatrix} A & * \\ C & * \end{pmatrix}\begin{pmatrix} |b\rangle \\ 0 \end{pmatrix} = \begin{pmatrix} A|b\rangle \\ C|b\rangle \end{pmatrix} = |0\rangle\ A|b\rangle + |1\rangle\ |\perp\rangle, \qquad |\perp\rangle := C|b\rangle.$$

Because the input's bottom block is zero, only the first block-*column* of U_A is probed; of that column the top block A produces the ancilla-$|0\rangle$ part $|0\rangle\ A|b\rangle$, while the lower-left block C dumps the leftover amplitude into the ancilla-$|1\rangle$ "junk" state $|\perp\rangle$ (unnormalized, and irrelevant once we post-select). Equivalently $(\langle 0| \otimes \mathbb{I})\, U_A\, (|0\rangle \otimes \mathbb{I}) = A$: sandwiching U_A between the ancilla flags $|0\rangle$ extracts exactly its top-left block. Measuring the ancilla and keeping the outcome 0 leaves the system in the normalized state $A|b\rangle / \|A|b\rangle\|$, which is obtained with success probability

$$p(0) = \|A|b\rangle\|^2 = \langle b|\, A^\dagger A\, |b\rangle. \tag{6.4}$$

Thus the "missing" scalar $\|A|b\rangle\|$ is recoverable from the measurement statistics; it depends only on A and $|b\rangle$, not on the irrelevant blocks of U_A. Because a block of a unitary can have operator norm at most 1, the embedding (6.2) requires $\|A\| \le 1$.

Rescaling and the general definition

Most matrices of interest have $\|A\| > 1$, or one simply cannot embed A exactly. Both obstacles are removed by allowing U_A to encode A/α for a *subnormalization factor* $\alpha \ge \|A\|$, using a ancilla qubits, and tolerating an error ϵ.

Definition 6.4 (Block encoding)**.** Let A be an n-qubit matrix and let $\alpha, \epsilon \ge 0$ and $a \in \mathbb{N}$. An $(a+n)$-qubit unitary U_A is an (α, a, ϵ)-*block-encoding* of A if

$$\left\| A - \alpha\, (\langle 0^a| \otimes \mathbb{I}_n)\, U_A\, (|0^a\rangle \otimes \mathbb{I}_n) \right\| \le \epsilon. \tag{6.5}$$

When $\epsilon = 0$ the encoding is *exact* and U_A is called an (α, a)-block-encoding. We write $U_A \in \mathrm{BE}_{\alpha,a}(A, \epsilon)$, and $\mathrm{BE}_{\alpha,a}(A) = \mathrm{BE}_{\alpha,a}(A, 0)$.

Equivalently, in the exact case the top-left n-qubit block of U_A equals A/α:

$$\left(\langle 0^a| \otimes \mathbb{I}_n\right) U_A \left(|0^a\rangle \otimes \mathbb{I}_n\right) = \frac{A}{\alpha}, \qquad U_A = \begin{pmatrix} A/\alpha & * \\ * & * \end{pmatrix}, \tag{6.6}$$

with 2^a block rows and columns. The three parameters are the currency of Part II:

- α (the *subnormalization factor*) rescales A into a block of a unitary and so obeys $\alpha \geq \|A\|$. It is not free: applying U_A to $|0^a\rangle |b\rangle$ produces $|0^a\rangle (A/\alpha)|b\rangle + |\perp\rangle$, so the flag $|0^a\rangle$ is read with probability $\|A|b\rangle\|^2/\alpha^2$. A larger α shrinks the success probability and, more importantly, multiplies the query cost of every downstream algorithm (QSVT, QLSP) by α. Keeping α close to $\|A\|$ is therefore a central design goal.
- a is the number of ancilla (flag) qubits; the useful subspace is flagged by $|0^a\rangle$.
- ϵ is the spectral-norm error, which propagates additively through the arithmetic of Chapter 7. The verification rule is worth stating separately, as it is used constantly.

Remark 6.5 (Verifying a block encoding entrywise). $U_A \in \mathrm{BE}_{1,a}(A)$ if and only if $\langle 0^a, i| U_A |0^a, j\rangle = A_{ij}$ for all $i, j \in [N]$, where $|0^a, j\rangle := |0^a\rangle |j\rangle$. When $U_A = U_B U_C$ factors, one may compute the matrix element by splitting the product, $\langle 0^a, i| U_A |0^a, j\rangle = (U_B^\dagger |0^a, i\rangle)^\dagger (U_C |0^a, j\rangle)$, evaluating the two states independently and taking their inner product. This is the workhorse calculation for every construction below.

Every contraction is block-encodable

Block encoding is not a restrictive assumption in principle—it is the way any contraction can be accessed—as the following construction shows.

Proposition 6.6 ($(1,1)$-block-encoding is universal for contractions)**.** *Every n-qubit matrix A with $\|A\| \leq 1$ admits an exact $(1,1)$-block-encoding.*

Proof. Let $A = W\Sigma V^\dagger$ be a singular value decomposition, with W, V unitary and $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_N)$, $\sigma_i \in [0,1]$ since $\|A\| \leq 1$. Because $\sigma_i \leq 1$, the diagonal matrix $\sqrt{\mathbb{I} - \Sigma^2}$ is real, and the $(n+1)$-qubit matrix

$$U_A = \begin{pmatrix} W & 0 \\ 0 & \mathbb{I}_n \end{pmatrix} \begin{pmatrix} \Sigma & \sqrt{\mathbb{I}_n - \Sigma^2} \\ \sqrt{\mathbb{I}_n - \Sigma^2} & -\Sigma \end{pmatrix} \begin{pmatrix} V^\dagger & 0 \\ 0 & \mathbb{I}_n \end{pmatrix} \tag{6.7}$$

is unitary: the outer factors are unitary, and the middle factor is unitary because each 2×2 block $\begin{pmatrix} \sigma_i & \sqrt{1-\sigma_i^2} \\ \sqrt{1-\sigma_i^2} & -\sigma_i \end{pmatrix}$ is a reflection (its columns are orthonormal). Its top-left block is $W\Sigma V^\dagger = A$, so $U_A \in \mathrm{BE}_{1,1}(A)$. □

Theorem 6.6 says block encoding costs nothing in principle. The content of Part II is that for *structured* A—sparse matrices (Section 6.5), sums and products of block encodings (Chapter 7), and matrices arising from physical Hamiltonians—one can build U_A with a polylog(N)-size circuit and a small factor α, so that the whole pipeline stays efficient.

Example 6.7 (Block-encoding a 2×2 matrix; reading off α)**.** Consider the single-qubit matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \|A\| = \frac{1+\sqrt{5}}{2} \approx 1.618,$$

whose spectral norm is the largest singular value (the golden ratio). Since $\|A\| > 1$, no exact $(1,\cdot)$-block-encoding exists; take $\alpha = 2 \geq \|A\|$ and $A/\alpha = \left(\begin{smallmatrix} 1/2 & 1/2 \\ 0 & 1/2 \end{smallmatrix}\right)$, which has $\|A/\alpha\| = \alpha^{-1}\|A\| \approx$

$0.809 < 1$. Its SVD gives a $(1,1)$-block-encoding of A/α by Theorem 6.6; regarded as a 2-qubit unitary acting on one flag qubit and one system qubit, this is a $(2,1)$-block-encoding of A. Concretely, with $b = 1/2$,

$$U_A = \begin{pmatrix} 1/2 & 1/2 & * & * \\ 0 & 1/2 & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix},$$

where the starred entries are fixed by demanding orthonormal columns. Applying U_A to $|0\rangle|b\rangle$ and post-selecting on the flag $|0\rangle$ returns $A|b\rangle / \|A|b\rangle\|$ with probability $\|A|b\rangle\|^2/\alpha^2$; had we been wasteful and chosen $\alpha = 10$, the same state would still emerge but the success probability—and hence the runtime—would drop by a factor 25. This is the practical meaning of "keep α close to $\|A\|$."

6.3 Matrix–vector multiplication: the first primitive

With both input models in hand we can already perform the most basic quantum numerical-linear-algebra operation: given a matrix A as a block encoding and a vector b as a state-preparation oracle, produce the normalized matrix–vector product $A|b\rangle / \|A|b\rangle\|$. This one primitive is the engine inside linear solvers, power iteration, and the read-out of a simulated state $e^{-iHt}|b\rangle$.

Proposition 6.8 (Applying a block encoding to a state)**.** *Let $U_A \in \mathrm{BE}_{\alpha,a}(A)$ and let O_b prepare $|b\rangle = O_b|0^n\rangle$. Then*

$$U_A\,(\mathbb{I}_a \otimes O_b)\,|0^a\rangle|0^n\rangle = |0^a\rangle\,\frac{A|b\rangle}{\alpha} + |\Phi^\perp\rangle, \qquad (|0^a\rangle\langle 0^a| \otimes \mathbb{I})\,|\Phi^\perp\rangle = 0,$$

and measuring the a-qubit flag returns $|0^a\rangle$ with probability $p = \|A|b\rangle\|^2/\alpha^2$, upon which the system register collapses to $A|b\rangle / \|A|b\rangle\|$.

Proof. O_b produces $|0^a\rangle|b\rangle$; applying U_A and using $(\langle 0^a| \otimes \mathbb{I})U_A(|0^a\rangle \otimes \mathbb{I}) = A/\alpha$ (Theorem 6.4) gives the stated split, whose flag-$|0^a\rangle$ part is $|0^a\rangle(A/\alpha)|b\rangle$. Its squared norm $\|(A/\alpha)|b\rangle\|^2 = \|A|b\rangle\|^2/\alpha^2$ is the probability of reading $|0^a\rangle$, and the post-measurement system state is the normalized $A|b\rangle / \|A|b\rangle\|$. □

Three features make this the template for all of Part II.

- *Input and output share one format.* The result is again an n-qubit state, like the input $|b\rangle$. Block encoding is thus both an *input* and an *output* model: the product feeds straight into another block encoding, so matrix–vector products chain into $A_k \cdots A_1|b\rangle$ with no read-out of intermediate vectors.
- *Cost is set by α and the output norm.* One run succeeds with probability $p = \|A|b\rangle\|^2/\alpha^2$, so naive repetition costs $\mathcal{O}(\alpha^2/\|A|b\rangle\|^2)$ queries. Amplitude amplification (Theorem 3.32) removes the square and brings this to $\mathcal{O}(\alpha/\|A|b\rangle\|)$ queries to U_A and O_b—the quadratic gain that makes the primitive worthwhile, and the quantitative reason to keep α small: every matrix–vector product pays for it linearly.
- *Only the direction is produced.* The scalar $\|A|b\rangle\|$ is not carried by the output state; it is recovered from the success probability as $\|A|b\rangle\| = \alpha\sqrt{p}$, exactly as the norm of a prepared vector had to be tracked classically in Section 6.1.

Example 6.9 (Matrix–vector product with the 2×2 encoding)**.** Take $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ with the $(2,1)$-block-encoding of Theorem 6.7 ($\alpha = 2$). For $|b\rangle = |0\rangle$ (so $O_b = \mathbb{I}$), $A|0\rangle = |0\rangle$ with $\||A|0\rangle\| = 1$, so the flag is read with probability $p = 1/\alpha^2 = 1/4$ and the output is $|0\rangle$. For $|b\rangle = |1\rangle$, $A|1\rangle = |0\rangle + |1\rangle$ has $\||A|1\rangle\| = \sqrt{2}$, giving $p = 2/4 = 1/2$ and output $(|0\rangle + |1\rangle)/\sqrt{2}$. The larger the output norm, the larger the success probability; and had we used the wasteful $\alpha = 10$ of Theorem 6.7, both probabilities would drop by a factor 25.

6.4 Adding and multiplying block encodings

Matrix–vector multiplication (Section 6.3) was the first operation on these input models; block encodings and state-preparation oracles also *combine* to make new ones. We record three such rules, each with a concrete instance: adding two block-encoded matrices, adding two prepared vectors, and multiplying two block-encoded matrices—so that the output is again a block encoding (or a state) ready for the next step. The general construction for an arbitrary number of terms—the linear combination of unitaries (LCU)—and its ℓ^1-norm cost are developed in Chapter 7.

Matrix addition

Proposition 6.10 (Sum of two block encodings)**.** *Let $U_A \in \mathrm{BE}_{\alpha,a}(A)$ and $U_B \in \mathrm{BE}_{\beta,a}(B)$ share an a-qubit flag. Adjoin one control qubit, let* PREP *act on it by* $\mathrm{PREP}|0\rangle = \frac{1}{\sqrt{\alpha+\beta}}(\sqrt{\alpha}\,|0\rangle + \sqrt{\beta}\,|1\rangle)$*, and set* $\mathrm{SEL} = |0\rangle\langle 0| \otimes U_A + |1\rangle\langle 1| \otimes U_B$*. Then*

$$W = (\mathrm{PREP}^\dagger \otimes \mathbb{I})\,\mathrm{SEL}\,(\mathrm{PREP} \otimes \mathbb{I}) \in \mathrm{BE}_{\alpha+\beta,\,a+1}(A+B).$$

A weighted sum $c_A A + c_B B$ ($c_A, c_B \geq 0$) follows by loading weights $\propto \sqrt{c_A\alpha}, \sqrt{c_B\beta}$, giving subnormalization $c_A\alpha + c_B\beta$.

Example 6.11 (Block-encoding $\frac{1}{2}(X+Z)$)**.** Both X and Z are unitary, hence trivially $(1,0)$-block-encodings of themselves. To form their average, take one control qubit with $\mathrm{PREP} = \mathsf{H}$ (so $\mathrm{PREP}|0\rangle = |+\rangle$, equal weights) and $\mathrm{SEL} = |0\rangle\langle 0| \otimes X + |1\rangle\langle 1| \otimes Z$. Then

$$(\langle 0| \otimes \mathbb{I})\,W\,(|0\rangle \otimes \mathbb{I}) = \tfrac{1}{2}X + \tfrac{1}{2}Z = \frac{X+Z}{2} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} = \frac{1}{\sqrt{2}}\,\mathsf{H},$$

the Hadamard matrix rescaled by $1/\sqrt{2}$. So $W \in \mathrm{BE}_{1,1}(\frac{1}{2}(X+Z))$: the subnormalization is $\alpha = 1$ (the weights sum to one), comfortably above $\left\|\frac{1}{2}(X+Z)\right\| = 1/\sqrt{2}$.

Vector addition

The same control-qubit trick adds *vectors* when the selected operators are state-preparation oracles (Theorem 6.1) rather than block encodings.

Proposition 6.12 (Sum of prepared vectors)**.** *Let $O_0, \ldots, O_{L-1}$ prepare normalized states $|u_j\rangle = O_j|0^n\rangle$, and let $c_j \geq 0$. With an $\lceil \log_2 L \rceil$-qubit control register,* $\mathrm{PREP}|0\rangle = \frac{1}{\sqrt{\|c\|_1}} \sum_j \sqrt{c_j}\,|j\rangle$ *and* $\mathrm{SEL} = \sum_j |j\rangle\langle j| \otimes O_j$*, applying* $\mathrm{SEL}\,(\mathrm{PREP} \otimes \mathbb{I})$ *to* $|0\rangle\,|0^n\rangle$*, then* $\mathrm{PREP}^\dagger$*, and post-selecting the control flag $|0\rangle$ yields the normalized $\sum_j c_j\,|u_j\rangle \,/\, \left\|\sum_j c_j\,|u_j\rangle\right\|$ with success probability $\left\|\sum_j c_j\,|u_j\rangle\right\|^2 / \|c\|_1^2$.*

Example 6.13 (Preparing $|+\rangle$ by adding $|0\rangle$ and $|1\rangle$)**.** Take $|u_0\rangle = |0\rangle$ ($O_0 = \mathbb{I}$) and $|u_1\rangle = |1\rangle$ ($O_1 = X$) with equal weights $c_0 = c_1 = 1$ (so $\|c\|_1 = 2$). One control qubit with PREP $= \mathsf{H}$ and SEL $= |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes X$ gives, after post-selecting the flag $|0\rangle$, the system state $(|0\rangle + |1\rangle)/\sqrt{2} = |+\rangle$, with probability $\| |0\rangle + |1\rangle \|^2 / \|c\|_1^2 = 2/4 = \frac{1}{2}$. This is the vector analogue of Theorem 6.11: a sum of *states* must be renormalized, exactly as a sum of matrices carries the subnormalization $\|c\|_1$.

Matrix–matrix multiplication

Proposition 6.14 (Product of two block encodings)**.** *Let $U_A \in \mathrm{BE}_{\alpha,a}(A)$ and $U_B \in \mathrm{BE}_{\beta,b}(B)$ act on* disjoint *ancilla registers (sizes a and b) and a shared system. With the joint flag $|0^a\rangle\,|0^b\rangle$,*

$$U_{AB} = (U_A \otimes \mathbb{I}_b)(\mathbb{I}_a \otimes U_B) \in \mathrm{BE}_{\alpha\beta,\, a+b}(AB),$$

because the two flags are independent and the blocks separate, $(\langle 0^a|\,\langle 0^b| \otimes \mathbb{I})\, U_{AB}\, (|0^a\rangle\,|0^b\rangle \otimes \mathbb{I}) = (A/\alpha)(B/\beta)$. The subnormalizations multiply and the ancillas add. (Reusing a single *flag fails: the junk branch of U_B re-enters the flagged block of U_A; disjoint flags prevent this.)*

Example 6.15 (Squaring $\frac{1}{2}(X+Z)$)**.** Take $M = \frac{1}{2}(X+Z)$ with the $(1,1)$-block-encoding W of Theorem 6.11, and multiply two copies on two disjoint flag qubits: $W_{M^2} = (W \otimes \mathbb{I})(\mathbb{I} \otimes W) \in \mathrm{BE}_{1,2}(M^2)$. Since X and Z anticommute ($\{X, Z\} = XZ + ZX = 0$) and $X^2 = Z^2 = \mathbb{I}$,

$$M^2 = \frac{(X+Z)^2}{4} = \frac{X^2 + XZ + ZX + Z^2}{4} = \frac{\mathbb{I} + 0 + \mathbb{I}}{4} = \frac{\mathbb{I}}{2},$$

so W_{M^2} is a $(1,2)$-block-encoding of $\frac{1}{2}\mathbb{I}$ (subnormalization $1 \cdot 1 = 1$, ancilla count $1 + 1 = 2$). Longer products $A_k \cdots A_1$ just iterate the rule, multiplying subnormalizations to $\prod_j \alpha_j$ and adding ancillas—an exponential ancilla growth that Chapter 7 tames with a compression gadget.

6.5 Block encoding of s-sparse matrices

We now give the canonical efficient construction: a block encoding of a sparse matrix built from two oracles—one that supplies entry *values*, one that supplies nonzero *locations*. This is the model under which almost all QNLA complexity is stated.

The entry and location oracles

Assume $\|A\|_{\max} := \max_{ij} |A_{ij}| \le 1$ (rescale A by $\|A\|_{\max}$ otherwise). The *entry oracle* encodes each matrix element in the amplitude of a signal qubit by a controlled rotation:

$$O_A\,|0\rangle\,|i\rangle\,|j\rangle = \left(A_{ij}\,|0\rangle + \sqrt{1 - |A_{ij}|^2}\;|1\rangle\right)|i\rangle\,|j\rangle\,. \tag{6.8}$$

(In practice a database oracle $\widetilde{O}_A\,|0^d\rangle\,|i\rangle\,|j\rangle = |\tilde{A}_{ij}\rangle\,|i\rangle\,|j\rangle$ returning a d-bit representation of A_{ij} is converted into (6.8) by a controlled rotation followed by uncomputation of the work register.) A matrix is *s-sparse* if every row and column has at most s nonzero entries; the *location oracle* lists them. For a 1-sparse matrix each column j has a single nonzero entry in row $c(j)$, and c is a permutation, so there is a unitary

$$O_c\,|j\rangle = |c(j)\rangle\,, \qquad O_c^\dagger\,|c(j)\rangle = |j\rangle\,. \tag{6.9}$$

For general s-sparse A, write $c(j,\ell)$ for the row of the ℓ-th nonzero entry in column j ($\ell \in [s]$, $s = 2^{s'}$), and assume a unitary

$$O_c \left|\ell\right\rangle \left|j\right\rangle = \left|\ell\right\rangle \left|c(j,\ell)\right\rangle . \tag{6.10}$$

Reversibility of O_c requires that, given the row $i = c(j,\ell)$ and the index ℓ, the column j be recoverable; a banded matrix with $c(j,\ell) = j + \ell - \ell_0 \pmod N$ satisfies this. The entry oracle carries the extra ℓ-register along,

$$O_A \left|0\right\rangle \left|\ell\right\rangle \left|j\right\rangle = \left(A_{c(j,\ell),j} \left|0\right\rangle + \sqrt{1 - \left|A_{c(j,\ell),j}\right|^2} \left|1\right\rangle\right) \left|\ell\right\rangle \left|j\right\rangle . \tag{6.11}$$

The one-sparse case

Proposition 6.16 (Block encoding of a 1-sparse matrix)**.** *Let A be 1-sparse with $\|A\|_{\max} \le 1$, accessed via O_c of (6.9) and O_A of (6.8). Then $U_A := (\mathbb{I} \otimes O_c)\, O_A$ is a $(1,1)$-block-encoding of A.*

Proof. Apply U_A to the flagged basis state $\left|0\right\rangle \left|j\right\rangle$ and take the overlap with $\left|0\right\rangle \left|i\right\rangle$:

$$\left\langle 0\right| \left\langle i\right| U_A \left|0\right\rangle \left|j\right\rangle = \left\langle 0\right| \left\langle i\right| \left(A_{c(j),j} \left|0\right\rangle + \sqrt{1 - \left|A_{c(j),j}\right|^2} \left|1\right\rangle\right) \left|c(j)\right\rangle = A_{c(j),j}\, \delta_{i,c(j)}.$$

The $\left|1\right\rangle$ branch is annihilated by $\left\langle 0\right|$, and $\delta_{i,c(j)}$ selects the unique nonzero row, so the overlap equals A_{ij} (which is $A_{c(j),j}$ when $i = c(j)$ and 0 otherwise). By Theorem 6.5, $U_A \in \mathrm{BE}_{1,1}(A)$. □

Example 6.17 (A 1-sparse worked encoding: the cyclic shift)**.** Let $N = 4$ and let $A = S$ be the cyclic shift $S\left|j\right\rangle = \left|(j+1) \bmod 4\right\rangle$, a permutation matrix; it is 1-sparse with all nonzero entries equal to 1, so $c(j) = (j+1) \bmod 4$ and $A_{c(j),j} = 1$. The entry oracle (6.8) is trivial here, $O_A \left|0\right\rangle \left|i\right\rangle \left|j\right\rangle = \left|0\right\rangle \left|i\right\rangle \left|j\right\rangle$ (all magnitudes are 1, so no rotation onto $\left|1\right\rangle$), and $O_c \left|j\right\rangle = \left|(j+1) \bmod 4\right\rangle = S\left|j\right\rangle$. Then $U_A = (\mathbb{I}\otimes O_c)O_A$ acts as $\left|0\right\rangle \left|j\right\rangle \mapsto \left|0\right\rangle \left|(j+1) \bmod 4\right\rangle$, and $\left\langle 0, i\right| U_A \left|0, j\right\rangle = \delta_{i,(j+1) \bmod 4} = S_{ij}$. Since S is already unitary, U_A block-encodes it with $\alpha = 1$ and the flag qubit is inert—a reassuring sanity check that a unitary is its own $(1,0)$-block-encoding (here padded to one idle ancilla).

Example 6.18 (Block encoding of a diagonal matrix)**.** For a diagonal $A = \mathrm{diag}(A_{00}, \ldots, A_{N-1,N-1})$ the location map is the identity, $c(j) = j$, so $O_c = \mathbb{I}$ and the entry oracle collapses to $O_A \left|0\right\rangle \left|i\right\rangle = \left(A_{ii} \left|0\right\rangle + \sqrt{1 - |A_{ii}|^2} \left|1\right\rangle\right) \left|i\right\rangle$. Taking $U_A = O_A$ gives $\left\langle 0, i\right| U_A \left|0, j\right\rangle = A_{ii}\delta_{ij}$, so $U_A \in \mathrm{BE}_{1,1}(A)$: a diagonal matrix is block-encoded by a single multiply-controlled rotation.

The general s-sparse case

For $s > 1$ we superpose over the s nonzero locations in each column. Let $D := \mathsf{H}^{\otimes s'}$ be the *diffusion* operator (the term is inherited from Grover's search, Chapter 3), acting on the s'-qubit ℓ-register as

$$D\left|0^{s'}\right\rangle = \frac{1}{\sqrt{s}} \sum_{\ell \in [s]} \left|\ell\right\rangle . \tag{6.12}$$

Theorem 6.19 (Block encoding of an s-sparse matrix)**.** *Let A be s-sparse with $\|A\|_{\max} \le 1$, accessed via the location oracle O_c of (6.10) and the entry oracle O_A of (6.11). Define*

$$U_A := (\mathbb{I} \otimes D \otimes \mathbb{I})\, (\mathbb{I} \otimes O_c)\, O_A\, (\mathbb{I} \otimes D \otimes \mathbb{I}), \tag{6.13}$$

where the two D's act on the ℓ-register and the middle O_c writes the row index. Then $U_A \in \mathrm{BE}_{s,\, s'+1}(A)$: it is an $(s, s'+1)$-block-encoding of A with subnormalization factor $\alpha = s$.

Proof. The ancilla registers are the signal qubit (call it $|0\rangle$) and the s'-qubit ℓ-register; the flag is $|0\rangle|0^{s'}\rangle$. Call $|0\rangle|0^{s'}\rangle|j\rangle$ the *source* and $|0\rangle|0^{s'}\rangle|i\rangle$ the *target*. Propagate the source through D, then O_A, then O_c:

$$\begin{aligned}
|0\rangle|0^{s'}\rangle|j\rangle &\xrightarrow{D} \frac{1}{\sqrt{s}}\sum_{\ell\in[s]}|0\rangle|\ell\rangle|j\rangle \\
&\xrightarrow{O_A} \frac{1}{\sqrt{s}}\sum_{\ell\in[s]}\Big(A_{c(j,\ell),j}|0\rangle + \sqrt{1-\left|A_{c(j,\ell),j}\right|^2}|1\rangle\Big)|\ell\rangle|j\rangle \\
&\xrightarrow{O_c} \frac{1}{\sqrt{s}}\sum_{\ell\in[s]}\Big(A_{c(j,\ell),j}|0\rangle + \sqrt{1-\left|A_{c(j,\ell),j}\right|^2}|1\rangle\Big)|\ell\rangle|c(j,\ell)\rangle\,.
\end{aligned}$$

Since only the flag component survives, evolve the target backward, equivalently apply the (Hermitian) D to it:

$$|0\rangle|0^{s'}\rangle|i\rangle \xrightarrow{D} \frac{1}{\sqrt{s}}\sum_{\ell'\in[s]}|0\rangle|\ell'\rangle|i\rangle\,.$$

The block matrix element is the inner product of the two, in which $\langle 0|$ kills the $|1\rangle$ branch and $\langle\ell'|\ell\rangle = \delta_{\ell\ell'}$, $\langle i|c(j,\ell)\rangle = \delta_{i,c(j,\ell)}$ collapse the double sum:

$$\langle 0|\langle 0^{s'}|\langle i|\,U_A\,|0\rangle|0^{s'}\rangle|j\rangle = \frac{1}{s}\sum_{\ell}A_{c(j,\ell),j}\,\delta_{i,c(j,\ell)} = \frac{1}{s}A_{ij},$$

because for each row i there is a unique ℓ with $c(j,\ell) = i$ (or the term is absent, giving $A_{ij} = 0$). Thus the top-left block is A/s, i.e. $U_A \in \mathrm{BE}_{s,s'+1}(A)$. □

The factor $\alpha = s$ is exactly the price of the two diffusion operators: each D spreads a unit amplitude over s locations with weight $1/\sqrt{s}$, and the two of them contribute $1/s$ to the surviving block. This is why sparsity controls cost in QNLA: an s-sparse matrix is block-encoded with $\alpha = s$, and downstream query counts scale linearly in α.

Remark 6.20 (More general location oracles). The single-oracle model (6.10) is restrictive: it demands that the row $c(j,\ell)$ be a fixed function of (j,ℓ) with a matching inverse, which is natural only for banded matrices. A more flexible model uses two oracles, $O_c|\ell\rangle|j\rangle = |c(j,\ell)\rangle|j\rangle$ and $O_r|\ell\rangle|i\rangle = |r(i,\ell)\rangle|i\rangle$, listing the ℓ-th nonzero in column j and row i respectively; inserting a SWAP between them yields $U_A \in \mathrm{BE}_{s,n+1}(A)$ by the same inner-product bookkeeping (this is the construction of Gilyén–Su–Low–Wiebe, up to two saved qubits). The value oracle is correspondingly controlled on both indices, $O_A|0\rangle|i\rangle|j\rangle = (A_{ij}|0\rangle + \sqrt{1-|A_{ij}|^2}|1\rangle)|i\rangle|j\rangle$; entries with $A_{ij} = 0$ never contribute, so the rotation there is immaterial.

6.6 Hermitian block encoding

The block encoding of Theorem 6.4 constrains only the top-left block of U_A; the unitary itself need not be Hermitian even when A is. Yet several downstream constructions—most importantly qubitization (Chapter 8)—are cleanest when the encoding unitary is *itself* Hermitian.

Definition 6.21 (Hermitian block encoding)**.** An (α,a,ϵ)-block-encoding U_A of A is a (α,a,ϵ)-*Hermitian-block-encoding* if $U_A = U_A^\dagger$. We write $U_A \in \mathrm{HBE}_{\alpha,a}(A,\epsilon)$, with $\mathrm{HBE}_{\alpha,a}(A) = \mathrm{HBE}_{\alpha,a}(A,0)$.

Two cautions frame the definition. First, A being Hermitian does *not* make a given block encoding Hermitian: the 1-sparse encoding $U_A = (\mathbb{I}\otimes O_c)O_A$ of Theorem 6.16 applied to a Hermitian diagonal A is generally not self-adjoint (Exercise 6.4). Second, a Hermitian U_A that is also unitary squares to the identity, $U_A^2 = \mathbb{I}$, so U_A is a *reflection*; its spectrum is $\{\pm 1\}$. This is precisely the structure qubitization exploits: a Hermitian block encoding of a Hermitian A makes the pair $(U_A,\ 2\,|0^a\rangle\langle 0^a| \otimes \mathbb{I} - \mathbb{I})$ generate an SU(2) on each eigenspace of A, turning matrix functions of A into single-qubit rotations. We defer that machinery to Chapter 8 and here only record how to *build* a Hermitian encoding for a Hermitian sparse matrix.

Theorem 6.22 (Hermitian block encoding of a Hermitian s-sparse matrix)**.** *Let A be Hermitian and s-sparse with $\|A\|_{\max} \le 1$. Suppose the locations are accessed through a single oracle $O_c\,|\ell\rangle\,|j\rangle = |c(j,\ell)\rangle\,|j\rangle$ (using Hermiticity, the ℓ-th nonzero of column j is also the ℓ-th nonzero of row j), and the entries through a square-root value oracle*

$$O_A\,|i\rangle\,|0\rangle\,|j\rangle = |i\rangle\left(\sqrt{A_{ij}}\;|0\rangle + \sqrt{1-|A_{ij}|}\;|1\rangle\right)|j\rangle\,, \tag{6.14}$$

where for complex $A_{ij} = |A_{ij}|\,\mathrm{e}^{\mathrm{i}\theta_{ij}}$ the branch $\sqrt{A_{ij}} = \sqrt{|A_{ij}|}\,\mathrm{e}^{\mathrm{i}\theta_{ij}/2}$ is chosen. Then there is a circuit built from D, O_c, O_A, a SWAP*, and their inverses that realizes $U_A \in \mathrm{HBE}_{s,\,n+2}(A)$.*

Proof idea. Use *two* signal qubits and two index registers, with computational states $|a\rangle\,|i\rangle\,|b\rangle\,|j\rangle$, $a,b \in \{0,1\}$, and let SWAP exchange the two $(a,i) \leftrightarrow (b,j)$ halves. Propagating the source $|0\rangle\,|0^n\rangle\,|0\rangle\,|j\rangle$ through D, O_c, O_A deposits a factor $\sqrt{A_{c(j,\ell),j}}$ on the surviving branch; the SWAP then makes the circuit symmetric under $i \leftrightarrow j$, and propagating the target $|0\rangle\,|0^n\rangle\,|0\rangle\,|i\rangle$ through the mirror-image gates deposits a matching $\sqrt{A_{c(i,\ell'),i}}$. Taking the inner product and using Hermiticity $A_{ij} = A_{ji}^*$,

$$\langle 0|\,\langle 0^n|\,\langle 0|\,\langle i|\;U_A\;|0\rangle\,|0^n\rangle\,|0\rangle\,|j\rangle = \frac{1}{s}\sqrt{A_{ij}}\sqrt{A_{ji}^*}\sum_{\ell,\ell'}\delta_{i,c(j,\ell)}\delta_{j,c(i,\ell')} = \frac{1}{s}A_{ij},$$

since $\sqrt{A_{ij}}\sqrt{A_{ji}^*} = \sqrt{A_{ij}}\sqrt{A_{ij}} = A_{ij}$ and each Kronecker delta is satisfied by a unique index. The symmetric placement of O_A and $O_A^\dagger$ about the SWAP makes the whole circuit equal to its own adjoint, so U_A is Hermitian; the factor $1/s$ from the two diffusion operators gives $\alpha = s$, and the register count is $n+2$. This is the Childs–Kothari–Somma quantum-walk construction. □

6.7 Block encoding beyond the computational basis

So far the useful block has been flagged by the *computational* state $|0^a\rangle$ on the ancilla register. Nothing in the algebra requires this: the flag can be any subspace, singled out by a projector. This generalization is what lets block encodings compose (Chapter 7) and underlies qubitization (Chapter 8).

Definition 6.23 (Projected (generalized) block encoding)**.** Let U be a unitary on a space $\mathcal{H}' \supseteq \mathcal{H}$, and let Π, Π' be orthogonal projectors on $\mathcal{H}'$ with images isomorphic to $\mathcal{H}$ via fixed isometries. Then U is an (α,ϵ)-*projected-block-encoding* of an operator A on $\mathcal{H}$ if

$$\left\| A - \alpha\,\Pi' U\,\Pi \right\| \le \epsilon, \tag{6.15}$$

where the norm is taken after identifying $\operatorname{im}\Pi$ and $\operatorname{im}\Pi'$ with $\mathcal{H}$.

The computational-basis Theorem 6.4 is the special case $\Pi = \Pi' = |0^a\rangle\langle 0^a| \otimes \mathbb{I}_n$ and $\Pi' U \Pi = (|0^a\rangle\langle 0^a| \otimes \mathbb{I}) U (|0^a\rangle\langle 0^a| \otimes \mathbb{I})$, whose nontrivial block is exactly $(\langle 0^a| \otimes \mathbb{I}) U (|0^a\rangle \otimes \mathbb{I})$. Allowing $\Pi \neq \Pi'$ accommodates rectangular A and asymmetric encodings; allowing a general Π lets the flag be prepared by *any* state-preparation unitary rather than by the bare $|0^a\rangle$.

Example 6.24 (LCU flag: a linear combination of unitaries)**.** Anticipating Chapter 7, suppose $A = \sum_{k=0}^{L-1} \beta_k V_k$ with $\beta_k > 0$ and each V_k unitary, and set $\alpha = \sum_k \beta_k$. Let PREP be the state-preparation unitary (Theorem 6.1) with $\mathrm{PREP}\,|0^a\rangle = \frac{1}{\sqrt{\alpha}} \sum_k \sqrt{\beta_k}\,|k\rangle$, and let $\mathrm{SEL} = \sum_k |k\rangle\langle k| \otimes V_k$ be the multiplexed "select" unitary. Then $U = (\mathrm{PREP}^\dagger \otimes \mathbb{I})\,\mathrm{SEL}\,(\mathrm{PREP} \otimes \mathbb{I})$ is an (α, a)-block-encoding of A, because

$$(\langle 0^a| \otimes \mathbb{I}) U (|0^a\rangle \otimes \mathbb{I}) = \sum_k \left(\frac{\sqrt{\beta_k}}{\sqrt{\alpha}}\right)^2 \alpha \cdot \frac{V_k}{1} = \frac{1}{\alpha} \sum_k \beta_k V_k = \frac{A}{\alpha}.$$

Here the flag subspace $\mathrm{im}\,\Pi$ is $|0^a\rangle$ *as seen through* PREP; the "beyond the computational basis" viewpoint of Theorem 6.23 makes this manifest, since $\Pi = \mathrm{PREP}\,|0^a\rangle\langle 0^a|\,\mathrm{PREP}^\dagger$ projects onto the prepared coefficient state. This LCU block encoding, with subnormalization $\alpha = \sum_k \beta_k$ the ℓ_1-norm of the coefficients, is the second pillar of QNLA and the subject of the next lecture.

Remark 6.25 (Why the projector form is the right abstraction)*.* The generalized form (6.15) exposes what block encoding really is: a unitary U together with a way of "entering" (Π) and "exiting" (Π') a distinguished subspace on which U acts as A/α. Products of block encodings multiply the operators (with the projectors composing), sums combine through an LCU flag, and—crucially for Chapters 8 and 9—the singular values of A become the parameters of the SU(2) rotations that U, Π, Π' generate on the two-dimensional invariant subspaces. Every algorithm in Part II is, at bottom, a statement about polynomials of the operator $\Pi' U \Pi$.

6.8 Quantum advantage: where it lives, and three caveats

It is worth stating plainly where the speedup in these operations comes from—and where it can silently evaporate. An n-qubit register stores $N = 2^n$ complex amplitudes, and a *structured* block encoding together with a state-preparation oracle applies A/α to $|b\rangle$ with a circuit of only polylog(N) gates (Section 6.5 does this for sparse A). Classically, multiplying a vector by a dense $N \times N$ matrix costs $\mathcal{O}(N^2)$, and by an s-sparse one $\mathcal{O}(sN)$— both at least linear in the exponentially large dimension. So matrix–vector multiplication, and likewise the sums and products of Section 6.4, act on 2^n-dimensional objects in time polylog(N): an *exponential advantage in the dimension*. Three caveats decide whether that advantage is real.

- *Input.* The speedup assumes the inputs are cheap to load. A generic vector needs $\Omega(2^n/n)$ gates to prepare (Theorem 6.2) and a generic matrix has no polylog block encoding; only structured data—sparse, efficiently integrable, or stored in QRAM—qualifies. If loading the data already costs $\mathcal{O}(N)$, there is no advantage.
- *Output.* The result $A\,|b\rangle \,/\, \|A\,|b\rangle\|$ is a quantum state of N amplitudes. Reading all of them out requires $\Omega(N)$ measurements, which again erases the gain. The advantage survives only when the sought answer is a *succinct summary* of the output—an expectation value $\langle\psi|\,M\,|\psi\rangle$, an overlap, a sample, a single bit—rather than the whole vector.
- *Normalization.* The subnormalization α (and, for the arithmetic of Chapter 7, the ℓ^1 factor $\|c\|_1$ and the product $\prod_j \alpha_j$) enters the success probability: a matrix–vector product needs $\mathcal{O}(\alpha/\,\|A\,|b\rangle\|)$ repetitions (Section 6.3). The speedup is thus exponential in the dimension but only polynomial in α, the condition number, and the accuracy.

In short, these primitives are exponential-in-dimension, not unconditional: the advantage is realized end to end only when structured inputs feed a polylog(N) computation whose output is a measurable quantity—as in the linear-system solver (Chapter 10), Hamiltonian simulation (Chapter 11), and ground-state energy estimation (Chapter 5). Reading the fine print on input and output is what separates a genuine quantum speedup from a mirage.

Exercises

Exercise 6.1. *Norm constraints on a block.* Let U_A be an exact (α, a)-block-encoding of A. Show that $\|A\| \le \alpha$, so that $\alpha \ge \|A\|$ is necessary. Conversely exhibit, for any A with $\|A\| \le \alpha$, a $(1, \cdot)$-block-encoding of A/α and hence an $(\alpha, \cdot)$-block-encoding of A (cf. Theorem 6.6). Explain why $\|A\|_{\max} \le 1$ does *not* by itself imply $\|A\| \le 1$, and give a 2×2 example where $\|A\| = s\, \|A\|_{\max}$ for $s = 2$.

Exercise 6.2. *Success probability and subnormalization.* Let U_A be an (α, a)-block-encoding of A and $|b\rangle$ an n-qubit state. Show that applying U_A to $|0^a\rangle |b\rangle$ and measuring the flag register yields $|0^a\rangle$ with probability $p = \|A|b\rangle\|^2 / \alpha^2$, and that conditioned on this outcome the system register holds $A|b\rangle / \|A|b\rangle\|$. Deduce how the expected number of repetitions needed to see the flag once scales with α, and comment on the practical cost of an overly large α (cf. Theorem 6.7).

Exercise 6.3. *Verifying a 1-sparse encoding.* Let A be the 4×4 matrix with $A_{c(j),j} = a_j$ in row $c(j) = (j+1) \bmod 4$ and zeros elsewhere, where $a_0, \dots, a_3 \in [0, 1]$. Write down the oracles O_c and O_A of (6.9)–(6.8), form $U_A = (\mathbb{I} \otimes O_c) O_A$, and verify directly that $\langle 0, i| U_A |0, j\rangle = A_{ij}$ for all i, j, confirming $U_A \in \mathrm{BE}_{1,1}(A)$.

Exercise 6.4. *A Hermitian matrix with a non-Hermitian encoding.* Take the Hermitian diagonal matrix $A = \mathrm{diag}(a_0, a_1)$ with $a_0 \ne a_1$ in $[0, 1]$, and its $(1, 1)$-block-encoding $U_A = O_A$ from Theorem 6.18 (a controlled rotation). Write U_A as an explicit 4×4 matrix and show that $U_A \ne U_A^\dagger$ in general, even though A is Hermitian. Explain in one sentence why this motivates the separate notion of a Hermitian block encoding (Theorem 6.21).

Exercise 6.5. *The sparsity factor $\alpha = s$.* In the proof of Theorem 6.19, track where each factor of $1/\sqrt{s}$ arises. Show that the two diffusion operators are jointly responsible for the block being A/s rather than A, and conclude that any construction based on (6.13) has subnormalization $\alpha = s$. What is the block-encoding factor of a dense matrix ($s = N$) built this way, and why does this make the construction useless for unstructured dense matrices?

Exercise 6.6. *An LCU block encoding of a Pauli sum.* Let $A = \frac{1}{2}(\mathsf{X} + \mathsf{Z})$ on a single qubit. Using Theorem 6.24 with $L = 2$, $\beta_0 = \beta_1 = \frac{1}{2}$, $V_0 = \mathsf{X}$, $V_1 = \mathsf{Z}$, write down PREP (a single-qubit rotation on the one flag qubit), SEL (a two-qubit controlled gate applying X or Z), and verify that $U = (\mathrm{PREP}^\dagger \otimes \mathbb{I})\mathrm{SEL}(\mathrm{PREP} \otimes \mathbb{I})$ is a $(1, 1)$-block-encoding of A. What is $\|A\|$, and is $\alpha = 1$ optimal here?

Exercise 6.7. *Encoding a Hermitian matrix on the unit circle.* Let $A \in \mathbb{C}^{N \times N}$ ($N = 2^n$) be Hermitian with entries on the complex unit circle, $A_{ij} = \mathrm{e}^{\mathrm{i}\theta_{ij}}$, $\theta_{ij} \in [0, 2\pi)$, accessible through a $2n$-qubit unitary V with

$$V |0^n\rangle |j\rangle = \frac{1}{\sqrt{N}} \sum_{i \in [N]} \mathrm{e}^{\mathrm{i}\theta_{ij}/2} |i\rangle |j\rangle , \qquad j \in [N].$$

Use V to build a block encoding U of A with n ancilla qubits (hint: the target register should be processed by a mirror-image copy of V, exploiting $\theta_{ji} = -\theta_{ij}$ from Hermiticity), and determine the subnormalization factor α.

Further reading

- Lin Lin, *qasc notes*, Ch. 6 (query model for matrix entries, block encoding, s-sparse and Hermitian block encodings, general query models).
- Lin Lin, *live notes*, Ch. 9 (block encoding, the LCU introduction, block encodings of matrix sums and products, and block encoding beyond the computational basis).
- Dong An, *Quantum Numerical Linear Algebra* tutorial talk (the vector and matrix input-model frames and the block-encoding definition).
- A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, *Quantum singular value transformation and beyond*, STOC 2019 (Lemma 48, the sparse-matrix block-encoding construction).

Chapter 7

Linear Combination of Unitaries and Block-Encoding Arithmetic

Learning objectives

- Implement a weighted sum of unitaries with the PREPARE–SELECT–PREPARE† (LCU) circuit and read off its block-encoding parameters.
- Realize matrix and vector addition via LCU as a block encoding with subnormalization equal to the ℓ^1-norm of the coefficients.
- Compose block encodings to multiply matrices, and explain why naive stacking fails while duplicate-ancilla and compression-gadget constructions succeed.
- Contrast the coherent LCU circuit with a hybrid importance-sampling estimator, and compare their costs.
- Track the ℓ^1 overhead $\|c\|_1$ through additions and the product $\prod \alpha_j$ through multiplications, and see when amplitude amplification removes the associated success-probability penalty.

The previous lecture (Chapter 6) delivered two input models: the state-preparation oracle for vectors and the block encoding for matrices. A single block encoding is, however, only as useful as the arithmetic we can build on top of it. Almost every matrix that arises in scientific computing is presented as a *combination* of simpler pieces—a Hamiltonian is a sum of local terms, a discretized differential operator is a sum of shifts, a propagator is a product of exponentials—so we need rules that turn block encodings of the pieces into a block encoding of the whole. Section 6.4 introduced these operations—matrix and vector addition, matrix–matrix multiplication—through one concrete example apiece; this lecture gives the *general* construction for arbitrarily many terms, together with its error propagation and ℓ^1 cost. We develop the single most important such rule, the *linear combination of unitaries* (LCU), and the composition calculus it generates: block encodings of matrix and vector *sums* (Section 7.2) and *products* (Section 7.3). LCU is the PREPARE–SELECT–PREPARE† sandwich already previewed in Theorem 6.24; here we derive it in full, quantify its success probability and repeat count, and note how oblivious amplitude amplification (Chapter 3) removes the subnormalization penalty when the target is (nearly) an isometry. We then contrast the coherent circuit with a purely classical *hybrid* alternative—importance sampling over the terms—whose appeal and whose variance we analyze in Section 7.4. The recurring theme, made precise in Section 7.5, is that the cost of LCU is governed by the ℓ^1-norm $\|c\|_1 = \sum_j |c_j|$ of the combination coefficients, exactly as the cost of a single block encoding was governed by its subnormalization factor α. Throughout, A is an n-qubit matrix ($N = 2^n$) and we reuse the block-encoding notation

$\mathrm{BE}_{\alpha,a}(A,\epsilon)$ of Theorem 6.4.

7.1 The LCU construction

Suppose a target operator is presented as a weighted sum of K unitaries,

$$T = \sum_{j=0}^{K-1} c_j\, U_j, \qquad U_j \in \mathrm{U}(2^n), \quad c_j > 0, \quad K = 2^a. \tag{7.1}$$

(We take the coefficients positive for now; a phase $c_j = |c_j|\,\mathrm{e}^{\mathrm{i}\phi_j}$ is absorbed into $U_j \mapsto \mathrm{e}^{\mathrm{i}\phi_j}U_j$, still unitary, so the general real or complex case reduces to this one.) The idea of LCU is to encode the coefficients in the amplitudes of an a-qubit *control* register, apply the U_j in superposition controlled on that register, and then coherently "fold" the superposition back so that the flag $|0^a\rangle$ picks out precisely T. The two ingredients are the SELECT and PREPARE oracles.

Definition 7.1 (SELECT and PREPARE oracles)**.** The *select oracle* for the family $\{U_j\}_{j\in[K]}$ is the $(a+n)$-qubit unitary

$$\mathrm{SEL} := \sum_{j\in[K]} |j\rangle\langle j| \otimes U_j, \tag{7.2}$$

which applies U_j to the system conditioned on the control register holding $|j\rangle$. The *prepare oracle* for the coefficients $c = (c_0,\dots,c_{K-1})$, $c_j \geq 0$, is any a-qubit unitary PREP with

$$\mathrm{PREP}\,|0^a\rangle = \frac{1}{\sqrt{\|c\|_1}} \sum_{j\in[K]} \sqrt{c_j}\;|j\rangle\,, \qquad \|c\|_1 = \sum_{j\in[K]} |c_j|\,. \tag{7.3}$$

Only the first column of PREP is fixed by (7.3); the remaining columns may be any completion to a unitary (its existence is guaranteed because the specified column is a unit vector). The following lemma is the workhorse of Part II.

Theorem 7.2 (Linear combination of unitaries)**.** *Let $T = \sum_{j\in[K]} c_j U_j$ with $c_j \geq 0$, $K = 2^a$, and let* SEL, PREP *be as in Theorem 7.1. Then*

$$W := (\mathrm{PREP}^\dagger \otimes \mathbb{I}_n)\,\mathrm{SEL}\,(\mathrm{PREP} \otimes \mathbb{I}_n) \tag{7.4}$$

is an exact $(\|c\|_1, a)$-block-encoding of T, i.e. $W \in \mathrm{BE}_{\|c\|_1,a}(T)$.

Proof. Fix an arbitrary n-qubit state $|\psi\rangle$ and propagate the flagged input $|0^a\rangle\,|\psi\rangle$ through the three factors. First PREP dresses the control register with the coefficient amplitudes,

$$(\mathrm{PREP} \otimes \mathbb{I})\,|0^a\rangle\,|\psi\rangle = \frac{1}{\sqrt{\|c\|_1}} \sum_j \sqrt{c_j}\;|j\rangle\,|\psi\rangle\,. \tag{7.5}$$

Next SEL applies U_j in each branch:

$$\mathrm{SEL}\,\frac{1}{\sqrt{\|c\|_1}} \sum_j \sqrt{c_j}\;|j\rangle\,|\psi\rangle = \frac{1}{\sqrt{\|c\|_1}} \sum_j \sqrt{c_j}\;|j\rangle\;U_j\,|\psi\rangle\,. \tag{7.6}$$

Finally $\mathrm{PREP}^\dagger$ acts on the control register. We do not need the full image; we need only the amplitude of the flag $|0^a\rangle$, obtained by projecting with $\langle 0^a|$. Using $\langle 0^a|\,\mathrm{PREP}^\dagger = (\mathrm{PREP}\,|0^a\rangle)^\dagger = \frac{1}{\sqrt{\|c\|_1}} \sum_j \sqrt{c_j}\;\langle j|$ and $\langle j'|j\rangle = \delta_{jj'}$,

$$(\langle 0^a| \otimes \mathbb{I})W(|0^a\rangle \otimes \mathbb{I})\,|\psi\rangle = \frac{1}{\|c\|_1} \sum_{j,j'} \sqrt{c_{j'}}\sqrt{c_j}\;\langle j'|j\rangle\;U_j\,|\psi\rangle = \frac{1}{\|c\|_1} \sum_j c_j U_j\,|\psi\rangle = \frac{T}{\|c\|_1}\,|\psi\rangle\,. \tag{7.7}$$

Since $|\psi\rangle$ was arbitrary, $(\langle 0^a| \otimes \mathbb{I})W(|0^a\rangle \otimes \mathbb{I}) = T/\|c\|_1$, which is exactly the exact-block-encoding relation (6.6) with subnormalization factor $\alpha = \|c\|_1$ and a ancilla qubits. □

Three remarks fix the meaning of the construction.

Remark 7.3 (Reading the circuit; why ℓ^1). The output of W on $|0^a\rangle|\psi\rangle$ is, keeping all branches,

$$W|0^a\rangle|\psi\rangle = \frac{1}{\|c\|_1}|0^a\rangle\, T|\psi\rangle + |\perp\rangle, \qquad (|0^a\rangle\langle 0^a| \otimes \mathbb{I})|\perp\rangle = 0, \tag{7.8}$$

so the flag $|0^a\rangle$ is measured with probability $\|T|\psi\rangle\|^2/\|c\|_1^2$, and conditioned on it the system holds the normalized $T|\psi\rangle/\|T|\psi\rangle\|$, exactly as in the general block-encoding picture (Section 6.2). The subnormalization is the ℓ^1-norm $\|c\|_1$, *not* $\|T\|$: because $\|T\| \le \sum_j c_j\|U_j\| = \|c\|_1$, LCU is never better than $\alpha = \|c\|_1$, and it is tight exactly when the U_j add up "in phase" (e.g. all $U_j = \mathbb{I}$). Reducing $\|c\|_1$—by choosing a more efficient decomposition of T into unitaries—is the central knob for controlling cost, and is the subject of Section 7.5.

Remark 7.4 (Cost is logarithmic in the number of terms). The register that carries the combination has only $a = \log_2 K$ qubits, so LCU can superpose an exponentially large number of terms with a linear number of ancillas. The genuine cost hides in the two oracles: PREP must load the $\sqrt{c_j}$ amplitudes (Theorem 6.1, subject to the same structure requirements as any state preparation) and SEL must apply a K-fold-multiplexed unitary. A direct implementation of either scales linearly in K; an *efficient* LCU therefore needs additional structure—for instance, the U_j being Pauli strings addressed by their binary label, so that SEL is a sequence of a controlled-Pauli layers.

Remark 7.5 (Unequal split of the coefficients). Nothing forces the "load" and "unload" oracles to be adjoints. Writing $c_j = \beta_j\gamma_j$ and using two preparations $\mathrm{PREP}|0^a\rangle = \frac{1}{\|\beta\|_2}\sum_j \beta_j|j\rangle$ and $\widetilde{\mathrm{PREP}}|0^a\rangle = \frac{1}{\|\gamma\|_2}\sum_j \gamma_j|j\rangle$, the sandwich $W = (\widetilde{\mathrm{PREP}}^\dagger \otimes \mathbb{I})\,\mathrm{SEL}\,(\mathrm{PREP} \otimes \mathbb{I})$ is a $(\|\beta\|_2\|\gamma\|_2, a)$-block-encoding of T. By Cauchy–Schwarz $\|c\|_1 = \sum_j|\beta_j\gamma_j| \le \|\beta\|_2\|\gamma\|_2$, so the symmetric choice $\beta_j = \gamma_j = \sqrt{c_j}$ (whence $\widetilde{\mathrm{PREP}} = \mathrm{PREP}$) minimizes the subnormalization at $\|c\|_1$. The freedom is occasionally useful when one of the two preparations is cheaper to build than its adjoint.

Example 7.6 (LCU of two Pauli matrices)**.** Take the single-qubit Hermitian operator $T = \frac{1}{2}\mathsf{X} + \frac{1}{2}\mathsf{Z}$, so $K = 2$, $a = 1$, $c_0 = c_1 = \frac{1}{2}$, $U_0 = \mathsf{X}$, $U_1 = \mathsf{Z}$, and $\|c\|_1 = 1$. The prepare oracle must satisfy $\mathrm{PREP}|0\rangle = \frac{1}{\sqrt{1}}(\sqrt{\frac{1}{2}}|0\rangle + \sqrt{\frac{1}{2}}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, which is the Hadamard gate $\mathrm{PREP} = \mathsf{H}$ (self-inverse, so $\mathrm{PREP}^\dagger = \mathsf{H}$). The select oracle is $\mathrm{SEL} = |0\rangle\langle 0| \otimes \mathsf{X} + |1\rangle\langle 1| \otimes \mathsf{Z}$, a controlled-$\mathsf{X}$ followed by a control-on-$|1\rangle$ Z. Then $W = (\mathsf{H} \otimes \mathbb{I})\,\mathrm{SEL}\,(\mathsf{H} \otimes \mathbb{I})$ and, by (7.7),

$$(\langle 0| \otimes \mathbb{I})W(|0\rangle \otimes \mathbb{I}) = \tfrac{1}{2}\mathsf{X} + \tfrac{1}{2}\mathsf{Z} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} = T,$$

so $W \in \mathrm{BE}_{1,1}(T)$. Here $\alpha = \|c\|_1 = 1$ is optimal: the two Paulis are orthogonal in Hilbert–Schmidt inner product, $\|T\| = \frac{1}{\sqrt{2}} \cdot \sqrt{2} = 1$ (its singular values are both $1/\sqrt{2} \cdot \sqrt{2}$; explicitly $T/\sqrt{2}$ is a reflection, $\|T\| = 1$), so no block encoding can do better than $\alpha = 1$. Applying W to $|0\rangle|\psi\rangle$ succeeds with probability $\|T|\psi\rangle\|^2$, which equals 1 for every $|\psi\rangle$ because T is (up to the scalar) unitary; LCU here needs no repetition.

Remark 7.7 (Oblivious amplitude amplification). When T is itself unitary (or an isometry), as in Theorem 7.6 with $\|c\|_1 = 1$, the flag $|0^a\rangle$ still appears only with probability $\|T|\psi\rangle\|^2/\|c\|_1^2$, which is < 1 whenever $\|c\|_1 > 1$. Naively one would amplify the flag with amplitude amplification

(Chapter 3), but the standard routine needs to reflect about the *input* state $|\psi\rangle$, which is unknown and possibly entangled. *Oblivious* amplitude amplification removes this obstacle: because W acts as the fixed 2×2 rotation $\left(\begin{smallmatrix}\sin\theta & \cos\theta\\ \cos\theta & -\sin\theta\end{smallmatrix}\right)$ (with $\sin\theta = \|T\| \,/\, \|c\|_1$ when T is proportional to an isometry) on the two-dimensional subspace spanned by $|0^a\rangle\,|\psi\rangle$ and $|\perp\rangle$ *for every* $|\psi\rangle$ simultaneously, one can reflect about $|0^a\rangle$ on the ancilla *alone* and drive the flag to certainty in $\mathcal{O}(\|c\|_1 \,/\, \|T\|)$ rounds, independently of $|\psi\rangle$. This is why a Hamiltonian written as $H = \sum_j c_j P_j$ with the P_j Pauli strings can be applied deterministically up to the ℓ^1 overhead. We use this repeatedly in the Taylor-series simulation method of Chapter 12.

Example 7.8 (LCU block encoding of the transverse-field Ising Hamiltonian)**.** The Hamiltonian of Theorem 5.1,

$$H = -\sum_{i=1}^{n-1} Z_i Z_{i+1} - g\sum_{i=1}^{n} X_i,$$

is *already* a linear combination of unitaries: each Z_iZ_{i+1} and each X_i is a Pauli string, hence unitary. There are $K = (n-1)+n = 2n-1$ terms, with coefficients $c_j = 1$ for the $n-1$ couplings and $c_j = g$ for the n field terms (the minus signs are phases $-P = \mathrm{e}^{\mathrm{i}\pi}P$ absorbed into the unitaries, and K is padded up to a power of two with zero-weight terms). By Theorem 7.2, with $a = \lceil \log_2(2n-1)\rceil$ control qubits, $\mathrm{PREP}\,|0^a\rangle = \frac{1}{\sqrt{\|c\|_1}}\sum_j \sqrt{c_j}\;|j\rangle$ loading the weights (amplitude 1 on each coupling label, $\sqrt{g}$ on each field label) and $\mathrm{SEL} = \sum_j |j\rangle\langle j| \otimes P_j$ applying the Pauli string P_j, the circuit $W = (\mathrm{PREP}^\dagger \otimes \mathbb{I})\,\mathrm{SEL}\,(\mathrm{PREP}\otimes\mathbb{I})$ is an exact block encoding of H with subnormalization

$$\alpha = \|c\|_1 = (n-1) + gn = \mathcal{O}(n).$$

This is the input model that feeds the ground-state-energy estimation of Section 5.1: QPE on $\mathrm{e}^{\mathrm{i}2\pi H/\alpha}$, or the eigenvalue filters built by QSVT in Chapter 9, run on W, and their cost scales linearly in α. The ℓ^1 growth is the generic local-Hamiltonian story—an n-qubit Hamiltonian with $\mathcal{O}(n)$ terms of $\mathcal{O}(1)$ weight block-encodes with $\alpha = \mathcal{O}(n)$—exactly the overhead budgeted in Section 7.5.

7.2 Matrix and vector addition

Section 6.4 introduced matrix and vector addition by two-term example; the LCU lemma of Section 7.1 promotes them to the general rule. It applies verbatim when the "unitaries" U_j are themselves block encodings or state-preparation oracles: the outer register of the block encoding merges into the LCU flag, and the surviving block is the desired sum—now for an *arbitrary* number of terms, and tracking the errors.

Proposition 7.9 (Sum of block-encoded matrices)**.** *Let $M = 2^m$ and let $A_0, \dots, A_{M-1}$ be n-qubit matrices, each supplied by a block encoding $U_{A_j} \in \mathrm{BE}_{\alpha_j,a}(A_j,\epsilon_j)$ with $\alpha_j \geq 0$. Set $\gamma := \sum_{j\in[M]}\alpha_j$. Let $\mathrm{SEL} = \sum_{j\in[M]} |j\rangle\langle j| \otimes U_{A_j}$ act on an m-qubit control register together with the shared a-qubit flag and system, and let* PREP *be any m-qubit unitary with $\mathrm{PREP}\,|0^m\rangle = \frac{1}{\sqrt{\gamma}}\sum_{j\in[M]}\sqrt{\alpha_j}\;|j\rangle$. Then*

$$W := (\mathrm{PREP}^\dagger \otimes \mathbb{I})\,\mathrm{SEL}\,(\mathrm{PREP}\otimes\mathbb{I}) \in \mathrm{BE}_{\gamma,\,a+m}\Big(\textstyle\sum_{j\in[M]} A_j,\ \sum_{j\in[M]}\epsilon_j\Big). \tag{7.9}$$

More generally, for a weighted sum $\sum_j c_j A_j$ with $c_j \geq 0$ one rescales the prepared amplitudes to $\sqrt{c_j\alpha_j}$ and obtains subnormalization $\sum_j c_j\alpha_j$.

Proof. Write $B_j := (\langle 0^a| \otimes \mathbb{I})U_{A_j}(|0^a\rangle \otimes \mathbb{I})$, the block of U_{A_j}, so that $\|A_j - \alpha_j B_j\| \leq \epsilon_j$ and (a block of a unitary) $\|B_j\| \leq 1$. The flag for W is the joint state $|0^m\rangle\,|0^a\rangle$. Running the three factors

exactly as in Theorem 7.2, but now with each branch's system operator being the block B_j rather than a full unitary, gives

$$(\langle 0^m| \langle 0^a| \otimes \mathbb{I}) W (|0^m\rangle |0^a\rangle \otimes \mathbb{I}) = \frac{1}{\gamma} \sum_{j\in[M]} \alpha_j B_j. \tag{7.10}$$

The only new point is that SEL leaves the a-qubit flag of U_{A_j} where it is, so the $|0^m\rangle |0^a\rangle$ projection extracts $\alpha_j B_j/\gamma$ in each branch; the $|j\rangle$-register bookkeeping is identical to (7.7). Comparing with the target $\gamma^{-1}\sum_j A_j$ and using the triangle inequality,

$$\Big\| \sum_j A_j - \gamma \cdot \tfrac{1}{\gamma} \sum_j \alpha_j B_j \Big\| = \Big\| \sum_j (A_j - \alpha_j B_j) \Big\| \le \sum_j \|A_j - \alpha_j B_j\| \le \sum_j \epsilon_j, \tag{7.11}$$

which is exactly the block-encoding statement (7.9). The weighted case follows by replacing α_j with $c_j\alpha_j$ in PREP. □

The subnormalization of a sum is the sum of the subnormalizations, and the errors add. This linear growth is benign for a fixed number of terms but is precisely the ℓ^1 overhead that we must budget for; see Section 7.5.

Remark 7.10 (Convex weights incur no subnormalization). When the weights form a convex combination the subnormalization penalty vanishes: for the average $\frac{1}{2}(A+B)$ of two exact encodings one takes $M=2$, PREP $= \mathsf{H}$, and weights $c_0 = c_1 = \frac{1}{2}$, so $\gamma = \sum_j c_j = 1$—averaging costs nothing. (Theorem 6.11 is exactly this circuit, block-encoding $\frac{1}{2}(X+Z)$.) The *unweighted* sum $A+B$ instead has $\gamma = 2$, reflecting that $\|A+B\|$ can reach $\|A\| + \|B\|$. In general $\gamma = \|c\|_1$ is the overhead that Section 7.5 analyzes.

Vector addition

The same construction adds *vectors* when the selected operators are state-preparation oracles rather than block encodings; Theorems 6.12 and 6.13 gave the two-term case $|0\rangle + |1\rangle \to |+\rangle$. For general nonnegative weights c_j, with SEL $= \sum_j |j\rangle\langle j| \otimes O_j$ and PREP $|0^m\rangle = \|c\|_1^{-1/2} \sum_j \sqrt{c_j}\, |j\rangle$, post-selecting the flag $|0^m\rangle$ leaves

$$\frac{1}{\|c\|_1} |0^m\rangle \sum_j c_j |u_j\rangle + |\perp\rangle,$$

seen with probability $\left\|\sum_j c_j |u_j\rangle\right\|^2 / \|c\|_1^2$. It is the primitive behind, for example, forming $\sum_k \hat{f}(k)\, \mathrm{e}^{\mathrm{i}kH} |\psi\rangle$ in Fourier-based matrix-function algorithms—a genuinely many-term superposition for which the general Theorem 7.9, not the two-term picture of Section 6.4, is what is needed.

7.3 Block-encoding multiplication

Addition composed block encodings side by side, sharing the system register. To *multiply* A and B we must feed the output of one into the input of the other—Theorem 6.15 did this for one concrete product (squaring $\frac{1}{2}(X+Z)$); here we give the general rule, its ancilla cost, and the reason naive stacking fails. The subtlety is that the block-encoding relation only holds on the flagged subspace, so we cannot simply apply U_A and then U_B on the same flag qubits.

Remark 7.11 (Why naive stacking fails). Suppose $U_A \in \mathrm{BE}_{\alpha,a}(A)$ and $U_B \in \mathrm{BE}_{\beta,a}(B)$ share the *same* a-qubit flag. Applying them in sequence gives

$$U_B U_A \left|0^a\right\rangle \left|\psi\right\rangle = U_B\Big(\tfrac{1}{\alpha}\left|0^a\right\rangle A\left|\psi\right\rangle + \left|\perp_A\right\rangle\Big),$$

and the $\left|\perp_A\right\rangle$ branch—which is *not* flagged by $\left|0^a\right\rangle$—is mapped by U_B back into the flagged subspace in an uncontrolled way. Concretely $(\langle 0^a| \otimes \mathbb{I})U_B U_A(|0^a\rangle \otimes \mathbb{I})$ picks up cross terms from $\left|\perp_A\right\rangle$ and does *not* equal $\frac{1}{\alpha\beta}BA$. The remedy is to keep the two flag registers *separate*, so that returning to the joint flag $\left|0^a\right\rangle\left|0^b\right\rangle$ forces *both* block encodings to have acted on their flagged subspaces.

Proposition 7.12 (Product of two block-encoded matrices)**.** *Let $U_A \in \mathrm{BE}_{\alpha,a}(A)$ and $U_B \in \mathrm{BE}_{\beta,b}(B)$ act on disjoint ancilla registers (of sizes a and b) and a common n-qubit system. Then*

$$U_{AB} := (U_A \otimes \mathbb{I}_b)(\mathbb{I}_a \otimes U_B) \in \mathrm{BE}_{\alpha\beta,\, a+b}(AB), \tag{7.12}$$

using $a+b$ ancilla qubits. (Here U_A acts on flag register a and the system, U_B on flag register b and the system, and the two flag registers do not overlap.)

Proof. The joint flag is $\left|0^a\right\rangle\left|0^b\right\rangle$. Because the two flag registers are disjoint, $(\langle 0^a| \otimes \mathbb{I})$ and $(\langle 0^b| \otimes \mathbb{I})$ commute past each other and the block of the product factorizes:

$$\begin{aligned}(\langle 0^a|\langle 0^b| \otimes \mathbb{I})U_{AB}(|0^a\rangle|0^b\rangle \otimes \mathbb{I}) &= (\langle 0^a| \otimes \mathbb{I})U_A(|0^a\rangle \otimes \mathbb{I})\ (\langle 0^b| \otimes \mathbb{I})U_B(|0^b\rangle \otimes \mathbb{I}) \\ &= \frac{A}{\alpha}\cdot\frac{B}{\beta} = \frac{AB}{\alpha\beta}.\end{aligned} \tag{7.13}$$

The middle step is the key: $\mathbb{I}_a \otimes U_B$ leaves flag register a untouched, so projecting it onto $\left|0^a\right\rangle$ before and after commutes through U_B; likewise $U_A \otimes \mathbb{I}_b$ leaves flag register b untouched. Thus the two blocks separate and multiply. Comparing with the exact-encoding relation (6.6) gives $U_{AB} \in \mathrm{BE}_{\alpha\beta,a+b}(AB)$. (With errors ϵ_A, ϵ_B the same telescoping bound as in Theorem 7.9 gives error $\alpha\beta$ times the product-rule expansion; the exact case suffices for us.) □

The product multiplies the subnormalizations, $\alpha\beta$, and *adds* the ancilla counts, $a+b$. Iterating (7.12) over L factors gives $\prod_j \alpha_j$ subnormalization and $\sum_j a_j$ ancillas—correct, but ancilla-hungry.

Remark 7.13 (Duplicate-ancilla vs. compression-gadget cost). Two refinements reduce the ancilla count for a product of L block encodings, each in $\mathrm{BE}_{\alpha,a}(A_i)$.

- *Duplicate ancilla.* A deferred-measurement trick records, in a single extra qubit, whether *all* flag registers returned to $\left|0^a\right\rangle$; one can then reuse the flag register across the L factors and needs only $a+(L-1)$ ancillas (one bookkeeping qubit per factor beyond the first) to obtain a unitary in $\mathrm{BE}_{\alpha^L,\, a+L-1}(A_L \cdots A_1)$.
- *Compression gadget.* The bookkeeping qubits are highly redundant—one only needs to know whether *any* factor left the flag subspace, a single bit of information that a counter can hold. The compression gadget replaces the $L-1$ bookkeeping qubits by a $\lceil \log_2(L+1) \rceil$-qubit counter, giving $\mathrm{BE}_{\alpha^L,\, a+\lceil \log_2(L+1)\rceil}(A_L \cdots A_1)$. This exponential saving in ancillas is what makes long products (e.g. Chebyshev/monomial expansions inside QSVT, Chapter 9) practical.

Both refinements leave the subnormalization at α^L. That exponential growth in L is intrinsic to the product rule and is *not* an artifact: even a matrix with spectral radius $\rho(A) < 1$, so that $\sup_L \left\|A^L\right\| \le C < \infty$, may have $\|A\| > 1$ and hence require $\alpha \ge \|A\| > 1$; encoding A^L this way carries subnormalization α^L even though $\left\|A^L\right\|$ stays bounded. Reconciling these two facts—bounded powers, blowing-up subnormalization—is exactly the difficulty that dedicated linear-ODE algorithms (Chapter 13) are built to resolve.

Example 7.14 (Two-term product)**.** Let A, B be 1-sparse with $\|A\|_{\max}, \|B\|_{\max} \leq 1$, block-encoded by $U_A \in \mathrm{BE}_{1,1}(A)$ and $U_B \in \mathrm{BE}_{1,1}(B)$ as in Theorem 6.16. Using *two* flag qubits (registers a and b), the circuit $U_{AB} = (U_A \otimes \mathbb{I})(\mathbb{I} \otimes U_B)$ is a $(1,2)$-block-encoding of AB: on flag $|0\rangle|0\rangle$ it returns $AB|\psi\rangle$ with the two flag qubits back in $|0\rangle$. If instead we tried to reuse a single flag qubit ($U_A U_B$ on the same ancilla), the $|1\rangle$ branch produced by U_B would be fed into U_A and re-enter the $|0\rangle$ block, corrupting the answer—the failure of Theorem 7.11. The extra ancilla is precisely the bookkeeping that keeps the two encodings from interfering.

7.4 Hybrid LCU and importance sampling

The circuit W of Theorem 7.2 realizes $T = \sum_j c_j U_j$ *coherently*: all K branches exist simultaneously in superposition, and a single run applies the whole combination. Often, though, we do not want the state $T|\psi\rangle$ itself but only an *expectation value* $u^\dagger O u$ of some observable O in the (unnormalized) output $u = T|u_0\rangle = \sum_j c_j U_j |u_0\rangle$. In that case a *hybrid* strategy—run one term at a time on the quantum computer, combine classically—becomes available and can trade circuit depth for repetitions.

Expanding the quadratic form,

$$u^\dagger O u = \sum_{j,j'} c_j c_{j'} \, \langle u_0| U_j^\dagger \, O \, U_{j'} |u_0\rangle \,. \tag{7.14}$$

Each summand $\langle u_0| U_j^\dagger O U_{j'} |u_0\rangle$ is an off-diagonal matrix element that a quantum computer can estimate with a *Hadamard test*: prepare $|u_0\rangle$, apply the controlled pair $U_j^\dagger O U_{j'}$ (or, for a single unitary O, the controlled $U_{j'}$ and U_j), and read the ancilla. No superposition over j is required—only two of the U's appear per run—so the circuit is as shallow as a single term, at the price of estimating an $\mathcal{O}(K^2)$-term sum.

Proposition 7.15 (Importance-sampling estimator for $u^\dagger O u$)**.** *Let $c_j \geq 0$, $\|c\|_1 = \sum_j c_j$, and define the probability distribution $p_j = c_j / \|c\|_1$ on $[K]$. Draw pairs (j, j') independently with $\mathbb{P}[(j,j')] = p_j p_{j'}$, and for each pair form the (Hadamard-test) unbiased estimate $\widehat{h}_{jj'}$ of $h_{jj'} := \langle u_0| U_j^\dagger O U_{j'} |u_0\rangle$. Then*

$$\widehat{\mathcal{E}} := \|c\|_1^2 \, \widehat{h}_{jj'} \qquad \text{is an unbiased estimator of } u^\dagger O u, \tag{7.15}$$

and averaging over R independent samples gives variance $\mathrm{Var}[\frac{1}{R}\sum \widehat{\mathcal{E}}] = \mathcal{O}(\|c\|_1^4 \, \|O\|^2 / R)$. Hence $\mathcal{O}(\|c\|_1^4 \, \|O\|^2 / \varepsilon^2)$ samples suffice to estimate $u^\dagger O u$ to additive error ε.

Proof. Unbiasedness: with $(j,j') \sim p_j p_{j'}$ and $\widehat{h}_{jj'}$ conditionally unbiased for $h_{jj'}$,

$$\mathbb{E}[\widehat{\mathcal{E}}] = \|c\|_1^2 \sum_{j,j'} p_j p_{j'} \, h_{jj'} = \|c\|_1^2 \sum_{j,j'} \frac{c_j}{\|c\|_1} \frac{c_{j'}}{\|c\|_1} h_{jj'} = \sum_{j,j'} c_j c_{j'} h_{jj'} = u^\dagger O u,$$

by (7.14). For the variance, each $|h_{jj'}| \leq \|O\|$ (since $\|U_j|u_0\rangle\| = 1$), so a single estimate obeys $\left|\widehat{\mathcal{E}}\right| \leq \|c\|_1^2 \, \|O\|$ (up to the $\mathcal{O}(1)$ Hadamard-test variance), giving $\mathrm{Var}[\widehat{\mathcal{E}}] = \mathcal{O}(\|c\|_1^4 \, \|O\|^2)$; the R-sample average divides this by R. Setting the standard deviation to ε yields $R = \mathcal{O}(\|c\|_1^4 \, \|O\|^2 / \varepsilon^2)$. □

Remark 7.16 (Coherent vs. hybrid: the trade-off). The two implementations occupy opposite ends of a depth–repetition trade-off.

- *Coherent LCU* (Theorem 7.2) uses one deep circuit containing all K terms via SEL, with an ancilla register and (if the state is wanted) $\mathcal{O}(\|c\|_1 / \|u\|)$ amplifications; it produces the quantum state $u/\|u\|$ and preserves coherence for further processing.

- *Hybrid importance sampling* (Theorem 7.15) uses many shallow circuits with at most two terms each and *no* multiplexed SEL, but pays a worse dependence on the ℓ^1-norm—$\|c\|_1^4$ in the sample count versus $\|c\|_1$ in the coherent repeat count—and only ever yields a scalar $u^\dagger O u$, not the state.

The hybrid route is attractive on near-term hardware, where circuit depth and clean ancillas are scarce, and when the final quantity of interest is a single expectation value; the coherent route wins asymptotically and whenever the output state must be fed into a downstream quantum subroutine.

7.5 Cost: the ℓ^1 overhead

Every construction above is governed by one scalar: the ℓ^1-norm of the combination coefficients. Collecting the accounting:

- *Addition* of block encodings with subnormalizations α_j produces subnormalization $\gamma = \sum_j \alpha_j$ (Theorem 7.9); for a weighted sum $\sum_j c_j A_j$ with $\|A_j\| \le 1$ the subnormalization is $\|c\|_1$. The number of times the coherent circuit must be run to see the flag is $\mathcal{O}(\gamma/\|T|\psi\rangle\|)$ (Theorem 7.3), reduced to $\mathcal{O}(\gamma/\|T\|)$ by oblivious amplitude amplification when T is proportional to an isometry (Theorem 7.7).
- *Multiplication* of L block encodings produces subnormalization $\prod_j \alpha_j$ (Theorem 7.12, Theorem 7.13). For $\alpha_j \equiv \alpha$ this is α^L, an exponential in the number of factors.
- *Hybrid sampling* costs $\mathcal{O}(\|c\|_1^4 \|O\|^2/\varepsilon^2)$ shallow runs (Theorem 7.15).

The moral is that $\|c\|_1$ is to LCU what α was to a single block encoding (Chapter 6): a subnormalization that shrinks the success probability quadratically and multiplies every downstream query count linearly (or, for products, exponentially in the number of factors). Minimizing it is therefore the principal design problem.

Remark 7.17 (Quantum advantage of the arithmetic—and its price). Set against classical cost, the win is the same exponential-in-dimension story as for a single block encoding (Section 6.8), now for *combinations*. Classically, forming $\sum_j c_j A_j |b\rangle$ for K structured $N \times N$ terms costs $\mathcal{O}(K\, sN)$, and a product $A_L \cdots A_1 |b\rangle$ costs $\mathcal{O}(L\, sN)$—linear in the exponentially large dimension $N = 2^n$. The LCU circuit instead uses $\mathcal{O}(K)$ calls to the per-term block encodings on only $\lceil \log_2 K \rceil$ control qubits, each call polylog(N) for structured terms, and the product rule chains L encodings with $\mathcal{O}(\log L)$ ancillas (the compression gadget, Theorem 7.13). So the arithmetic runs in polylog(N) time in the dimension—an exponential advantage—while the *price* of combining is paid in the subnormalization: $\|c\|_1$ for a sum, $\prod_j \alpha_j$ for a product, each shrinking the success probability quadratically and multiplying the downstream query count (linearly for sums, exponentially in the number of factors). The input and output caveats of Section 6.8 apply unchanged: the advantage is genuine only when the terms are efficiently encodable and the final read-out is a succinct summary. This is exactly the regime of the quantum linear-system solver (Chapter 10) and Hamiltonian simulation (Chapter 11), where LCU supplies the arithmetic and the answer is an expectation value or a sampled bit, not the full solution vector.

Example 7.18 (The ℓ^1-norm depends on the decomposition)**.** A given T can be written as a linear combination of unitaries in many ways, and the resulting $\|c\|_1$ varies. Consider the diagonal matrix $T = \mathrm{diag}(1, 1, \ldots, 1, -1)$ on n qubits (a single sign flip on the last basis state). One decomposition writes $T = \mathbb{I} - 2\,|N-1\rangle\langle N-1|$; since $|N-1\rangle\langle N-1|$ is not unitary, expand it into Paulis, $|N-1\rangle\langle N-1| = \prod_k \frac{\mathbb{I}-Z_k}{2}$, whose expansion has 2^n Pauli terms each with coefficient 2^{-n}, giving $\|c\|_1 = \mathbb{I}\text{-term} + 2 \cdot 2^{-n} \cdot 2^n = 1 + 2 = 3$ after cancellation—but the naive term-by-term bound is $\|c\|_1 = 1 + 2\sum_k \ldots$, exponentially worse if one is careless. A far better decomposition

recognizes T itself as a single *reflection* (it is unitary and Hermitian), so $T = 1 \cdot T$ with $\|c\|_1 = 1$ and no LCU overhead at all. The lesson: before invoking LCU, check whether the target is already (a scalar times) a unitary, and otherwise seek the decomposition with the smallest ℓ^1-norm. This is why Hamiltonians are grouped into as few Pauli terms as possible, and why product formulas (Chapter 11) sometimes beat LCU for simulation.

Remark 7.19 (Where LCU sits in the toolbox). LCU and single-matrix block encoding (Chapter 6) together generate the full arithmetic of block encodings: sums, weighted sums, and products of encodable matrices are all encodable, with the bookkeeping rules of this lecture. What LCU does *not* directly give is a general *function* $f(A)$ beyond polynomials expressible as short LCUs (e.g. truncated Chebyshev or Fourier series). For that we need qubitization and quantum signal processing (Chapter 8) and the quantum singular value transformation (Chapter 9), which realize essentially arbitrary bounded polynomials of A with subnormalization independent of the polynomial degree—often dramatically better than the $\|c\|_1$ of the corresponding LCU. We will return to compare the two approaches once QSVT is in hand.

Exercises

Exercise 7.1. *Deriving the LCU block.* Let $T = \sum_{j\in[K]} c_j U_j$ with $c_j \geq 0$, $K = 2^a$, and let $W = (\mathrm{PREP}^\dagger \otimes \mathbb{I})\,\mathrm{SEL}\,(\mathrm{PREP} \otimes \mathbb{I})$ as in Theorem 7.2. Reproduce the three-line computation (7.5)–(7.7) to show $(\langle 0^a| \otimes \mathbb{I})W(|0^a\rangle \otimes \mathbb{I}) = T/\|c\|_1$. Then show that if the input flag is $|k\rangle \neq |0^a\rangle$ instead of $|0^a\rangle$, the surviving $\langle 0^a|$-block is $\frac{1}{\|c\|_1}\sum_j \sqrt{c_j}\,\overline{(\mathrm{PREP})_{jk}}\,\sqrt{c_j}\,U_j$ in general—i.e. only the $|0^a\rangle$ column of PREP matters for the encoded operator.

Exercise 7.2. *Optimality of* $\|c\|_1$. Using Theorem 7.5, show that for any factorization $c_j = \beta_j\gamma_j$ the asymmetric circuit block-encodes T with subnormalization $\|\beta\|_2\,\|\gamma\|_2$, and prove via Cauchy–Schwarz that this is minimized, at value $\|c\|_1$, by $\beta_j = \gamma_j = \sqrt{c_j}$. Conclude that among all PREPARE–SELECT–UNPREPARE circuits for a fixed $\{U_j\}$, the symmetric one has the smallest subnormalization.

Exercise 7.3. *LCU of three Paulis.* Consider $T = \mathsf{X} + \mathsf{Y} + 2\mathsf{Z}$ on one qubit. Determine $\|c\|_1$, the number of control qubits a, and write down explicitly the prepare state $\mathrm{PREP}\,|0^a\rangle$ and the select oracle $\mathrm{SEL} = \sum_j |j\rangle\langle j| \otimes U_j$ (you may use a 2-qubit control register with one label unused, and set the coefficient of the unused label to 0). State the resulting block-encoding parameters (α, a) and compute $\|T\|$ to check that $\alpha = \|c\|_1$ overshoots $\|T\|$ here.

Exercise 7.4. *Block-encoding a weighted sum.* Let $U_A \in \mathrm{BE}_{\alpha_A,a}(A)$ and $U_B \in \mathrm{BE}_{\alpha_B,a}(B)$ share a flag register. Following Theorem 7.9, construct a circuit that block-encodes $C = A+B$ and identify its subnormalization and ancilla count. Then do the same for a general $\alpha_A \neq \alpha_B$, writing down the required PREP amplitudes $\propto (\sqrt{\alpha_A}, \sqrt{\alpha_B})$, and confirm the subnormalization is $\alpha_A + \alpha_B$.

Exercise 7.5. *Why naive multiplication fails.* Let $U_A \in \mathrm{BE}_{1,1}(A)$ and $U_B \in \mathrm{BE}_{1,1}(B)$ share a *single* flag qubit. Write $U_B\,|0\rangle\,|\psi\rangle = |0\rangle\,B\,|\psi\rangle + |1\rangle\,|\chi\rangle$ and $U_A(|0\rangle\,B\,|\psi\rangle + |1\rangle\,|\chi\rangle)$, and show by an explicit 2×2 example ($A = B = \frac{1}{2}\mathsf{X}$, say, so $\|A\| = \|B\| = \frac{1}{2}$) that $(\langle 0| \otimes \mathbb{I})U_AU_B(|0\rangle \otimes \mathbb{I}) \neq AB$ in general because of the $|1\rangle\,|\chi\rangle$ cross term. Then verify that placing U_A, U_B on *disjoint* flag qubits (Theorem 7.12) restores $(\langle 00| \otimes \mathbb{I})U_{AB}(|00\rangle \otimes \mathbb{I}) = AB$.

Exercise 7.6. *Subnormalization of a power.* Let A be block-encoded by $U_A \in \mathrm{BE}_{\alpha,a}(A)$ with $\alpha = \|A\|$. Using Theorem 7.12 iteratively (or the compression gadget of Theorem 7.13), give a

block encoding of A^L and its subnormalization. Exhibit a 2×2 matrix A with $\rho(A) < 1$ but $\|A\| > 1$ (hint: a nonnormal matrix such as $\left(\begin{smallmatrix} 0 & 2 \\ 0 & 0 \end{smallmatrix}\right)$ scaled so that $\rho = 0$, or a Jordan-type block), and explain why the subnormalization α^L grows without bound even though $\left\|A^L\right\|$ does not.

Exercise 7.7. *Cost of the hybrid estimator.* In the setting of Theorem 7.15, suppose $K = 100$ terms with all $c_j = 1$ (so $\|c\|_1 = 100$) and a bounded observable $\|O\| \le 1$. How many shallow Hadamard-test runs R are needed to estimate $u^\dagger O u$ to additive error $\varepsilon = 10^{-2}$? Compare with the coherent-LCU repeat count $\mathcal{O}(\|c\|_1 / \|u\|)$ (assume $\|u\| = \Theta(\|c\|_1)$) and comment on which implementation you would choose (i) on a shallow near-term device and (ii) if the state $u/\|u\|$ is needed for a subsequent quantum subroutine.

Further reading

- Lin Lin, *qasc notes*, Sec. 7.3 (linear combination of unitaries).
- Lin Lin, *live notes*, Ch. 9 (LCU lemma, block encodings of matrix additions and multiplications, and worked examples including the highly oscillatory integral and Fourier/eigenvalue transformations).
- Dong An, *Quantum Numerical Linear Algebra* tutorial talk (LCU as PREPARE–SELECT–UNPREPARE, matrix and vector addition, matrix multiplication by duplicate ancillas and the compression gadget, and the hybrid importance-sampling estimator).
- G. H. Low and I. L. Chuang, *Hamiltonian simulation by qubitization*, Quantum **3**, 163 (2019), and A. M. Childs and N. Wiebe, *Hamiltonian simulation using linear combinations of unitary operations*, Quantum Inf. Comput. **12**, 901 (2012), for the original LCU and oblivious amplitude amplification arguments.

Chapter 8

Qubitization and Quantum Signal Processing

Learning objectives

- Derive Chebyshev transformations from a 2×2 rotation.
- Qubitize Hermitian and general block encodings.
- State the quantum signal processing (QSP) theorem.
- Survey methods for finding phase factors.

The block-encoding calculus of Chapter 7 let us add and multiply matrices, and hence realize any *polynomial* of a block-encoded matrix A as a short linear combination of unitaries—at the price of an ℓ^1 overhead that grows with the degree. This lecture develops the far more economical route. The key observation, due to Low and Chuang, is that a single block encoding U_A of a Hermitian A is secretly a direct sum of 2×2 rotations, one per eigenvalue: each eigenvector $|v_j\rangle$ spans, together with a companion "perpendicular" state, a two-dimensional invariant subspace on which U_A acts as a fixed rotation by angle $\arccos \lambda_j$. This is *qubitization* (Section 8.2). Once the eigenvalues sit inside independent qubits, powers of the (reflected) block encoding compute Chebyshev polynomials $T_k(\lambda_j)$ exactly (Section 8.1), and interleaving single-qubit Z-rotations between the block encodings realizes an essentially arbitrary bounded polynomial $P(A)$ with degree equal to the number of queries and subnormalization independent of the degree. The precise class of achievable polynomials, and the phase factors that select them, is the content of *quantum signal processing* (Section 8.4). We treat the Hermitian and general (non-Hermitian) block encodings separately (Sections 8.2 and 8.3), because the alternation of U_A and $U_A^\dagger$ forced by a non-Hermitian encoding is exactly what makes the even/odd parity distinction of the QSP theorem appear. We close (Section 8.5) with a survey of how the phase factors are computed in practice—a problem once thought hard, now routinely solved. Throughout, A is an n-qubit Hermitian matrix with $\|A\| \leq 1$, block-encoded as in Chapter 6; the singular-value generalization to non-Hermitian A (QSVT) is the subject of Chapter 9.

8.1 Chebyshev polynomials from a 2×2 block

Everything in this lecture rests on one elementary fact about rotations. Recall the *Chebyshev polynomials of the first and second kind*, defined for $\lambda = \cos\theta$ by

$$T_k(\lambda) = \cos(k\theta) = \cos\big(k \arccos\lambda\big), \qquad U_{k-1}(\lambda) = \frac{\sin(k\theta)}{\sin\theta} = \frac{\sin\big(k\arccos\lambda\big)}{\sqrt{1-\lambda^2}}. \tag{8.1}$$

They satisfy the three-term recurrences

$$T_0 = 1, \quad T_1(\lambda) = \lambda, \quad T_{k+1}(\lambda) = 2\lambda T_k(\lambda) - T_{k-1}(\lambda), \tag{8.2}$$

and the same recurrence holds for U_k with $U_0 = 1$, $U_1 = 2\lambda$. In particular T_k has parity $k \bmod 2$ and $|T_k(\lambda)| \le 1$ on $[-1, 1]$.

Lemma 8.1 (Chebyshev polynomials from a rotation)**.** *Fix $-1 \le \lambda \le 1$, write $\lambda = \cos\theta$ with $0 \le \theta \le \pi$, and set*

$$U(\lambda) = \begin{pmatrix} \lambda & \sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & -\lambda \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad O(\lambda) := U(\lambda)Z. \tag{8.3}$$

Then $O(\lambda)$ is the planar rotation by θ,

$$O(\lambda) = \begin{pmatrix} \lambda & -\sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & \lambda \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \tag{8.4}$$

and its powers carry Chebyshev polynomials in every entry:

$$O(\lambda)^k = \begin{pmatrix} \cos(k\theta) & -\sin(k\theta) \\ \sin(k\theta) & \cos(k\theta) \end{pmatrix} = \begin{pmatrix} T_k(\lambda) & -\sqrt{1-\lambda^2}\,U_{k-1}(\lambda) \\ \sqrt{1-\lambda^2}\,U_{k-1}(\lambda) & T_k(\lambda) \end{pmatrix}. \tag{8.5}$$

Proof. The matrix $U(\lambda)$ is a reflection ($U(\lambda) = U(\lambda)^\dagger$ and $U(\lambda)^2 = \mathbb{I}$ since $\lambda^2 + (1-\lambda^2) = 1$), and right-multiplication by Z negates the second column, giving (8.4); the identification with $\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ is immediate from $\lambda = \cos\theta$, $\sqrt{1-\lambda^2} = \sin\theta$ (with $\sin\theta \ge 0$ on $[0, \pi]$). A rotation by θ raised to the k-th power is a rotation by $k\theta$, which is (8.5) after substituting $\cos(k\theta) = T_k(\lambda)$ and $\sin(k\theta) = \sin\theta\, U_{k-1}(\lambda) = \sqrt{1-\lambda^2}\,U_{k-1}(\lambda)$ from (8.1). □

Example 8.2 (The rotation picture with numbers: $\lambda = \frac{1}{2}$)**.** Take $\lambda = \frac{1}{2}$, so $\theta = \arccos\frac{1}{2} = 60^\circ$ and $\sqrt{1-\lambda^2} = \frac{\sqrt{3}}{2}$. Then

$$O(\tfrac{1}{2}) = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \qquad \text{(rotation by } 60^\circ\text{)},$$

and its powers are rotations by $k \cdot 60^\circ$, whose $(1,1)$-entries read off the Chebyshev values directly:

$$[O^2]_{11} = \cos 120^\circ = -\tfrac{1}{2} = T_2(\tfrac{1}{2}), \quad [O^3]_{11} = \cos 180^\circ = -1 = T_3(\tfrac{1}{2}), \quad [O^6]_{11} = \cos 360^\circ = 1 = T_6(\tfrac{1}{2}),$$

each matching the recurrence (8.2): $T_2(\frac{1}{2}) = 2(\frac{1}{2})^2 - 1 = -\frac{1}{2}$ and $T_3(\frac{1}{2}) = 2 \cdot \frac{1}{2} \cdot (-\frac{1}{2}) - \frac{1}{2} = -1$. Raising the single matrix $O = U(\lambda)Z$ to the k-th power *is* evaluating the degree-k Chebyshev polynomial at λ—no polynomial arithmetic, just a rotation iterated. Qubitization replaces $\frac{1}{2}$ by an eigenvalue λ_j of A and runs this one rotation on every eigenspace at once.

The content of Theorem 8.1 is worth restating in plain terms: the $(1,1)$-entry of $O(\lambda)^k$ is exactly $T_k(\lambda)$. If we could somehow substitute a matrix A for the scalar λ—replacing the 2×2 block by a block encoding of A, and the reflection Z by an appropriate reflection on the ancilla—then the analogous power would be a block encoding of $T_k(A)$, with the degree-k Chebyshev polynomial computed at unit subnormalization and with no LCU whatsoever. For a Hermitian $A=\sum_j \lambda_j |v_j\rangle\langle v_j|$ the substitution is literally the spectral one, $T_k(A)=\sum_j T_k(\lambda_j)|v_j\rangle\langle v_j|$, applied eigenvalue by eigenvalue. Making this precise is the business of qubitization.

8.2 Qubitization of Hermitian matrices

Let A be Hermitian with $\|A\|\le 1$ and spectral decomposition $A=\sum_j \lambda_j |v_j\rangle\langle v_j|$, and suppose first that we hold a *Hermitian* block encoding $U_A\in \mathrm{HBE}_{1,m}(A)$ (Theorem 6.21), i.e. $U_A=U_A^\dagger$ and $(\langle 0^m|\otimes\mathbb{I})U_A(|0^m\rangle\otimes\mathbb{I})=A$. Write $\Pi:=|0^m\rangle\langle 0^m|\otimes\mathbb{I}$ for the projector onto the flagged subspace.

Lemma 8.3 (Invariant two-dimensional subspaces)**.** *For each eigenvector $|v_j\rangle$ of A there is a normalized state $|\perp_j\rangle$ with $\Pi|\perp_j\rangle=0$ such that $\mathcal{H}_j:=\mathrm{span}\{|0^m\rangle|v_j\rangle,|\perp_j\rangle\}$ is invariant under U_A, and in the ordered basis $\mathcal{B}_j=\{|0^m\rangle|v_j\rangle,|\perp_j\rangle\}$*

$$[U_A]_{\mathcal{B}_j}=\begin{pmatrix}\lambda_j & \sqrt{1-\lambda_j^2}\\ \sqrt{1-\lambda_j^2} & -\lambda_j\end{pmatrix},\qquad [\Pi]_{\mathcal{B}_j}=\begin{pmatrix}1&0\\0&0\end{pmatrix}. \tag{8.6}$$

Proof. Applying U_A to the flagged eigenstate and using $\Pi U_A\Pi=|0^m\rangle\langle 0^m|\otimes A$,

$$U_A|0^m\rangle|v_j\rangle=|0^m\rangle A|v_j\rangle+|e_j^\perp\rangle=\lambda_j|0^m\rangle|v_j\rangle+|e_j^\perp\rangle, \tag{8.7}$$

where $|e_j^\perp\rangle:=(\mathbb{I}-\Pi)U_A|0^m\rangle|v_j\rangle$ satisfies $\Pi|e_j^\perp\rangle=0$. Since U_A is unitary the left side is a unit vector, so $\big\||e_j^\perp\rangle\big\|^2=1-\lambda_j^2$ and we may write $|e_j^\perp\rangle=\sqrt{1-\lambda_j^2}\,|\perp_j\rangle$ with $|\perp_j\rangle$ normalized (if $\lambda_j^2=1$ then $|e_j^\perp\rangle=0$ and $\mathcal{H}_j$ is one-dimensional; take $|\perp_j\rangle$ to be any completing unit vector). This gives the first column of (8.6). For the second, use $U_A=U_A^\dagger$: applying U_A to (8.7) and $U_A^2=\mathbb{I}$,

$$|0^m\rangle|v_j\rangle=\lambda_j U_A|0^m\rangle|v_j\rangle+\sqrt{1-\lambda_j^2}\,U_A|\perp_j\rangle=\lambda_j^2|0^m\rangle|v_j\rangle+\lambda_j\sqrt{1-\lambda_j^2}\,|\perp_j\rangle+\sqrt{1-\lambda_j^2}\,U_A|\perp_j\rangle, \tag{8.8}$$

so solving for $U_A|\perp_j\rangle$,

$$U_A|\perp_j\rangle=\sqrt{1-\lambda_j^2}\,|0^m\rangle|v_j\rangle-\lambda_j|\perp_j\rangle, \tag{8.9}$$

which is the second column of (8.6). Hence $\mathcal{H}_j$ is invariant and U_A restricts to the stated reflection. The matrix of Π is immediate since $\Pi|0^m\rangle|v_j\rangle=|0^m\rangle|v_j\rangle$ and $\Pi|\perp_j\rangle=0$. □

The name *qubitization* reflects (8.6): an arbitrarily large block encoding U_A is block-diagonalized into N independent "qubits", one 2×2 reflection per eigenvalue, and these blocks are precisely the reflection $U(\lambda_j)$ of Theorem 8.1. Two eigenvectors with $\lambda_j^2=1$ require no companion state; all other subspaces are genuinely two-dimensional and mutually orthogonal (the $|\perp_j\rangle$ are orthogonal because they live in distinct eigenspaces of A propagated by U_A).

To turn each reflection into the *rotation* $O(\lambda_j)$ of (8.4) we need the ancilla reflection $Z_\Pi:=2\Pi-\mathbb{I}$, which acts as $+1$ on the flagged subspace and -1 on its complement, so $[Z_\Pi]_{\mathcal{B}_j}=\mathrm{diag}(1,-1)=Z$. Then $O:=U_AZ_\Pi$ restricts to $[O]_{\mathcal{B}_j}=[U_A]_{\mathcal{B}_j}Z=O(\lambda_j)$, exactly the rotation of Theorem 8.1. This is the substitution promised in Section 8.1.

Theorem 8.4 (Qubitization: Chebyshev polynomials of a Hermitian matrix)**.** *Let $U_A \in \mathrm{HBE}_{1,m}(A)$ with A Hermitian, $\|A\| \le 1$, and let $Z_\Pi = 2\Pi - \mathbb{I}$ with $\Pi = |0^m\rangle\langle 0^m| \otimes \mathbb{I}$. Then for every $k \ge 0$,*

$$(U_A Z_\Pi)^k \in \mathrm{BE}_{1,m}(T_k(A)), \qquad \textit{i.e.} \qquad \Pi\,(U_A Z_\Pi)^k\,\Pi = |0^m\rangle\langle 0^m| \otimes T_k(A). \tag{8.10}$$

Proof. Each $\mathcal{H}_j$ is invariant under both U_A (Theorem 8.3) and Z_Π (which is diagonal in $\mathcal{B}_j$), hence under $O = U_A Z_\Pi$, and $[O]_{\mathcal{B}_j} = O(\lambda_j)$. By Theorem 8.1, $[O^k]_{\mathcal{B}_j} = O(\lambda_j)^k$ has $(1,1)$-entry $T_k(\lambda_j)$, i.e.

$$\langle 0^m|\,\langle v_j|\,O^k\,|0^m\rangle\,|v_j\rangle = T_k(\lambda_j). \tag{8.11}$$

For $j \neq j'$ the subspaces $\mathcal{H}_j, \mathcal{H}_{j'}$ are orthogonal, so $\langle 0^m|\,\langle v_{j'}|\,O^k\,|0^m\rangle\,|v_j\rangle = 0$. The states $\{|0^m\rangle\,|v_j\rangle\}_j$ form an orthonormal basis of the range of Π, so (8.11) says $\Pi O^k \Pi = |0^m\rangle\langle 0^m| \otimes \sum_j T_k(\lambda_j)\,|v_j\rangle\langle v_j| = |0^m\rangle\langle 0^m| \otimes T_k(A)$, which is (8.10). □

The reflection Z_Π is cheap: for $m = 1$ it is a single Pauli Z, and in general it is a multiply-controlled phase implementable with one clean ancilla and an m-fold Toffoli. One step of qubitization is therefore "U_A then Z_Π", and k steps give $T_k(A)$ with a *single* query per Chebyshev degree and unit subnormalization—compare the growing ℓ^1 cost of building $T_k(A)$ as an LCU of monomials in Chapter 7. Because $\{T_k\}_{k\ge 0}$ is a basis of the polynomials, one could now form any polynomial of A by an LCU of the qubitized Chebyshev blocks (Theorem 7.9); QSP will do the same job below with a single ancilla and no LCU.

Example 8.5 (A degree-2 Chebyshev block)**.** Take $A = \frac{1}{2}\mathsf{X}$ on one qubit, with the Hermitian block encoding $U_A \in \mathrm{HBE}_{1,1}(A)$ of Theorem 6.22, so $m = 1$ and $Z_\Pi = \mathsf{Z}$ on the flag qubit. The two eigenvalues are $\lambda_\pm = \pm\frac{1}{2}$ (eigenvectors $|\pm\rangle$), and Theorem 8.4 gives $(U_A\mathsf{Z})^2 \in \mathrm{BE}_{1,1}(T_2(A))$ with $T_2(A) = 2A^2 - \mathbb{I} = 2\cdot\frac{1}{4}\mathbb{I} - \mathbb{I} = -\frac{1}{2}\mathbb{I}$. One checks directly that $T_2(\lambda_\pm) = 2\cdot\frac{1}{4} - 1 = -\frac{1}{2}$, consistent with the spectral formula $T_2(A) = \sum_\pm T_2(\lambda_\pm)\,|\pm\rangle\langle\pm| = -\frac{1}{2}\mathbb{I}$. Two queries to U_A thus block-encode $-\frac{1}{2}\mathbb{I}$ exactly.

8.3 Qubitization of general matrices

Hermitian block encodings of Hermitian matrices exist (Theorem 6.22) but may demand extra structure and oracles. In general the block encoding U_A of a Hermitian A is *not* itself Hermitian, and the single relation (8.7) no longer closes on a two-dimensional invariant subspace, because we cannot re-apply U_A and use $U_A^2 = \mathbb{I}$. The remedy is to alternate U_A and $U_A^\dagger$. The result is that U_A maps one two-dimensional subspace to a *second* one, and $U_A^\dagger$ maps it back.

Lemma 8.6 (Qubitization with a general block encoding)**.** *Let $U_A \in \mathrm{BE}_{1,m}(A)$ with A Hermitian, $\|A\| \le 1$, and $\Pi = |0^m\rangle\langle 0^m| \otimes \mathbb{I}$. For each eigenvector $|v_j\rangle$ there are normalized states $|\perp_j\rangle, |\perp'_j\rangle$ (both annihilated by Π) such that, with $\mathcal{B}_j = \{|0^m\rangle\,|v_j\rangle, |\perp_j\rangle\}$ and $\mathcal{B}'_j = \{|0^m\rangle\,|v_j\rangle, |\perp'_j\rangle\}$,*

$$[U_A]^{\mathcal{B}'_j}_{\mathcal{B}_j} = \begin{pmatrix} \lambda_j & \sqrt{1-\lambda_j^2} \\ \sqrt{1-\lambda_j^2} & -\lambda_j \end{pmatrix} = [U_A^\dagger]^{\mathcal{B}_j}_{\mathcal{B}'_j}. \tag{8.12}$$

Consequently $\mathcal{H}_j = \operatorname{span}\mathcal{B}_j$ is invariant under $\widetilde{O} := U_A^\dagger Z_\Pi U_A Z_\Pi$, with

$$[\widetilde{O}]_{\mathcal{B}_j} = \begin{pmatrix} \lambda_j & -\sqrt{1-\lambda_j^2} \\ \sqrt{1-\lambda_j^2} & \lambda_j \end{pmatrix}^2 = O(\lambda_j)^2. \tag{8.13}$$

Proof. As in (8.7), $U_A|0^m\rangle|v_j\rangle = \lambda_j|0^m\rangle|v_j\rangle + \sqrt{1-\lambda_j^2}\,|\perp_j'\rangle$ defines the normalized $|\perp_j'\rangle$; this is the first column of (8.12). Because A is Hermitian, $U_A^\dagger$ is also a block encoding of A ($\Pi U_A^\dagger \Pi = (\Pi U_A \Pi)^\dagger = |0^m\rangle\langle 0^m| \otimes A$), so likewise $U_A^\dagger|0^m\rangle|v_j\rangle = \lambda_j|0^m\rangle|v_j\rangle + \sqrt{1-\lambda_j^2}\,|\perp_j\rangle$ defines $|\perp_j\rangle$. Applying U_A to this and using $U_A U_A^\dagger = \mathbb{I}$ gives, exactly as in (8.8)– (8.9), $U_A|\perp_j\rangle = \sqrt{1-\lambda_j^2}\,|0^m\rangle|v_j\rangle - \lambda_j|\perp_j'\rangle$, the second column of (8.12); the identity $[U_A^\dagger]_{\mathcal{B}_j'}^{\mathcal{B}_j} = ([U_A]_{\mathcal{B}_j}^{\mathcal{B}_j'})^\dagger$ equals the same real symmetric matrix. Thus U_A carries $\mathcal{H}_j \to \mathcal{H}_j'$ and $U_A^\dagger$ carries $\mathcal{H}_j' \to \mathcal{H}_j$; each of Z_Π fixes the shared vector $|0^m\rangle|v_j\rangle$ and flips the perpendicular one. Composing the four factors of $\widetilde{O} = U_A^\dagger Z_\Pi U_A Z_\Pi$ maps $\mathcal{H}_j \to \mathcal{H}_j$: the inner $Z_\Pi U_A Z_\Pi$ carries $\mathcal{H}_j$ to $\mathcal{H}_j'$ acting as $Z\,[U_A]_{\mathcal{B}_j}^{\mathcal{B}_j'} Z$ up to the shared vector, and the outer $U_A^\dagger$ returns to $\mathcal{H}_j$. In the basis $\mathcal{B}_j$ the composite matrix is $\left([U_A]_{\mathcal{B}_j}^{\mathcal{B}_j'} Z\right)^2 = \left(U(\lambda_j)Z\right)^2 = O(\lambda_j)^2$, which is (8.13). □

Theorem 8.7 (Chebyshev polynomials from a general block encoding)**.** *With the notation of Theorem 8.6, for every $k \geq 1$,*

$$\left(U_A^\dagger Z_\Pi U_A Z_\Pi\right)^k \in \mathrm{BE}_{1,m}(T_{2k}(A)), \qquad U_A Z_\Pi \left(U_A^\dagger Z_\Pi U_A Z_\Pi\right)^k \in \mathrm{BE}_{1,m}(T_{2k+1}(A)). \tag{8.14}$$

Hence the even Chebyshev polynomial $T_{2k}(A)$ is obtained by k alternating pairs $U_A^\dagger Z_\Pi$, $U_A Z_\Pi$, and the odd one $T_{2k+1}(A)$ by one extra $U_A Z_\Pi$.

Proof. By (8.13), $[\widetilde{O}^k]_{\mathcal{B}_j} = O(\lambda_j)^{2k}$, whose $(1,1)$-entry is $T_{2k}(\lambda_j)$ (Theorem 8.1); orthogonality of the $\mathcal{H}_j$ and completeness of $\{|0^m\rangle|v_j\rangle\}$ then give $\Pi\widetilde{O}^k\Pi = |0^m\rangle\langle 0^m| \otimes T_{2k}(A)$ as in Theorem 8.4. For the odd case, one more $U_A Z_\Pi$ sends $\mathcal{H}_j \to \mathcal{H}_j'$; since the two bases share the vector $|0^m\rangle|v_j\rangle$, the surviving flagged block is the $(1,1)$-entry of $O(\lambda_j)^{2k+1}$, namely $T_{2k+1}(\lambda_j)$, giving the second encoding in (8.14). □

The lesson is that a non-Hermitian block encoding produces Chebyshev polynomials of A just as before, but the fundamental building block is a $U_A, U_A^\dagger$ *pair*, and even/odd degrees must be assembled separately. This alternation is not an artifact: it is the same structural feature that forces the parity condition in the QSP theorem, to which we now turn. (When U_A happens to be Hermitian, $U_A^\dagger = U_A$ and Theorem 8.7 collapses to Theorem 8.4.)

8.4 Quantum signal processing

Chebyshev polynomials are only the phase-free special case. Interleaving single-qubit Z-rotations $\mathrm{e}^{\mathrm{i}\phi_j Z}$ between the qubitization steps produces a far richer family. Because every operation preserves the two-dimensional subspaces $\mathcal{H}_j$, the entire question reduces, by qubitization, to a statement about SU(2): what 2×2 matrix polynomials arise from a product of the fixed rotation $O(\lambda)$ of (8.4) and free Z-rotations? This is *quantum signal processing*.

Definition 8.8 (QSP sequence)**.** For phase factors $\Phi = (\phi_0, \phi_1, \ldots, \phi_d) \in \mathbb{R}^{d+1}$ and $\lambda \in [-1,1]$, define the *QSP unitary*

$$U_\Phi(\lambda) := \mathrm{e}^{\mathrm{i}\phi_0 Z} \prod_{j=1}^{d} \left[O(\lambda)\,\mathrm{e}^{\mathrm{i}\phi_j Z}\right], \qquad O(\lambda) = \begin{pmatrix} \lambda & -\sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & \lambda \end{pmatrix}. \tag{8.15}$$

The number of signal operators $O(\lambda)$ is d, the "query depth". The following theorem characterizes exactly the polynomials that appear in the top-left entry.

Theorem 8.9 (Quantum signal processing)**.** *There exist phase factors* $\Phi \in \mathbb{R}^{d+1}$ *with*

$$U_\Phi(\lambda) = \begin{pmatrix} P(\lambda) & -Q(\lambda)\sqrt{1-\lambda^2} \\ Q^*(\lambda)\sqrt{1-\lambda^2} & P^*(\lambda) \end{pmatrix} \tag{8.16}$$

if and only if $P, Q \in \mathbb{C}[\lambda]$ *satisfy*
(1) $\deg P \le d$ *and* $\deg Q \le d-1$*;*
(2) P *has parity* $d \bmod 2$ *and* Q *has parity* $(d-1) \bmod 2$*;*
(3) $|P(\lambda)|^2 + (1-\lambda^2)\,|Q(\lambda)|^2 = 1$ *for all* $\lambda \in [-1,1]$.
Here $\deg Q = -1$ *means* $Q \equiv 0$*, and* P^*, Q^* *denote coefficientwise complex conjugation.*

Proof of the forward direction (⇒). We prove by induction on d that $U_\Phi(\lambda)$ has the form (8.16) with P, Q satisfying (1)–(2); condition (3) then follows for free, since $U_\Phi(\lambda)$ is a product of unitaries, hence unitary, and comparing $|\cdot|^2$ of its first column gives $|P|^2 + (1-\lambda^2)\,|Q|^2 = 1$.

Base case $d = 0$. Here $U_\Phi(\lambda) = \mathrm{e}^{\mathrm{i}\phi_0 Z} = \mathrm{diag}(\mathrm{e}^{\mathrm{i}\phi_0}, \mathrm{e}^{-\mathrm{i}\phi_0})$, so $P(\lambda) = \mathrm{e}^{\mathrm{i}\phi_0}$ (degree 0, even) and $Q \equiv 0$, satisfying (1)–(2).

Inductive step. Suppose the length-$(d-1)$ product $U_{(\phi_0,\dots,\phi_{d-1})}(\lambda)$ equals (8.16) with P, Q of degrees $\le d-1, \le d-2$ and parities $(d-1) \bmod 2, (d-2) \bmod 2$. Appending one more block multiplies on the right by $O(\lambda)\,\mathrm{e}^{\mathrm{i}\phi_d Z}$:

$$\begin{pmatrix} P & -Q\sqrt{1-\lambda^2} \\ Q^*\sqrt{1-\lambda^2} & P^* \end{pmatrix} \begin{pmatrix} \lambda & -\sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & \lambda \end{pmatrix} \begin{pmatrix} \mathrm{e}^{\mathrm{i}\phi_d} & 0 \\ 0 & \mathrm{e}^{-\mathrm{i}\phi_d} \end{pmatrix}. \tag{8.17}$$

Carrying out the first product, the new top-left entry (before the phase) is $\lambda P - (1-\lambda^2)Q$ and the new top-right entry is $-\sqrt{1-\lambda^2}\,(P + \lambda Q)$; after the diagonal phase the matrix is

$$\begin{pmatrix} \mathrm{e}^{\mathrm{i}\phi_d}\big(\lambda P - (1-\lambda^2)Q\big) & -\mathrm{e}^{-\mathrm{i}\phi_d}\sqrt{1-\lambda^2}\,(P + \lambda Q) \\ \mathrm{e}^{\mathrm{i}\phi_d}\sqrt{1-\lambda^2}\,(P^* + \lambda Q^*) & \mathrm{e}^{-\mathrm{i}\phi_d}\big(\lambda P^* - (1-\lambda^2)Q^*\big) \end{pmatrix}. \tag{8.18}$$

This has the form (8.16) with

$$\widetilde{P} = \mathrm{e}^{\mathrm{i}\phi_d}(\lambda P - (1-\lambda^2)Q), \qquad \widetilde{Q} = \mathrm{e}^{-\mathrm{i}\phi_d}(P + \lambda Q). \tag{8.19}$$

Degrees: $\deg \widetilde{P} \le \max(1 + \deg P, 2 + \deg Q) \le d$ and $\deg \widetilde{Q} \le \max(\deg P, 1 + \deg Q) \le d-1$, giving (1). Parities: multiplying by λ or $(1-\lambda^2)\cdot$(something of parity $d-2$) flips or preserves parity so that $\widetilde{P}$ has parity $d \bmod 2$ and $\widetilde{Q}$ has parity $(d-1) \bmod 2$, giving (2). This completes the induction, and (3) follows from unitarity as noted. □

The converse—that *every* P, Q satisfying (1)–(3) is realized by some Φ—is proved by the reverse induction: given P, Q of degree ℓ, condition (3) forces the two leading coefficients to have equal modulus, and a single phase $\mathrm{e}^{2\mathrm{i}\phi_d} =$ (ratio of leading coefficients) peels off one degree, reducing to a shorter QSP problem; iterating recovers all ϕ_j. We refer to the reading for the details (it is essentially the Gilyén–Su–Low–Wiebe argument), and record the practically important consequences.

Corollary 8.10 (QSP for real target polynomials)**.** *Let* $p \in \mathbb{R}[\lambda]$ *have* $\deg p = d$*, parity* $d \bmod 2$*, and* $|p(\lambda)| \le 1$ *on* $[-1,1]$*. Then there exist* P, Q *satisfying* (1)–(3) *of Theorem 8.9 with* $\mathrm{Re}\,P = p$*, and correspondingly phase factors* Φ *with* $\mathrm{Re}[U_\Phi(\lambda)_{11}] = p(\lambda)$.

The corollary is what one uses in practice: almost every application (matrix inversion, Hamiltonian simulation, eigenvalue filtering) starts from a real target polynomial p, scales it so that $|p| \le 1$,

and then produces P with $\operatorname{Re} P = p$. The real part is extracted by one extra Hadamard-sandwiched ancilla, exactly the "P_{Re}" construction of qubitization: running U_Φ and $U_{-\Phi}$ (which block-encodes P^*) in an LCU with equal weights gives $\frac{1}{2}(P + P^*) = \operatorname{Re} P = p$. Lifting from the scalar λ to a Hermitian A is now automatic: replace $O(\lambda)$ by $O = U_A Z_\Pi$ (Hermitian encoding) or by the alternating $U_A, U_A^\dagger$ pattern (general encoding), and each $\mathrm{e}^{\mathrm{i}\phi_j Z}$ by the controlled rotation $\mathrm{e}^{\mathrm{i}\phi_j Z_\Pi}$ on the flag; since every factor preserves the subspaces $\mathcal{H}_j$, the flagged block is $P(A) = \sum_j P(\lambda_j)\,|v_j\rangle\langle v_j|$. Thus d queries to U_A block-encode $P(A)$ with unit subnormalization. This is the *quantum eigenvalue transformation*; its singular-value extension to non-Hermitian A is QSVT (Chapter 9).

Example 8.11 (QSP realizations of low-degree polynomials)**.** *(a) T_d via all-zero phases.* Setting $\Phi = (0, \ldots, 0)$ in (8.15) gives $U_\Phi(\lambda) = O(\lambda)^d$, whose $(1,1)$-entry is $T_d(\lambda)$ by Theorem 8.1. Thus $P = T_d$, $Q = U_{d-1}$: the Chebyshev polynomials are the phase-free QSP polynomials, consistent with Theorem 8.4.

(b) Degree 1. For $d = 1$, $U_\Phi(\lambda) = \mathrm{e}^{\mathrm{i}\phi_0 Z} O(\lambda) \mathrm{e}^{\mathrm{i}\phi_1 Z}$ has top-left entry

$$P(\lambda) = \mathrm{e}^{\mathrm{i}(\phi_0+\phi_1)}\lambda,$$

so $\Phi = (0,0)$ realizes $P(\lambda) = \lambda = T_1(\lambda)$ (real, odd), while $\Phi = (\frac{\pi}{4}, \frac{\pi}{4})$ realizes $P(\lambda) = \mathrm{i}\lambda$; in every case $|P(\lambda)| = |\lambda| \le 1$ and $\deg P = 1$, matching (1)–(3) with $Q = \mathrm{e}^{\mathrm{i}(\phi_1 - \phi_0)}$ a constant.

(c) Degree 2*, target T_2, symmetric phases.* A more symmetric choice realizing $T_2(\lambda) = 2\lambda^2 - 1$ uses the eigenvalue-transform convention $U_A \mathrm{e}^{\mathrm{i}e_j Z}$ (with U_A the reflection, not O): the phases $e = (\frac{\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4})$ yield $(-\mathrm{i})^2 U_{e,11} = 2\lambda^2 - 1 = T_2$. Direct multiplication of $\mathrm{e}^{\mathrm{i}\frac{\pi}{4}Z} U_A \mathrm{e}^{\mathrm{i}\frac{\pi}{2}Z} U_A \mathrm{e}^{\mathrm{i}\frac{\pi}{4}Z}$ confirms the top-left entry equals $\mathrm{i}\,(2\lambda^2 - 1)$, whose $(-\mathrm{i})^2 = -1$ times the leading sign returns T_2; equivalently in the O-convention $\Phi = (0,0,0)$ gives $O(\lambda)^2$ with $(1,1)$-entry $2\lambda^2 - 1$. These small cases are the ones we verify by hand (and, in the companion notebook, numerically: the QSP unitary equals the claimed polynomial to machine precision).

8.5 Finding phase factors

QSP is only useful if the phase factors Φ realizing a desired p can actually be computed. When QSP was introduced this was considered a serious obstacle; it is now, for all practical purposes, solved. We survey the three main families of methods; all take as input a real target p of degree d, parity $d \bmod 2$, with $\|p\|_\infty < 1$ on $[-1,1]$, and return $\Phi \in \mathbb{R}^{d+1}$ (of which only $\lceil (d+1)/2 \rceil$ are free, by parity).

Direct (root-finding) methods. The converse proof of Theorem 8.9 is constructive: given P, Q one strips one degree at a time, each step reading off a phase from the ratio of leading coefficients. The missing ingredient is Q, obtained from the *completion* step—given P with $|P| \le 1$, find Q with $|P|^2 + (1-\lambda^2)\,|Q|^2 = 1$ by factoring the degree-$2d$ polynomial $1 - |P(\lambda)|^2$ over its roots and grouping them into a Fejér–Riesz square. This is the Remez-exchange / root-finding approach: exact in principle, but the root-finding is numerically delicate in high precision because the roots of $1 - P^2$ cluster and the extraction is ill-conditioned, requiring high-precision arithmetic for large d.

Optimization-based methods. Rather than solve for Φ exactly, one minimizes the least-squares mismatch between the QSP output and the target on a Chebyshev grid. Fixing the parity, only $\widetilde{d} = \lceil (d+1)/2 \rceil$ phases are free, and p is determined by its values at $\widetilde{d}$ points, which one takes to

be the positive Chebyshev nodes $\lambda_k = \cos\!\left(\frac{(2k-1)\pi}{4\widetilde{d}}\right)$. Minimizing

$$F(\Phi) = \frac{1}{\widetilde{d}} \sum_{k=1}^{\widetilde{d}} \Big(\mathrm{Re}\big[U_\Phi(\lambda_k)_{11}\big] - p(\lambda_k) \Big)^2 \tag{8.20}$$

over Φ drives F to 0 at a valid phase set. The remarkable empirical (and now proven) fact is that starting from the symmetric ansatz $\Phi_0 = (\frac{\pi}{4}, 0, \ldots, 0, \frac{\pi}{4})$ this nonconvex problem behaves almost convexly near the solution, so quasi-Newton descent converges rapidly and stably even in double precision for degrees in the thousands.

Symmetric QSP and fixed-point iteration. Restricting to *symmetric* phase sequences ($\phi_j = \phi_{d-j}$) makes $U_\Phi(\lambda)_{11}$ automatically real with the correct parity, halving the unknowns and turning the completion problem into a well-conditioned fixed-point / Newton iteration on the phases directly (no explicit Q needed). Related developments recast phase-factor finding as a nonlinear Fourier transform, for which fast and provably stable algorithms exist. In practice one calls a mature routine (e.g. the `QSPPACK` optimization solver or a symmetric-QSP Newton solver) as a black box: supply the coefficients of p, receive Φ, and the QSP circuit of Section 8.4 then block-encodes $p(A)$ with d queries.

The upshot for the rest of Part II is that we may treat "block-encode $p(A)$ for a bounded degree-d polynomial p" as a primitive costing $\mathcal{O}(d)$ queries to U_A, one extra ancilla, and d classically-precomputed single-qubit rotations, with phase factors supplied by one of the solvers above. This primitive, and its singular-value form, is the engine of quantum linear algebra: matrix inversion, Hamiltonian simulation, and eigenvalue filtering are all instances of choosing the right p, as Chapters 9 and 10 will show.

Exercises

Exercise 8.1. *Chebyshev from a rotation.* Verify (8.4), i.e. that $O(\lambda) = U(\lambda)Z$ equals the planar rotation by $\theta = \arccos\lambda$. Using $O(\lambda)^k = \left(\begin{smallmatrix} \cos k\theta & -\sin k\theta \\ \sin k\theta & \cos k\theta \end{smallmatrix}\right)$ and the addition formulas, prove the recurrence $T_{k+1}(\lambda) = 2\lambda T_k(\lambda) - T_{k-1}(\lambda)$ directly from (8.1). Compute $O(\lambda)^3$ and read off $T_3(\lambda)$ and $U_2(\lambda)$ explicitly.

Exercise 8.2. *The qubitized block, by hand.* Let $A = \frac{1}{2}\mathsf{X}$ with the Hermitian block encoding $U_A \in \mathrm{HBE}_{1,1}(A)$ of Theorem 6.22. Diagonalize A, write down $\mathcal{B}_\pm = \{|0\rangle|\pm\rangle, |\perp_\pm\rangle\}$ and the 2×2 matrices $[U_A]_{\mathcal{B}_\pm}$, and verify (8.10) for $k = 1, 2$ by checking that $\Pi(U_A\mathsf{Z})^k\Pi = |0\rangle\langle 0| \otimes T_k(A)$ with $T_1(A) = A$ and $T_2(A) = -\frac{1}{2}\mathbb{I}$ (cf. Theorem 8.5).

Exercise 8.3. *Why Z_Π turns a reflection into a rotation.* Show that $[U_A]_{\mathcal{B}_j}$ in (8.6) is a reflection ($([U_A]_{\mathcal{B}_j})^2 = \mathbb{I}$, determinant -1) while $[U_A Z_\Pi]_{\mathcal{B}_j}$ is a rotation (determinant $+1$). Explain why one *cannot* obtain $T_k(A)$ from powers of U_A alone, and why exactly one reflection Z_Π per query is the minimal fix.

Exercise 8.4. *General block encoding: even vs. odd.* Suppose $U_A \in \mathrm{BE}_{1,m}(A)$ is *not* Hermitian, A Hermitian. Using Theorem 8.6, write out the sequence of $U_A, U_A^\dagger, Z_\Pi$ operators that block-encodes $T_4(A)$ and the one that block-encodes $T_3(A)$, and count the number of queries to U_A (and to $U_A^\dagger$) in each. Confirm that the query count equals the Chebyshev degree.

Exercise 8.5. *QSP degree-1 polynomials.* For $d = 1$ compute $U_\Phi(\lambda) = \mathrm{e}^{\mathrm{i}\phi_0 Z} O(\lambda) \mathrm{e}^{\mathrm{i}\phi_1 Z}$ in closed form and confirm $P(\lambda) = \mathrm{e}^{\mathrm{i}(\phi_0+\phi_1)}\lambda$, $Q(\lambda) = \mathrm{e}^{\mathrm{i}(\phi_1-\phi_0)}$. Check conditions (1)–(3) of Theorem 8.9. Which Φ gives $P(\lambda) = \lambda$? Which gives $P(\lambda) = \mathrm{i}\lambda$? Can any degree-1 QSP sequence produce $P(\lambda) = \frac{1}{2}\lambda$ (a *subnormalized* target), and if not, why does that not contradict Theorem 8.10?

Exercise 8.6. *The QSP induction.* Carry out the inductive step of Theorem 8.9 in detail: starting from (8.16) of degree $d-1$, multiply by $O(\lambda)\mathrm{e}^{\mathrm{i}\phi_d Z}$ and derive the update (8.19) $\widetilde{P} = \mathrm{e}^{\mathrm{i}\phi_d}(\lambda P - (1-\lambda^2)Q)$, $\widetilde{Q} = \mathrm{e}^{-\mathrm{i}\phi_d}(P + \lambda Q)$. Verify the degree and parity claims, and check directly that $\left|\widetilde{P}\right|^2 + (1-\lambda^2)\left|\widetilde{Q}\right|^2 = |P|^2 + (1-\lambda^2)\,|Q|^2$, so unitarity (condition (3)) is preserved.

Exercise 8.7. *Phase factors for T_2.* Verify by direct 2×2 multiplication that, in the O-convention, $\Phi = (0,0,0)$ gives $U_\Phi(\lambda) = O(\lambda)^2$ with $(1,1)$-entry $T_2(\lambda) = 2\lambda^2 - 1$. Then, using the relation $\mathrm{e}^{\mathrm{i}\frac{\pi}{2}Z} = \mathrm{i}Z$ and $O(\lambda) = U_A(\lambda)Z$, translate to the eigenvalue-transform convention $e = (\frac{\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4})$ acting on the reflection $U_A(\lambda)$, and confirm that the top-left entry equals $(-\mathrm{i})^2 T_2(\lambda)$ up to the overall phase $(-\mathrm{i})^d$ (cf. Theorem 8.11(c)).

Further reading

- Lin Lin, *qasc notes*, Ch. 7 (Sec. 7.1 qubitization with Hermitian block encoding, Sec. 7.4 qubitization with general block encoding, Sec. 7.5 quantum eigenvalue transformation, Sec. 7.6 quantum signal processing and the QSP theorem for real polynomials).
- Lin Lin, *live notes*, Ch. 10 (qubitization, Chebyshev eigenvalue and singular-value transformations, cosine–sine decomposition) and Ch. 12 (quantum signal processing, symmetric QSP, and phase-factor finding).
- Dong An, *Quantum Numerical Linear Algebra* tutorial talk (the 2×2 Chebyshev construction $O = U(\lambda)Z$, the qubitization invariant subspaces and reflection $Z_\Pi = 2\Pi - \mathbb{I}$, the QSP theorem, and a survey of phase-factor algorithms).
- G. H. Low and I. L. Chuang, *Hamiltonian simulation by qubitization*, Quantum **3**, 163 (2019); A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, *Quantum singular value transformation and beyond*, Proc. STOC 2019, 193–204, for the original QSP/QSVT theorems.
- J. Wang, Y. Dong, and L. Lin, *On the energy landscape of symmetric quantum signal processing*, Quantum **6**, 850 (2022), and R. Chao, D. Ding, A. Gilyén, C. Huang, and M. Szegedy, *Finding angles for quantum signal processing with machine precision*, arXiv:2003.02831, for phase-factor finding.

Chapter 9

Quantum Singular Value Transformation and Matrix Functions

Learning objectives

- Extend QSP to the singular value transformation of general matrices.
- State the real-polynomial QSVT theorem.
- Realize matrix functions (inverse, exponential, filters) via QSVT.
- Connect QSVT to LCU as two routes to matrix functions.

Qubitization and quantum signal processing (Chapter 8) taught us to apply a bounded degree-d polynomial to the *eigenvalues* of a Hermitian block-encoded matrix using $\mathcal{O}(d)$ queries and unit subnormalization. But the input models of Chapter 6 routinely produce block encodings of *non-Hermitian* matrices—a sparse coefficient matrix of a linear system, a nonsymmetric generator of a differential equation, a rectangular data matrix—for which "eigenvalue" is not even the right notion. The correct object is the *singular value decomposition*, and the correct generalization of the eigenvalue transformation is the *quantum singular value transformation* (QSVT), due to Gilyén, Su, Low, and Wiebe: a single degree-d polynomial applied to the *singular values* of A, again with $\mathcal{O}(d)$ queries to the block encoding U_A and its adjoint $U_A^\dagger$. Remarkably, the circuit is *identical* to the general-encoding qubitization of Section 8.3—the alternation of U_A and $U_A^\dagger$ that we were forced into by non-Hermiticity is exactly what threads a degree through the singular values—so QSVT requires almost no new machinery beyond a change of viewpoint from eigenvectors to left/right singular vectors (Section 9.1). Once the transformation is set up we record the working theorem for real target polynomials (Section 9.2), and then harvest the three applications that make QSVT the "grand unification" of quantum algorithms: matrix inversion $x \mapsto 1/(\kappa x)$ (the engine of the quantum linear systems problem, Chapter 10), Hamiltonian simulation $x \mapsto \mathrm{e}^{-\mathrm{i}xt}$ via the Jacobi–Anger expansion (Chapter 12), and spectral filtering by an approximate sign or rectangle function (Section 9.3). We then compare QSVT with the LCU route of Chapter 7—coherent versus ℓ^1-weighted, singular value versus eigenvalue—to see when each is preferable (Section 9.4), and close with the practical question of computing the phase factors (Section 9.5). Throughout, $A \in \mathbb{C}^{N\times N}$ with $N = 2^n$ is accessed by a $(1, m)$-block encoding $U_A \in \mathrm{BE}_{1,m}(A)$, so that $\|A\| \le 1$ and all singular values lie in $[0, 1]$; we reuse the QSP theorem (Theorem 8.9) and its real-polynomial corollary (Theorem 8.10) without reproof.

9.1 From eigenvalue to singular value transformation

Singular value decomposition and the three matrix functions. Let $A \in \mathbb{C}^{N\times N}$ have singular value decomposition

$$A = W\Sigma V^\dagger = \sum_{i\in[N]} \sigma_i \, |w_i\rangle\langle v_i| \,, \qquad 0 \le \sigma_i \le 1, \tag{9.1}$$

where $\{|v_i\rangle\}$ (right singular vectors) and $\{|w_i\rangle\}$ (left singular vectors) are orthonormal bases and $\Sigma = \mathrm{diag}(\sigma_0, \ldots, \sigma_{N-1})$. Equivalently $A\,|v_i\rangle = \sigma_i\,|w_i\rangle$ and $A^\dagger\,|w_i\rangle = \sigma_i\,|v_i\rangle$. Given a scalar function $f\colon \mathbb{R} \to \mathbb{C}$, there are *three* natural matrix functions built from (9.1), depending on which singular basis is used on the two sides.

Definition 9.1 (Generalized matrix functions)**.** With A as in (9.1) and $f(\Sigma) := \mathrm{diag}(f(\sigma_0), \ldots, f(\sigma_{N-1}))$, define

$$f^{(\mathrm{SV})}(A) := W f(\Sigma) V^\dagger, \qquad f^{\triangleright}(A) := W f(\Sigma) W^\dagger, \qquad f^{\triangleleft}(A) := V f(\Sigma) V^\dagger. \tag{9.2}$$

We call $f^{(\mathrm{SV})}(A)$ the (two-sided) *singular value transformation*, and $f^{\triangleright}, f^{\triangleleft}$ the *left* and *right* generalized matrix functions; all three are loosely referred to as singular value transformations.

The three coincide when A is Hermitian and f has definite parity; in general they differ, and QSVT will produce $f^{(\mathrm{SV})}(A)$ for *odd* f and $f^{\triangleleft}(A)$ (equivalently $f^{\triangleright}(A)$) for *even* f. Two identities record their basic algebra.

Proposition 9.2 (Relations among the generalized functions)**.** $f^{(\mathrm{SV})}(A^\dagger) = \left(f^{(\mathrm{SV})}(A)\right)^\dagger$, $f^{\triangleleft}(A) = f^{\triangleright}(A^\dagger)$*, and, for even* f*,*

$$f^{\triangleleft}(A) = f(\sqrt{A^\dagger A}), \qquad f^{\triangleright}(A) = f(\sqrt{AA^\dagger}). \tag{9.3}$$

Proof. The first two are immediate from (9.2) and the SVD of $A^\dagger = V\Sigma W^\dagger$. For (9.3), note $A^\dagger A = V\Sigma^2 V^\dagger$, so $\sqrt{A^\dagger A} = V\Sigma V^\dagger$ is Hermitian with eigenvalue decomposition sharing the right singular vectors and eigenvalues σ_i; hence $f(\sqrt{A^\dagger A}) = V f(\Sigma) V^\dagger = f^{\triangleleft}(A)$, and symmetrically for $AA^\dagger = W\Sigma^2 W^\dagger$. □

Qubitization of a general matrix, in the singular basis. The construction of Section 8.3 carries over verbatim once eigenvectors are replaced by singular vectors. Applying $U_A \in \mathrm{BE}_{1,m}(A)$ to the flagged right singular state gives, with $\Pi = |0^m\rangle\langle 0^m| \otimes \mathbb{I}$,

$$U_A\,|0^m\rangle\,|v_i\rangle = \sigma_i\,|0^m\rangle\,|w_i\rangle + \sqrt{1-\sigma_i^2}\;|\perp_i'\rangle\,, \qquad \Pi\,|\perp_i'\rangle = 0, \tag{9.4}$$

because $\Pi U_A \Pi = |0^m\rangle\langle 0^m| \otimes A$ and $A\,|v_i\rangle = \sigma_i\,|w_i\rangle$. Since $U_A^\dagger \in \mathrm{BE}_{1,m}(A^\dagger)$ and $A^\dagger\,|w_i\rangle = \sigma_i\,|v_i\rangle$, the same step run backward defines a companion state $|\perp_i\rangle$ via $U_A^\dagger\,|0^m\rangle\,|w_i\rangle = \sigma_i\,|0^m\rangle\,|v_i\rangle + \sqrt{1-\sigma_i^2}\;|\perp_i\rangle$. Set

$$\mathcal{B}_i = \{|0^m\rangle\,|v_i\rangle\,, |\perp_i\rangle\}, \qquad \mathcal{B}_i' = \{|0^m\rangle\,|w_i\rangle\,, |\perp_i'\rangle\}, \tag{9.5}$$

spanning two-dimensional subspaces $\mathcal{H}_i, \mathcal{H}_i'$. Exactly as in Theorem 8.6,

$$[U_A]_{\mathcal{B}_i}^{\mathcal{B}_i'} = \begin{pmatrix} \sigma_i & \sqrt{1-\sigma_i^2} \\ \sqrt{1-\sigma_i^2} & -\sigma_i \end{pmatrix} = [U_A^\dagger]_{\mathcal{B}_i'}^{\mathcal{B}_i}, \qquad [Z_\Pi]_{\mathcal{B}_i} = [Z_\Pi]_{\mathcal{B}_i'} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{9.6}$$

so U_A maps $\mathcal{H}_i \to \mathcal{H}_i'$ and $U_A^\dagger$ maps $\mathcal{H}_i' \to \mathcal{H}_i$, and $Z_\Pi = 2\Pi - \mathbb{I}$ fixes the shared flagged vector while flipping the perpendicular one. The qubitization iterate $\widetilde{O} = U_A^\dagger Z_\Pi U_A Z_\Pi$ therefore restricts to $O(\sigma_i)^2$ on $\mathcal{H}_i$, with $O(\sigma_i)$ the planar rotation of (8.4). Reading off the $(1,1)$-entries of the powers gives the singular value analogue of Theorem 8.7.

Theorem 9.3 (Chebyshev singular value transformation)**.** *Let $U_A \in \mathrm{BE}_{1,m}(A)$ and $Z_\Pi = 2\Pi - \mathbb{I}$ with $\Pi = |0^m\rangle\langle 0^m| \otimes \mathbb{I}$. Then for every $k \geq 1$,*

$$\left(U_A^\dagger Z_\Pi U_A Z_\Pi\right)^k \in \mathrm{BE}_{1,m}(T_{2k}^{\triangleleft}(A)), \qquad U_A Z_\Pi \left(U_A^\dagger Z_\Pi U_A Z_\Pi\right)^k \in \mathrm{BE}_{1,m}(T_{2k+1}^{(\mathrm{SV})}(A)), \tag{9.7}$$

where $T_{2k}^{\triangleleft}(A) = V T_{2k}(\Sigma) V^\dagger$ and $T_{2k+1}^{(\mathrm{SV})}(A) = W T_{2k+1}(\Sigma) V^\dagger$.

Proof. By (9.6) the composite $\widetilde{O}$ acts on $\mathcal{H}_i$ as $\left([U_A]_{\mathcal{B}_i}^{\mathcal{B}_i'} Z\right)^2 = O(\sigma_i)^2$, whose k-th power $O(\sigma_i)^{2k}$ has $(1,1)$-entry $T_{2k}(\sigma_i)$ (Theorem 8.1); since the $\mathcal{H}_i$ are orthogonal and $\{|0^m\rangle\,|v_i\rangle\}$ span the range of Π, $\Pi \widetilde{O}^k \Pi = |0^m\rangle\langle 0^m| \otimes \sum_i T_{2k}(\sigma_i)\,|v_i\rangle\langle v_i| = |0^m\rangle\langle 0^m| \otimes V T_{2k}(\Sigma) V^\dagger$, the even (left/right) case with base vector $|v_i\rangle$ mapped to itself. Prepending one $U_A Z_\Pi$ carries $\mathcal{H}_i \to \mathcal{H}_i'$; the surviving flagged block is the $(1,1)$-entry of $O(\sigma_i)^{2k+1}$, namely $T_{2k+1}(\sigma_i)$, and the base vector is now $|0^m\rangle\,|w_i\rangle$, giving $\sum_i T_{2k+1}(\sigma_i)\,|w_i\rangle\langle v_i| = W T_{2k+1}(\Sigma) V^\dagger$. □

Two structural facts are worth isolating. First, *parity is forced*: an even degree returns a one-sided transformation ($V \cdots V^\dagger$), an odd degree returns the two-sided $W \cdots V^\dagger$. This is not a defect but the honest reflection of the fact that singular values are nonnegative and a matrix function must respect the two singular bases. Second, when we only care about $f^{(\mathrm{SV})}(A)$ for a general f, we approximate an *odd* extension of f (odd about the origin) so that the polynomial has odd parity; for $f^{\triangleleft}(A)$ we approximate an *even* extension. Because all $\sigma_i \geq 0$, evaluating the extended polynomial on $[0,1]$ recovers f on the singular values regardless of the extension's behavior on $[-1,0)$.

The Hermitian special case. When A is Hermitian with eigendecomposition $A = \sum_i \lambda_i\,|u_i\rangle\langle u_i|$, the SVD is $\sigma_i = |\lambda_i|$, $|v_i\rangle = |u_i\rangle$, $|w_i\rangle = \mathrm{sgn}(\lambda_i)\,|u_i\rangle$. For an *even* polynomial P, $P(\lambda_i) = P(\sigma_i)$, so $P^{\triangleleft}(A) = P(A)$; for an *odd* polynomial, $P(\lambda_i) = \mathrm{sgn}(\lambda_i) P(\sigma_i)$, so $P^{(\mathrm{SV})}(A) = P(A)$. Thus *for a definite-parity polynomial the QSVT and the QET of Chapter 8 produce the same operator*: eigenvalue transformation and singular value transformation are two views of one circuit. The distinction becomes real only when A is non-Hermitian, or when P has indefinite parity (then the two objects genuinely differ and cannot be produced by a single definite-parity QSVT sweep).

9.2 Real-polynomial QSVT

Lifting the scalar QSP identity to a matrix is now automatic: replace the signal rotation $O(\lambda)$ by the qubitization iterate built from $U_A, U_A^\dagger$, and each Z-rotation $\mathrm{e}^{\mathrm{i}\phi_j Z}$ by the flag rotation $\mathrm{e}^{\mathrm{i}\phi_j Z_\Pi}$. Because every factor preserves the subspaces $\mathcal{H}_i$ and acts there as the corresponding 2×2 QSP factor at $\lambda = \sigma_i$, the flagged block of the whole product is the QSP polynomial evaluated singular value by singular value. Concretely, for even degree d,

$$U_\Phi = \mathrm{e}^{\mathrm{i}\phi_0 Z_\Pi} \prod_{j=1}^{d/2} \left[\left(U_A^\dagger Z_\Pi\right) \mathrm{e}^{\mathrm{i}\phi_{2j-1} Z_\Pi} \left(U_A Z_\Pi\right) \mathrm{e}^{\mathrm{i}\phi_{2j} Z_\Pi} \right] \tag{9.8}$$

block-encodes $P^{\triangleleft}(A)$ for an even P, and for odd degree d the analogous product ending in one extra $U_A Z_\Pi$ block-encodes $P^{(\mathrm{SV})}(A)$ for an odd P. As in Section 8.4, the interleaved reflections

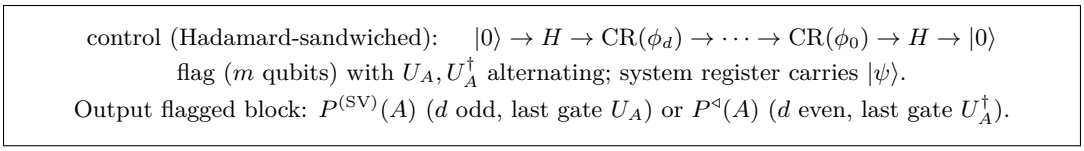


Figure 9.1: Schematic of the real-polynomial QSVT circuit U_Φ of Theorem 9.4. It is the general-encoding qubitization of Section 8.3 with interleaved flag rotations $\mathrm{CR}(\phi_j) \equiv \mathrm{e}^{\mathrm{i}\phi_j Z_\Pi}$ (in the C-convention) and one extra control ancilla to extract $\mathrm{Re}\, P$.

Z_Π are absorbed into the flag rotations by a $\pm\pi/2$ shift of the phases (the "C-convention"), so the physical circuit is: d alternating calls to $U_A/U_A^\dagger$, $d+1$ controlled flag rotations $\mathrm{CR}(\phi_j)$ (each an m-qubit-controlled single-qubit Z-rotation), and one extra ancilla. This is Theorem 8.9 transported to the block encoding, and it holds for the full class of complex definite-parity P of Theorem 8.9. In practice we want a *real* target, extracted by the same $\frac{1}{2}(P + P^*)$ trick as in QSP.

Theorem 9.4 (QSVT with real polynomials)**.** *Let $A \in \mathbb{C}^{N\times N}$ be encoded by $U_A \in \mathrm{BE}_{1,m}(A)$, and let $P \in \mathbb{R}[x]$ have degree d, parity $d \bmod 2$, and $|P(x)| \le 1$ on $[-1,1]$ (the conditions of Theorem 8.10). Then there is a phase sequence $\Phi \in \mathbb{R}^{d+1}$ such that the QSVT circuit U_Φ (Figure 9.1) is a $(1, m+1)$-block encoding of $P^{(\mathrm{SV})}(A) = WP(\Sigma)V^\dagger$ when d is odd, and of $P^{\triangleleft}(A) = VP(\Sigma)V^\dagger$ when d is even. The circuit uses $U_A, U_A^\dagger$, an m-qubit controlled-NOT, and single-qubit rotation gates a total of $\mathcal{O}(d)$ times, with one additional ancilla qubit.*

Proof sketch. By the cosine–sine decomposition of Section 8.3, in the singular basis U_Φ is a direct sum of the 2×2 QSP unitaries $U_\Phi(\sigma_i)$ over i (plus inert 1×1 blocks when the flagged and unflagged ranks differ). By Theorem 8.9 the $(1,1)$-entry of $U_\Phi(\sigma_i)$ is a complex P of the stated degree and parity, so the flagged block of U_Φ is $\sum_i P(\sigma_i)(\mathrm{base})_i$, i.e. $P^{(\mathrm{SV})}$ (odd d) or $P^{\triangleleft}$ (even d). To extract the *real* part $P_{\mathrm{Re}} = \mathrm{Re}\, P$ without a second ancilla, sandwich U_Φ between Hadamards on a control qubit and negate all phases in the $|1\rangle$ branch: the $|1\rangle$ branch block-encodes $(U_\Phi)^*$, whose flagged block is $P^*(\Sigma)$, and the Hadamard interference leaves $\frac{1}{2}(P + P^*)(\Sigma) = P_{\mathrm{Re}}(\Sigma)$ in the flagged $|0\rangle$ outcome. Since running $U_{-\Phi}$ is the same circuit with the signal qubit initialized in $|1\rangle$, no new gates are needed; the phase factors Φ exist by Theorem 8.10. Query count and ancilla count are read off the circuit. □

Matrix dilation: QSVT from QET. There is an instructive second route that reduces QSVT to the Hermitian QET of Chapter 8. Form the Hermitian *dilation*

$$D(A) = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix}, \qquad |z_i^\pm\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle\,|v_i\rangle \pm |1\rangle\,|w_i\rangle), \qquad D(A)\,|z_i^\pm\rangle = \pm\sigma_i\,|z_i^\pm\rangle\,, \tag{9.9}$$

so the eigenvalues of $D(A)$ are $\pm\sigma_i$. A block encoding $U_{D(A)}$ is built from a controlled $U_A/U_A^\dagger$ with one extra ancilla. Applying the QET of $D(A)$ with a polynomial f yields, in the 2×2 ancilla block,

$$f(D(A)) = \begin{pmatrix} f^{\triangleleft}_{\mathrm{even}}(A) & f^{(\mathrm{SV})}_{\mathrm{odd}}(A^\dagger) \\ f^{(\mathrm{SV})}_{\mathrm{odd}}(A) & f^{\triangleright}_{\mathrm{even}}(A) \end{pmatrix}, \tag{9.10}$$

i.e. the even part of f produces the one-sided functions on the diagonal and the odd part produces the two-sided $f^{(\mathrm{SV})}$ off-diagonal. In particular, for odd f, $f(D(A))\,|0\rangle\,|\psi\rangle = |1\rangle\,f^{(\mathrm{SV})}(A)\,|\psi\rangle$: the signal qubit deterministically flips to $|1\rangle$ and the system holds $f^{(\mathrm{SV})}(A)\,|\psi\rangle$. The dilation

viewpoint makes the parity bookkeeping of (9.7) transparent and is how one often analyzes QLSP (Chapter 10); the direct circuit of Theorem 9.4 is more economical because it never doubles the matrix.

9.3 Applications: inversion, exponential, filtering

Every application below is an instance of one recipe: pick the target f, find a bounded real polynomial P of odd (or even) parity approximating f on the relevant sub-interval of $[0,1]$, and invoke Theorem 9.4. The degree d is the query cost, so the game is polynomial approximation theory.

Matrix inversion: $f(x) = 1/(\kappa x)$

Let A be invertible with condition number $\kappa = \|A\|\,\|A^{-1}\|$; after the normalization $\|A\| \le 1$, the singular values lie in $[1/\kappa, 1]$. Then $A^{-1} = V\Sigma^{-1}W^\dagger = g^{(\mathrm{SV})}(A^\dagger)$ with $g(x) = 1/x$, an odd function. Since $1/x$ is unbounded and singular at 0, we cannot approximate it on all of $[-1,1]$; we only need it on the spectral support $[1/\kappa, 1]$ (and, by oddness, on $[-1, -1/\kappa]$). Choose an odd polynomial P with

$$\Big|P(x) - \frac{1}{\eta\kappa x}\Big| \le \epsilon \quad \text{for all } x \in [-1, -\tfrac{1}{\kappa}] \cup [\tfrac{1}{\kappa}, 1], \qquad |P(x)| \le 1 \text{ on } [-1,1], \tag{9.11}$$

where the subnormalization $\eta \ge 1$ (say $\eta = 4/3$) merely rescales so that $|P| \le 1$ globally, as Theorem 8.10 requires. A polynomial of degree

$$d = \mathcal{O}\Big(\kappa \log \tfrac{1}{\epsilon}\Big) \tag{9.12}$$

achieves (9.11); the linear scaling in κ is essentially optimal and matches the classical intuition that resolving $1/x$ down to $x = 1/\kappa$ requires resolution $\sim \kappa$. Theorem 9.4 then block-encodes $P^{(\mathrm{SV})}(A^\dagger) \approx \frac{1}{\eta\kappa}A^{-1}$ with $\mathcal{O}(\kappa \log(1/\epsilon))$ queries. This is the primitive that solves the quantum linear systems problem: prepare $|b\rangle$, apply the block encoding, and amplitude-amplify the flagged outcome; the full complexity accounting (including the overlap-dependent success probability and the $\mathcal{O}(\kappa)$ rounds of amplitude amplification) is the subject of Chapter 10. The worked example at the end of the section constructs the approximating polynomial explicitly.

Hamiltonian simulation: $f(x) = \mathrm{e}^{-\mathrm{i}xt}$

For a Hermitian $A = H$ with $\|H\| \le 1$, time evolution is the eigenvalue transformation $\mathrm{e}^{-\mathrm{i}Ht} = \sum_i \mathrm{e}^{-\mathrm{i}\lambda_i t}\,|u_i\rangle\langle u_i|$. Split into real and imaginary parts $\cos(Ht)$ and $\sin(Ht)$, each of definite parity, and approximate the scalars $\cos(tx), \sin(tx)$ on $[-1,1]$ by their truncated *Jacobi–Anger* expansions,

$$\cos(tx) = J_0(t) + 2\sum_{k\ge 1}(-1)^k J_{2k}(t)\,T_{2k}(x), \qquad \sin(tx) = 2\sum_{k\ge 0}(-1)^k J_{2k+1}(t)\,T_{2k+1}(x), \tag{9.13}$$

where J_ν are Bessel functions of the first kind. Because $|J_\nu(t)|$ decays super-exponentially once $\nu \gtrsim \mathrm{e}t/2$, truncating at

$$d = \mathcal{O}\Big(t + \frac{\log(1/\epsilon)}{\log(\mathrm{e} + \log(1/\epsilon)/t)}\Big) = \mathcal{O}\Big(t + \log\tfrac{1}{\epsilon}\Big) \tag{9.14}$$

gives a uniform ϵ-approximation. Each truncated series is a bounded real even/odd polynomial (rescaled by $\frac{1}{2}$ so the sum of |coefficients| does not exceed 1), so QET realizes $\cos(Ht)$ and $\sin(Ht)$, and one LCU combines them into $\mathrm{e}^{-\mathrm{i}Ht}$. The additive t term is the optimal "no fast-forwarding"

scaling and the additive $\log(1/\epsilon)$ is the optimal precision dependence—QSVT is thus an optimal Hamiltonian-simulation primitive, developed in full in Chapter 12. (For a general non-Hermitian block encoding one can use the same Jacobi–Anger polynomials on the singular values, or the QETU circuit of the reading, which applies $F(\cos(H/2))$ directly from the evolution oracle $\mathrm{e}^{-\mathrm{i}H}$.)

Example 9.5 (Degree of a Hamiltonian-simulation polynomial, with numbers)**.** Fix an evolution time $t = 50$ (in units where $\|H\| \le 1$) and target error $\epsilon = 10^{-6}$. The Jacobi–Anger coefficients $J_\nu(t)$ in (9.13) stay $\mathcal{O}(1)$ for $\nu \lesssim t$ and begin a super-exponential decay once $\nu \gtrsim \frac{\mathrm{e}}{2}t \approx 68$; a handful of further terms buys each additional digit. Truncating the series at

$$d \approx \tfrac{\mathrm{e}}{2}\, t + \mathcal{O}\big(\log \tfrac{1}{\epsilon}\big) \approx 68 + (\text{a few}) \approx 90$$

Chebyshev terms already gives $\|\cos(Ht) - P_{\cos}(H)\| \le \epsilon$ (and likewise for sin). So QSVT block-encodes $\mathrm{e}^{-\mathrm{i}Ht}$ at $t = 50$ to six digits with only ~ 90 queries to the block encoding of H—linear in t and essentially independent of the dimension 2^n. For contrast, a first-order product formula (Chapter 11) needs $\mathcal{O}(t^2/\epsilon)$ steps to the same accuracy, astronomically more at these parameters; this gap is the practical payoff of the optimal $\mathcal{O}(t + \log(1/\epsilon))$ scaling (9.14).

Filtering: eigenstate and energy filters

Many algorithms need a *spectral filter*: a matrix that is ≈ 1 on part of the spectrum and ≈ 0 elsewhere, used to project (approximately) onto a target eigenspace, prepare ground states, or estimate energies. The building block is a polynomial approximation to the *sign function* $\mathrm{sgn}(x)$, which in turn gives a shifted step (rectangle). Concretely, an odd polynomial P_{sgn} with

$$\big|P_{\mathrm{sgn}}(x) - \mathrm{sgn}(x)\big| \le \epsilon \quad \text{for } |x| \in [\Delta, 1], \qquad |P_{\mathrm{sgn}}| \le 1 \text{ on } [-1, 1], \tag{9.15}$$

exists with degree $d = \mathcal{O}\big(\Delta^{-1}\log(1/\epsilon)\big)$, where 2Δ is the width of the "dead zone" around 0 that we decline to resolve (equivalently the spectral gap we assume). Such P_{sgn} is built from a rescaled error function $\mathrm{erf}(kx)$ with $k \sim \Delta^{-1}\sqrt{\log(1/\epsilon)}$, itself expanded in Chebyshev polynomials. From the sign we assemble:

- an *approximate rectangle* on $[a, b]$ via $R(x) = \frac{1}{2}\big[P_{\mathrm{sgn}}(x-a) - P_{\mathrm{sgn}}(x-b)\big]$ (an even or indefinite-parity polynomial, handled by the parity splitting of Section 9.2);
- an *eigenstate filter* $R(H)$ that suppresses eigencomponents outside $[a, b]$, so that $R(H)\,|\psi\rangle$ retains only the target band—the basis of ground-state preparation and of the linear-systems eigenvalue filter of Lin–Tong;
- a *fixed-point amplitude amplification / uniform singular value amplification* primitive, obtained by choosing P_{sgn} to send a small known overlap $a \ge \delta$ to ≈ 1.

Proposition 9.6 (Fixed-point amplitude amplification via QSVT)**.** *Let U_A be an n-qubit unitary and Π' a projector with $\Pi' U_A\,|\varphi_0\rangle = a\,|\psi\rangle$, $a \ge \delta > 0$. Then there is an $(n+1)$-qubit QSVT circuit U_Φ with global-phase-invariant distance $D_p\big(|0\rangle\,|\psi\rangle\,,\, U_\Phi\,|0\rangle\,|\varphi_0\rangle\big) \le \epsilon$, using $U_A, U_A^\dagger$ and controlled reflections $\mathcal{O}\big(\delta^{-1}\log(1/\epsilon)\big)$ times.*

Proof. In the bases $\mathcal{B} = \{|\varphi_0\rangle, \dots\}$, $\mathcal{B}' = \{|\psi\rangle, \dots\}$, U_A has the flagged 1×1 "singular value" $a = \langle\psi|\,U_A\,|\varphi_0\rangle$ (cf. the Grover/reflection picture of Chapter 3). Choose an odd real polynomial F with $|F(x) - 1| \le \epsilon^2/2$ for $x \in [\delta, 1]$ and $|F| \le 1$; by (9.15) this is a shifted sign approximation with degree $\mathcal{O}(\delta^{-1}\log(1/\epsilon))$. QSVT then block-encodes $F(a)\,|\psi\rangle\langle\varphi_0|$, and the output overlap with $|0\rangle\,|\psi\rangle$ is $|F(a)| \ge 1 - \epsilon^2/2$, whence $D_p = \sqrt{2(1 - |F(a)|)} \le \epsilon$. Unlike standard amplitude amplification this never overshoots and needs only the lower bound δ, not the exact value of a. □

Choosing $A = R_{\psi_0}$ (the Grover diffusion reflection) and $a = 1/\sqrt{N}$ recovers unstructured search: $\deg F = \mathcal{O}(\sqrt{N}\log(1/\epsilon))$ finds the marked state to precision ϵ with the Grover speedup and no overshoot. The same construction with A a general block encoding *amplifies its singular values uniformly* toward $\|A\|$ (uniform singular value amplification), a subroutine we reuse when converting a loosely subnormalized block encoding to a tight one.

Example 9.7 (Approximating $1/x$ on $[1/\kappa, 1]$)**.** We make (9.11) concrete. A standard construction starts from the integral identity, valid for $x \neq 0$,

$$\frac{1}{x} = \int_0^\infty \mathrm{e}^{-x^2t^2/2}\sqrt{\tfrac{2}{\pi}}\,\operatorname{sgn}(x)\,x\,\mathrm{d}(\dots) \tag{9.16}$$

but the cleaner route (Childs–Kothari–Somma) is: on $|x| \geq 1/\kappa$ write $\frac{1}{x} = \frac{1-(1-x^2)^b}{x}$ up to error $(1-\kappa^{-2})^b \leq \mathrm{e}^{-b/\kappa^2}$, choose $b = \mathcal{O}(\kappa^2\log(1/\epsilon))$, and note that $\frac{1-(1-x^2)^b}{x}$ is an *odd* polynomial of degree $2b-1$ in x that is bounded by $\mathcal{O}(\kappa)$ on $[-1,1]$. Truncating its Chebyshev expansion and discarding the negligible high-degree tail reduces the degree to $\mathcal{O}(\kappa\log(1/\epsilon))$ while keeping the error ϵ on $[1/\kappa, 1]$; finally divide by $\eta\kappa$ (with $\eta = 4/3$) so that $|P| \leq 1$ on all of $[-1,1]$. The resulting odd P satisfies (9.11), and by Theorem 9.4 the QSVT circuit block-encodes $P^{(\mathrm{SV})}(A^\dagger) = VP(\Sigma)V^\dagger \approx \frac{1}{\eta\kappa}A^{-1}$ using $\mathcal{O}(\kappa\log(1/\epsilon))$ queries. This is precisely the polynomial whose phase factors one obtains from a QSP solver (Section 9.5); the companion notebook plots P against $1/(\eta\kappa x)$ and checks the error on $[1/\kappa, 1]$.

9.4 QSVT versus LCU

We now have two general-purpose ways to realize a matrix function $f(A)$ from a block encoding: build it as a *linear combination of unitaries* (Chapter 7), or apply *QSVT* (this chapter). They are complementary, and the choice matters in practice.

What each transforms. LCU forms $\sum_\ell c_\ell U_\ell$; when the U_ℓ are qubitized Chebyshev blocks $T_k(A)$ this realizes any polynomial $\sum_k c_k T_k(A)$—an *eigenvalue* transformation for Hermitian A (and only awkwardly a singular value transformation, via the dilation (9.9)). QSVT natively realizes the *singular value* transformation of a general matrix, and coincides with the eigenvalue transformation in the Hermitian definite-parity case. So for a non-Hermitian A with an intrinsic $f(\sigma_i)$ target, QSVT is the natural tool.

Subnormalization: coherent versus ℓ^1. This is the decisive difference. QSVT produces $P(A)$ at *unit* subnormalization (a $(1, m+1)$-block encoding) using d queries, independent of the size of the coefficients of P. LCU produces $\sum_\ell c_\ell U_\ell$ at subnormalization $\|c\|_1 = \sum_\ell |c_\ell|$: the success probability of the flagged outcome scales like $1/\|c\|_1^2$, so a target polynomial with a large ℓ^1 coefficient norm (a generic degree-d polynomial has $\|c\|_1$ growing with d) costs an $\mathcal{O}(\|c\|_1)$ overhead in amplitude amplification on top of the query count. QSVT avoids this overhead entirely because qubitization threads the degree through a single coherent product rather than summing independent branches. Concretely, to realize $T_d(A)$: LCU of monomials pays $\|c\|_1 \sim 2^d$; QSVT (or a single qubitization sweep) pays 1.

Ancillas, generality, and when LCU wins. QSVT needs one extra ancilla and a classically precomputed phase sequence; LCU needs $\lceil \log_2 L \rceil$ ancillas for L terms and a prep/select pair. When the function is already a *short* LCU with small $\|c\|_1$—for example $\mathrm{e}^{-\mathrm{i}Ht} = \cos(Ht) - \mathrm{i}\sin(Ht)$ as

a two-term combination of QSVT-built factors, or a sum of a handful of Hamiltonian-simulation terms (LCHS, Chapter 13)—LCU is simplest and its ℓ^1 cost is negligible. LCU is also the right frame when the terms are genuinely distinct unitaries (a Pauli/LCU Hamiltonian) rather than powers of one block encoding. The two techniques compose: the standard pattern is "QSVT to build each definite-parity piece at unit subnormalization, then a low-weight LCU to add the pieces," which is exactly how $\mathrm{e}^{-\mathrm{i}Ht}$ and indefinite-parity $f(A)$ are assembled (Sections 9.2 and 9.3).

	QSVT	**LCU**
transforms	singular values (eigenvalues if Hermitian, definite parity)	eigenvalues (Chebyshev/monomial basis)
subnormalization	1 (coherent)	$\|c\|_1$ (ℓ^1 overhead)
queries	$\mathcal{O}(d)$ to $U_A, U_A^\dagger$	$\mathcal{O}(L)$ selects + AA overhead $\mathcal{O}(\|c\|_1)$
ancillas	$+1$	$+\lceil \log_2 L \rceil$
best when	high-degree single-A polynomial, non-Hermitian A	short sum, distinct unitaries, small $\|c\|_1$

9.5 Phase-factor computation

QSVT is only useful if the phase sequence $\Phi \in \mathbb{R}^{d+1}$ realizing a target real polynomial P can actually be computed. The problem inherits directly from QSP (Section 8.5)—QSVT adds no new phase-finding difficulty, only the deterministic C-convention $\pm\pi/2$ shift of Section 9.2—so we recall the three families and what is specific to matrix-function targets.

Direct (root-finding) methods. Given the real target P_{Re}, one first *completes* it: find Q with $|P|^2 + (1-x^2)\,|Q|^2 = 1$ (with $\mathrm{Re}\,P = P_{\mathrm{Re}}$) by factoring the degree-$2d$ polynomial $1 - P_{\mathrm{Re}}^2$ into Fejér–Riesz square form over its roots, then strips one degree per phase from the ratio of leading coefficients (the constructive converse of Theorem 8.9). Exact in principle; the root extraction is ill-conditioned for large d and needs high-precision arithmetic. Prony-type and capitalization variants improve stability.

Optimization-based methods. Fixing the parity leaves $\widetilde{d} = \lceil (d+1)/2 \rceil$ free phases; minimize the least-squares mismatch

$$F(\Phi) = \frac{1}{\widetilde{d}} \sum_{k=1}^{\widetilde{d}} \Big(\mathrm{Re}\big[U_\Phi(x_k)_{11}\big] - P_{\mathrm{Re}}(x_k) \Big)^2, \qquad x_k = \cos\Big(\tfrac{(2k-1)\pi}{4\widetilde{d}}\Big), \tag{9.17}$$

over Φ on the positive Chebyshev nodes. Started from the symmetric ansatz $\Phi_0 = (\frac{\pi}{4}, 0, \ldots, 0, \frac{\pi}{4})$ this nonconvex objective behaves almost convexly near the solution, so quasi-Newton descent converges in double precision even for d in the thousands—the workhorse for the inversion and filtering polynomials of Section 9.3.

Symmetric QSP and the nonlinear Fourier transform. Restricting to symmetric sequences $\phi_j = \phi_{d-j}$ makes $U_\Phi(x)_{11}$ automatically real of the correct parity, halves the unknowns, and turns completion into a well-conditioned fixed-point/Newton iteration with no explicit Q. A recent and conceptually clean viewpoint recasts phase-factor finding as an (inverse) *nonlinear Fourier*

transform, for which fast, provably stable algorithms exist. In practice one calls a mature solver (e.g. `QSPPACK`'s optimization routine or a symmetric-QSP Newton solver) as a black box: input the Chebyshev coefficients of P_{Re}, receive Φ; the QSVT circuit of Theorem 9.4 then block-encodes $P^{(\mathrm{SV})}(A)$ with $\mathcal{O}(d)$ queries. For the rest of Part II we therefore treat "block-encode a bounded degree-d singular value transformation" as a primitive costing $\mathcal{O}(d)$ queries, one ancilla, and d classically precomputed rotations.

Exercises

Exercise 9.1. *The three generalized functions.* For $A = W\Sigma V^\dagger$ prove Theorem 9.2: $f^{(\mathrm{SV})}(A^\dagger) = (f^{(\mathrm{SV})}(A))^\dagger$, $f^{\triangleleft}(A) = f^{\triangleright}(A^\dagger)$, and, for even f, $f^{\triangleleft}(A) = f(\sqrt{A^\dagger A})$, $f^{\triangleright}(A) = f(\sqrt{AA^\dagger})$. Give an explicit 2×2 non-Hermitian A for which $f^{\triangleleft}(A) \neq f^{\triangleright}(A)$ when $f(x) = x^2$.

Exercise 9.2. *Parity and which function you get.* Using Theorem 9.3, explain why an even-degree QSVT sweep produces the one-sided $P^{\triangleleft}(A) = VP(\Sigma)V^\dagger$ while an odd-degree sweep produces the two-sided $P^{(\mathrm{SV})}(A) = WP(\Sigma)V^\dagger$. For a Hermitian $A = \sum_i \lambda_i |u_i\rangle\langle u_i|$ and P odd, show $P^{(\mathrm{SV})}(A) = P(A)$ (eigenvalue transformation), and for P even show $P^{\triangleleft}(A) = P(A)$; identify where $\mathrm{sgn}(\lambda_i)$ enters.

Exercise 9.3. *The Chebyshev singular value block.* Let $A \in \mathrm{BE}_{1,m}(A)$ be a general (non-Hermitian) encoding. Write out the operator sequences that block-encode $T_2^{\triangleleft}(A)$ and $T_3^{(\mathrm{SV})}(A)$, count the queries to U_A and to $U_A^\dagger$ in each, and verify the counts equal the Chebyshev degree. Using $T_2(x) = 2x^2 - 1$, express $T_2^{\triangleleft}(A)$ in terms of $A^\dagger A$.

Exercise 9.4. *Matrix dilation.* Verify (9.9): with $|z_i^{\pm}\rangle = \frac{1}{\sqrt{2}}(|0\rangle |v_i\rangle \pm |1\rangle |w_i\rangle)$, show $D(A) |z_i^{\pm}\rangle = \pm\sigma_i |z_i^{\pm}\rangle$. Deduce (9.10), and conclude that for an odd f, $f(D(A)) |0\rangle |\psi\rangle = |1\rangle f^{(\mathrm{SV})}(A) |\psi\rangle$ so that measuring the signal qubit yields outcome 1 with certainty.

Exercise 9.5. *Inversion degree.* Following Theorem 9.7, show that $g_b(x) = \frac{1-(1-x^2)^b}{x}$ is an odd polynomial of degree $2b-1$, that $|g_b(x) - 1/x| \leq \mathrm{e}^{-b/\kappa^2}/|x|$ for $|x| \geq 1/\kappa$, and hence that $b = \mathcal{O}(\kappa^2 \log(1/\epsilon))$ suffices for pointwise error ϵ on $[1/\kappa, 1]$. Explain how truncating the Chebyshev expansion of g_b reduces the effective degree to $\mathcal{O}(\kappa \log(1/\epsilon))$, and why the extra $\eta = 4/3$ factor is needed for Theorem 8.10.

Exercise 9.6. *Jacobi–Anger truncation.* From (9.13) and the bound $|J_\nu(t)| \leq (t/2)^\nu/\nu!$, show that truncating the $\cos(tx)$ series after the T_{2K} term incurs uniform error $\mathcal{O}((t/2)^{2K}/(2K)!)$ on $[-1, 1]$, and deduce that $K = \mathcal{O}(t + \log(1/\epsilon))$ gives error ϵ. Why do we rescale the truncated series by $\frac{1}{2}$ before feeding it to Theorem 9.4?

Exercise 9.7. *Fixed-point amplitude amplification.* Prove Theorem 9.6: given $\Pi' U_A |\varphi_0\rangle = a |\psi\rangle$ with $a \geq \delta$, and an odd F with $|F(x) - 1| \leq \epsilon^2/2$ on $[\delta, 1]$ and $|F| \leq 1$, show that the QSVT circuit outputs a state at global-phase-invariant distance $\sqrt{2(1 - |F(a)|)} \leq \epsilon$ from $|0\rangle |\psi\rangle$. Specialize to Grover search ($a = 1/\sqrt{N}$) and read off the degree $\mathcal{O}(\sqrt{N} \log(1/\epsilon))$; contrast with standard Grover's overshooting behavior.

Exercise 9.8. *QSVT versus LCU for T_d.* Compare the two ways to block-encode $T_d(A)$ for Hermitian A: (i) a single qubitization/QSVT sweep, and (ii) an LCU of the monomials $A^0, \ldots, A^d$ with $T_d(x) = \sum_k c_k x^k$. Estimate $\|c\|_1$ for T_d (recall the leading coefficient of T_d is 2^{d-1}), and hence the amplitude-amplification overhead of the LCU route. State the subnormalization and query count of each and explain why QSVT is preferred for large d.

Further reading

- Lin Lin, *qasc notes*, Ch. 8 (generalized matrix functions, qubitization of general matrices and the Chebyshev singular value transformation, the QSVT circuit and real-polynomial QSVT theorem, matrix dilation, application to linear systems, and Grover/fixed-point amplitude amplification revisited).

- Lin Lin, *live notes*, Ch. 13 (derivation of QSVT from the cosine–sine decomposition, real-polynomial singular value transformation, QSVT beyond the computational basis, fixed-point amplitude amplification and uniform singular value amplification, Gibbs-state preparation, eigenvalue transformation with Hamiltonian-evolution oracles (QETU), and perturbation theory of singular value transformations).

- Dong An, *Quantum Numerical Linear Algebra* tutorial talk (the singular value vs. eigenvalue transformation picture, the main result "block-encode $p(A)$ with $\mathcal{O}(d)$ queries," and the applications $f(x) = 1/(\kappa x)$ (linear systems), $f(x) = \mathrm{e}^{-\mathrm{i}xt}$ (Hamiltonian simulation), filtering, and amplitude amplification $f(x) = qx/2$ with $\deg p \sim q \log(1/\epsilon)$).

- A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, *Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics*, Proc. STOC 2019, 193–204 (arXiv:1806.01838), for the original QSVT theorem and the polynomial-approximation corollaries used here; and J. Martyn, Z. Rossi, A. Tan, and I. Chuang, *Grand unification of quantum algorithms*, PRX Quantum **2**, 040203 (2021), for the unifying picture.

- A. M. Childs, R. Kothari, and R. D. Somma, *Quantum algorithm for systems of linear equations with exponentially improved dependence on precision*, SIAM J. Comput. **46**, 1920–1950 (2017), for the explicit $1/x$ polynomial; and L. Lin and Y. Tong, *Optimal quantum eigenstate filtering with application to solving quantum linear systems*, Quantum **4**, 361 (2020), for eigenstate/energy filtering.

Chapter 10

Quantum Linear Systems

Learning objectives

- Formulate the quantum linear system problem (QLSP) and its parameters.
- Solve QLSP with HHL and with QSVT.
- Explain adiabatic and Zeno-based near-optimal solvers.
- Compare the complexity landscape across methods.

The quantum numerical linear algebra of the last four chapters culminates in the oldest and most-studied "killer application" of the whole program: solving a linear system $A|x\rangle \propto |b\rangle$ on a quantum computer. The input models of Chapter 6, the LCU arithmetic of Chapter 7, the qubitization of Chapter 8, and above all the matrix-inversion polynomial $x \mapsto 1/(\kappa x)$ of Chapter 9 were each introduced with this target in mind. What makes the problem subtle is not implementing A^{-1} on paper—that is a single QSVT sweep—but controlling the cost as a function of two adversarial parameters: the target error ϵ and, decisively, the *condition number* $\kappa = \|A\| \, \|A^{-1}\|$, which measures how nearly singular A is and which no generic quantum solver can escape (Section 10.1). We tell the story historically and by increasing sophistication. The Harrow–Hassidim–Lloyd (HHL) algorithm (Section 10.2) makes the mechanism transparent—phase-estimate the eigenvalues of A, rotate each by $1/\lambda$, uncompute, postselect—and exposes the two sources of inefficiency (precision-limited QPE and small success probability) that drive its $\mathcal{O}(\kappa^2/\epsilon)$ cost. QSVT (Section 10.3) replaces the eigenvalue arithmetic by a direct degree-$\mathcal{O}(\kappa \log(1/\epsilon))$ polynomial, improving the ϵ dependence to logarithmic. Both, even after amplitude amplification, retain a quadratic κ dependence in the worst case; the adiabatic and quantum Zeno solvers (Section 10.4) recast QLSP as tracing a zero-eigenvalue eigenpath from $|b\rangle$ to $|x\rangle$ and reach the near-optimal and optimal $\mathcal{O}(\kappa \,\text{polylog})$ and $\mathcal{O}(\kappa \log(1/\epsilon))$ scalings. We close (Section 10.5) with the complexity landscape and with the practical fact that κ, not the exponential Hilbert-space dimension, governs everything—motivating preconditioning, whose quantum incarnation (block-encoding normalization, the $(A+B)$ pipeline, and multilevel BPX preconditioners) is where genuine speedups over classical iterative solvers are won or lost. Throughout, $A \in \mathbb{C}^{N\times N}$ with $N = 2^n$ is accessed by a block encoding $U_A \in \mathrm{BE}_{\alpha,m}(A)$ and $|b\rangle$ by a state-preparation unitary U_b with $U_b|0^n\rangle = |b\rangle$.

10.1 Problem setup: condition number and error

The problem and its output model. Classically, given $A \in \mathbb{C}^{N\times N}$ and $b \in \mathbb{C}^N$, one seeks $x = A^{-1}b$. The *quantum* linear system problem asks only for the *direction* of the solution, encoded

as a normalized state.

Definition 10.1 (Quantum linear system problem)**.** Let $A \in \mathbb{C}^{N\times N}$ be invertible, accessed by a block encoding $U_A \in \mathrm{BE}_{\alpha,m}(A)$, and let $|b\rangle$ be a normalized state prepared by $U_b|0^n\rangle = |b\rangle$. Given a tolerance $\epsilon \in (0,1)$, the *QLSP* is to prepare a state $|\tilde{x}\rangle$ with

$$\||\tilde{x}\rangle - |x\rangle\| \le \epsilon, \qquad |x\rangle = \frac{A^{-1}|b\rangle}{\|A^{-1}|b\rangle\|}. \tag{10.1}$$

The *query complexity* is the number of calls to $U_A, U_A^\dagger$ and to $U_b, U_b^\dagger$. A solver may fail with probability bounded away from 1, signaling failure through a flag qubit.

Two normalizations are standard. First, we always rescale so that $\|A\| \le 1$; after the block-encoding subnormalization this places the singular values of A in $[1/\kappa, 1]$, where

$$\kappa = \kappa(A) := \|A\| \left\|A^{-1}\right\| = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} \ge 1 \tag{10.2}$$

is the condition number. Second, without loss of generality we may take A *Hermitian positive definite*: a general A is handled by the Hermitian dilation

$$\widetilde{A} = \begin{pmatrix} 0 & A \\ A^\dagger & 0 \end{pmatrix}, \qquad \widetilde{A}\begin{pmatrix} 0 \\ x \end{pmatrix} = \begin{pmatrix} b \\ 0 \end{pmatrix}, \tag{10.3}$$

which has the same singular values (hence the same κ) and returns the solution x in the lower block (Chapter 9, (9.9)). We freely use both viewpoints below.

Why κ and ϵ drive the cost. The map $x \mapsto 1/x$ must be resolved on the whole spectral interval $[1/\kappa, 1]$. Near the small singular value $\sigma_{\min} = 1/\kappa$ the inverse is as large as κ, so any device that outputs A^{-1} at unit subnormalization can only encode A^{-1}/κ: the solution appears with amplitude $\sim 1/\kappa$ inside a larger state, and *postselecting* it succeeds with probability $\sim 1/\kappa^2$ in the worst case. This is the origin of every κ (or κ^2) in the complexities below. The error ϵ enters because the inverse must be approximated; how gently it enters (as $1/\epsilon$ or $\log(1/\epsilon)$) is exactly what distinguishes the methods. That κ cannot be removed is a theorem, not an artifact of any particular algorithm.

Proposition 10.2 (Condition-number lower bound)**.** *Any quantum algorithm that solves the QLSP of Theorem 10.1 for all inputs with a given κ must make $\Omega(\kappa)$ queries to the oracle for A.*

Proof idea. One embeds a Hamiltonian-simulation instance into a linear system so that a sublinear-in-κ QLSP solver would simulate $\mathrm{e}^{-\mathrm{i}Ht}$ for time $t \sim \kappa$ with $o(\kappa)$ queries, violating the "no fast-forwarding" theorem of Chapter 12 for generic Hamiltonians. Thus the linear scaling in κ of the optimal solvers (Section 10.4) is best possible for generic A; special structure (e.g. positive definiteness, exploited classically by conjugate gradient to get $\sqrt{\kappa}$) can beat it only in restricted settings. □

Reading out the answer. The QLSP outputs the *normalized* state $|x\rangle$; the norm $\|A^{-1}|b\rangle\|$ is recovered from the success probability of the flagged block encoding, since a QLSP solver realizes $\widetilde{U} \in \mathrm{BE}_{\gamma,m}(A^{-1}, \epsilon)$ and the unflagged vector $|\tilde{x}\succ\rangle = \langle 0^m|\widetilde{U}|0^m\rangle|b\rangle$ satisfies $\|x - \gamma|\tilde{x}\succ\rangle\| \le \epsilon$, so $\gamma\||\tilde{x}\succ\rangle\| \approx \|x\|$. Downstream one typically wants an observable $\langle x|O|x\rangle$, estimated by a Hadamard or swap test on $|x\rangle$, or an overlap $\langle y\rangle x$; these are the "end-to-end" tasks in which a QLSP solver is a subroutine. Crucially, the quantum solver never writes out the N entries of x; the hoped-for exponential advantage lives entirely in the ability to prepare $|x\rangle$ and sample a *scalar* functional of it in time $\mathrm{poly}(n)$ rather than $\mathrm{poly}(N)$.

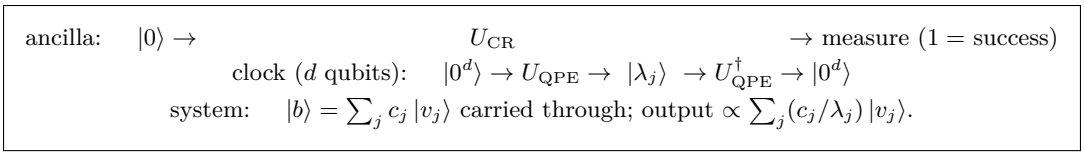


Figure 10.1: The HHL circuit: QPE writes the eigenphases into the clock register, a controlled rotation attaches amplitude C/λ_j to $|1\rangle$, uncomputation clears the clock, and postselecting $|1\rangle$ leaves $C\,A^{-1}|b\rangle$ in the system register.

10.2 Phase estimation and HHL

The Harrow–Hassidim–Lloyd algorithm was the first quantum QLSP solver. Take A Hermitian with eigendecomposition $A|v_j\rangle = \lambda_j |v_j\rangle$, accessed through the Hamiltonian-evolution oracle $U = \mathrm{e}^{\mathrm{i}2\pi A}$, and expand $|b\rangle = \sum_j c_j |v_j\rangle$. Then

$$A^{-1}|b\rangle = \sum_j \frac{c_j}{\lambda_j}|v_j\rangle\,, \tag{10.4}$$

so it suffices to multiply the j-th eigencomponent by $1/\lambda_j$. HHL does this in four steps, sketched in the circuit of Figure 10.1.

The four steps. Assume for the analysis that each $\lambda_j \in (0,1)$ has an exact d-bit binary expansion (this idealization isolates the mechanism; the finite-precision reality is the point of Theorem 10.3).

1. *Phase estimation* (Chapter 4) writes each eigenphase into a d-qubit register:
$$U_{\mathrm{QPE}}|0^d\rangle|b\rangle = \sum_j c_j |\lambda_j\rangle|v_j\rangle\,. \tag{10.5}$$
2. A *controlled rotation* on a fresh ancilla, driven by the eigenvalue register through coherent classical arithmetic $\lambda \mapsto \arcsin(C/\lambda)$, produces
$$U_{\mathrm{CR}}|0\rangle|\lambda_j\rangle = \Big(\sqrt{1-\tfrac{C^2}{\lambda_j^2}}\,|0\rangle + \tfrac{C}{\lambda_j}|1\rangle\Big)|\lambda_j\rangle\,, \tag{10.6}$$
where $C \le \lambda_{\min} = 1/\kappa$ is a normalization constant chosen so $|C/\lambda_j| \le 1$.
3. *Uncomputation* applies $U_{\mathrm{QPE}}^\dagger$, returning the eigenvalue register to $|0^d\rangle$ and leaving
$$U_{\mathrm{HHL}}|0\rangle|b\rangle = \sum_j c_j \Big(\sqrt{1-\tfrac{C^2}{\lambda_j^2}}\,|0\rangle + \tfrac{C}{\lambda_j}|1\rangle\Big)|v_j\rangle\,. \tag{10.7}$$
4. *Postselection* measures the ancilla; outcome 1 yields the unnormalized vector $|\tilde{x}\succ\rangle = \sum_j (C/\lambda_j)c_j|v_j\rangle = C\,A^{-1}|b\rangle$, i.e. the target solution up to the global factor C.

Success probability and cost. With $\|A\| = 1$ so that $\lambda_{\min} = 1/\kappa$, the largest admissible constant is $C = 1/\kappa$. The success probability is the squared norm of $|\tilde{x}\succ\rangle$,

$$p(1) = \||\tilde{x}\succ\rangle\|^2 = C^2\left\|A^{-1}|b\rangle\right\|^2 = \frac{\xi^2}{\kappa^2}, \qquad \xi := \left\|A^{-1}|b\rangle\right\| \in [1,\kappa], \tag{10.8}$$

using $\|A^{-1}|b\rangle\| \ge \|A^{-1}\|^{-1}\cdots$; the trivial bound $\xi \ge 1$ gives $p(1) = \Omega(\kappa^{-2})$ (worst case), while an $\Omega(1)$ overlap of $|b\rangle$ with the smallest-eigenvalue eigenvector gives $\xi = \Omega(\kappa)$ and $p(1) = \Omega(1)$ (best

case). A single run is dominated by QPE, whose cost to resolve $\lambda_{\min} = \Theta(1/\kappa)$ to the required precision is $\mathcal{O}(\kappa/\epsilon)$ (see Theorem 10.3). Repeating for success costs $1/p(1) = \mathcal{O}(\kappa^2/\xi^2)$ rounds, so the total query complexity is

$$\mathcal{O}\Big(\frac{\kappa}{\epsilon} \cdot \frac{\kappa^2}{\xi^2}\Big) = \mathcal{O}\Big(\frac{\kappa^3}{\xi^2 \epsilon}\Big), \tag{10.9}$$

i.e. $\mathcal{O}(\kappa/\epsilon)$ in the best case and $\mathcal{O}(\kappa^3/\epsilon)$ in the worst case. *Amplitude amplification* (Chapter 3) replaces the $1/p(1)$-repetition overhead by its square root, boosting the worst case to $\mathcal{O}(\kappa^2/\epsilon)$; *variable-time* amplitude amplification (a refinement that amplifies the many well-conditioned branches early and the few ill-conditioned ones late) further reduces the κ^2 to κ, at the cost of a more delicate analysis.

Remark 10.3 (Multiplicative accuracy and the $1/\epsilon$ in QPE). Because the estimated eigenvalue $\tilde{\lambda}_j = \lambda_j(1+e_j)$ enters through $1/\tilde{\lambda}_j$, a *multiplicative* error $|e_j| \le \epsilon/4$ is needed to guarantee (10.1): writing $|\tilde{x}\succ\rangle - C\,x = C\sum_j c_j(1/\tilde{\lambda}_j - 1/\lambda_j)\,|v_j\rangle$ and bounding $\left|1/\tilde{\lambda}_j - 1/\lambda_j\right| = |e_j|\,/(\lambda_j\,|1+e_j|) \le (\epsilon/2)/\lambda_j$ gives $\||\tilde{x}\succ\rangle - Cx\| \le (C\epsilon/2)\,\|x\|$, whence $\||\tilde{x}\rangle - |x\rangle\| \le \epsilon$. Since $\lambda_{\min} = \Theta(1/\kappa)$, resolving it to multiplicative precision ϵ means additive precision ϵ/κ, so $T_{\mathrm{QPE}} = \Omega(\kappa/\epsilon)$. This inverse-linear dependence on ϵ is the chief weakness of HHL and the reason to prefer QSVT, which needs only $\log(1/\epsilon)$.

10.3 QSVT solver

QSVT (Chapter 9) inverts A without ever estimating an eigenvalue: it applies a single bounded polynomial directly to the singular values. From the SVD $A = W\Sigma V^\dagger$,

$$A^{-1} = V\Sigma^{-1}W^\dagger = g^{(\mathrm{SV})}(A^\dagger), \qquad g(x) = \tfrac{1}{x}\ (\text{odd}), \tag{10.10}$$

so we approximate the odd function $1/x$ on the spectral support $[1/\kappa, 1]$ by a bounded odd polynomial.

Theorem 10.4 (QSVT linear-system solver)**.** *Let $U_A \in \mathrm{BE}_{\alpha,m}(A)$ with condition number κ (singular values in $[1/\kappa, 1]$ after normalization), let U_b prepare $|b\rangle$, and set $\xi = \|A^{-1}\,|b\rangle\| \in [1,\kappa]$. There is a QSVT solver producing $|\tilde{x}\rangle$ with $\||\tilde{x}\rangle - |x\rangle\| \le \epsilon$ and constant success probability, using*

$$\mathcal{O}\Big(\frac{\kappa^2}{\xi}\log\tfrac{\kappa}{\epsilon}\Big) \ \text{queries to } U_A, U_A^\dagger, \qquad \mathcal{O}\Big(\frac{\kappa}{\xi}\Big) \ \text{queries to } U_b. \tag{10.11}$$

In the worst case $\xi = 1$ this is $\mathcal{O}(\kappa^2\log(\kappa/\epsilon))$; in the best case $\xi = \Theta(\kappa)$ it is $\mathcal{O}(\kappa\log(\kappa/\epsilon))$.

Proof sketch. By Theorem 9.7 there is an odd polynomial P of degree $d = \mathcal{O}(\kappa\log(1/\epsilon'))$ with $|P(x) - 1/(\eta\kappa x)| \le \epsilon'$ on $[1/\kappa, 1]$ and $|P| \le 1$ on $[-1,1]$ (here $\eta = \Theta(1)$ enforces the global bound). Theorem 9.4 block-encodes $P^{(\mathrm{SV})}(A^\dagger) = VP(\Sigma)V^\dagger \approx (\eta\kappa)^{-1}A^{-1}$ with $\mathcal{O}(d)$ queries. Applying it to $|0^m\rangle\,|b\rangle$ and reading the flagged block yields the unnormalized $|\tilde{x}\succ\rangle \approx (\eta\kappa)^{-1}A^{-1}\,|b\rangle$, of norm $\approx \xi/(\eta\kappa)$, so the flag succeeds with probability $\approx (\xi/\eta\kappa)^2$. Amplitude amplification (Chapter 3) boosts this to $\Omega(1)$ in $\mathcal{O}(\eta\kappa/\xi)$ rounds, each calling $U_A, U_A^\dagger$ order d times and $U_b, U_b^\dagger$ order once, giving (10.11). To meet the final tolerance ϵ in (10.1) one sets the polynomial precision $\epsilon' = \Theta(\xi\epsilon/\kappa)$, which only affects the logarithm; hence the $\log(\kappa/\epsilon)$. □

The inversion polynomial, made explicit. The workhorse construction (Childs–Kothari–Somma) starts from the smooth odd extension

$$g_b(x) = \frac{1-(1-x^2)^b}{x}, \tag{10.12}$$

an odd polynomial of degree $2b-1$ that agrees with $1/x$ up to $(1-x^2)^b/|x| \le \mathrm{e}^{-bx^2}/|x|$; choosing $b = \mathcal{O}(\kappa^2 \log(1/\epsilon))$ makes the error ϵ on $[1/\kappa, 1]$. Truncating the Chebyshev expansion of g_b and discarding its negligible high-degree tail reduces the effective degree to $d = \mathcal{O}(\kappa \log(1/\epsilon))$; dividing by $\eta\kappa$ (with $\eta = 4/3$) enforces $|P| \le 1$. The resulting phase factors come from a QSP solver (Section 9.5); a convex-optimization construction yields the same degree with an equioscillating error and is what production code uses. The essential point is the *linear-in-κ, logarithmic-in-ϵ* degree—an exponential improvement in ϵ over HHL, and the reason QSVT is the default solver whenever the block encoding is available.

Example 10.5 (Poisson's equation: $\kappa \sim N^{2/d}$)**.** Consider the one-dimensional Poisson problem $-u''(r) = b(r)$ on $[0,1]$ with $u(0) = u(1) = 0$, discretized by central differences on a uniform grid $r_i = ih$, $h = 1/(N+1)$. The Laplacian becomes the tridiagonal Toeplitz matrix

$$A = \frac{1}{h^2}\,\mathrm{tridiag}(-1,\, 2,\, -1) \in \mathbb{R}^{N\times N}. \tag{10.13}$$

A Toeplitz tridiagonal matrix $\mathrm{tridiag}(b, a, b)$ has eigenvalues $\lambda_k = a + 2|b|\cos(\frac{k\pi}{N+1})$; with $a = 2/h^2$, $b = -1/h^2$ the extreme eigenvalues are $\lambda_{\max} \approx 4/h^2$ and $\lambda_{\min} \approx \pi^2$, so

$$\kappa(A) = \frac{\lambda_{\max}}{\lambda_{\min}} \approx \frac{4}{h^2\pi^2} = \mathcal{O}(N^2). \tag{10.14}$$

Thus even the QSVT solver costs $\widetilde{\mathcal{O}}(\kappa^2) = \widetilde{\mathcal{O}}(N^4)$ (worst case) or $\widetilde{\mathcal{O}}(\kappa) = \widetilde{\mathcal{O}}(N^2)$, with only $\log(1/\epsilon)$ precision dependence—no advantage over classical sparse solvers for a fixed 1D grid. The situation changes in *high dimension*: the d-dimensional Laplacian is $A_d = \sum_{i=1}^{d} \mathbb{I}^{\otimes(i-1)} \otimes A \otimes \mathbb{I}^{\otimes(d-i)}$ with $\|A_d\| \approx 4d/h^2$ and $\left\|A_d^{-1}\right\|^{-1} \approx d\pi^2$, so $\kappa(A_d) \approx \kappa(A) = \mathcal{O}(N^2)$ is *independent of d*. Since the total grid has N^d points, a solver whose cost is $\mathrm{poly}(d)\cdot\widetilde{\mathcal{O}}(\kappa)$ can outperform classical methods whose per-iteration cost $\mathcal{O}(N^d)$ is exponential in d—provided the evolution oracle and state preparation are efficient and one wants only a scalar functional of the solution. This dimensional scaling $\kappa \sim N^{2/d}$ (here $N_{\mathrm{tot}} = N^d$, so $\kappa \sim N_{\mathrm{tot}}^{2/d}$) is the standard lens through which PDE-derived QLSP instances are judged.

10.4 Adiabatic and quantum Zeno solvers

Both HHL and QSVT, even with amplitude amplification, retain a quadratic κ dependence in the worst case, because the solution norm $\left\|A^{-1}|b\rangle\right\|$ can be as small as the subnormalization $1/\kappa$. The near-optimal and optimal solvers avoid postselection entirely by recasting QLSP as an *eigenpath-tracing* problem: build a family of Hamiltonians whose zero-eigenstate deforms continuously from $|b\rangle$ to $|x\rangle$, and follow it either by slow (adiabatic) evolution or by repeated projective measurement (Zeno).

QLSP as a zero-eigenvalue eigenpath. Take A Hermitian positive definite with spectrum in $[1/\kappa, 1]$, and set $Q_b = \mathbb{I} - |b\rangle\langle b|$. Define

$$H_0 = \begin{pmatrix} 0 & Q_b \\ Q_b & 0 \end{pmatrix}, \qquad H_1 = \begin{pmatrix} 0 & AQ_b \\ Q_b A & 0 \end{pmatrix}. \tag{10.15}$$

The null space of H_1 is spanned by $|0\rangle|x\rangle$ (with $A|x\rangle \propto |b\rangle$, since $Q_b A|x\rangle \propto Q_b|b\rangle = 0$) and $|1\rangle|b\rangle$; the null space of H_0 is spanned by $|0\rangle|b\rangle$ and $|1\rangle|b\rangle$. Interpolating

$$H(f) = (1-f)H_0 + fH_1, \qquad f = f(s)\colon [0,1] \to [0,1],\ f(0) = 0,\ f(1) = 1, \tag{10.16}$$

with scheduling function f, the state $|0\rangle|b\rangle$ (a null vector of H_0) is deformed into $|0\rangle|x\rangle$ (a null vector of H_1), which encodes the solution. The key structural fact is that the relevant gap never closes and shrinks only near the end.

Lemma 10.6 (Gap of the interpolating Hamiltonian)**.** *For $A \succeq \kappa^{-1}\mathbb{I}$ the spectral gap $\Delta(f)$ of $H(f)$ around its zero eigenvalue satisfies*

$$\Delta(f) \geq 1 - f + f/\kappa \ \geq\ 1/\kappa. \tag{10.17}$$

Proof. Write $H(f) = \left(\begin{smallmatrix} 0 & MQ_b \\ Q_b M & 0 \end{smallmatrix}\right)$ with $M = (1-f)\mathbb{I} + fA$. An eigenvalue λ of $H(f)$ satisfies $0 = \det(\lambda^2\mathbb{I} - Q_b M^2 Q_b)$ (block elimination, using that $\lambda\mathbb{I}$ commutes with the off-diagonal blocks). On the subspace orthogonal to $|b\rangle$, $Q_b M^2 Q_b$ acts as M^2, so the smallest nonzero λ^2 obeys

$$\Delta^2(f) = \inf_{\langle b|\varphi\rangle = 0,\ \||\varphi\rangle\| = 1} \langle\varphi| M^2 |\varphi\rangle \ \geq\ \inf_{\||\varphi\rangle\| = 1} \langle\varphi| M^2 |\varphi\rangle = \big((1-f) + f/\kappa\big)^2, \tag{10.18}$$

since the eigenvalues of $M = (1-f)\mathbb{I} + fA$ lie in $[(1-f) + f/\kappa,\ (1-f) + f]$. Taking square roots gives (10.17); the minimum over $f \in [0,1]$ is $1/\kappa$ at $f = 1$. □

Adiabatic solvers: AQC(p) and AQC(exp). Evolving $\frac{\mathrm{i}}{T}\partial_s |\psi_T(s)\rangle = H(f(s))|\psi_T(s)\rangle$ from $|\psi_T(0)\rangle = |0\rangle|b\rangle$, the adiabatic theorem guarantees that $|\psi_T(1)\rangle$ lands in the null space of H_1—the solution—provided T is large relative to the gap. The naive bound scales the runtime as $T = \mathcal{O}(\Delta_*^{-3}/\epsilon) = \mathcal{O}(\kappa^3/\epsilon)$, ruinous for small gaps. Two ideas fix this.

- *Gap-adapted schedule* AQC(p): choose $\dot f(s) \propto \Delta^p(f(s))$ so the evolution slows exactly where the gap is small. For $1 < p < 2$ this yields $T = \mathcal{O}(\kappa/\epsilon)$—linear in κ, but still $1/\epsilon$.
- *Boundary-cancelling schedule* AQC(exp): choose

$$f(s) = c_e^{-1}\int_0^s \exp\!\Big(-\tfrac{1}{\tau(1-\tau)}\Big)\mathrm{d}\tau, \qquad c_e = \int_0^1 \exp\!\Big(-\tfrac{1}{\tau(1-\tau)}\Big)\mathrm{d}\tau, \tag{10.19}$$

  which makes *all* derivatives $H^{(k)}(0) = H^{(k)}(1) = 0$ vanish at the endpoints. Integration by parts then suppresses the adiabatic error faster than any polynomial in $1/T$, giving runtime

$$T = \mathcal{O}\Big(\kappa\ \mathrm{polylog}\,\tfrac{\kappa}{\epsilon}\Big) \tag{10.20}$$

  (near-optimal in κ, exponentially good in ϵ). The continuous evolution (10.16) must still be discretized, e.g. by a truncated Dyson series (Chapter 13), contributing only polylogarithmic overhead.

The quantum Zeno / eigenstate-filtering solver. A digital alternative discretizes the schedule at $f_j = f(j/M)$ and, instead of simulating a continuous evolution, *projects* onto the successive zero-eigenspaces by block-encoded spectral projectors P_{f_j}. Frequent projection freezes the state on the eigenpath (the quantum Zeno effect: many measurements suppress transitions), and only the last projection determines the final accuracy. Two ingredients make this optimal.

First, the schedule

$$f(s) = \frac{1-\kappa^{-s}}{1-\kappa^{-1}} \tag{10.21}$$

equalizes the eigenpath length across steps: by perturbation theory $\|\partial_f |x(f)\rangle\| \le 2/\Delta_*(f)$ with $\Delta_*(f) = 1 - f + f/\kappa$, so the total path length is

$$L(0,1) \le \int_0^1 \frac{2}{1-f+f/\kappa}\,\mathrm{d}f = \frac{2\log\kappa}{1-1/\kappa} = \mathcal{O}(\log\kappa), \tag{10.22}$$

and the schedule concentrates steps where the gap is small. With $M = \Theta(\log^2\kappa)$ steps the consecutive overlaps satisfy $|\langle x(f_j)\rangle\, x(f_{j-1})| \ge 1 - \frac{1}{2}L_*(f_{j-1}, f_j)^2$, so all M projections succeed with probability $\Omega(1)$.

Second, each projector P_{f_j} is realized by an *eigenstate filter*: an even polynomial approximating the rectangle that is ≈ 1 at eigenvalue 0 and ≈ 0 beyond the gap $\Delta_*(f_j)$, built by QSVT with degree $\mathcal{O}(\Delta_*(f_j)^{-1}\log(1/\epsilon_j))$ (Section 9.3, the Lin–Tong filter). Summing the per-step costs along the schedule,

$$\sum_{j=1}^{M-1} \frac{\log(1/\epsilon')}{1-f(s_j)+f(s_j)/\kappa} \;\lesssim\; M\log\tfrac{1}{\epsilon'}\int_0^1 \kappa^s\,\mathrm{d}s = M\log\tfrac{1}{\epsilon'}\,\frac{\kappa}{\log\kappa}, \tag{10.23}$$

and taking $M = \Theta(\log\kappa)$ (with fixed-point amplitude amplification to keep each step's success near 1) plus a final projector of precision ϵ gives the following optimal scaling.

Theorem 10.7 (Quantum Zeno / eigenstate-filtering solver)**.** *Let A have singular values in $[1/\kappa, 1]$, accessed by $U_A \in \mathrm{BE}_{\alpha,m}(A)$, and let U_b prepare $|b\rangle$. Then a QZE-based solver prepares $|\tilde{x}\rangle$ with $|\langle\tilde{x}|\, x\rangle| \ge 1-\epsilon$ and constant success probability, using*

$$\mathcal{O}(\kappa\log(1/\epsilon)) \tag{10.24}$$

queries to U_A and U_b, up to $\log\log\kappa$ factors.

Proof idea. Combine Theorem 10.6, the equalized schedule (10.21), the path-length bound (10.22), and the eigenstate-filter cost. The first $M-1$ projectors cost $\mathcal{O}(\kappa(\log\log\kappa)^2)$ in total by (10.23) with $M = \Theta(\log\kappa)$ and fixed-point amplification holding each step's fidelity at $1 - 1/M$; the final projector costs $\mathcal{O}(\kappa\log(1/\epsilon))$. Summing gives (10.24). The matching lower bound $\Omega(\kappa)$ is Theorem 10.2, so the scaling is optimal in κ and in ϵ. A closely related *discrete* adiabatic theorem (Costa et al.) achieves the same $\mathcal{O}(\kappa\log(1/\epsilon))$ without the extra $\log\log\kappa$. □

10.5 Complexity landscape and preconditioning

The landscape. The methods above form a clean progression in the two parameters κ and ϵ. The table below collects the query complexities to the oracle for A (worst case, before/after amplitude amplification as noted).

Method	**Queries to A**	**Precision dep.**
HHL	$\mathcal{O}(\kappa^2/\epsilon)$ (with AA)	$1/\epsilon$
LCU (Fourier/Chebyshev)	$\mathcal{O}(\kappa^2\,\mathrm{polylog}(\kappa/\epsilon))$	polylog
QSVT	$\mathcal{O}(\kappa^2\log(\kappa/\epsilon))$	$\log(1/\epsilon)$
AQC(p)	$\mathcal{O}((\kappa/\epsilon)\log(\kappa/\epsilon))$	$1/\epsilon$
AQC(exp)	$\mathcal{O}(\kappa\,\mathrm{polylog}(\kappa/\epsilon))$	polylog
QZE + eigenfilter	$\mathcal{O}(\kappa\log(1/\epsilon))$	$\log(1/\epsilon)$
discrete AQC (optimal)	$\mathcal{O}(\kappa\log(1/\epsilon))$	$\log(1/\epsilon)$

Table: QLSP query complexity in the worst case $\xi = \|A^{-1}|b\rangle\| = 1$. Best case ($\xi = \Theta(\kappa)$) removes one factor of κ from the postselection methods. The optimal $\mathcal{O}(\kappa\log(1/\epsilon))$ matches the lower bound $\Omega(\kappa)$ of Theorem 10.2.

Two lessons. First, the frontier moved from HHL's κ^2/ϵ to the optimal $\kappa\log(1/\epsilon)$ by two independent improvements—better ϵ *dependence* (QSVT's polynomial replacing QPE's arithmetic) and better κ *dependence* (eigenpath methods replacing postselection). Second, even the optimal solver is $\Omega(\kappa)$: unlike classical conjugate gradient's $\mathcal{O}(\sqrt{\kappa})$ for positive-definite systems, generic quantum solvers do not achieve $\sqrt{\kappa}$, and no amount of algorithmic cleverness removes the linear κ. Hence, whenever κ is large, the decisive move is to *reduce* κ *itself*—preconditioning.

Why κ, not N, is the enemy. The comparison with classical iterative solvers is instructive. For a d-sparse Hermitian positive-definite A, one matrix–vector product costs $\mathcal{O}(dN)$ flops, and steepest descent needs $\mathcal{O}(\kappa\log(1/\epsilon))$ iterations, conjugate gradient $\mathcal{O}(\sqrt{\kappa}\log(1/\epsilon))$; total classical cost $\mathcal{O}(dN\sqrt{\kappa}\log(1/\epsilon))$. The quantum solver's cost is $\widetilde{\mathcal{O}}(\kappa)$ *queries*, each of which may be a $\mathrm{poly}(n)$-gate circuit when U_A is efficiently implementable—so the potential exponential advantage is in replacing the $\mathcal{O}(N)$ per-step cost by $\mathrm{poly}(\log N)$, *not* in the κ dependence, which is if anything worse than CG. The advantage is real only when (i) U_A and U_b are cheap, (ii) one wants a scalar functional of $|x\rangle$ rather than all of x, and (iii) κ is controlled. The Poisson example (Theorem 10.5) with $\kappa \sim N^{2/d}$ shows all three tensions at once.

Preconditioning: classical and quantum. Left-preconditioning replaces $Ax = b$ by $MAx = Mb$; it helps when $\kappa(MA) \ll \kappa(A)$. Classical choices—diagonal (Jacobi), incomplete factorizations, sparse approximate inverse (SPAI)—have quantum analogues (SPAI, circulant preconditioners), but a genuine obstacle appears: on a quantum computer every block-encoded matrix has spectral norm ≤ 1, so multiplying by M can only *shrink* the state, and recovering the norm costs $\mathcal{O}(\kappa)$ amplitude-amplification rounds. Applying two preconditioners therefore incurs a *subnormalization* $\kappa(M)\kappa(M_r) \propto \kappa(A)$, which can eat the very benefit sought. Concretely, the complexity of a quantum solve is governed not by $\kappa(S)$ alone but by $\tilde{\gamma}(U_S)\,\kappa(S)$, the product of the block-encoding subnormalization and the condition number. A clean pipeline where this works: if $A+B$ is the target with A easily invertible and of large norm and $B, A^{-1}, (A+B)^{-1}$ moderate, then $\kappa(\mathbb{I} + A^{-1}B) = \mathcal{O}(1)$, and one composes block encodings along

$$A^{-1} \;\to\; A^{-1}B \;\to\; \mathbb{I} + A^{-1}B \;\to\; (\mathbb{I} + A^{-1}B)^{-1} \;\to\; (\mathbb{I} + A^{-1}B)^{-1}A^{-1} = (A+B)^{-1}, \quad (10.25)$$

requiring only matrix addition, multiplication (Chapter 7), and fast inversion of the well-conditioned factor.

Optimal multilevel (BPX) preconditioning. For elliptic PDEs the sharpest tool is a *multilevel* preconditioner. Discretizing $-\operatorname{div}(A\nabla u) = f$ by finite elements on a hierarchy of grids $G_1 \subset \cdots \subset G_L$ with $h_\ell = 2^{-\ell}$ gives a stiffness matrix S with $\kappa(S) = \mathcal{O}(h^{-2}) = \mathcal{O}(N^{2/d})$. The Bramble–Pasciak–Xu (BPX) preconditioner P is a hierarchical decomposition across levels; under the standard Jackson (direct) and Bernstein (inverse) estimates,

$$\inf_{v_\ell \in V_\ell} \|u - v_\ell\|_{L^2} \le 2^{-\ell} C_1 \|u\|_{H^1}, \qquad \|v_\ell\|_{H^1} \le 2^{\ell} C_2 \|v_\ell\|_{L^2}, \tag{10.26}$$

the symmetrically preconditioned matrix satisfies $\kappa(P^{1/2} S P^{1/2}) = \mathcal{O}(1)$, *independent of* L. Because the preconditioned system PS itself need not be well conditioned as a product, one *splits* $P = FF^{\mathsf{T}}$ over a multilevel generating system $\mathcal{F} = \{2^{-\ell(2-d)/2} \lambda_j^{(\ell)}\}$ and solves the frame system $F^{\mathsf{T}} S F\, y = F^{\mathsf{T}} r$, $c = Fy$. On Cartesian grids the resulting block encoding of $F^{\mathsf{T}} S F$ has subnormalization $\mathcal{O}(dL)$ and $\kappa = \mathcal{O}(1)$, so the quantum finite-element solve attains query complexity near-optimal in κ and ϵ—turning a $\kappa \sim N^{2/d}$ instance into an $\mathcal{O}(1)$-conditioned one and making the QLSP speedup accessible for elliptic PDEs. This is the state of the art for reconciling the $\Omega(\kappa)$ lower bound with practical PDE problems.

Exercises

Exercise 10.1. *Condition number and normalization.* Let $A \in \mathbb{C}^{N\times N}$ be invertible with $\|A\| \le 1$. Show that the singular values of A lie in $[1/\kappa, 1]$ where $\kappa = \|A\|\,\|A^{-1}\|$, and that the Hermitian dilation $\widetilde{A} = \left(\begin{smallmatrix} 0 & A \\ A^\dagger & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm\sigma_i$ and the same condition number. Verify that $\widetilde{A}(0,x)^{\mathsf{T}} = (b,0)^{\mathsf{T}}$ is equivalent to $Ax = b$.

Exercise 10.2. *HHL success probability.* From (10.7) with $C = 1/\kappa$, show the postselection success probability is $p(1) = \left\|A^{-1}|b\rangle\right\|^2/\kappa^2 = \xi^2/\kappa^2$. Deduce $p(1) = \Omega(\kappa^{-2})$ using only $\xi \ge 1$, and $p(1) = \Omega(1)$ if $|b\rangle$ has $\Omega(1)$ overlap with the eigenvector of the smallest eigenvalue. Explain how amplitude amplification turns the κ^2 repetition overhead into κ.

Exercise 10.3. *Multiplicative precision in HHL.* Suppose the eigenvalue estimates satisfy $\tilde{\lambda}_j = \lambda_j(1+e_j)$ with $|e_j| \le \epsilon/4$. Following Theorem 10.3, show $\||\tilde{x}\rangle - |x\rangle\| \le \epsilon$, and conclude that because $\lambda_{\min} = \Theta(1/\kappa)$ the QPE runtime must be $\Omega(\kappa/\epsilon)$. Contrast with the $\log(1/\epsilon)$ dependence of the QSVT solver.

Exercise 10.4. *The QSVT inversion polynomial.* For $g_b(x) = (1-(1-x^2)^b)/x$ of (10.12), show g_b is an odd polynomial of degree $2b-1$ and that $|g_b(x) - 1/x| \le \mathrm{e}^{-bx^2}/|x|$ for $x \ne 0$. Deduce that $b = \mathcal{O}(\kappa^2 \log(1/\epsilon))$ gives error ϵ on $[1/\kappa, 1]$, and explain why Chebyshev truncation reduces the effective QSVT degree to $\mathcal{O}(\kappa \log(1/\epsilon))$. Why must one divide by $\eta\kappa$ with $\eta > 1$?

Exercise 10.5. *Gap of the interpolating Hamiltonian.* Prove Theorem 10.6: for $A \succeq \kappa^{-1}\mathbb{I}$, the gap of $H(f) = (1-f)H_0 + fH_1$ around its zero eigenvalue is $\Delta(f) \ge 1 - f + f/\kappa$. (Use the block-elimination identity $\det(\lambda^2\mathbb{I} - Q_b M^2 Q_b) = 0$ with $M = (1-f)\mathbb{I} + fA$, and restrict to the subspace orthogonal to $|b\rangle$.) Where is the gap smallest, and what value does it take?

Exercise 10.6. *Eigenpath length and the Zeno schedule.* For the schedule $f(s) = (1-\kappa^{-s})/(1-\kappa^{-1})$ of (10.21) and the bound $\|\partial_f |x(f)\rangle\| \le 2/(1-f+f/\kappa)$, show that the total eigenpath length is $L(0,1) \le 2\log\kappa/(1-1/\kappa) = \mathcal{O}(\log\kappa)$. Explain why this schedule allocates roughly equal path length per step, and why that yields the $\mathcal{O}(\kappa\log(1/\epsilon))$ query complexity of Theorem 10.7.

Exercise 10.7. *Poisson condition number.* Using the eigenvalues $\lambda_k = a + 2\,|b|\cos(k\pi/(N+1))$ of a Toeplitz tridiagonal matrix, derive $\kappa(A) \approx 4/(h^2\pi^2) = \mathcal{O}(N^2)$ for the 1D Laplacian (10.13). Show that the d-dimensional Laplacian $A_d = \sum_i \mathbb{I}^{\otimes(i-1)} \otimes A \otimes \mathbb{I}^{\otimes(d-i)}$ has $\kappa(A_d) \approx \kappa(A)$ independent of d, and explain why this makes high-dimensional Poisson a candidate for quantum advantage while the 1D case is not.

Exercise 10.8. *The $(A+B)$ preconditioner.* Suppose A is easily invertible with large $\|A\|$, and $B, \|A^{-1}\|, \|(A+B)^{-1}\|$ are moderate. Show $\kappa(\mathbb{I} + A^{-1}B) = \mathcal{O}(1)$ and verify the identity $(\mathbb{I} + A^{-1}B)^{-1}A^{-1} = (A+B)^{-1}$. Using the block-encoding arithmetic of Chapter 7, describe the pipeline (10.25) and identify at which step the well-conditioned inversion (cheap) versus the ill-conditioned one (expensive) occurs. Why does naive left-preconditioning MA incur a subnormalization $\propto \kappa(M)$?

Further reading

- Lin Lin, *live notes*, Ch. 19 (QLSP problem setup and output model; the HHL algorithm via QPE and controlled rotation, its success probability and multiplicative-accuracy requirement; the QSVT solver and Theorem 10.4; Poisson's equation and fermionic Green's functions; the quantum adiabatic near-optimal solver AQC(exp); and the quantum Zeno / eigenstate-filtering optimal solver, Theorem 10.7).

- Dong An, *Quantum Numerical Linear Algebra* tutorial talk (the QLSP definition and parameters ϵ, κ; QSVT for $f(x) = 1/(\kappa x)$ with degree $\mathcal{O}(\kappa \log(1/\epsilon))$; HHL as QPE + controlled rotation with cost $\mathcal{O}(\kappa^2/\epsilon)$; the LCU Fourier/Chebyshev approaches; the AQC(p)/AQC(exp) schedules and their $\mathcal{O}(\kappa/\epsilon)$ and $\mathcal{O}(\kappa$ polylog) costs; the complexities comparison table; and preconditioning, including SPAI and the $(A+B)$ pipeline).

- A. W. Harrow, A. Hassidim, and S. Lloyd, *Quantum algorithm for linear systems of equations*, Phys. Rev. Lett. **103**, 150502 (2009) (arXiv:0811.3171), the original HHL algorithm.

- A. M. Childs, R. Kothari, and R. D. Somma, *Quantum algorithm for systems of linear equations with exponentially improved dependence on precision*, SIAM J. Comput. **46**, 1920–1950 (2017) (arXiv:1511.02306), for the LCU solver and the explicit $1/x$ polynomial; D. An and L. Lin, *Quantum linear system solver based on time-optimal adiabatic quantum computing and quantum approximate optimization algorithm*, ACM Trans. Quantum Comput. **3**, 5 (2022) (arXiv:1909.05500), for AQC(p) and AQC(exp); L. Lin and Y. Tong, *Optimal polynomial based quantum eigenstate filtering with application to solving quantum linear systems*, Quantum **4**, 361 (2020) (arXiv:1910.14596), for the QZE/eigenstate-filtering optimal solver; and P. C. S. Costa et al., *Optimal scaling quantum linear-systems solver via discrete adiabatic theorem*, PRX Quantum **3**, 040303 (2022) (arXiv:2111.08152).

- Quantum preconditioning notes (SuzhouNotes, "Quantum preconditioning") for the block-encoding normalization/subnormalization accounting, the QLSP definition in terms of $\tilde{\gamma} \cdot \kappa$, and the optimal multilevel (BPX) preconditioner following J. H. Bramble, J. E. Pasciak, and J. Xu, *Parallel multilevel preconditioners*, Math. Comp. **55**, 1–22 (1990).

Part III

Differential Equations and Simulation

Chapter 11

Hamiltonian Simulation I: Product Formulas

Learning objectives

- Split a Hamiltonian and build first- and second-order product formulas.
- Construct higher-order Suzuki formulas by recursion.
- State commutator-type error bounds and their scaling.

Hamiltonian simulation—preparing $|\psi(t)\rangle = \mathrm{e}^{-\mathrm{i}Ht}\,|\psi_0\rangle$ from an initial state $|\psi_0\rangle$ and a Hamiltonian H—is the task that first motivated quantum computers (Feynman, 1982) and remains the cleanest example of a provable exponential speedup. Equivalently, we are solving the linear Schrödinger ODE $\partial_t\psi = -\mathrm{i}H\psi$, $\psi(0) = \psi_0$, on a Hilbert space of dimension $N = 2^n$ where classical vectors do not fit in memory. This chapter develops the oldest and most widely deployed family of simulation algorithms, the *product formulas* (or Trotter methods). Their idea is disarmingly simple: if $H = \sum_\gamma H_\gamma$ is a sum of terms each of which can be exponentiated exactly and cheaply, approximate the joint evolution $\mathrm{e}^{-\mathrm{i}Ht}$ by an ordered product of the individual evolutions $\mathrm{e}^{-\mathrm{i}H_\gamma \Delta t}$ over short time steps. The subtlety—and the entire technical content of the chapter—is that the H_γ do not commute, so the product is only approximate, and one must control the error as a function of the time t, the tolerance ϵ, and the (non)commutativity of the terms. We build the first- and second-order formulas from the Lie–Trotter product (Sections 11.1 and 11.2), generate arbitrary even order by the Suzuki recursion (Section 11.3), and then prove the commutator-type error bounds (Section 11.4) that explain why product formulas, despite their pessimistic worst-case operator-norm scaling, are so effective in practice for the geometrically local Hamiltonians of physics. We close (Section 11.5) with randomized and structure-exploiting variants—qDRIFT, random permutations, and the real-space/Fourier splitting for kinetic-plus-potential Hamiltonians—and place product formulas in the landscape of methods whose optimal $\mathcal{O}(t + \log(1/\epsilon))$ scaling is the subject of Chapter 12. Throughout, $\|\cdot\|$ is the spectral (operator) norm and $\Delta t = t/L$ is the Trotter step for L steps.

11.1 Trotter splitting

The splitting hypothesis. Write the Hamiltonian as a sum of Γ Hermitian terms,

$$H = \sum_{\gamma=1}^{\Gamma} H_\gamma, \tag{11.1}$$

where each $\mathrm{e}^{-\mathrm{i}H_\gamma s}$ can be implemented exactly by a short circuit for any s. This is the case for essentially every Hamiltonian that arises in practice: a *k-local* spin Hamiltonian is a sum of Pauli strings, each a tensor product of at most k single-qubit Paulis, and $\mathrm{e}^{-\mathrm{i}\theta P}$ for a Pauli string P is a single rotation conjugated by Clifford gates; an electronic-structure Hamiltonian in second quantization is a sum of $\mathcal{O}(n^4)$ fermionic terms, each mapped to a Pauli string by Jordan–Wigner or Bravyi–Kitaev; and a differential operator such as $H = -\Delta + V$ splits into a kinetic part diagonal in the Fourier basis and a potential part diagonal in real space (Theorem 11.4). If the H_γ *commuted* we would be done, since $\mathrm{e}^{-\mathrm{i}Ht} = \prod_\gamma \mathrm{e}^{-\mathrm{i}H_\gamma t}$ exactly; the whole difficulty is that in general $[H_\gamma, H_{\gamma'}] \neq 0$.

The Lie–Trotter product formula. For two terms $H = H_1 + H_2$ (the general case is analogous), the classical Lie product formula asserts that the error of a single ordered product vanishes in the limit of infinitely many steps:

$$\mathrm{e}^{-\mathrm{i}Ht} = \lim_{L\to\infty} \Big(\mathrm{e}^{-\mathrm{i}\frac{t}{L}H_1}\,\mathrm{e}^{-\mathrm{i}\frac{t}{L}H_2}\Big)^L. \tag{11.2}$$

The identity holds for any bounded operators H_1, H_2 (and, with care, for many unbounded ones), and is proved by expanding both sides to first order in $1/L$: the $\mathcal{O}(1/L)$ terms match, and the mismatch first appears at $\mathcal{O}(1/L^2)$, of which there are L copies, so the accumulated error is $\mathcal{O}(1/L) \to 0$. Taking L *finite* and writing $\Delta t = t/L$ turns (11.2) into an algorithm: the *first-order Trotter formula*

$$S_1(\Delta t) := \mathrm{e}^{-\mathrm{i}\Delta t\, H_1}\,\mathrm{e}^{-\mathrm{i}\Delta t\, H_2} \tag{11.3}$$

is applied L times to approximate $\mathrm{e}^{-\mathrm{i}Ht} = S_1(\Delta t)^L + \text{error}$. Each application is a circuit of $\mathcal{O}(\Gamma)$ elementary evolutions, so the total gate count is $\mathcal{O}(L\Gamma)$ and the entire cost analysis reduces to bounding how large L must be.

Local and global error. Expanding (11.3) and $\mathrm{e}^{-\mathrm{i}\Delta t\, H}$ to second order in Δt, the zeroth- and first-order terms agree and the leading discrepancy is the commutator:

$$\mathrm{e}^{-\mathrm{i}\Delta t\, H} - S_1(\Delta t) = \tfrac{1}{2}(\Delta t)^2[H_1, H_2] + \mathcal{O}(\Delta t^3), \qquad \big\|\mathrm{e}^{-\mathrm{i}\Delta t\, H} - S_1(\Delta t)\big\| = \mathcal{O}(\Delta t^2). \tag{11.4}$$

This is the *local* (one-step) error. Errors compose sub-additively for products of near-unitaries: if $\|U_j - V_j\| \le \delta$ for unitaries U_j, V_j, then $\big\|\prod_{j=1}^L U_j - \prod_{j=1}^L V_j\big\| \le L\delta$ (a telescoping estimate we prove in Theorem 11.1). Hence the *global* error after L steps is

$$\Big\|\mathrm{e}^{-\mathrm{i}Ht} - S_1(\Delta t)^L\Big\| = \mathcal{O}(L\,\Delta t^2) = \mathcal{O}\Big(\frac{t^2}{L}\Big), \tag{11.5}$$

so reaching spectral-norm accuracy ϵ requires

$$L = \mathcal{O}\Big(\frac{t^2}{\epsilon}\Big) \tag{11.6}$$

steps. The quadratic dependence on t and inverse-linear dependence on ϵ are the signatures of the first-order method; both are improved below.

Lemma 11.1 (Telescoping of product errors)**.** *Let $U_1, \dots, U_L$ and $V_1, \dots, V_L$ be unitaries with $\|U_j - V_j\| \le \delta_j$. Then $\big\|\prod_{j=1}^L U_j - \prod_{j=1}^L V_j\big\| \le \sum_{j=1}^L \delta_j$.*

Proof. Insert a telescoping sum $\prod_j U_j - \prod_j V_j = \sum_{k=1}^L \big(\prod_{j<k} U_j\big)(U_k - V_k)\big(\prod_{j>k} V_j\big)$, take norms, and use unitary invariance $\|WMW'\| = \|M\|$ for unitary W, W' together with the triangle inequality. Each summand contributes $\|U_k - V_k\| \le \delta_k$. □

11.2 First- and second-order formulas

Why symmetrize. The first-order formula (11.3) is not time-symmetric: $S_1(\Delta t)\,S_1(-\Delta t) \neq \mathbb{I}$. Time-reversal symmetry is the structural feature that kills even-order error terms. Define the *second-order* (Strang, or symmetric Trotter) formula by conjugating a full step of H_1 with two half-steps of H_2,

$$S_2(\Delta t) := \mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2}\,\mathrm{e}^{-\mathrm{i}\Delta t\,H_1}\,\mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2}. \tag{11.7}$$

By construction $S_2(\Delta t) = S_2(-\Delta t)^{-1}$ (read the product backward and negate Δt), i.e. S_2 is *self-inverse under time reversal*. This forces its error expansion to contain only odd powers of Δt, so the leading error jumps from $\mathcal{O}(\Delta t^2)$ to $\mathcal{O}(\Delta t^3)$:

$$\left\| \mathrm{e}^{-\mathrm{i}\Delta t\,H} - S_2(\Delta t) \right\| = \mathcal{O}(\Delta t^3). \tag{11.8}$$

Proposition 11.2 (Symmetric formulas have odd error expansions)**.** *Let $S(\Delta t)$ be a product formula satisfying $S(\Delta t)S(-\Delta t) = \mathbb{I}$ and $S(\Delta t) = \mathrm{e}^{-\mathrm{i}\Delta t\,H} + \mathcal{O}(\Delta t^2)$. Then its error $E(\Delta t) := S(\Delta t) - \mathrm{e}^{-\mathrm{i}\Delta t\,H}$ satisfies $E(\Delta t) = -\mathrm{e}^{-2\mathrm{i}\Delta t\,H}E(-\Delta t) + \mathcal{O}(\Delta t^{?})$ so that the leading term is odd; consequently the first nonvanishing error of S_2 is $\mathcal{O}(\Delta t^3)$.*

Proof idea. Write $S(\Delta t) = \exp(-\mathrm{i}\Delta t\,H + \sum_{k\geq 2}(\Delta t)^k E_k)$ for operators E_k (a valid single-exponential form by Baker–Campbell–Hausdorff, since S is a product of exponentials). The condition $S(\Delta t)S(-\Delta t) = \mathbb{I}$ means the exponents of $S(\Delta t)$ and $S(-\Delta t)$ are negatives; substituting the series and matching powers forces $E_k = 0$ for every *even* k. Hence the leading surviving correction is $E_3(\Delta t)^3$, giving (11.8). The same argument shows that any symmetric formula has error $\mathcal{O}(\Delta t^{2\ell+1})$ for some integer $\ell \geq 1$, which is what the Suzuki recursion exploits. □

Global cost of the second-order formula. By Theorem 11.1 the global error is $\mathcal{O}(L\,\Delta t^3) = \mathcal{O}(t^3/L^2)$, so accuracy ϵ needs

$$L = \mathcal{O}\Big(\frac{t^{3/2}}{\epsilon^{1/2}}\Big). \tag{11.9}$$

Compared with (11.6) this improves both the time scaling ($t^{3/2}$ vs t^2) and the precision scaling ($\epsilon^{-1/2}$ vs ϵ^{-1}) at the cost of one extra half-step of H_2 per step—and, crucially, when the steps are chained the adjacent half-steps merge, $\mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2}\mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2} = \mathrm{e}^{-\mathrm{i}\Delta t\,H_2}$, so a run of L symmetric steps costs only one extra H_2-evolution overall, essentially the same per-step cost as S_1. For this reason S_2 is the default low-order product formula. The generalization to Γ terms is the obvious palindromic order,

$$S_2(\Delta t) = \Big(\prod_{\gamma=1}^{\Gamma} \mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_\gamma}\Big)\Big(\prod_{\gamma=\Gamma}^{1} \mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_\gamma}\Big), \tag{11.10}$$

which is again manifestly symmetric.

Example 11.3 (Transverse-field Ising model)**.** The one-dimensional transverse-field Ising model with open boundary is

$$H = \underbrace{-\sum_{i=1}^{n-1} Z_i Z_{i+1}}_{H_1} \underbrace{- g\sum_{i=1}^{n} X_i}_{H_2}, \qquad Z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right). \tag{11.11}$$

Both halves are sums of *mutually commuting* Paulis, so each exponentiates as a product of two-qubit and one-qubit rotations,

$$\mathrm{e}^{-\mathrm{i}\Delta t\, H_1} = \prod_{i=1}^{n-1} \mathrm{e}^{\mathrm{i}\Delta t\, Z_i Z_{i+1}}, \qquad \mathrm{e}^{-\mathrm{i}\Delta t\, H_2} = \prod_{i=1}^{n} \mathrm{e}^{\mathrm{i}\Delta t\, g X_i}, \tag{11.12}$$

each gate acting on at most two neighboring qubits. A single S_2 step is therefore a constant-depth circuit of $\mathcal{O}(n)$ nearest-neighbor gates. We return to this model in Theorem 11.7 to see that its *commutator* structure makes the true cost far better than the naive operator-norm bound suggests.

Example 11.4 (Real-space Schrödinger operator)**.** Let $H = -\Delta_r + V(r) = H_1 + H_2$ on $\Omega = [0,1]^d$ with periodic boundary conditions, discretized on a uniform grid. The kinetic term $H_1 = -\Delta_r$ is diagonal in the Fourier basis, so $\mathrm{e}^{-\mathrm{i}\Delta t\, H_1} = \mathrm{QFT}^\dagger\, D_{\mathrm{kin}}\, \mathrm{QFT}$ with D_{kin} a diagonal phase encoding the momenta $\sum_j p_j^2$; the potential $H_2 = V(r)$ is diagonal in real space, so $\mathrm{e}^{-\mathrm{i}\Delta t\, H_2} = D_V$ is a single diagonal unitary. A Trotter step is thus "QFT—multiply by kinetic phases—inverse QFT—multiply by potential phases," the quantum split-step (pseudospectral) method, at a per-step cost of $\mathcal{O}(dn^2)$ gates dominated by the QFTs. This is the workhorse for simulating quantum dynamics in first quantization and reappears in the PDE algorithms of Chapter 13.

11.3 Higher-order Suzuki recursion

Fractal (Suzuki–Trotter) construction. Higher order is obtained by a beautiful recursion of Suzuki: build the order-$(2k+2)$ symmetric formula out of five copies of the order-$2k$ one, with cleverly chosen substep lengths that cancel the next error term. Starting from the second-order S_2 of (11.7), define recursively

$$S_{2k+2}(\Delta t) = S_{2k}(p_k \Delta t)^2\, S_{2k}((1-4p_k)\Delta t)\, S_{2k}(p_k \Delta t)^2, \qquad p_k = \frac{1}{4 - 4^{1/(2k+1)}}. \tag{11.13}$$

The magic number p_k is exactly the value that makes the five substeps sum correctly, $4p_k + (1-4p_k) = 1$, while forcing the leading $\mathcal{O}(\Delta t^{2k+1})$ error of S_{2k} to cancel between the forward substeps (length $p_k \Delta t > 0$) and the single backward-ish substep (length $(1-4p_k)\Delta t < 0$, since $p_k > 1/4$). Because each S_{2k} factor is symmetric, so is S_{2k+2}; by Theorem 11.2 its error is again odd, and the construction guarantees

$$\left\| \mathrm{e}^{-\mathrm{i}\Delta t\, H} - S_{2k}(\Delta t) \right\| = \mathcal{O}(\Delta t^{2k+1}). \tag{11.14}$$

Cost of the $2k$-th order formula. Unrolling the recursion, S_{2k} is a product of $2 \cdot 5^{k-1}$ copies of S_2 (equivalently $\Upsilon_k = 2 \cdot 5^{k-1}\Gamma$ elementary exponentials for Γ terms), so its per-step gate cost grows exponentially in the order k. Against this, the global error of L steps is $\mathcal{O}(L\, \Delta t^{2k+1}) = \mathcal{O}(t^{2k+1}/L^{2k})$, so

$$L = \mathcal{O}\Big(t^{\frac{2k+1}{2k}}\, \epsilon^{-\frac{1}{2k}}\Big) = \mathcal{O}\big(t^{1+\frac{1}{2k}}\, \epsilon^{-\frac{1}{2k}}\big). \tag{11.15}$$

As $k \to \infty$ the time scaling approaches the optimal linear-in-t and the precision scaling approaches ϵ^{-0}, but the prefactor 5^{k-1} makes very high order counterproductive. Balancing the 5^k prefactor against the $\epsilon^{-1/(2k)}$ tail by choosing $k = \Theta(\sqrt{\log(1/\epsilon)})$ yields the best product-formula scaling

$$\text{gates} = \mathcal{O}\big(\Gamma\, t \cdot (t/\epsilon)^{o(1)}\big) = \Gamma\, t\, \mathrm{e}^{\mathcal{O}(\sqrt{\log(t/\epsilon)})}, \tag{11.16}$$

nearly linear in t but *super*-polynomial (though sub-polynomial in $1/\epsilon$) in the precision—strictly worse than the $\mathcal{O}(t + \log(1/\epsilon))$ achieved by the qubitization and LCU methods of Chapter 12. The

practical sweet spot for near-term work is fourth order ($k = 2$), i.e. S_4, which offers a good balance of accuracy and circuit depth.

Remark 11.5 (Real vs. complex coefficients). The Suzuki recursion necessarily uses a *negative* substep coefficient $1 - 4p_k < 0$; there is a no-go theorem (Sheng; Suzuki) that no product formula of order ≥ 3 can have all-positive coefficients. Negative times are fine for a Hamiltonian (run the evolution backward), but forbid the same trick for imaginary-time evolution $\mathrm{e}^{-\beta H}$ or for Lindbladians, where "negative time" is unphysical—one reason imaginary-time and open-system simulation (Chapter 14) need different tools.

11.4 Commutator error bounds

The Duhamel argument. The estimates (11.4) and (11.8) were stated as $\mathcal{O}(\Delta t^p)$; we now make the constants explicit and, more importantly, show they are controlled by the *commutators* of the terms rather than by their norms. This distinction is what makes product formulas competitive for physical systems. Consider the first-order formula $\tilde{u}(t) = \mathrm{e}^{-\mathrm{i}H_1 t}\mathrm{e}^{-\mathrm{i}H_2 t}$ (one step, so $t = \Delta t$). Differentiating,

$$\begin{aligned} \mathrm{i}\partial_t \tilde{u}(t) &= H_1\, \mathrm{e}^{-\mathrm{i}H_1 t}\mathrm{e}^{-\mathrm{i}H_2 t} + \mathrm{e}^{-\mathrm{i}H_1 t} H_2\, \mathrm{e}^{-\mathrm{i}H_2 t} \\ &= (H_1 + H_2)\tilde{u}(t) + [\mathrm{e}^{-\mathrm{i}H_1 t},\, H_2]\mathrm{e}^{-\mathrm{i}H_2 t}, \end{aligned} \tag{11.17}$$

with $\tilde{u}(0) = \mathbb{I}$. Writing $u(t) = \mathrm{e}^{-\mathrm{i}Ht}$ and applying Duhamel's principle (the variation-of-constants formula for the inhomogeneous term),

$$\tilde{u}(t) = u(t) - \mathrm{i}\int_0^t \mathrm{e}^{-\mathrm{i}H(t-s)}\,[\mathrm{e}^{-\mathrm{i}H_1 s}, H_2]\mathrm{e}^{-\mathrm{i}H_2 s}\,\mathrm{d}s, \tag{11.18}$$

so, using unitarity to drop the outer factors,

$$\|\tilde{u}(t) - u(t)\| \leq \int_0^t \|[\mathrm{e}^{-\mathrm{i}H_1 s}, H_2]\|\,\mathrm{d}s. \tag{11.19}$$

Bounding the inner commutator. Set $G(s) = [\mathrm{e}^{-\mathrm{i}H_1 s}, H_2]\mathrm{e}^{\mathrm{i}H_1 s} = \mathrm{e}^{-\mathrm{i}H_1 s}H_2\mathrm{e}^{\mathrm{i}H_1 s} - H_2$, so $G(0) = 0$ and $\mathrm{i}\partial_s G(s) = \mathrm{e}^{-\mathrm{i}H_1 s}[H_1, H_2]\mathrm{e}^{\mathrm{i}H_1 s}$. Integrating and taking norms gives $\left\|[\mathrm{e}^{-\mathrm{i}H_1 s}, H_2]\right\| = \|G(s)\| \leq s\,\|[H_1, H_2]\|$. Substituting into (11.19),

$$\left\|\mathrm{e}^{-\mathrm{i}\Delta t\, H_1}\mathrm{e}^{-\mathrm{i}\Delta t\, H_2} - \mathrm{e}^{-\mathrm{i}\Delta t\, H}\right\| \leq \int_0^{\Delta t} s\,\|[H_1, H_2]\|\,\mathrm{d}s = \frac{(\Delta t)^2}{2}\,\|[H_1, H_2]\|\,. \tag{11.20}$$

This is the *commutator-norm* error estimate. Bounding $\|[H_1, H_2]\| \leq 2\,\|H_1\|\,\|H_2\| \leq 2\nu^2$ with $\nu = \max_\gamma \|H_\gamma\|$ recovers the coarser *operator-norm* estimate $\frac{1}{2}\Delta t^2 \cdot 2\nu^2 = \Delta t^2\nu^2$. The difference is decisive for local Hamiltonians: many pairs of terms commute, so $\|[H_1, H_2]\|$ can be far smaller than ν^2.

Theorem 11.6 (Product-formula error, commutator form)**.** *Let $H = \sum_{\gamma=1}^{\Gamma} H_\gamma$ and let $S_p(\Delta t)$ be the order-p product formula ($p = 1, 2, \dots$). Then the one-step error obeys*

$$\left\|S_p(\Delta t) - \mathrm{e}^{-\mathrm{i}\Delta t\, H}\right\| \leq C_p\,(\Delta t)^{p+1} \sum_{\gamma_1,\dots,\gamma_{p+1}} \left\|[H_{\gamma_{p+1}}, [\cdots [H_{\gamma_2}, H_{\gamma_1}]\cdots]]\right\|, \tag{11.21}$$

for a constant C_p depending only on p, where the sum runs over the terms of H and the summand is a nested commutator of order p. Consequently the L-step error is $\mathcal{O}(L\,\Delta t^{p+1}\,\alpha_{\mathrm{comm}})$ with $\alpha_{\mathrm{comm}} =$

$\sum \|[\cdots]\|$ *the sum of nested-commutator norms, and the gate complexity to simulate to time* t *within* ϵ *is*

$$\mathcal{O}\Big(\Gamma\,\frac{(\alpha_{\mathrm{comm}}\,t^{p+1})^{1/p}}{\epsilon^{1/p}}\Big). \tag{11.22}$$

Proof idea. Iterate the Duhamel identity (11.18) p times: each pass differentiates one more exponential and produces one more commutator, so after p iterations the remainder integrand is a sum of p-fold nested commutators of the H_γ, integrated over a p-simplex of volume $(\Delta t)^{p+1}/(p+1)!$. The symmetry of S_p (for even p) kills the lower-order terms exactly as in Theorem 11.2, leaving the stated leading term; the constant C_p collects the simplex volume and the number of orderings. Summing over $L = t/\Delta t$ steps and solving $L\,\Delta t^{p+1}\alpha_{\mathrm{comm}} = \epsilon$ for L gives (11.22). This is the Childs–Su–Tran–Wiebe–Zhu "theory of Trotter error" bound; its power is that α_{comm}, not ν, sets the cost. □

Example 11.7 (TFIM: commutator vs. operator norm)**.** For the TFIM of Theorem 11.3, $\|H_1\|, \|H_2\| = \mathcal{O}(n)$, so the operator-norm bound uses $\nu^2 = \mathcal{O}(n^2)$ and predicts $L = \mathcal{O}(n^2t^2/\epsilon)$ (first order). But $[Z_iZ_{i+1}, X_k] \neq 0$ only when $k \in \{i, i+1\}$, so in $[H_1, H_2] = \sum_{i,k}[Z_iZ_{i+1}, -gX_k]$ only $\mathcal{O}(n)$ of the $\mathcal{O}(n^2)$ pairs survive, each of norm $\mathcal{O}(1)$; hence $\|[H_1, H_2]\| = \mathcal{O}(n)$. The commutator bound therefore gives $L = \mathcal{O}(n\,t^2/\epsilon)$ (first order), a factor-n improvement, and the same locality gain compounds at higher order. This is the general lesson: for a *geometrically local* Hamiltonian on n qubits the number of noncommuting term-pairs is $\mathcal{O}(n)$ rather than $\mathcal{O}(n^2)$, so Trotter cost scales like n (extensive), and product formulas can even be *asymptotically optimal* for such systems.

Example 11.8 (Schrödinger operator: commutator gain)**.** For $H = -\Delta + V$ in $d = 1$ on an N-point grid, $\|H_1\| = \mathcal{O}(N^2)$ (the discrete Laplacian has eigenvalues up to $\mathcal{O}(N^2)$) while $\|V\| = \mathcal{O}(1)$, so $\nu^2 = \mathcal{O}(N^4)$ and the operator-norm bound gives $L = \mathcal{O}(N^2t^2/\epsilon)$. But the commutator acts as a lower-order differential operator: for smooth ψ,

$$[H_1, H_2]\psi = \Big[-\tfrac{\mathrm{d}^2}{\mathrm{d}r^2}, V\Big]\psi = -V''\psi - 2V'\psi', \tag{11.23}$$

so $\|[H_1, H_2]\| \le \|V''\| + 2\,\|V'\|\,\|\partial_r\| = \mathcal{O}(N)$ (only the first-derivative term carries a factor N). Thus the commutator bound gives $L = \mathcal{O}(\sqrt{N}\,t/\sqrt{\epsilon})$ at first order after the Δt^2 improvement is propagated—an order-of-magnitude reduction that reflects the physical smoothness of the potential.

11.5 Randomized and sparse variants

Random orderings. The deterministic product formula fixes one ordering of the Γ terms; the *ordering* itself is a free choice that can be randomized to cancel error in expectation. In the simplest scheme one applies the Γ term-evolutions in a uniformly random permutation each step. Because the leading first-order error is a sum of commutators whose *sign* depends on the ordering, averaging over the permutation and its reverse cancels the $\mathcal{O}(\Delta t^2)$ term *on average*, so the random formula behaves like a second-order deterministic one at first-order cost. Formally one bounds the error in the diamond norm of the mixed (channel-averaged) evolution and gains a $\sqrt{\Gamma}$ or Γ factor over the worst-case deterministic bound; this is the Childs–Ostrander–Su randomized-Trotter analysis.

qDRIFT. A more radical randomization abandons the fixed step structure entirely. Write $H = \sum_\gamma h_\gamma H_\gamma$ with $\|H_\gamma\| = 1$, $h_\gamma > 0$, and set $\lambda = \sum_\gamma h_\gamma$. *qDRIFT* (Campbell) implements a random

product of N_{q} elementary rotations, at each of which a single term H_γ is drawn with probability h_γ/λ and evolved for the *fixed* small angle $\tau = \lambda t/N_{\mathrm{q}}$:

$$\mathcal{E}(t) = \Big(\mathbb{E}_{\gamma\sim h_\gamma/\lambda} \mathrm{e}^{-\mathrm{i}\tau H_\gamma}\Big)^{N_{\mathrm{q}}}, \qquad N_{\mathrm{q}} = \mathcal{O}\Big(\frac{\lambda^2 t^2}{\epsilon}\Big). \tag{11.24}$$

The channel $\mathcal{E}$ approximates $\mathrm{e}^{-\mathrm{i}Ht}(\cdot)\mathrm{e}^{\mathrm{i}Ht}$ in diamond norm. Its striking feature is that the gate count (11.24) depends only on the 1-norm $\lambda = \sum_\gamma h_\gamma$ and *not* on the number of terms Γ nor on any commutator structure. This makes qDRIFT the method of choice for Hamiltonians with very many small terms and no exploitable locality—most notably electronic-structure Hamiltonians with $\mathcal{O}(n^4)$ terms of widely varying weight, where a handful of large h_γ dominate λ. The tradeoff is the t^2/ϵ scaling of a first-order method: qDRIFT wins on term count, not on time or precision.

Sparse and structured splittings. When H is d-sparse (each row has at most d nonzeros, accessed by the sparse-access oracle of Chapter 6), one can color the graph of H into $\mathcal{O}(d^2)$ classes of mutually commuting 1-sparse Hermitian matrices, each of which is a permutation-times-phase that exponentiates exactly; product formulas over the color classes then simulate H with $\mathcal{O}(d^2\,\|H\|_{\max}\,t)$-type cost, the original Aharonov–Ta-Shma / Berry–Ahokas–Cleve–Sanders route to sparse Hamiltonian simulation. For the kinetic-plus-potential splitting of Theorem 11.4 the two "color classes" are simply the Fourier-diagonal and real-diagonal parts, and the QFT sandwich is the exact exponentiation—no coloring needed.

Where product formulas sit. Product formulas are simple, ancilla-free (no block encoding, no LCU), and—by Theorem 11.6—often *optimal* for geometrically local lattice models, where the commutator sum is extensive and the naive operator-norm pessimism does not apply. Their weakness is the precision scaling: even the best Suzuki order gives (11.16), sub-polynomial but super-logarithmic in $1/\epsilon$. When ϵ must be very small, or when H is dense with no locality to exploit, the block-encoding methods of Chapter 12—qubitization, LCU/Taylor-series, and the quantum-signal-processing simulation—achieve the optimal $\mathcal{O}(\|H\|\,t + \log(1/\epsilon))$ query complexity, matching the no-fast-forwarding lower bound (Section 12.5). The two families are complementary, and modern simulations often combine them.

Exercises

Exercise 11.1. *Local error of the first-order formula.* By expanding both $\mathrm{e}^{-\mathrm{i}\Delta t\,H}$ and $S_1(\Delta t) = \mathrm{e}^{-\mathrm{i}\Delta t\,H_1}\mathrm{e}^{-\mathrm{i}\Delta t\,H_2}$ to third order in Δt, verify (11.4): $\mathrm{e}^{-\mathrm{i}\Delta t\,H} - S_1(\Delta t) = \frac{1}{2}(\Delta t)^2[H_1, H_2] + \mathcal{O}(\Delta t^3)$. Then use Theorem 11.1 to obtain the global bound $\big\|\mathrm{e}^{-\mathrm{i}Ht} - S_1(\Delta t)^L\big\| = \mathcal{O}(t^2/L)$ and hence $L = \mathcal{O}(t^2/\epsilon)$.

Exercise 11.2. *Symmetry kills even orders.* Show that the second-order formula $S_2(\Delta t) = \mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2}\mathrm{e}^{-\mathrm{i}\Delta t\,H_1}\mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2}$ satisfies $S_2(\Delta t)S_2(-\Delta t) = \mathbb{I}$. Assuming the single-exponential form $S_2(\Delta t) = \exp(-\mathrm{i}\Delta t\,H + \sum_{k\ge 2}(\Delta t)^k E_k)$, deduce that $E_k = 0$ for all even k, and conclude the local error is $\mathcal{O}(\Delta t^3)$.

Exercise 11.3. *Merging half-steps.* Show that for a chain of L symmetric steps the interior half-steps merge, $S_2(\Delta t)^L = \mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2}\big(\mathrm{e}^{-\mathrm{i}\Delta t\,H_1}\mathrm{e}^{-\mathrm{i}\Delta t\,H_2}\big)^{L-1}\mathrm{e}^{-\mathrm{i}\Delta t\,H_1}\mathrm{e}^{-\mathrm{i}\frac{\Delta t}{2}H_2}$, so that a length-$L$ symmetric simulation costs only one more H_2-exponential than the corresponding first-order run. Count the elementary exponentials of both and compare.

Exercise 11.4. *The Suzuki coefficient.* Verify that the substep lengths in (11.13) sum correctly, $4p_k + (1-4p_k) = 1$, and that $p_k = 1/(4-4^{1/(2k+1)}) > 1/4$, so the middle substep runs backward in time. Using (11.14), show the number of steps needed for accuracy ϵ is $L = \mathcal{O}(t^{1+1/(2k)}\epsilon^{-1/(2k)})$, and that the per-step gate count is $2\cdot 5^{k-1}$ times that of S_2.

Exercise 11.5. *Commutator error bound via Duhamel.* Fill in the Duhamel argument of Section 11.4: derive (11.17), integrate to (11.18), and using $G(s) = \mathrm{e}^{-\mathrm{i}H_1 s}H_2\mathrm{e}^{\mathrm{i}H_1 s} - H_2$ show $\left\|[\mathrm{e}^{-\mathrm{i}H_1 s}, H_2]\right\| \le s\,\|[H_1,H_2]\|$. Conclude $\left\|\mathrm{e}^{-\mathrm{i}\Delta t\,H_1}\mathrm{e}^{-\mathrm{i}\Delta t\,H_2} - \mathrm{e}^{-\mathrm{i}\Delta t\,H}\right\| \le \frac{1}{2}(\Delta t)^2\,\|[H_1,H_2]\|$, and recover the operator-norm bound $\le \Delta t^2\nu^2$ with $\nu = \max\{\|H_1\|, \|H_2\|\}$.

Exercise 11.6. *Commutator scaling for local models.* For the TFIM $H = -\sum_i Z_iZ_{i+1} - g\sum_i X_i$ on n qubits, show $\|H_1\|, \|H_2\| = \mathcal{O}(n)$ so the operator-norm bound gives $L = \mathcal{O}(n^2t^2/\epsilon)$, but $\|[H_1,H_2]\| = \mathcal{O}(n)$ because only $\mathcal{O}(n)$ term-pairs fail to commute, giving the commutator bound $L = \mathcal{O}(nt^2/\epsilon)$. Explain why this factor-n gain is generic for geometrically local Hamiltonians.

Exercise 11.7. *qDRIFT is term-count independent.* For $H = \sum_\gamma h_\gamma H_\gamma$ with $\|H_\gamma\| = 1$ and $\lambda = \sum_\gamma h_\gamma$, explain why the qDRIFT gate count $N_{\mathrm{q}} = \mathcal{O}(\lambda^2t^2/\epsilon)$ of (11.24) is independent of the number of terms Γ and of the commutators, and contrast this with the deterministic first-order bound $L = \mathcal{O}(\Gamma(\alpha_{\mathrm{comm}}t^2/\epsilon))$. For which Hamiltonians (locality vs. term count vs. weight distribution) does each method win?

Exercise 11.8. *Fourth-order versus second-order in practice.* Suppose you must simulate to time $t = 100$ with $\epsilon = 10^{-6}$. Using (11.9) and (11.15), estimate the step counts L for S_2 ($k=1$) and S_4 ($k=2$), and include the per-step factors (1 vs. $2\cdot 5^1 = 10$). Which formula uses fewer total elementary exponentials here, and how does the crossover shift as $\epsilon \to 0$?

Further reading

- Lin Lin, *qasc notes*, Ch. 5 (Hamiltonian simulation by product formulas: Lie–Trotter and Strang splittings, Suzuki higher-order recursion, and the commutator-type error analysis).
- Lin Lin, *live notes*, Ch. 15 (operator splitting: first- and second-order Trotter, the Duhamel/-commutator error bound, and the transverse-field Ising and real-space Schrödinger examples).
- A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, *Theory of Trotter error with commutator scaling*, Phys. Rev. X **11**, 011020 (2021) (arXiv:1912.08854), for the tight nested-commutator bound (11.21) and its optimality for local Hamiltonians.
- M. Suzuki, *Fractal decomposition of exponential operators with applications to many-body theories and Monte Carlo simulations*, Phys. Lett. A **146**, 319–323 (1990), for the recursion (11.13); and E. Campbell, *Random compiler for fast Hamiltonian simulation*, Phys. Rev. Lett. **123**, 070503 (2019) (arXiv:1811.08017), for qDRIFT (11.24); with A. M. Childs, A. Ostrander, and Y. Su, *Faster quantum simulation by randomization*, Quantum **3**, 182 (2019) (arXiv:1805.08385), for random orderings.
- D. Brody, *The Joy of Quantum Computing*, Princeton Univ. Press, 2025, Ch. 9 (simulating quantum dynamics: why the Schrödinger evolution $\mathrm{e}^{-\mathrm{i}Ht}$ is the native task of a quantum computer, and the intuition behind splitting a Hamiltonian into simulable pieces) — `material/Brody_The Joy of Quantum Computing_Princeton_2025.pdf`.

Chapter 12

Hamiltonian Simulation II: Block-Encoding Methods

Learning objectives

- Simulate time-independent Hamiltonians with QSVT at optimal query cost.
- Apply the truncated Taylor-series method with oblivious amplitude amplification.
- Handle time-dependent Hamiltonians via the truncated Dyson series and interaction picture.
- State and motivate the no-fast-forwarding theorem.

The product formulas of Chapter 11 take as their raw material the individual term-evolutions $\mathrm{e}^{-\mathrm{i}H_\gamma s}$ and stitch them together. This chapter takes a different input: a *block encoding* of H (Chapter 6), i.e. a unitary U_H whose top-left corner equals H/α for some $\alpha \geq \|H\|$. From this single oracle we build $\mathrm{e}^{-\mathrm{i}Ht}$ directly, using the quantum singular value transformation (QSVT, Chapter 9) and the linear-combination-of-unitaries primitive (LCU, Chapter 7). The payoff is a query complexity

$$\mathcal{O}\Big(\alpha t + \log(1/\epsilon)\Big), \tag{12.1}$$

which is *additive* in the time and the log-inverse-precision—exponentially better in ϵ than the best product formula (11.16), and, as we prove in Section 12.5, optimal: no algorithm can do better than linear in αt. The trade-off is that block-encoding methods treat H as a monolithic object and cannot, by themselves, exploit the commutator structure or geometric locality that makes product formulas so efficient for lattice models; the two families are complementary.

We proceed in five steps. Section 12.1 builds $\mathrm{e}^{-\mathrm{i}Ht}$ from a block encoding by approximating cos and sin through the Jacobi–Anger expansion, running each through QSVT, and combining them by LCU—the route to the optimal scaling (12.1). Section 12.2 presents the older but conceptually transparent truncated Taylor-series method, an LCU of the powers $(-\mathrm{i}Ht)^k/k!$ cleaned up by oblivious amplitude amplification. Section 12.3 extends the setting to time-dependent $H(t)$, where the naive matrix exponential fails and the time-ordered (Dyson) exponential takes its place. Section 12.4 implements the truncated Dyson series as an LCU over a clock register and develops the interaction-picture method, which fast-forwards a large but easy term. Section 12.5 proves the no-fast-forwarding theorem, closing the loop by showing that (12.1) cannot be beaten in general. Throughout, $\|\cdot\|$ is the spectral norm, $U_H \in \mathrm{BE}_{\alpha,m}(H)$ denotes an (α, m)-block encoding of H on m ancillas, and we write $U_H \in \mathrm{BE}_{\alpha,m}(H, \epsilon)$ when the encoded operator is within ϵ of H in spectral norm.

12.1 QSVT and optimal-scaling simulation

Reduction to $\alpha = 1$. Given $U_H \in \mathrm{BE}_{\alpha,m}(H)$, the encoded operator is $A := H/\alpha$ with $\|A\| \le 1$. Since $\mathrm{e}^{-\mathrm{i}Ht} = \mathrm{e}^{-\mathrm{i}A(\alpha t)}$, the normalization α is absorbed into a rescaled time $\tau := \alpha t$, and we may assume $\|H\| \le 1$ at the price of simulating for the longer (dimensionless) time τ. Every query complexity below is stated in $\tau = \alpha t$, which is why α appears in (12.1). A block-encoding error is also harmless: if $U_{\tilde H} \in \mathrm{BE}_{1,m}(\tilde H)$ with $\|\tilde H - H\| \le \epsilon'$, Duhamel's formula (variation of constants) gives

$$\mathrm{e}^{-\mathrm{i}\tilde H t} - \mathrm{e}^{-\mathrm{i}Ht} = -\mathrm{i}\int_0^t \mathrm{e}^{-\mathrm{i}H(t-s)}(\tilde H - H)\,\mathrm{e}^{-\mathrm{i}\tilde H s}\,\mathrm{d}s, \qquad \left\|\mathrm{e}^{-\mathrm{i}\tilde H t} - \mathrm{e}^{-\mathrm{i}Ht}\right\| \le \epsilon' t, \tag{12.2}$$

so choosing $\epsilon' = \epsilon/(2t)$ makes the encoding error contribute at most $\epsilon/2$. We therefore assume an exact $U_H \in \mathrm{BE}_{1,m}(H)$ from now on.

The strategy: $\mathrm{e}^{-\mathrm{i}Ht} = \cos(Ht) - \mathrm{i}\sin(Ht)$. QSVT (Chapter 9) implements $f(A)$ for a *real* polynomial f of definite parity satisfying $|f(x)| \le 1$ on $[-1,1]$, using $\deg f$ applications of $U_H, U_H^\dagger$. The target $\mathrm{e}^{-\mathrm{i}tx}$ is complex, so we split it into its real and imaginary parts,

$$\mathrm{e}^{-\mathrm{i}Ht} = \cos(Ht) - \mathrm{i}\sin(Ht), \tag{12.3}$$

where $\cos(tx)$ is real and *even* and $\sin(tx)$ is real and *odd*—exactly the two parity classes QSVT handles. We approximate each by a polynomial, realize each by a QSVT circuit, and add them with a single LCU ancilla.

Jacobi–Anger expansion. The efficient polynomial approximation of the trigonometric functions is the Fourier–Chebyshev (Jacobi–Anger) series. With T_k the degree-k Chebyshev polynomial and J_ν the Bessel function of the first kind,

$$\cos(tx) = J_0(t) + 2\sum_{k=1}^{\infty}(-1)^k J_{2k}(t)\,T_{2k}(x), \qquad \sin(tx) = 2\sum_{k=0}^{\infty}(-1)^k J_{2k+1}(t)\,T_{2k+1}(x). \tag{12.4}$$

These converge extraordinarily fast because the Bessel coefficients decay super-exponentially once the index exceeds t: for $m \ge 0$,

$$|J_m(t)| \le \frac{1}{m!}\Big(\frac{t}{2}\Big)^m, \tag{12.5}$$

so truncating each series at Chebyshev degree $2K$ (resp. $2K{+}1$) leaves a tail bounded by $2\sum_{k>2K}|J_k(t)| = \mathcal{O}\big((\mathrm{e}t/(2K{+}1))^{2K+1}\mathrm{e}^{t/2}\big)$. The Bessel index must clear t before the factorial dominates; beyond that point each extra degree buys a constant-factor reduction in error. This gives the following truncation lemma.

Lemma 12.1 (Jacobi–Anger truncation degree)**.** *For any $t > 0$ and $\epsilon \in (0, \mathrm{e}^{-1})$ there is a truncation order*

$$K = \Theta\Big(t + \frac{\log(1/\epsilon)}{\log(\mathrm{e} + \log(1/\epsilon)/t)}\Big) = \mathcal{O}\big(t + \log(1/\epsilon)\big) \tag{12.6}$$

such that the degree-$2K$ truncation $C(x)$ of $\cos(tx)$ and the degree-$(2K{+}1)$ truncation $S(x)$ of $\sin(tx)$ obey $\sup_{x\in[-1,1]}|\cos(tx) - C(x)| \le \epsilon$ and $\sup_{x\in[-1,1]}|\sin(tx) - S(x)| \le \epsilon$.

Proof sketch. Bound the truncated tail by (12.5); summing the geometric-like remainder shows the error is controlled by a model inequality of the form $(a/K)^K \le \epsilon$ with $a = \Theta(t)$. That inequality is solved exactly by the Lambert-W function: its unique positive root is $K^* = \ln(1/\epsilon)/W(\ln(1/\epsilon)/a)$, and the elementary estimate $\ln(1+z) \ge z/(1+z)$ applied at $K = a + \ln(1/\epsilon)$ gives the crude bound $K^* \le a + \ln(1/\epsilon) = \mathcal{O}(t + \log(1/\epsilon))$; a sharper analysis of W yields the refined logarithmic denominator in (12.6). □

Assembling the circuit. Choose K by Theorem 12.1 so that $C(x), S(x)$ each approximate their targets to precision $\epsilon/2$. Because a QSVT-realizable polynomial must satisfy $|p(x)| \le 1$ on $[-1,1]$ but $\max|C|, \max|S|$ can slightly exceed 1, rescale by $\beta := 1+\epsilon$ so that $\beta^{-1}C$ and $\beta^{-1}S$ are bounded by 1. The combined error is

$$\max_{x\in[-1,1]} \left|(C(x) - \mathrm{i}S(x)) - \mathrm{e}^{-\mathrm{i}tx}\right| \le \sqrt{(\epsilon/2)^2 + (\epsilon/2)^2} = \epsilon/\sqrt{2} < \epsilon. \tag{12.7}$$

Feeding the (even) phase factors for $\beta^{-1}C$ and the (odd) phase factors for $\beta^{-1}S$ into QSVT produces block encodings $U_C \in \mathrm{BE}_{\beta,m+1}(\cos(Ht), \epsilon/2)$ and $U_S \in \mathrm{BE}_{\beta,m+1}(\sin(Ht), \epsilon/2)$. One further LCU ancilla combines them as $C(H) - \mathrm{i}S(H)$: the prepare step creates $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and the select step applies U_C conditioned on $|0\rangle$ and (the $-$i-phased) U_S conditioned on $|1\rangle$. The result is $U_\Phi \in \mathrm{BE}_{2\beta,m+2}(\mathrm{e}^{-\mathrm{i}Ht}, \mathcal{O}(\epsilon))$. Only the sin branch needs a *controlled* U_H; the cos branch uses uncontrolled queries.

Theorem 12.2 (Optimal time-independent simulation, Low–Chuang)**.** *From $U_H \in \mathrm{BE}_{1,m}(H)$ and for any $\epsilon \in (0,1)$, $t > 0$, one can construct a block encoding $U_\Phi \in \mathrm{BE}_{2(1+\epsilon),m+2}(\mathrm{e}^{-\mathrm{i}Ht}, \epsilon)$ using*

$$\mathcal{O}\Big(t + \frac{\log(1/\epsilon)}{\log(\mathrm{e} + \log(1/\epsilon)/t)}\Big) = \mathcal{O}(t + \log(1/\epsilon)) \tag{12.8}$$

applications of U_H, $U_H^\dagger$, plus one controlled U_H. Restoring the normalization α, the cost is $\mathcal{O}(\alpha t + \log(1/\epsilon))$ queries to a block encoding $U_H \in \mathrm{BE}_{\alpha,m}(H)$. This matches the lower bound of Section 12.5 and is therefore optimal.

Proof. Immediate from Theorem 12.1 and the LCU combination above: each of U_C, U_S costs $2K = \mathcal{O}(t + \log(1/\epsilon))$ queries, and the LCU doubles the count and adds two ancillas. The error (12.7) propagates through the (unitary) LCU circuit to give the stated ϵ. □

Cleaning up the subnormalization. The block encoding U_Φ has subnormalization $2\beta = 2(1+\epsilon)$, so a bare measurement of the $|0^{m+2}\rangle$ flag succeeds with amplitude $\approx 1/2$. Because the target $\mathrm{e}^{-\mathrm{i}Ht}$ is itself *unitary*, we may deterministically boost the amplitude to 1 with a single round of *oblivious* amplitude amplification (Chapter 7)—"oblivious" because it works for the unknown input state $|\psi\rangle$ without measuring it. Using U_Φ twice and $U_\Phi^\dagger$ once yields $U \in \mathrm{BE}_{1,m+2}(\mathrm{e}^{-\mathrm{i}Ht}, \mathcal{O}(\epsilon))$, at the cost of 3 controlled U_H applications. To simulate for a longer time $T = Lt$ one simply concatenates L copies of U, with error growing linearly as $\mathcal{O}(L\epsilon)$.

Contrast with product formulas. Two structural differences with Chapter 11 stand out. First, precision: QSVT achieves $\log(1/\epsilon)$ cost where even arbitrarily-high-order Suzuki formulas pay $(1/\epsilon)^{o(1)}$ (equation (11.16)). Second, structure: the product formula bound (11.22) is governed by nested-commutator norms α_{comm}, which for a geometrically local Hamiltonian are far smaller than $\alpha = \|H\|$; the block-encoding cost (12.1) sees only α and cannot exploit that locality. Thus for a dense Hamiltonian, or when ϵ is tiny, QSVT wins; for a local lattice model at moderate precision, a Trotter formula may still be cheaper. Modern simulations often combine both.

Example 12.3 (Truncation degree for $\|H\|\,t = 4\pi$)**.** Take $\|H\|\,t = \tau = 4\pi \approx 12.57$ and target $\epsilon = 10^{-3}$. The Bessel coefficient $J_m(\tau)$ becomes negligible once $m \gtrsim \tau$; numerically one finds that Chebyshev degree $d = 2K \approx 24$ already gives a QSVT representation error near 10^{-2}, while $d \approx 50$ pushes the uniform error on $[-1, 1]$ below 10^{-6}. The crude bound of Theorem 12.1 predicts $K \lesssim \tau + \ln(1/\epsilon) \approx 12.57 + 6.9 \approx 19.5$, i.e. Chebyshev degree ≈ 39, comfortably consistent with the empirical $d \approx 24$–50 range: past the threshold $m \approx \tau$, each additional pair of Chebyshev degrees cuts the error by a constant factor, so the degree grows only additively in $\log(1/\epsilon)$. This is the hallmark additive scaling of (12.1).

12.2 Truncated Taylor series

An LCU of powers. Predating the QSP methods, the truncated Taylor-series method of Berry–Childs–Cleve–Kothari–Somma offers a different and very concrete route to nearly-optimal simulation, and is the natural warm-up for the time-dependent Dyson method of Section 12.4. Assume $\|H\| \le 1$. For a short step Δt, truncate the Taylor series of the propagator after order $K - 1$:

$$\mathrm{e}^{-\mathrm{i}H\Delta t} \approx \sum_{k=0}^{K-1} \frac{(-\mathrm{i}H\Delta t)^k}{k!}. \tag{12.9}$$

This is a *linear combination of the unitaries* implicit in the powers of H: each $(-\mathrm{i}H)^k$ is a product of k copies of the block encoding U_H (with a global phase), and the coefficients $c_k = (\Delta t)^k/k!$ are nonnegative. We implement it with the standard LCU pair

$$U_{\mathrm{SEL}} = \sum_{k=0}^{K-1} |k\rangle\langle k| \otimes U_k, \qquad V_{\mathrm{PREP}}\,|0\rangle = \frac{1}{\sqrt{\|c\|_1}} \sum_{k=0}^{K-1} \sqrt{c_k}\;|k\rangle\,, \tag{12.10}$$

where U_k is a block encoding of $(-\mathrm{i}H)^k$ and $\|c\|_1 = \sum_k c_k$ is the coefficient 1-norm.

Choosing the step so the 1-norm is 2. The LCU subnormalization equals $\|c\|_1$. Picking $\Delta t = \ln 2$ makes the full (untruncated) series sum to $\sum_{k\ge 0}(\Delta t)^k/k! = \mathrm{e}^{\Delta t} = 2$, so $\|c\|_1 \le 2$ for any finite K, with $|\,\|c\|_1 - 2\,|$ controlled by the Taylor remainder. The LCU therefore produces a block encoding of $\mathrm{e}^{-\mathrm{i}H\Delta t}$ with subnormalization ≤ 2. Because the target is unitary, one round of oblivious amplitude amplification (two uses of the LCU walk and one of its inverse) removes the factor of 2, yielding

$$W \in \mathrm{BE}_{1,a}(\mathrm{e}^{-\mathrm{i}H\Delta t}, \epsilon'), \qquad a = \mathcal{O}(\log K + m). \tag{12.11}$$

Concatenating segments and choosing K. For total time $t = L\Delta t$ we chain L segments, $\mathrm{e}^{-\mathrm{i}Ht} \approx W^L$, with error growing linearly: $\|(\langle 0^a|\,W\,|0^a\rangle)^L - \mathrm{e}^{-\mathrm{i}Ht}\| \le L\epsilon'$, so we take $\epsilon' = \epsilon/L = \epsilon\Delta t/t$. The per-segment truncation error is the Taylor tail

$$\Big\|\sum_{k=K}^{\infty} \frac{(-\mathrm{i}H\Delta t)^k}{k!}\Big\| \le \frac{(\|H\|\,\Delta t)^K}{K!}\,\mathrm{e}^{\|H\|\Delta t} \le \Big(\frac{\mathrm{e}\,\|H\|\,\Delta t}{K}\Big)^K \mathrm{e}^{\|H\|\Delta t} \le \epsilon', \tag{12.12}$$

using $K! \ge (K/\mathrm{e})^K$. With $\|H\| \le 1$ and $\Delta t = \ln 2$ the base is $\mathcal{O}(1)$, so the model inequality $(a/K)^K \le \epsilon'$ with $a = \mathcal{O}(1)$ has (via the Lambert-W estimate of Theorem 12.1) the solution

$$K = \mathcal{O}\Big(\frac{\log(1/\epsilon')}{\log\log(1/\epsilon')}\Big) = \mathcal{O}\Big(\frac{\log(t/\epsilon)}{\log\log(t/\epsilon)}\Big). \tag{12.13}$$

Because there are $L = \mathcal{O}(t)$ segments (each of fixed duration $\ln 2$), each costing $\mathcal{O}(K)$ queries to U_H, the total query complexity is

$$\mathcal{O}\Big(t\,\frac{\log(t/\epsilon)}{\log\log(t/\epsilon)}\Big), \tag{12.14}$$

with $\mathcal{O}(m + \log\log(t/\epsilon))$ ancilla qubits. This is *near*-optimal: linear in t, but with an extra $\log(t/\epsilon)/\log\log(t/\epsilon)$ factor that the QSVT method of Theorem 12.2 removes. Its virtue is conceptual transparency and its role as the template for the Dyson method.

Example 12.4 (Taylor order at $\epsilon = 10^{-10}$)**.** For a segment with $\epsilon' = 10^{-10}$, the estimate (12.13) gives $K \approx \log(10^{10})/\log\log(10^{10}) = \ln(10^{10})/\ln(\ln 10^{10}) \approx 23.0/\ln(23.0) \approx 23.0/3.14 \approx 7.3$, so a truncation order $K \approx 8$ already delivers ten digits of accuracy per segment—the doubly-logarithmic denominator makes the required order essentially flat in ϵ. Contrast the product-formula precision cost $(1/\epsilon)^{1/(2k)}$ of (11.15), which at $\epsilon = 10^{-10}$ and fourth order ($k = 2$) inflates the step count by $(10^{10})^{1/4} \approx 316$.

12.3 Time-dependent Hamiltonians

Why the matrix exponential fails. Many important settings—adiabatic algorithms, control fields, the interaction picture below—involve a Hamiltonian $H(t)$ that changes in time. The Schrödinger equation is now $\partial_t |\psi(t)\rangle = -\mathrm{i}H(t)\,|\psi(t)\rangle$. When H commutes with itself at different times, $[H(s), H(s')] = 0$, the solution is the naive exponential of the integrated Hamiltonian; e.g. for $H(t) = (T - t)\mathbb{I} + tX$,

$$U(t) = \mathrm{e}^{-\mathrm{i}\int_0^t H(s)\,\mathrm{d}s} = \mathrm{e}^{-\mathrm{i}(\mathbb{I}(Tt - t^2/2) + t^2 X/2)}, \tag{12.15}$$

which one verifies by differentiation. But this breaks the moment the Hamiltonian fails to commute with itself: for $H(t) = (T - t)Z + tX$ one has $[H(t), H(t')] \neq 0$ for $t \neq t'$, and

$$U(t) \neq \mathrm{e}^{-\mathrm{i}\int_0^t H(s)\,\mathrm{d}s}, \tag{12.16}$$

because the chain rule no longer lets us differentiate $\mathrm{e}^{-\mathrm{i}\int_0^t H(s)\mathrm{d}s}$ into $-\mathrm{i}H(t)$ times itself. Time dependence introduces genuinely new features; the correct object is the *time-ordered* exponential.

Theorem 12.5 (Ordered exponential and Dyson series)**.** *Let $A\colon [0,t] \to \mathbb{C}^{d\times d}$ be continuous. The operator ODE $\partial_t U(t) = A(t)U(t)$ with $U(0) = \mathbb{I}$ has a unique solution, the* ordered operator exponential*, given by the convergent series*

$$U(t) = \mathcal{T}\mathrm{e}^{\int_0^t A(s)\mathrm{d}s} = \mathbb{I} + \sum_{n=1}^{\infty} \int_0^t \int_0^{s_n} \cdots \int_0^{s_2} A(s_n)\cdots A(s_1)\,\mathrm{d}s_1 \cdots \mathrm{d}s_n. \tag{12.17}$$

Proof. By the fundamental theorem of calculus, $U(t) = \mathbb{I} + \int_0^t A(s)U(s)\,\mathrm{d}s$. Iterating this integral equation p times reproduces the first p terms of (12.17) plus a remainder $R_{p+1}(t) = \int_0^t \int_0^{s_{p+1}} \cdots \int_0^{s_2} A(s_{p+1})\cdots A(s_1)U$ Let $M = \sup_{s\in[0,t]} \|A(s)\| < \infty$. Since $\|U(s)\| \le \mathrm{e}^{\int_0^s \|A(\tau)\|\mathrm{d}\tau} \le \mathrm{e}^{Mt}$, submultiplicativity and the volume $t^{p+1}/(p+1)!$ of the ordered simplex give $\|R_{p+1}(t)\| \le \mathrm{e}^{Mt}(Mt)^{p+1}/(p+1)! \to 0$ as $p \to \infty$. Uniqueness is standard Picard–Lindelöf. When $A(t) \equiv A$ is constant the nested integrals collapse to $A^n t^n/n!$ and (12.17) is the ordinary Taylor series e^{At}—the Dyson series is precisely the time-dependent analogue. □

Time ordering. An equivalent bookkeeping uses the *time-ordering* symbol $\mathcal{T}$, which sorts a product of operators into nonincreasing time order: $\mathcal{T}[A(s_1)A(s_2)] = A(s_1)A(s_2)$ if $s_1 \geq s_2$ and $A(s_2)A(s_1)$ otherwise. Since each of the $n!$ orderings of $(s_1, \dots, s_n)$ contributes the same integral over the full cube, $\int_{[0,t]^n} \mathcal{T}[A(s_1)\cdots A(s_n)]\mathrm{d}s = n! \int_{\text{simplex}} A(s_n)\cdots A(s_1)\mathrm{d}s$, whence the compact form

$$U(t) = \mathcal{T}\mathrm{e}^{\int_0^t A(s)\mathrm{d}s} = \mathbb{I} + \sum_{n=1}^{\infty} \frac{1}{n!} \int_{[0,t]^n} \mathcal{T}[A(s_1)\cdots A(s_n)]\,\mathrm{d}s_1 \cdots \mathrm{d}s_n. \tag{12.18}$$

The symbol $\mathcal{T}$ is a *prescription*, not an operator on the Hilbert space; the Dyson-series form (12.17) is usually cleaner to manipulate, but the time-ordered form organizes the terms we must implement in the algorithm.

Block-encoding a time-dependent Hamiltonian. To simulate $H(t)$ we must block encode not just the operator but also its time argument. Discretize $[0, T]$ into 2^{n_t} steps of size $\Delta t = T/2^{n_t}$ and round to the piecewise-constant $\tilde{H}(t) = H(t_\ell)$, $t_\ell = \ell\Delta t$, $\ell = \lfloor t/\Delta t \rfloor$. By Taylor's theorem the local error is $\|H(t) - \tilde{H}(t)\| \leq \max_s \|H'(s)\|\,\Delta t$, and the induced error in the ordered exponential over $[0, T]$ is

$$\left\| \mathcal{T}\mathrm{e}^{-\mathrm{i}\int_0^T H(s)\mathrm{d}s} - \mathcal{T}\mathrm{e}^{-\mathrm{i}\int_0^T \tilde{H}(s)\mathrm{d}s} \right\| \leq \max_s \left\|H'(s)\right\|\, T\,\Delta t. \tag{12.19}$$

To keep this below ϵ_T it suffices to take

$$n_t = \mathcal{O}\Big(\log \frac{\max_s \|H'(s)\|\, T^2}{\epsilon_T}\Big), \tag{12.20}$$

so the time register grows only *logarithmically* in the variation $\max_s \|H'(s)\|$ and in $1/\epsilon_T$. The discretized Hamiltonian is then encoded via a *clock register* $|\ell\rangle$ tensored with the system block encoding.

Definition 12.6 (HAM-T oracle)**.** For $H(t) \in \mathbb{C}^{2^n \times 2^n}$ and normalization $\alpha_T > 0$, the oracle HAM-T on $n_t + m + n$ qubits block encodes the block-diagonal operator indexed by the clock,

$$(\mathbb{I} \otimes \langle 0^m| \otimes \mathbb{I})\,\text{HAM-T}\,(\mathbb{I} \otimes |0^m\rangle \otimes \mathbb{I}) = \sum_{\ell=0}^{2^{n_t}-1} |\ell\rangle\langle\ell| \otimes \frac{H(\ell T/2^{n_t})}{\alpha_T}. \tag{12.21}$$

Standard-form construction. To build HAM-T, write $H(t) = \sum_{j=0}^{M-1} c_j(t) U_j(t)$ with $c_j(t) \geq 0$ and $U_j(t)$ unitary (signs and phases absorbed into the U_j). A time-aware SELECT applies $U_j(\ell T/2^{n_t})$ conditioned on the clock $|\ell\rangle$ and index $|j\rangle$, and a PREP-T oracle prepares the coefficients $|\ell\rangle|0\rangle \mapsto |\ell\rangle \sum_j \sqrt{c_j(\ell T/2^{n_t})/\alpha}\,|j\rangle$ with $\alpha \geq \max_\ell \sum_j c_j$; then HAM-T $=$ (PREP-T$^\dagger \otimes \mathbb{I}$) SELECT (PREP-T $\otimes \mathbb{I}$) by the LCU lemma of Chapter 7. Time-dependent phases $\mathrm{e}^{\pm\mathrm{i}\omega t_\ell}$ are realized by a Hermitian clock operator $\mathcal{C} = \sum_\ell (\ell/2^{n_t})\,|\ell\rangle\langle\ell|$: the unitary $\mathrm{e}^{\pm\mathrm{i}\omega T\mathcal{C}}$ applies the phase $\mathrm{e}^{\pm\mathrm{i}\omega\ell T/2^{n_t}} = \mathrm{e}^{\pm\mathrm{i}\omega t_\ell}$ to $|\ell\rangle$ by phase kickback, and factors into n_t single-qubit R_z rotations on the clock.

Example 12.7 (Clock encoding of $H(t) = X + \sin(\omega t)Z$)**.** The coefficient $\sin(\omega t)$ changes sign, so we split it into complex exponentials to restore nonnegative coefficients:

$$H(t) = X + \tfrac{1}{2}(-\mathrm{i}\mathrm{e}^{\mathrm{i}\omega t})Z + \tfrac{1}{2}(\mathrm{i}\mathrm{e}^{-\mathrm{i}\omega t})Z, \tag{12.22}$$

so the coefficients are $c = [1, \tfrac{1}{2}, \tfrac{1}{2}]$ and the unitaries are $[X, -\mathrm{i}\mathrm{e}^{\mathrm{i}\omega t}Z, \mathrm{i}\mathrm{e}^{-\mathrm{i}\omega t}Z]$. A uniform PREP-T (Hadamards on the index register) suffices, the clock phases $\mathrm{e}^{\pm\mathrm{i}\omega T\mathcal{C}}$ are applied inside SELECT by

phase kickback, and the $\pm\mathrm{i}$ factors are set by a fixed single-qubit gate (a phase gate conjugated by X). This small example already shows the two ingredients: nonnegative LCU coefficients plus a clock that carries the time dependence.

12.4 Truncated Dyson series and interaction picture

Dyson as an LCU over the clock. With HAM-T in hand we implement the truncated Dyson series just as we did the truncated Taylor series, except that the powers H^k are replaced by time-ordered products $\tilde{H}(s_k)\cdots\tilde{H}(s_1)$ integrated over the ordered simplex. Truncating (12.17) at order K and using piecewise-constancy turns each n-fold integral into a finite sum over clock values,

$$\mathcal{T}\mathrm{e}^{-\mathrm{i}\int_0^T \tilde{H}(s)\mathrm{d}s} \approx \mathbb{I} + \sum_{k=1}^{K} \frac{(-\mathrm{i})^k}{k!} \sum_{s_1,\dots,s_k\in S} \mathcal{T}\big[\tilde{H}(s_k)\cdots\tilde{H}(s_1)\big](\Delta t)^k, \tag{12.23}$$

where $S = \{\ell\Delta t\}$. The domain grows exponentially in k, but—exactly as in the Taylor method—the LCU cost depends only on the coefficient 1-norm, not on the number of terms, so implementation is not exponential in k. The one genuinely new ingredient is time ordering: each order-k term needs its own k clock registers, and a reversible sorting network arranges the factors $\tilde{H}(s_k)\cdots\tilde{H}(s_1)$ into nonincreasing time order before applying them.

Lemma 12.8 (Short-segment Dyson simulation)**.** *Let $H\colon[0,T]\to\mathbb{C}^{2^n\times 2^n}$ be differentiable with a standard-form LCU decomposition, and let $T\in[0,\ln 2/\alpha_T]$. There is an algorithm implementing V with, for some $\beta\le 2+\mathcal{O}(\epsilon)$,*

$$\big\|\beta(\langle 0|\otimes\mathbb{I})V(|0\rangle\otimes\mathbb{I}) - \mathcal{T}\mathrm{e}^{-\mathrm{i}\int_0^T H(s)\mathrm{d}s}\big\| \le \epsilon, \tag{12.24}$$

using n_t satisfying (12.20) and $\mathcal{O}(\log(1/\epsilon)/\log\log(1/\epsilon))$ queries to HAM-T.

Proof sketch. Split the error into discretization and truncation parts. The discretization part is $\le\epsilon/2$ by (12.19) once n_t obeys (12.20) with $\epsilon_T=\epsilon/2$. The truncation part is the Dyson tail, bounded exactly as the Taylor tail: at order K it is $\le(\alpha_T T)^{K+1}/(K+1)!$. Choosing Δt so that $\alpha_T T=\ln 2$ makes the base $\mathcal{O}(1)$, and the Lambert-W estimate of Theorem 12.1 gives $K=\mathcal{O}(\log(\alpha_T T/\epsilon)/\log\log(\alpha_T T/\epsilon))=\mathcal{O}(\log(1/\epsilon)/\log\log(1/\epsilon))$. Each Dyson order costs one HAM-T query, so K queries suffice. □

Theorem 12.9 (Long-time Dyson simulation)**.** *Under the hypotheses of Theorem 12.8, for any $\epsilon,T>0$ there is a unitary U with $\|(\langle 0|\otimes\mathbb{I})U(|0\rangle\otimes\mathbb{I}) - \mathcal{T}\mathrm{e}^{-\mathrm{i}\int_0^T H(s)\mathrm{d}s}\|\le\epsilon$ using n_t time qubits from (12.20) and $\widetilde{\mathcal{O}}(\alpha_T T\log(1/\epsilon))$ queries to* HAM-T.

Proof sketch. Divide $[0,T]$ into $r-1$ segments of duration $\ln 2/\alpha_T$ (so $\alpha_T T_{\mathrm{seg}}=\ln 2$) plus one shorter final segment. Each full segment has success probability $\approx 1/4$ and is boosted to ≈ 1 by one round of oblivious amplitude amplification; the final segment is amplified by fixed-point amplitude amplification (Chapter 3). Concatenating the $r=\mathcal{O}(\alpha_T T)$ segment circuits, each costing $\mathcal{O}(\log(1/\epsilon)/\log\log(1/\epsilon))$ queries, gives the stated $\widetilde{\mathcal{O}}(\alpha_T T\log(1/\epsilon))$. By allowing segment durations to adapt to a locally smaller $\alpha_T(t)$, the count sharpens to $\widetilde{\mathcal{O}}(\int_0^T\alpha_T(t)\mathrm{d}t\cdot\log(1/\epsilon))$. □

This is the near-optimal time-dependent result $\mathcal{O}(\max_t\|H(t)\|\,T\log(\cdots))$ quoted in the Dong An talk. It is a log factor short of the qubitization/QSVT optimum because time-dependent Hamiltonians have no global eigenbasis: qubitization builds a walk operator whose spectrum is that of a *fixed* H, and one cannot define a single walk operator that encodes an entire time-varying evolution. Bridging this gap remains open.

The interaction picture: fast-forwarding an easy term. Given the log overhead, why prefer Dyson methods? Because for a *time-independent* $H = \alpha A + \beta B$ with $\|A\|, \|B\| \le 1$ and $\beta \gg \alpha$, the dynamics is dominated by the large, "easy" term B. Pass to the frame rotating with B: define $|\psi(t)\rangle_I := \mathrm{e}^{\mathrm{i}\beta Bt}|\psi(t)\rangle$.

Proposition 12.10 (Interaction Hamiltonian)**.** *If* $\partial_t|\psi(t)\rangle = -\mathrm{i}(\alpha A + \beta B)|\psi(t)\rangle$ *and* $|\psi(t)\rangle_I = \mathrm{e}^{\mathrm{i}\beta Bt}|\psi(t)\rangle$*, then* $\partial_t|\psi(t)\rangle_I = -\mathrm{i}H_I(t)|\psi(t)\rangle_I$ *with*

$$H_I(t) = \alpha\,\mathrm{e}^{\mathrm{i}\beta Bt} A\,\mathrm{e}^{-\mathrm{i}\beta Bt}, \qquad \|H_I(t)\| = \alpha\,\|A\| \le \alpha. \tag{12.25}$$

Proof. By the product rule, $\partial_t|\psi(t)\rangle_I = \mathrm{i}\beta B\mathrm{e}^{\mathrm{i}\beta Bt}|\psi\rangle + \mathrm{e}^{\mathrm{i}\beta Bt}(-\mathrm{i})(\alpha A + \beta B)|\psi\rangle = -\mathrm{i}\alpha\mathrm{e}^{\mathrm{i}\beta Bt}A|\psi\rangle = -\mathrm{i}\alpha\mathrm{e}^{\mathrm{i}\beta Bt}A\mathrm{e}^{-\mathrm{i}\beta Bt}|\psi\rangle_I$; the βB terms cancel. Unitary conjugation preserves the spectral norm, giving $\|H_I(t)\| = \alpha\,\|A\| \le \alpha$. □

The interaction Hamiltonian has norm governed by the *small* coefficient α, not β. Applying Theorem 12.9 to H_I gives a query count of $\widetilde{\mathcal{O}}(\alpha T\log(1/\epsilon))$—the large term B no longer appears in the leading cost. The catch is that $H_I(t)$ oscillates rapidly, $H_I'(t) = \mathrm{i}\alpha\beta\mathrm{e}^{\mathrm{i}\beta Bt}[B,A]\mathrm{e}^{-\mathrm{i}\beta Bt}$, so the variation scale is set by β; but by (12.20) the time register grows only *logarithmically* in that variation, so it costs little. The remaining requirement is that we can implement $\mathrm{e}^{\mathrm{i}\beta Bt}$ efficiently—i.e. that B can be *fast-forwarded* (simulated in time sublinear in βt), which holds for one-sparse B (Pauli strings), Fourier-diagonal kinetic terms, and other special cases.

Example 12.11 (Particle in a periodic potential)**.** Discretize $H = \frac{p^2}{2m} + \omega\cos^2(x)$ on a grid of $2N+1$ points. The kinetic term is circulant—diagonal in the Fourier basis, $\frac{p^2}{2m} = \mathrm{QFT}^\dagger D_{\mathrm{kin}}\mathrm{QFT}$—and the potential is diagonal in position; the block-encoding normalization is $\alpha = \mathcal{O}(N^2/(mL^2) + |\omega|)$. Direct QSVT/Taylor simulation (Sections 12.1 and 12.2) then costs $\widetilde{\mathcal{O}}((N^2/(mL^2)+|\omega|)T\log(1/\epsilon))$ queries, which *diverges* as we refine the grid ($N \to \infty$) to reduce discretization error. Now take the interaction picture with respect to the large kinetic term B (fast-forwardable via the QFT). Then $\|H_I\|$ is set by the *potential* coefficient $|\omega|$, and Theorem 12.9 gives $\widetilde{\mathcal{O}}(|\omega|T\log(1/\epsilon))$ queries—independent of N. The interaction picture makes an otherwise grid-limited simulation feasible even for exponentially fine grids, and underlies leading query complexities for chemistry and quantum field theory.

12.5 No-fast-forwarding theorem

Can we beat linear in αt? Some Hamiltonians *can* be fast-forwarded: $\mathrm{e}^{-\mathrm{i}Zt}$ is one single-qubit rotation for any t, cost independent of t; more generally any Hamiltonian diagonalized by an efficiently-implementable transform can be evolved in time independent of t. Could this be general? A heuristic recursion says no: if every Hamiltonian could be simulated in $\mathcal{O}(\sqrt{\int_0^T \alpha_T})$ queries, then a quantum computer simulating a quantum computer simulating … would compress any T-time dynamics to $T^{o(1)}$, collapsing all quantum computation to near-constant time—absurd. The rigorous statement comes from query complexity.

Theorem 12.12 (No fast-forwarding, Berry–Ahokas–Cleve–Sanders)**.** *There is a family of Hamiltonians H (specifically a 2-sparse, banded operator G_x with $\alpha = \|G_x\| = \Theta(n)$) for which any quantum algorithm implementing a channel W with induced trace distance $D_{\mathrm{Tr}}(W, \mathrm{e}^{-\mathrm{i}Ht}) \le 1/4$ must make $\Omega(\alpha t)$ queries to the block-encoding oracle. Consequently the linear-in-αt scaling of Theorem 12.2 is optimal, and no method can be uniformly fast-forwarding.*

Proof sketch. The argument reduces *parity* to Hamiltonian simulation; since computing the parity of an n-bit string s requires $\Omega(n)$ oracle queries (Chapter 3), a sub-linear-in-αt simulator would violate that bound.

The angular-momentum walk. Let J_x be the spin-$n/2$ angular-momentum operator with nonzero entries $\langle j+1| J_x |j\rangle = \frac{1}{2}\sqrt{(n-j)(j+1)}$. Using the $\mathfrak{su}(2)$ algebra $[J_x, J_z] = -\mathrm{i}J_y$, etc., a Heisenberg-picture computation gives $\mathrm{e}^{-\mathrm{i}J_x\pi} J_z \mathrm{e}^{\mathrm{i}J_x\pi} = \cos(\pi)J_z - \sin(\pi)J_y = -J_z$, whence $\mathrm{e}^{-\mathrm{i}J_x\pi}|0\rangle = |n\rangle$: evolving for time π walks the state from the bottom of the ladder to the top. Now build a "dual-rail" variant G_x that additionally toggles a parity qubit according to the bits of s: its lower-diagonal entries are $\langle j+1|\langle m \oplus s_j| G_x |j\rangle |m\rangle = \frac{1}{2}\sqrt{(n-j)(j+1)}$. On the permuted basis G_x is unitarily equivalent to $J_x \otimes \mathbb{I}$, so

$$\mathrm{e}^{-\mathrm{i}G_x\pi}|0\rangle|0\rangle = |n\rangle\left|\bigoplus_j s_j\right\rangle = |n\rangle\left|\sum_j s_j \bmod 2\right\rangle : \tag{12.26}$$

simulating G_x for time π deposits the parity of s in the parity register.

Query cost of the block encoding. G_x is 2-sparse and banded with largest entry $\mathcal{O}(n)$, so it is block-encoded (via the sparse-access construction of Chapter 6) with normalization $\alpha = \max_j \frac{1}{2}\sqrt{(n-j)(j+1)} = \Theta(n)$, using $\mathcal{O}(1)$ queries to the string oracle U_s per block-encoding call. Hence $\alpha t = \alpha\pi = \Theta(n)$.

Contradiction. Suppose a simulator existed using $o(\alpha t) = o(n)$ block-encoding queries to reach trace distance $\leq 1/4$. Composing with (12.26) would compute the parity of s using $o(n)$ queries to U_s within trace distance $1/4$—but no quantum algorithm can compute parity in $o(n)$ queries (Chapter 3). Therefore $\Omega(\alpha t)$ queries are necessary, and G_x cannot be fast-forwarded. □

Interpretation. The lower bound is not an artifact of a clumsy encoding: G_x is genuinely as hard as parity, and parity is genuinely $\Omega(n)$. The theorem thus pins down the landscape of Chapters 11 and 12. The optimal query complexity of any general-purpose Hamiltonian simulator is $\Theta(\alpha t + \log(1/\epsilon))$: QSVT (Theorem 12.2) achieves it, the no-fast-forwarding bound forbids beating the αt term, and the $\log(1/\epsilon)$ term is likewise information-theoretically necessary. Product formulas undercut α by α_{comm} for local Hamiltonians but pay a super-logarithmic precision price; block-encoding methods pay optimally in ϵ but see only α. Fast-forwarding is possible only for *structured* Hamiltonians (a known diagonalizing transform, commuting terms, or the interaction-picture reduction of a fast-forwardable dominant term), never in general.

Exercises

Exercise 12.1. *Block-encoding error is benign.* Starting from $U_{\tilde{H}} \in \mathrm{BE}_{1,m}(\tilde{H})$ with $\|\tilde{H} - H\| \leq \epsilon'$, derive the Duhamel identity (12.2) by differentiating $g(s) = \mathrm{e}^{-\mathrm{i}H(t-s)}\mathrm{e}^{-\mathrm{i}\tilde{H}s}$, and conclude $\|\mathrm{e}^{-\mathrm{i}\tilde{H}t} - \mathrm{e}^{-\mathrm{i}Ht}\| \leq \epsilon' t$. Hence show $\epsilon' = \epsilon/(2t)$ suffices to make the encoding error contribute at most $\epsilon/2$ to the simulation.

Exercise 12.2. *Parity split for QSVT.* Verify that $\cos(tx)$ is even and $\sin(tx)$ is odd in x, and that both are real. Explain why this parity/reality structure is exactly what real-QSVT requires (Chapter 9), and why the imaginary unit in $\mathrm{e}^{-\mathrm{i}Ht} = \cos(Ht) - \mathrm{i}\sin(Ht)$ forces the two-branch LCU combination rather than a single QSVT circuit.

Exercise 12.3. *Jacobi–Anger truncation and the Bessel tail.* Using the bound $|J_m(t)| \leq \frac{1}{m!}(t/2)^m$ of (12.5), show the degree-$2K$ truncation error of $\cos(tx)$ in (12.4) is $\mathcal{O}((\mathrm{e}t/(2K+1))^{2K+1}\mathrm{e}^{t/2})$. Explain qualitatively why the required degree behaves as $K = \mathcal{O}(t + \log(1/\epsilon))$: identify the threshold

index beyond which the factorial dominates, and why each further degree then gives constant-factor error decay.

Exercise 12.4. *Taylor order at high precision.* For a truncated-Taylor segment with $\|H\| \le 1$, $\Delta t = \ln 2$, verify $\|c\|_1 = \sum_{k=0}^{K-1} (\Delta t)^k/k! \le 2$ and derive the tail bound (12.12). Using $K! \ge (K/\mathrm{e})^K$ and the Lambert-W estimate, show $K = \mathcal{O}(\log(1/\epsilon')/\log\log(1/\epsilon'))$, and estimate K numerically for $\epsilon' = 10^{-8}$ (compare Theorem 12.4).

Exercise 12.5. *Dyson series for constant A.* Show that when $A(t) \equiv A$ is time-independent, the Dyson series (12.17) collapses to the ordinary Taylor series e^{At} by evaluating the ordered-simplex integrals $\int_0^t \int_0^{s_n} \cdots \int_0^{s_2} \mathrm{d}s_1 \cdots \mathrm{d}s_n = t^n/n!$. Then verify the remainder bound $\|R_{p+1}(t)\| \le \mathrm{e}^{Mt}(Mt)^{p+1}/(p+1)!$ used in the convergence proof of Theorem 12.5.

Exercise 12.6. *The interaction-picture equation.* Prove Theorem 12.10: for $H = \alpha A + \beta B$ and $|\psi(t)\rangle_I = \mathrm{e}^{\mathrm{i}\beta Bt}|\psi(t)\rangle$, show $\partial_t |\psi\rangle_I = -\mathrm{i}H_I(t)|\psi\rangle_I$ with $H_I(t) = \alpha \mathrm{e}^{\mathrm{i}\beta Bt} A \mathrm{e}^{-\mathrm{i}\beta Bt}$, and hence $\|H_I(t)\| = \alpha\|A\| \le \alpha$. Compute $H_I'(t)$ and explain, using (12.20), why its β-scale oscillation costs only $\mathcal{O}(\log\beta)$ extra clock qubits.

Exercise 12.7. *Angular-momentum walk and the no-fast-forwarding reduction.* Using the $\mathfrak{su}(2)$ relations $[J_x, J_y] = \mathrm{i}J_z$, $[J_y, J_z] = \mathrm{i}J_x$, $[J_z, J_x] = \mathrm{i}J_y$, expand $\mathrm{e}^{-\mathrm{i}J_x\pi} J_z \mathrm{e}^{\mathrm{i}J_x\pi}$ as a nested-commutator (Hadamard) series and sum it to $-J_z$, thereby proving $\mathrm{e}^{-\mathrm{i}J_x\pi}|0\rangle = |n\rangle$. Explain how the dual-rail modification G_x deposits the parity $\bigoplus_j s_j$ as in (12.26), and why $\alpha = \|G_x\| = \Theta(n)$ with $\alpha t = \Theta(n)$ then yields the $\Omega(\alpha t)$ lower bound via the $\Omega(n)$ parity bound of Chapter 3.

Further reading

- Lin Lin, *live notes*, Ch. 14 (block-encoding-based Hamiltonian simulation: QSP/QSVT with optimal query complexity, the truncated Taylor-series method, time-ordered exponentials and the Dyson series, block encoding of time-dependent operators, the truncated Dyson method, the interaction picture, and the no-fast-forwarding theorem).
- Dong An, *Quantum Numerical Linear Algebra* talk (Hamiltonian simulation: the $\mathrm{e}^{-\mathrm{i}\|H\|Tx} = \cos - \mathrm{i}\sin$ decomposition, the Jacobi–Anger polynomial approximation, the optimal $\mathcal{O}(\|H\|T + \log(1/\epsilon))$ QSVT scaling, and the near-optimal truncated-Dyson result for time-dependent $H(t)$).
- G. H. Low and I. L. Chuang, *Optimal Hamiltonian simulation by quantum signal processing*, Phys. Rev. Lett. **118**, 010501 (2017) (arXiv:1606.02685), for Theorem 12.2.
- D. W. Berry, A. M. Childs, R. Cleve, R. Kothari, and R. D. Somma, *Simulating Hamiltonian dynamics with a truncated Taylor series*, Phys. Rev. Lett. **114**, 090502 (2015) (arXiv:1412.4687), for Section 12.2.
- G. H. Low and N. Wiebe, *Hamiltonian simulation in the interaction picture*, arXiv:1805.00675 (2018); and M. Kieferová, A. Scherer, and D. W. Berry, *Simulating the dynamics of time-dependent Hamiltonians with a truncated Dyson series*, Phys. Rev. A **99**, 042314 (2019) (arXiv:1805.00582), for Section 12.4.
- D. W. Berry, G. Ahokas, R. Cleve, and B. C. Sanders, *Efficient quantum algorithms for simulating sparse Hamiltonians*, Commun. Math. Phys. **270**, 359–371 (2007) (arXiv:quant-ph/0508139), for the no-fast-forwarding theorem (12.26) of Section 12.5.

Chapter 13

Quantum Algorithms for Linear Differential Equations

Learning objectives

- Reduce linear ODEs and semi-discretized PDEs to problems solvable on a quantum computer, and identify the norm-growth obstruction absent from Hamiltonian simulation.
- Compare the linear-system, time-marching, and LCHS approaches and their scalings in $\alpha_A T$ and the norm-growth factors q, Q.
- Explain Schrödingerization: the warped phase transform that lifts a non-unitary linear ODE to a unitary Schrödinger evolution in one extra variable.
- Work the heat equation as a running example through several of these methods.

The previous two chapters simulated the Schrödinger equation $\partial_t |\psi\rangle = -\mathrm{i}H|\psi\rangle$, whose propagator $\mathrm{e}^{-\mathrm{i}Ht}$ is unitary and therefore norm-preserving. This chapter attacks the more general linear differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t}x(t) = A(t)\,x(t) + b(t), \qquad t \in [0,T], \qquad x(0) = x_0, \tag{13.1}$$

with $A(t) \in \mathbb{C}^{d\times d}$ not necessarily anti-Hermitian. Two things change at once. First, the propagator $\mathcal{T}\mathrm{e}^{\int_0^t A(s)\mathrm{d}s}$ is no longer unitary, so the "easy" route of the last chapter—block encode e^{At} directly and clean up with oblivious amplitude amplification (Chapter 12)—fails except for very short times, because that clean-up step requires the encoded operator to be close to unitary. Second, the *magnitude* $\|x(t)\|$ carries physical information (temperature decays, an absorbed wave loses norm), so the output model must track it. As we shall see, this magnitude reappears in every complexity estimate as a norm-growth factor.

We develop four algorithms. Section 13.1 fixes the problem, input, and output models and records Duhamel's principle, which reduces the inhomogeneous problem to the homogeneous one. Section 13.2 presents the historically first approach: discretize time, stack the whole solution history into one large linear system $\mathcal{A}\mathbf{x} = \mathbf{b}$, and solve it with a QLSP solver (Chapter 10); the cost is linear in $\alpha_A T$ but carries a condition-number and success-probability analysis. Section 13.3 adapts the classical time-marching viewpoint—advance one copy of the state step by step—using uniform singular value amplification; this trades the history state's ancilla and initial-state overhead for a quadratic dependence on $\alpha_A T$ and a growth factor Q. Section 13.4 exploits dissipative structure: writing $A = L + \mathrm{i}H$ with $L \succeq 0$, the LCHS identity expresses the non-unitary e^{-At} as an integral of *unitary* evolutions $\mathrm{e}^{-\mathrm{i}(H+kL)t}$, implemented by LCU (Chapter 7) plus Hamiltonian

simulation (Chapters 11 and 12), and recovers near-optimal cost in both oracles. Section 13.5 presents Schrödingerization, the PDE-native cousin of LCHS: a warped phase transform in one extra variable p turns the non-unitary dynamics into a genuine Schrödinger equation that is simulated unitarily and then projected. Section 13.6 runs the heat equation through these methods and compares the resulting resource scalings. Throughout, $\|\cdot\|$ is the spectral norm, $\mathcal{T}$ the time-ordering symbol of Section 12.3, and $\widetilde{\mathcal{O}}$ suppresses polylogarithmic factors.

13.1 Problem setup and semi-discretization

Semi-discretization of PDEs. The equations we care about often begin as partial differential equations. The prototypical example is the heat equation $\partial_t u = \partial_{rr} u$ on $r \in [0,1]$ with homogeneous Dirichlet data. Discretizing space on a uniform grid $r_j = jh$, $h = 1/(N+1)$, $j = 1, \dots, N$, and approximating the second derivative by the central difference $u''(r_j) \approx (u_{j-1} - 2u_j + u_{j+1})/h^2$ turns the PDE into the linear ODE system

$$\frac{\mathrm{d}}{\mathrm{d}t} u(t) = L\, u(t), \qquad L = \frac{1}{h^2} \begin{pmatrix} -2 & 1 & & \\ 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -2 \end{pmatrix}, \tag{13.2}$$

where $L \in \mathbb{R}^{N \times N}$ is the discrete Laplacian: symmetric and negative definite, with $\|L\| \approx 4/h^2 \approx 4N^2$. This "method of lines" is generic: a linear PDE, once its spatial operator is discretized on d^{th}-dimensional grid of N points per axis, becomes an ODE of the form (13.1) with A inheriting the sparsity and the norm of the discretized operator. The exponent of N in $\|A\|$ (here N^2) is what makes these problems demanding and is the origin of most of the cost below.

Duhamel's principle. For the homogeneous problem ($b = 0$) the solution is $x(t) = U(t,0)x_0$, where $U(t,s)$ is the propagator solving $\partial_t U(t,s) = A(t)U(t,s)$, $U(s,s) = \mathbb{I}$; explicitly $U(t,s) = \mathcal{T}\mathrm{e}^{\int_s^t A(\tau)\mathrm{d}\tau}$ (Theorem 12.5). The inhomogeneity is handled once and for all by the following classical identity.

Proposition 13.1 (Duhamel's principle)**.** *With $U(t,s)$ as above, the solution of (13.1) is*

$$x(t) = U(t,0)\, x_0 + \int_0^t U(t,s)\, b(s)\, \mathrm{d}s. \tag{13.3}$$

Proof. Write $R(t)$ for the right-hand side and differentiate, using $\partial_t U(t,s) = A(t)U(t,s)$ and $U(t,t) = \mathbb{I}$:

$$R'(t) = A(t)U(t,0)x_0 + U(t,t)b(t) + \int_0^t A(t)U(t,s)b(s)\,\mathrm{d}s = A(t)R(t) + b(t).$$

Since $R(0) = x_0$, uniqueness of solutions (Picard–Lindelöf, valid because A, b are Lipschitz) gives $R \equiv x$. □

Because the integral in (13.3) is a numerical quadrature of propagators $U(t,s)$ applied to $b(s)$, a block encoding of U together with LCU produces the inhomogeneous term. *We therefore restrict attention to the homogeneous problem $\dot{x} = A(t)x$* from now on. Equivalently, one may augment the system: with $\tilde{x} = (x; 1)$ and $\tilde{A} = \left(\begin{smallmatrix} A & b \\ 0 & 0 \end{smallmatrix}\right)$ the inhomogeneous equation becomes homogeneous in one higher dimension.

Input model. The initial state is prepared by a unitary $U_{x_0}|0\rangle = |x_0\rangle$ (assume $\|x_0\| = 1$). The matrix $A(t)$ is accessed, exactly as the HAM-T oracle of Theorem 12.6, through a block encoding on a time grid $\{t_k\}_{k=0}^{N_t-1}$, $N_t = 2^{n_t}$:

$$(\langle 0^{m_A}| \otimes \mathbb{I}_t \otimes \mathbb{I}_s)\, U_A\, (|0^{m_A}\rangle \otimes \mathbb{I}_t \otimes \mathbb{I}_s) = \frac{1}{\alpha_A} \sum_{k=0}^{N_t-1} |k\rangle\langle k| \otimes A(t_k), \tag{13.4}$$

with subnormalization $\alpha_A \geq \max_t \|A(t)\|$. The inhomogeneity, when present, is accessed by an analogous oracle U_b.

Output model and the norm-growth obstruction. Unlike Hamiltonian simulation, the solution norm can swing over many orders of magnitude. The scalar example $\dot{x} = -Zx$ on $\mathbb{C}^2$ has $x(t) = (x_0^{(0)}\mathrm{e}^{-t}, x_0^{(1)}\mathrm{e}^{t})$: one component decays, the other grows, so the *normalized* state and its norm are both nontrivial functions of t. We therefore define the quantum solution to be a block encoding of the *unnormalized* vector $x(T)$,

$$(\langle 0^{m_{\mathrm{sol}}}| \otimes \mathbb{I}_s)\, U_{\mathrm{sol}}\, (|0^{m_{\mathrm{sol}}}\rangle\, |x_0\rangle) = \frac{1}{\alpha_{\mathrm{sol}}}\, x(T), \tag{13.5}$$

from which $\|x(T)\|$ is read off through the success probability. The obstruction that rules out the naive strategy is this: a truncated Taylor block encoding of e^{At} has subnormalization $\approx \mathrm{e}^{\alpha t}$, growing *exponentially* in t; and oblivious amplitude amplification cannot rescue it because that primitive requires the encoded operator to be near-unitary, i.e. A near-anti-Hermitian. The remedy of every method below is to avoid exponentiating A as a monolithic block. The relevant norm-growth quantities are

$$q := \frac{\max_{0\leq t\leq T} \|x(t)\|}{\|x(T)\|}, \qquad Q := \frac{\sup_{0=t_0<\cdots<t_L=T} \prod_l \|U(t_{l+1}, t_l)\|}{\|x(T)\|}, \tag{13.6}$$

which quantify how much the algorithm must "fight" against the decay of the target relative to intermediate states. Both equal 1 when $\|x(t)\|$ is constant (e.g. A anti-Hermitian), and both blow up when $x(T)$ is much smaller than earlier states.

13.2 Linear-system-of-equations method

The history state. The oldest quantum algorithm for (13.1) (Berry, 2014) reformulates the whole evolution as one linear system. On a uniform grid $t_k = k\,\Delta t$, $\Delta t = T/N_t$, write $x_k = x(t_k)$ and encode the entire trajectory in a single *history state*

$$X(T) = \frac{1}{\sqrt{N_t}} \sum_{k=1}^{N_t} |t_k\rangle \otimes x_k, \tag{13.7}$$

the Feynman–Kitaev history vector. The forward Euler discretization $x_{k+1} - x_k = \Delta t\,(A_k x_k + b_k)$ (with $A_k = A(t_k)$) is precisely the statement that $X(T)$ solves a block-bidiagonal linear system $\mathcal{A}\mathbf{x} = \mathbf{b}$, with

$$\mathcal{A} = \begin{pmatrix} \mathbb{I} & & & \\ -(\mathbb{I}+\Delta t\, A_1) & \mathbb{I} & & \\ & \ddots & \ddots & \\ & & -(\mathbb{I}+\Delta t\, A_{N_t-1}) & \mathbb{I} \end{pmatrix}, \tag{13.8}$$

and right-hand side carrying the initial data $(\mathbb{I} + \Delta t\, A_0)x_0 + \Delta t\, b_0$ in its first block. Compactly,

$$\mathcal{A} = \mathbb{I}_t \otimes \mathbb{I}_s - \sum_{k=1}^{N_t-1} |t_{k+1}\rangle\langle t_k| \otimes (\mathbb{I}_s + \Delta t\, A_k), \tag{13.9}$$

so a block encoding of $\mathcal{A}$ needs one query to U_A (the shift $|t_k\rangle \mapsto |t_{k+1}\rangle$ is an adder circuit) and has subnormalization at most $2 + \alpha_A \Delta t = \mathcal{O}(1)$ once $\Delta t = \mathcal{O}(\alpha_A^{-1})$. Crucially, this route never forms the matrix power $(\mathbb{I} + \Delta t A)^{N_t}$ explicitly—the difficulty flagged in Chapter 9—because the QLSP solver inverts $\mathcal{A}$ rather than multiplying its blocks.

Padding for the final state. The final value x_{N_t} appears in (13.7) with weight $1/\sqrt{N_t}$, so measuring it directly is inefficient. Append N_t copies of x_{N_t} using one ancilla qubit, $\tilde{X} = |0\rangle \otimes \mathbf{x} + |1\rangle \otimes \mathbf{y}$ with $\mathbf{y} = (x_{N_t}, \dots, x_{N_t})$, enforced by additional rows $-x_{N_t-1+j} + x_{N_t+j} = 0$. Measuring the ancilla in $|1\rangle$ and succeeding with probability

$$\frac{\|x_{N_t}\|^2}{\|x_{N_t}\|^2 + \frac{1}{N_t}\sum_k \|x_k\|^2} \approx \frac{1}{1+\chi^2}, \qquad \chi^2 = \frac{\frac{1}{T}\int_0^T \|x(t)\|^2\,\mathrm{d}t}{\|x(T)\|^2}, \tag{13.10}$$

projects onto (a superposition dominated by) the final state; here $\chi \le q$. When $\|x(t)\| \equiv 1$ the success probability is exactly $1/2$.

Proposition 13.2 (Condition number of the block-bidiagonal system)**.** *Let $\{B_k\}_{k=1}^{m-1}$ be $n \times n$ blocks and $\mathcal{A}$ the block lower-bidiagonal matrix with $\mathbb{I}$ on the diagonal and $-B_k$ on the subdiagonal. Put $\zeta_0 = 1$ and $\zeta_k = \sup_{1\le l\le m-k} \|B_{l+k-1}\cdots B_l\|$. Then*

$$\kappa(\mathcal{A}) \le (\zeta_0 + \zeta_1) \sum_{k=0}^{m-1} \zeta_k. \tag{13.11}$$

Proof. Write $\mathcal{A} = \mathcal{B}_0 - \mathcal{B}_1$ with $\mathcal{B}_0 = \mathbb{I}$ and $\mathcal{B}_1$ the strictly-lower part holding the B_k. Since $\mathcal{B}_1^\dagger \mathcal{B}_1$ is block-diagonal with blocks $B_l^\dagger B_l$, we have $\|\mathcal{B}_1\| \le \sup_l \|B_l\| = \zeta_1$, hence $\|\mathcal{A}\| \le \zeta_0 + \zeta_1$. The inverse has the explicit lower-triangular form $\mathcal{A}^{-1} = \sum_{k=0}^{m-1} \mathcal{B}^{(k)}$, where the k-th sub-block-diagonal $\mathcal{B}^{(k)}$ collects the ordered products $B_{l+k-1}\cdots B_l$. The same block-diagonal argument gives $\left\|\mathcal{B}^{(k)}\right\| \le \zeta_k$, so by the triangle inequality $\left\|\mathcal{A}^{-1}\right\| \le \sum_{k=0}^{m-1} \zeta_k$. Multiplying the two bounds yields (13.11). □

Applying Theorem 13.2 with $B_k = \mathbb{I} + \Delta t\, A_k \approx \mathrm{e}^{\Delta t A_k}$ and $\zeta = \sup_{0\le s\le t\le T} \|U(t,s)\|$ gives $\zeta_k \le \zeta$ for all k, so $\kappa(\mathcal{A}) = \mathcal{O}(N_t\zeta) = \mathcal{O}(T\alpha_A\zeta)$. When the dynamics is contractive ($\left\|\mathrm{e}^{At}\right\| \le C_1 \mathrm{e}^{-C_2 t}$) the geometric series telescopes and $\left\|\mathcal{A}^{-1}\right\| = \mathcal{O}(1/\Delta t)$ independently of N_t, so $\kappa(\mathcal{A}) = \mathcal{O}(1/\Delta t)$. If instead A has an eigenvalue with positive real part, e^{At} and hence $\kappa(\mathcal{A})$ grow without bound in T: the condition number correctly diagnoses exponentially growing solutions.

Theorem 13.3 (Linear-system solver)**.** *For the homogeneous equation with $A(t)$ accessed by (13.4) on a grid with $0 < t_{k+1} - t_k \le \ln 2/\alpha_A$, using short-time Dyson propagators (Theorem 12.8) for the diagonal blocks, there is a quantum algorithm that prepares $|x(T)\rangle$ to error ϵ with success probability $\Omega(1)$ using*

$$\widetilde{\mathcal{O}}(\alpha_A T\, \zeta\, q) \tag{13.12}$$

queries to U_A and U_{x_0} (and their inverses), where ζ and q are as above.

Proof sketch. By Theorem 13.2, $\kappa(\mathcal{A}) = \mathcal{O}(N_t\zeta) = \mathcal{O}(T\alpha_A\zeta)$ with $N_t = \Theta(T\alpha_A)$ steps of size $\ln 2/\alpha_A$. A near-optimal QLSP solver (Chapter 10) inverts $\mathcal{A}$ in $\mathcal{O}(\kappa \operatorname{polylog}(\kappa/\epsilon))$ queries; perturbation theory for linear systems fixes the per-block accuracy $\epsilon' = \mathcal{O}(\epsilon/T\alpha_A\zeta)$, giving $\mathcal{O}(\log(1/\epsilon'))$ Dyson queries per block. Amplitude amplification of the padded-state success probability (13.10) contributes the factor q. □

The scaling is *linear* in $\alpha_A T$. Its drawback is that the QLSP solver queries U_A and U_{x_0} together, so the initial-state cost is also $\widetilde{\mathcal{O}}(\alpha_A T\zeta q)$; when U_{x_0} is expensive this is wasteful, and the next two methods improve it to $\mathcal{O}(q)$.

13.3 Time-marching method

Why naive marching fails. The classical instinct is to march: prepare x_0, apply the short-time propagator $B = \mathbb{I} + A\Delta t$ (block-encoded from U_A) once per step, and reach x_{N_t}. Each application is a block encoding with subnormalization $1 + \alpha_A\Delta t$, so a single measurement of the success flag succeeds with amplitude $\leq (1+\alpha_A\Delta t)^{-1} \|x_k\| / \|x_{k-1}\|$. Over all N_t steps the probability of success multiplies to

$$\prod_{k=1}^{N_t} \frac{1}{(1+\alpha_A\Delta t)^2} \frac{\|x_k\|^2}{\|x_{k-1}\|^2} \approx \mathrm{e}^{-2\alpha_A T} \frac{\|x(T)\|^2}{\|x_0\|^2}, \tag{13.13}$$

which decays *exponentially* in $\alpha_A T$—even for Hamiltonian simulation ($A = -\mathrm{i}H$), where $\|x(T)\| = \|x_0\|$, the factor $\mathrm{e}^{-2\|H\|T}$ kills the run. For Hamiltonian simulation the fix was oblivious amplitude amplification, which works only because $\mathrm{e}^{-\mathrm{i}H\Delta t}$ is unitary; for genuinely non-unitary B we need a different tool.

Uniform singular value amplification. The remedy (Fang–Lin–Tong 2023) is to amplify each short-time propagator up to its operator norm before chaining. Build a short-time Dyson block encoding $V_k \in \mathrm{BE}(\Xi_k)$ of $\Xi_k = U(t_{k+1}, t_k)$ with $\mathcal{O}(\log(1/\epsilon'))$ queries and subnormalization $\Theta(1)$, then apply *uniform singular value amplification* (USVA, Chapter 9) to produce $V_{\Phi_k} \in \mathrm{BE}_{\|\Xi_k\|(1+\delta)}(\Xi_k, \epsilon')$ using $\mathcal{O}(\delta^{-1}\log(1/\epsilon'))$ applications of V_k. Chaining the N_t amplified blocks as a product of block encodings gives cumulative subnormalization $\prod_k \|\Xi_k\| (1+\delta)^{N_t} \leq 2\prod_k \|\Xi_k\| = 2Q\|x(T)\|$ once $\delta N_t = \ln 2$, so the overall success probability is $\Omega(Q^{-2})$.

Theorem 13.4 (Time-marching solver)**.** *Under the input model (13.4) with steps $\leq \ln 2/\alpha_A$, there is a quantum algorithm preparing $|x(T)\rangle$ to error ϵ with success probability $\Omega(1)$ using*

$$\widetilde{\mathcal{O}}(Q\,\alpha_A^2 T^2) \tag{13.14}$$

queries to U_A and only $\mathcal{O}(Q)$ applications of U_{x_0}, where Q is the growth factor of (13.6).

Proof sketch. With $N_t = \mathcal{O}(T\alpha_A)$ steps and $\delta = \Theta(N_t^{-1})$, each amplified block costs $\mathcal{O}(\delta^{-1}\log^2(1/\epsilon')) = \mathcal{O}(T\alpha_A \log^2(T\alpha_A Q/\epsilon))$ queries. Summed over the N_t blocks this is $\mathcal{O}(T^2\alpha_A^2 \log^2(\cdots))$. Choosing $\epsilon' = \mathcal{O}(\epsilon/N_t Q)$ controls error amplification through the non-unitary chain. Amplitude amplification of the $\Omega(Q^{-2})$ success probability multiplies the query count by Q, giving (13.14); because U_{x_0} enters only inside the amplitude amplification, it is queried $\mathcal{O}(Q)$ times. □

The trade-off relative to Theorem 13.3 is stark: time marching is *quadratic* in $\alpha_A T$ (worse) but needs only $\mathcal{O}(Q)$ initial-state preparations (better). It wins precisely when U_{x_0} dominates the cost.

13.4 Linear combination of Hamiltonian simulation (LCHS)

Dissipative structure. Consider $\dot{x} = -A(t)x$ (the minus sign emphasizes dissipation) and split A into its Hermitian and anti-Hermitian parts via the Cartesian decomposition

$$A = L + \mathrm{i}H, \qquad L = \frac{A + A^\dagger}{2} = L^\dagger, \qquad H = \frac{A - A^\dagger}{2\mathrm{i}} = H^\dagger. \tag{13.15}$$

Assume the *positivity condition* $L(t) \succeq 0$ for all t. Then $\left\| \mathcal{T} \mathrm{e}^{-\int_s^t A(\tau)\mathrm{d}\tau} \right\| \leq 1$, so the solution norm is non-increasing and $\max_t \|x(t)\| = \|x_0\|$, whence $q = \|x_0\| \,/\, \|x(T)\|$. This regime covers many open-system and non-Hermitian problems (e.g. complex absorbing potentials, where $L = V_I \succeq 0$ is a damping potential).

The LCHS identity. The key is a complex-analytic representation of the non-unitary e^{-Lt} (and, when $H \neq 0$, of e^{-At}) as a superposition of unitary evolutions. The Cauchy kernel gives the two warm-up cases: for $L = 0$ ($A = \mathrm{i}H$), $\int_{\mathbb{R}} \frac{1}{\pi(1+k^2)}\mathrm{d}k = 1$ trivially yields $\mathrm{e}^{-\mathrm{i}Ht} = \int_{\mathbb{R}} \frac{1}{\pi(1+k^2)} \mathrm{e}^{-\mathrm{i}Ht}\mathrm{d}k$; and for $H = 0$ ($A = L \succeq 0$), the Fourier representation $\int_{\mathbb{R}} \frac{1}{\pi(1+k^2)} \mathrm{e}^{-\mathrm{i}kx}\mathrm{d}k = \mathrm{e}^{-|x|}$ combined with the spectral mapping theorem gives $\mathrm{e}^{-Lt} = \int_{\mathbb{R}} \frac{1}{\pi(1+k^2)} \mathrm{e}^{-\mathrm{i}kLt}\mathrm{d}k$. The general statement writes the Cauchy weight as $\frac{1}{\pi(1+k^2)} = \frac{f(k)}{1-\mathrm{i}k}$ and generalizes both to $A = L + \mathrm{i}H$.

Theorem 13.5 (Linear combination of Hamiltonian simulation)**.** *Let $f(z)$ be analytic on the lower half-plane $\{\mathrm{Im}\, z < 0\}$ and continuous on $\{\mathrm{Im}\, z \leq 0\}$, satisfy a mild decay bound $|z|^\alpha |f(z)| \leq \mathrm{const}$ for $\mathrm{Im}\, z \leq 0$ and some $\alpha > 0$, and normalize as $\int_{\mathbb{R}} \frac{f(k)}{1-\mathrm{i}k}\mathrm{d}k = 1$. If $A(t) = L(t) + \mathrm{i}H(t)$ with $L(t) \succeq 0$ for all $0 \leq t \leq T$, then*

$$\mathcal{T} \mathrm{e}^{-\int_0^T A(s)\mathrm{d}s} = \int_{\mathbb{R}} \frac{f(k)}{1 - \mathrm{i}k}\, \mathcal{T} \mathrm{e}^{-\mathrm{i}\int_0^T \left(kL(s) + H(s)\right)\mathrm{d}s}\, \mathrm{d}k. \tag{13.16}$$

Proof sketch. Let $O_L(t) = \mathcal{T} \mathrm{e}^{-\int_0^t A(s)\mathrm{d}s}$ and let $O_R(t)$ be the right-hand side of (13.16) with upper limit t. Both satisfy $O(0) = \mathbb{I}$: for O_R this is the normalization $\int_{\mathbb{R}} \frac{f(k)}{1-\mathrm{i}k}\mathrm{d}k = 1$. Differentiating O_R under the (Cauchy principal value) integral and using $\partial_t \mathcal{T} \mathrm{e}^{-\mathrm{i}\int_0^t (kL+H)} = -\mathrm{i}(kL(t) + H(t))\mathcal{T} \mathrm{e}^{-\mathrm{i}\int_0^t (kL+H)}$, then rewriting $-\mathrm{i}kL - \mathrm{i}H = (-L - \mathrm{i}H) + (1 - \mathrm{i}k)L$, splits the derivative into $-A(t)O_R(t)$ plus a remainder proportional to $L(t)\, \mathrm{P} \int_{\mathbb{R}} f(k)\, \mathcal{T} \mathrm{e}^{-\mathrm{i}\int_0^t (kL+H)}\mathrm{d}k$. The integrand of the remainder is analytic in the lower half-plane, and when $L(t) \succeq \lambda_0 > 0$ a contour argument shows the remainder vanishes; hence $O_R' = -AO_R$. Since O_L, O_R solve the same ODE with the same initial condition, uniqueness gives $O_L = O_R$ for $\lambda_0 > 0$, and the limit $\lambda_0 \to 0$ completes the proof. □

From identity to algorithm. Two design choices make (13.16) implementable. First, the *kernel*: the Cauchy choice $f(k) = \frac{1}{\pi(1+\mathrm{i}k)}$ decays only like $1/k$, giving a cusp in the Fourier transform and a large truncation range. The near-optimal kernel

$$f(k) = \frac{1}{2\pi}\, \mathrm{e}^{-2^\beta}\, \mathrm{e}^{(1+\mathrm{i}k)^\beta}, \qquad \beta \in (0, 1), \tag{13.17}$$

still satisfies $2\pi f(-\mathrm{i}) = 1$ but decays faster than any polynomial, so $\frac{f(k)}{1-\mathrm{i}k} = \mathcal{O}(\mathrm{e}^{-C|k|^\beta})$ and the integral may be truncated to $|k| \leq K = \mathcal{O}(\log^{1/\beta}(1/\epsilon))$. Second, *quadrature*: replace the truncated integral by a finite sum $\sum_j \omega_j\, \mathcal{T} \mathrm{e}^{-\mathrm{i}\int_0^T (k_j L + H)}$, a genuine linear combination of Hamiltonian simulations.

Its LCU cost is set by the coefficient 1-norm $\sum_j |\omega_j| \approx \int_{\mathbb{R}} \frac{|f(k)|}{\sqrt{1+k^2}}\mathrm{d}k = \mathcal{O}(1)$, *independent* of T, α_A, ϵ. Each Hamiltonian simulation problem $\mathrm{e}^{-\mathrm{i}\int_0^T (k_j L + H)}$ has $\max_t \|k_j L(t) + H(t)\| = \mathcal{O}(\alpha_A \log^{1/\beta}(1/\epsilon))$ and is implemented by the truncated Dyson method (Section 12.4).

Theorem 13.6 (LCHS solver)**.** *For $\dot{x} = -A(t)x$ with $L(t) \succeq 0$ and any $\beta \in (0,1)$, there is a quantum algorithm preparing $|x(T)\rangle$ to error ϵ with success probability $\Omega(1)$ using*

$$\widetilde{\mathcal{O}}\Big(q\,\alpha_A T\,(\log(1/\epsilon))^{1+1/\beta}\Big) \tag{13.18}$$

queries to U_A (and its inverse), and $\mathcal{O}(q)$ applications of U_{x_0}, where $q = \|x_0\| \,/\, \|x(T)\|$.

Proof sketch. Truncating the kernel to $|k| \le K = \mathcal{O}(\log^{1/\beta}(1/\epsilon))$ and quadrature give an LCU whose 1-norm is $\mathcal{O}(1)$. Each of the $\mathcal{O}(K)$ Hamiltonian evolutions is simulated by the truncated Dyson series (Theorem 12.9) at cost $\widetilde{\mathcal{O}}(\alpha_A T \log(1/\epsilon))$ queries to U_A; the extra factor $K^{1/\beta}$ enters through the norm $\max_t K\,\|L\| + \|H\|$, giving $\widetilde{\mathcal{O}}(\alpha_A T(\log(1/\epsilon))^{1+1/\beta})$ per solve. Amplitude amplification of the $\Omega(q^{-2})$ success probability multiplies by q; U_{x_0} enters only there, so it is queried $\mathcal{O}(q)$ times. □

LCHS thus achieves near-optimal $\alpha_A T$ scaling (linear, like the linear-system method) and optimal $\mathcal{O}(q)$ initial-state cost (like time marching)—the best of both. Its price is the dissipativity requirement $L \succeq 0$, which the other two methods do not need.

Example 13.7 (Complex absorbing potentials)**.** In molecular scattering one replaces the real potential by $V_R - \mathrm{i}V_I$ with $V_I \ge 0$ an absorbing (damping) layer, giving $\mathrm{i}\partial_t u = (-\frac{1}{2}\Delta + V_R - \mathrm{i}V_I)u$, i.e. $\dot{u} = -Au$ with $A = V_I + \mathrm{i}(-\frac{1}{2}\Delta + V_R)$. Here the Hermitian part is $H = -\frac{1}{2}\Delta + V_R$ with $\|H\| = \mathcal{O}(N^2 + \max_t \|V_R\|)$ and the anti-Hermitian generator has $L = V_I \succeq 0$: exactly the LCHS regime. LCHS prepares $u(T)/\|u(T)\|$ to error ϵ using $\widetilde{\mathcal{O}}(q\,\alpha_A T)$ queries with $\alpha_A = \mathcal{O}(N^2 + \max_t \|V_R\| + \|V_I\|)$ and $q = \|u_0\| \,/\, \|u(T)\|$; the physical interest is precisely the pre-escape regime where $\|u(T)\|$ is not too small, so $q = \mathcal{O}(1)$.

13.5 Schrödingerization

The warped phase transform. Schrödingerization (Jin–Liu–Yu 2022) is the PDE-native cousin of LCHS: instead of a spectral integral, it introduces one extra continuous variable that turns dissipative dynamics into a genuine Schrödinger equation. Consider first the heat equation $\partial_t u = \partial_{rr} u$. Introduce $p \ge 0$ and define the *warped phase transform*

$$w(t, r, p) = \mathrm{e}^{-p}\, u(t, r), \qquad p \ge 0. \tag{13.19}$$

Since e^{-p} is an eigenfunction of $-\partial_p$ with eigenvalue 1, the diffusion term becomes an advection in p: $\partial_t w = \partial_{rr} u \cdot \mathrm{e}^{-p} = -\partial_p \partial_{rr} w$. The trick generalizes to any A. For a general linear ODE $\dot{u} = Au$ with $A = A_1 + \mathrm{i}A_2$ ($A_1 = A_1^\dagger \preceq 0$ for stability, $A_2 = A_2^\dagger$), the same substitution $v(t,p) = \mathrm{e}^{-p}u(t)$ for $p > 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t} v(t,p) = A\, v(t,p) = -A_1\, \partial_p v + \mathrm{i}A_2\, v, \tag{13.20}$$

using $A_1 v = A_1 \mathrm{e}^{-p} u = -\partial_p(\mathrm{e}^{-p}) A_1 u \to -A_1 \partial_p v$. The right-hand side is now *first order* in p (a transport term $-A_1\partial_p$) plus a Hermitian term $\mathrm{i}A_2$. Symmetrically extend the initial data to $p < 0$ as $v(0,p) = \mathrm{e}^{-|p|} u_0$; because the transport moves the profile leftward and $\mathrm{e}^{-|p|}$ decays, the region $p > 0$ is unaffected by the extension, and one recovers

$$u(t) = \int_0^\infty v(t,p)\,\mathrm{d}p, \qquad \text{or simply} \qquad u(t) = \mathrm{e}^{p_*} v(t, p_*) \quad (p_* > 0). \tag{13.21}$$

Fourier spectral discretization in p. The transport operator $-\partial_p$ is diagonalized by the Fourier transform in p: with η the Fourier variable, $-\mathrm{i}\partial_p \to \eta$. Applying $\mathcal{F}_p$ to (13.20),

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{v}(t,\eta) = -\mathrm{i}(\eta A_1 - A_2)\tilde{v}(t,\eta) = -\mathrm{i}\,\mathcal{H}(\eta)\,\tilde{v}(t,\eta), \qquad \mathcal{H}(\eta) = \eta A_1 - A_2, \tag{13.22}$$

and $\mathcal{H}(\eta) = \mathcal{H}(\eta)^\dagger$ is Hermitian for every η (both A_1, A_2 are Hermitian and $\eta \in \mathbb{R}$). Equation (13.22) is a bona fide Schrödinger equation: its propagator $\mathrm{e}^{-\mathrm{i}\mathcal{H}(\eta)t}$ is unitary. Embedding $\tilde{v}(0,\cdot)$ into amplitudes, the whole evolution is the single unitary $\mathrm{e}^{-\mathrm{i}(A_1\otimes\hat{\eta} - A_2\otimes\mathbb{I})t}$ on the joint system-plus-p register, where $\hat{\eta}$ is the (diagonal) quadrature operator. Discretizing $p \in [-L, R]$ on $N_p = 2^{n_p}$ points and applying the DFT F_p realizes $\hat{\eta} \to D_\mu = \mathrm{diag}(\eta_k)$ with $|D_\mu|_{\max} \lesssim 1/\Delta p$.

The algorithm. The recipe (Figure 13.1) is: (i) form the product state $|w(0)\rangle = |u_0\rangle \otimes |\mathrm{e}^{-|p|}\rangle$; (ii) apply the QFT F_p on the p-register; (iii) evolve under the unitary $\mathrm{e}^{-\mathrm{i}(A_1\otimes D_\mu - A_2\otimes\mathbb{I})t}$ by any Hamiltonian simulation method of Chapters 11 and 12; (iv) apply the inverse QFT; (v) measure/project the p-register onto a mode $p_k > 0$ and read out $u(t) = \mathrm{e}^{p_k} w(t, p_k)$. Step (v) succeeds with probability $\mathrm{e}^{-2p_k}\,\|u(T)\|^2 \,/\, \|w(0)\|^2$, the Schrödingerization analogue of the norm-growth factor. Sparse-access to A is inherited by $\mathcal{H}$: if A has sparsity $s(A)$ and max-entry $\|A\|_{\max}$, then $s(\mathcal{H}) = \mathcal{O}(s(A))$ and $\|\mathcal{H}\|_{\max} = \mathcal{O}(\|A\|_{\max}/\Delta p)$, and the Berry–Childs–Kothari sparse Hamiltonian simulation lemma (near-optimal, $\widetilde{\mathcal{O}}(\tau)$ queries with $\tau = s\,\|\mathcal{H}\|_{\max}\,t$; Chapter 11) applies directly.

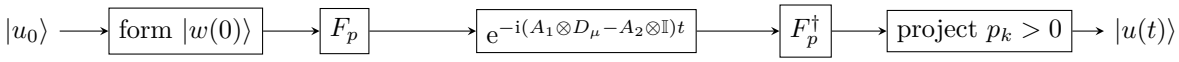


Figure 13.1: The Schrödingerization pipeline: warp the state into the extra p-register, QFT into the η-basis, evolve unitarily under the Hermitian $\mathcal{H}(\eta) = \eta A_1 - A_2$, inverse-QFT, and project onto a mode with $p_k > 0$ to recover $u(t) = \mathrm{e}^{p_k} w(t, p_k)$.

Theorem 13.8 (Schrödingerization gate complexity)**.** *Let $A \in \mathbb{C}^{N_A \times N_A}$ arise from a d-dimensional discretization ($N_A \sim 2^{m_d}$, $m_d = dm$), split as $A = A_1 + \mathrm{i}A_2$ with $A_1 \preceq 0$. Using m_p qubits on the p-register, the Schrödingerization method simulates $\dot{u} = Au$ to time T with gate complexity*

$$N_{\text{gates}} = (m_d + m_p)\,\widetilde{\mathcal{O}}\Big(\frac{s(A)\;\|A\|_{\max}\;T}{\Delta p}\Big) + \mathcal{O}(m_p \log m_p). \tag{13.23}$$

In the special case where A_1 is diagonalized in the momentum basis and A_2 is diagonal, a first-order time-splitting scheme achieves $N_{\text{gates}} = \frac{T}{\Delta t}\,\mathcal{O}(dm \log m + m_p \log m_p)$.

Proof sketch. The Hamiltonian $\mathcal{H} = A_1 \otimes D_\mu - A_2 \otimes \mathbb{I}$ has sparsity $s(\mathcal{H}) \le s(A_1) + s(A_2) = \mathcal{O}(s(A))$ and max-entry $\|\mathcal{H}\|_{\max} \le \|A_1\|_{\max}\,|D_\mu|_{\max} + \|A_2\|_{\max} = \mathcal{O}(\|A\|_{\max}/\Delta p)$, using $|D_\mu|_{\max} \lesssim 1/\Delta p$. Feeding $\tau = s(\mathcal{H})\,\|\mathcal{H}\|_{\max}\,T = \widetilde{\mathcal{O}}(s(A)\,\|A\|_{\max}\,T/\Delta p)$ into the sparse simulation lemma (Chapter 11) and adding two QFTs on the p-register ($\mathcal{O}(m_p \log m_p)$ gates) gives (13.23). When A_1 is momentum-diagonal and A_2 diagonal, the two factors of the split evolution are each diagonalized by a QFT, so each step costs $\mathcal{O}(dm \log m + m_p \log m_p)$ gates and there are $T/\Delta t$ steps. □

The relationship to LCHS is now transparent: both express non-unitary dynamics through unitary evolutions parameterized by an extra real variable (η in Schrödingerization, k in LCHS), and both read out the answer by projecting/weighting that variable. LCHS phrases this in the language of block encodings and quadrature; Schrödingerization phrases it as an honest higher-dimensional Schrödinger PDE with a spectral discretization—often the more natural choice when one already has a grid-based simulator and Fourier tooling.

13.6 Example: the heat equation

We now run the one-dimensional heat equation

$$\partial_t u(r,t) = \partial_{rr} u(r,t), \qquad r \in [0,1], \qquad u(0,t) = u(1,t) = 0, \tag{13.24}$$

through the methods above and compare the resource scaling. Central differencing on N interior points gives $\dot{u} = -Au$ with $A = -L \in \mathbb{R}^{N\times N}$ the (negated) discrete Laplacian of (13.2): Hermitian, positive definite, tridiagonal, with $\|A\| \approx 4N^2$. Numerical stability of the explicit time-stepping requires $\Delta t = \mathcal{O}(\|A\|^{-1}) = \mathcal{O}(N^{-2})$, so the associated linear system has size NN_t with $N_t = T/\Delta t = \mathcal{O}(N^2 T)$.

Linear-system method. Since A is Hermitian with all eigenvalues positive, e^{-At} is contractive and $\zeta = \mathcal{O}(1)$; with $\alpha_A = \mathcal{O}(N^2)$, Theorem 13.3 gives query complexity $\widetilde{\mathcal{O}}(q\,N^2 T)$, where $q = \|u_0\| \,/\, \|u(T)\|$. (This matches the qasc estimate $\mathcal{O}(N^4\epsilon^{-1})$ once one substitutes the forward-Euler condition number $\kappa = \mathcal{O}(1/\Delta t) = \mathcal{O}(N^2)$ and the first-order accuracy $\epsilon = \mathcal{O}(\Delta t)$ into the HHL cost $\mathcal{O}(\kappa^2\epsilon^{-1})$; the near-optimal QLSP solver removes one factor of κ.)

LCHS. Because A is Hermitian and positive definite, $A = L$ with $H = 0$, so the LCHS identity reduces to $\mathrm{e}^{-At} = \int_{\mathbb{R}} \frac{f(k)}{1-\mathrm{i}k} \mathrm{e}^{-\mathrm{i}ktA} \mathrm{d}k$. Theorem 13.6 then gives query complexity $\widetilde{\mathcal{O}}(q\,N^2 T)$—the same $\alpha_A T$ dependence as the linear system method, but with the initial-state oracle queried only $\mathcal{O}(q)$ times.

Time-marching. The time-marching solver of Theorem 13.4 costs $\widetilde{\mathcal{O}}(Q\,N^4 T^2)$ queries, the extra factor of $N^2 T$ reflecting its quadratic $\alpha_A T$ dependence; it too queries U_{x_0} only $\mathcal{O}(Q)$ times.

Schrödingerization. Warping to $w = \mathrm{e}^{-p} u$ turns (13.24) into $\partial_t w = -\partial_p \partial_{rr} w$; after $\mathcal{F}_p$ the Hamiltonian is $\mathcal{H} = A \otimes D_\eta$, self-adjoint. Since A is tridiagonal, $\mathcal{H}$ is simulated by the finite-difference shift-operator circuits of Section 13.5: with $\gamma_0 = 1/(h^2 R)$, the evolution factorizes over the p-modes, $U_{\mathrm{heat}}(\tau) = \sum_k (\mathrm{e}^{\mathrm{i}H_0\tau})^{k-N_p/2} \otimes |k\rangle\langle k|$ with $H_0 = \gamma_0(S^- + S^+ - 2\mathbb{I})$, and each nearest-neighbor hopping term $\mathrm{e}^{\mathrm{i}\gamma_0\tau(S_j^- + S_j^+)}$ becomes a Bell-basis-conjugated controlled-R_z gate. Theorem 13.8 then gives gate complexity $\frac{T}{\Delta t}\,\mathcal{O}(m \log m + m_p \log m_p)$ with $m = \log N$ qubits per axis: polylogarithmic in N per step, at the cost of the extra p-register and the e^{-2p_k} readout probability.

The higher-dimensional payoff. In one dimension a super-polynomial quantum advantage in N is unlikely (classical sparse solvers are efficient). The prospect improves in d dimensions: for $\partial_t u = \Delta u$ on $[0,1]^d$ the discrete Laplacian has $\|A\| = \Theta(dN^2)$, growing only *linearly* in d while the classical state space grows as N^d. This linear-in-d scaling of α_A is the structural source of potential exponential advantage—subject, as always, to the caveats of input preparation, output readout, and the hardness of the best classical solver.

Exercises

Exercise 13.1. *Duhamel and the augmented system.* (a) Prove Theorem 13.1 by differentiating the right-hand side of (13.3) and invoking uniqueness. (b) Show that the inhomogeneous equation (13.1) is equivalent to the homogeneous equation $\dot{\tilde{x}} = \tilde{A}\tilde{x}$ with $\tilde{x} = (x; 1)$ and $\tilde{A} = \left(\begin{smallmatrix} A & b \\ 0 & 0 \end{smallmatrix}\right)$, and explain why this justifies restricting the algorithms to the homogeneous case.

Exercise 13.2. *Norm growth and the output model.* For $\dot{x} = -Zx$ on $\mathbb{C}^2$ with $x_0 = (a, b)$, compute $x(t)$, $\|x(t)\|$, and the normalized $|x(t)\rangle$. Evaluate the norm-growth factor $q = \max_t \|x(t)\| \,/\, \|x(T)\|$ over $[0, T]$ and explain why it grows exponentially in T whenever $b \neq 0$. Contrast with the anti-Hermitian case $\dot{x} = -\mathrm{i}Zx$, where $q = 1$.

Exercise 13.3. *The big linear system.* Starting from the forward-Euler recursion $x_{k+1} - x_k = \Delta t\, A_k x_k$, derive the block-bidiagonal matrix (13.8) and its compact form (13.9). Verify that a block encoding of $\mathcal{A}$ can be built from one query to U_A with subnormalization $\mathcal{O}(1)$ when $\Delta t = \mathcal{O}(\alpha_A^{-1})$.

Exercise 13.4. *Condition number bound.* Prove Theorem 13.2: (a) show $\|\mathcal{A}\| \leq \zeta_0 + \zeta_1$ by bounding the block-diagonal $\mathcal{B}_1^\dagger \mathcal{B}_1$; (b) verify the explicit lower-triangular formula for $\mathcal{A}^{-1}$ in terms of ordered products $B_{l+k-1} \cdots B_l$, and conclude $\|\mathcal{A}^{-1}\| \leq \sum_k \zeta_k$. For the time-independent, contractive case $\left\|\mathrm{e}^{At}\right\| \leq C_1 \mathrm{e}^{-C_2 t}$, show $\kappa(\mathcal{A}) = \mathcal{O}(1/\Delta t)$ independent of N_t.

Exercise 13.5. *Why naive marching fails.* Derive the multiplicative success probability (13.13) for step-by-step application of the block-encoded propagator $B = \mathbb{I} + A\Delta t$. Specialize to Hamiltonian simulation $A = -\mathrm{i}H$ and show the probability decays as $\mathrm{e}^{-2\|H\|T}$ even though $\|x(T)\| = \|x_0\|$. Explain in one sentence why oblivious amplitude amplification rescues this case but not a general non-unitary B.

Exercise 13.6. *LCHS warm-up cases.* (a) Using $\int_{\mathbb{R}} \frac{1}{\pi(1+k^2)} \mathrm{e}^{-\mathrm{i}kx} \mathrm{d}k = \mathrm{e}^{-|x|}$, prove $\mathrm{e}^{-Lt} = \int_{\mathbb{R}} \frac{1}{\pi(1+k^2)} \mathrm{e}^{-\mathrm{i}kLt} \mathrm{d}k$ for $L = L^\dagger \succeq 0$ via the spectral mapping theorem. (b) Verify $2\pi f(-\mathrm{i}) = 1$ and the normalization $\int_{\mathbb{R}} \frac{f(k)}{1-\mathrm{i}k} \mathrm{d}k = 1$ for the near-optimal kernel (13.17), and explain why its super-polynomial decay lets the k-integral be truncated to $|k| \leq K = \mathcal{O}(\log^{1/\beta}(1/\epsilon))$.

Exercise 13.7. *Schrödingerization of the heat equation.* (a) With $w(t, r, p) = \mathrm{e}^{-p} u(t, r)$, verify $\partial_t w = -\partial_p \partial_{rr} w$ for $\partial_t u = \partial_{rr} u$. (b) Apply the Fourier transform in p ($-\mathrm{i}\partial_p \to \eta$) to obtain the Schrödinger equation $\partial_t \tilde{w} = \mathrm{i}\eta\, \partial_{rr} \tilde{w}$, and confirm that $\eta\, \partial_{rr}$ is self-adjoint under Dirichlet boundary conditions so the evolution is unitary. (c) Explain how $u(t, r)$ is recovered from a measured mode $p_k > 0$ and why the success probability carries the factor e^{-2p_k}.

Further reading

- Lin Lin, *qasc notes*, Ch. 4.5–4.6 (the linear-system-of-equations method with forward/backward Euler, the Gershgorin condition-number analysis, padded history states for observables, and the heat equation as the worked example).
- Lin Lin, *live notes*, Ch. 20 (problem setup and Duhamel's principle; the linear-system, time-marching, and LCHS methods with full complexity and condition-number analysis; the heat equation and complex-absorbing-potential examples; the norm-growth lower bound via reduction to unstructured search).
- D. W. Berry, *High-order quantum algorithm for solving linear differential equations*, J. Phys. A **47**, 105301 (2014) (arXiv:1010.2745), for the original history-state / linear-system approach of Section 13.2.
- D. Fang, L. Lin, and Y. Tong, *Time-marching based quantum solvers for time-dependent linear differential equations*, Quantum **7**, 955 (2023) (arXiv:2208.06941), for the time-marching method of Section 13.3.

- D. An, J.-P. Liu, and L. Lin, *Linear combination of Hamiltonian simulation for nonunitary dynamics with optimal state preparation cost*, Phys. Rev. Lett. **131**, 150603 (2023) (arXiv:2303.01029); and D. An, A. M. Childs, and L. Lin, *Quantum algorithm for linear non-unitary dynamics with near-optimal dependence on all parameters*, arXiv:2312.03916 (2023), for LCHS (Section 13.4).

- S. Jin, N. Liu, and Y. Yu, *Quantum simulation of partial differential equations via Schrodingerization*, Phys. Rev. Lett. **133**, 230602 (2024) (arXiv:2212.13969); and S. Jin and N. Liu, *Quantum simulation of discrete linear dynamical systems and simple iterative methods via Schrodingerization*, arXiv:2304.02865 (2023), for the warped phase transform, spectral discretization, and finite-difference circuits of Section 13.5.

Chapter 14

Open Quantum Systems and Nonlinear PDEs

Learning objectives

- Describe Lindblad dynamics and its dilation to a unitary.
- Prepare thermal and ground states dissipatively.
- Outline how nonlinear PDEs are linearized (Carleman, Cole–Hopf) for quantum solvers.

The last three chapters kept the dynamics *linear*: the Schrödinger equation (Chapters 11 and 12) and the general linear ODE $\dot{x} = Ax + b$ (Chapter 13). This chapter relaxes both idealizations, in two independent directions that nonetheless share a common strategy—*embed the hard problem inside a larger linear or unitary one*. The first half treats *open* quantum systems, whose state is a density operator ρ evolving under a non-unitary but completely positive semigroup; the governing equation is the Lindblad master equation, the continuous-time analogue of the CPTP channels of Chapter 2. We simulate it by dilating its non-unitary generator to a unitary on a larger space (Stinespring, Theorem 2.12) and by operator splitting and a truncated Dyson series, all reusing the Hamiltonian simulation of Chapters 11 and 12. We then turn dissipation into a resource: by engineering the jump operators so that a target thermal or ground state is the *unique fixed point*, Lindblad dynamics prepares that state, an open-system cousin of the adiabatic and Zeno solvers of Chapter 10. The second half attacks *nonlinear* PDEs. Because quantum mechanics is linear, a nonlinear evolution must first be *linearized*: we present Carleman linearization, which embeds a quadratic ODE into an infinite linear system on the tensor powers of the state and truncates it (solving the result with the methods of Chapter 13), and the exact Cole–Hopf transformation, which maps the viscous Burgers equation onto the heat equation. Throughout, $\|\cdot\|$ is the spectral norm, $\|\cdot\|_1$ the trace norm, $\|\cdot\|_\diamond$ the diamond norm of Chapter 2, and $\mathcal{D}(\mathcal{H})$ the set of density operators on $\mathcal{H}$.

14.1 Lindblad dynamics

The master equation. An open system interacts with environmental degrees of freedom that are not modeled explicitly. In the Markovian regime—environmental correlations decay fast on the system's timescale—the state $\rho(t) \in \mathcal{D}(\mathbb{C}^d)$ obeys the Gorini–Kossakowski–Sudarshan–Lindblad

(GKSL) master equation

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \underbrace{-\mathrm{i}[H,\rho]}_{\mathcal{L}_H(\rho)} + \underbrace{\sum_{j=1}^{J}\Big(L_j\rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j,\rho\}\Big)}_{\mathcal{L}_D(\rho)} =: \mathcal{L}(\rho), \tag{14.1}$$

where $H = H^\dagger$ is the system Hamiltonian, the $L_j \in \mathbb{C}^{d\times d}$ are *jump operators* encoding the coupling to the environment, $[A,B] = AB-BA$, and $\{A,B\} = AB+BA$. The generator $\mathcal{L}$ is the *Lindbladian*; its coherent part $\mathcal{L}_H$ reproduces the Liouville–von Neumann equation, while its dissipative part $\mathcal{L}_D$ is genuinely irreversible. The formal solution is the one-parameter semigroup

$$\rho(t) = \Phi_t(\rho(0)), \qquad \Phi_t := \mathrm{e}^{\mathcal{L}t}, \quad t \geq 0. \tag{14.2}$$

When all $L_j = 0$ this collapses to $\rho(t) = \mathrm{e}^{-\mathrm{i}Ht}\rho(0)\mathrm{e}^{\mathrm{i}Ht}$, closed-system unitary evolution. The value of the Lindblad form is that Φ_t is always a legitimate channel.

Theorem 14.1 ($\mathrm{e}^{\mathcal{L}t}$ is a quantum channel)**.** *For every $t \geq 0$ the map $\Phi_t = \mathrm{e}^{\mathcal{L}t} : L(\mathbb{C}^d) \to L(\mathbb{C}^d)$ defined by (14.1)–(14.2) is completely positive and trace preserving (CPTP), i.e. a quantum channel in the sense of Theorem 2.4. The family $\{\Phi_t\}_{t\geq 0}$ forms a semigroup: $\Phi_0 = \mathrm{id}$ and $\Phi_{t+s} = \Phi_t\Phi_s$.*

Proof. The semigroup laws are immediate from $\mathrm{e}^{\mathcal{L}\cdot 0} = \mathrm{id}$ and $\mathrm{e}^{\mathcal{L}(t+s)} = \mathrm{e}^{\mathcal{L}t}\mathrm{e}^{\mathcal{L}s}$. For the CPTP claim, fix a small step $\Delta t = t/N_t$ and construct a genuine channel Φ with $\left\|\mathrm{e}^{\mathcal{L}\Delta t} - \Phi\right\|_\diamond \leq C_1\Delta t^2$; we exhibit such a Φ explicitly in Section 14.2 (equation (14.5)). Granting this, let $C_2 = \sup_{0\leq s\leq t}\left\|\mathrm{e}^{\mathcal{L}s}\right\|_\diamond$. Because the diamond norm is submultiplicative and $\|\Phi\|_\diamond = 1$, the telescoping identity

$$\mathrm{e}^{\mathcal{L}t} - \Phi^{N_t} = \sum_{k=0}^{N_t-1} \mathrm{e}^{\mathcal{L}(N_t-1-k)\Delta t}\left(\mathrm{e}^{\mathcal{L}\Delta t} - \Phi\right)\Phi^k$$

gives $\left\|\mathrm{e}^{\mathcal{L}t} - \Phi^{N_t}\right\|_\diamond \leq C_2C_1N_t\Delta t^2 = C_1C_2\,t\,\Delta t \to 0$ as $N_t \to \infty$. Thus $\mathrm{e}^{\mathcal{L}t}$ is a diamond-norm limit of the channels Φ^{N_t} (a composition of channels is a channel). Since the set of CPTP maps is closed in the diamond norm, $\mathrm{e}^{\mathcal{L}t}$ is itself CPTP. □

The converse is the structural GKSL theorem: in finite dimension, (14.1) is the *most general* generator of a norm-continuous semigroup of channels.

Proposition 14.2 (GKSL characterization)**.** *Let $\{\Phi_t\}_{t\geq 0}$ be a norm-continuous semigroup of quantum channels on $L(\mathbb{C}^d)$ with generator $\mathcal{G}(\rho) = \lim_{t\to 0^+}(\Phi_t(\rho)-\rho)/t$. Then there exist a Hermitian H and operators $L_1,\dots,L_J$ such that $\mathcal{G}$ has the form of the right-hand side of (14.1).*

Proof sketch. For small t the channel Φ_t has a Kraus form $\Phi_t(\rho) = \sum_\alpha K_\alpha(t)\rho K_\alpha(t)^\dagger$ with $\sum_\alpha K_\alpha(t)^\dagger K_\alpha(t) = \mathbb{I}$. Expanding the Choi matrix to first order in t, one may choose the Kraus operators so that one of them is close to the identity, $K_0(t) = \mathbb{I} + t\mathcal{G}_0 + o(t)$, while the rest are $\mathcal{O}(\sqrt{t})$, namely $K_j(t) = \sqrt{t}\,L_j + \mathcal{O}(t)$. The trace-preservation condition, expanded to first order, forces $\mathcal{G}_0 + \mathcal{G}_0^\dagger + \sum_j L_j^\dagger L_j = 0$, so $\mathcal{G}_0 = -\mathrm{i}H - \frac{1}{2}\sum_j L_j^\dagger L_j$ for a Hermitian H. Substituting back and collecting the $\mathcal{O}(t)$ terms of $(\Phi_t(\rho)-\rho)/t$ reproduces (14.1). □

The adjoint (Heisenberg) equation. Expectation values of an observable O evolve through the *adjoint* Lindbladian $\mathcal{L}^\dagger$, defined by $\mathrm{Tr}[\mathcal{L}(\rho)\,O] = \mathrm{Tr}[\rho\,\mathcal{L}^\dagger(O)]$ under the Hilbert–Schmidt pairing $\langle X|Y\rangle = \mathrm{Tr}[X^\dagger Y]$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{Tr}[\rho(t)O] = \mathrm{Tr}[\rho(t)\,\mathcal{L}^\dagger(O)], \qquad \mathcal{L}^\dagger(O) = \mathrm{i}[H,O] + \sum_{j=1}^{J}\Big(L_j^\dagger O L_j - \tfrac{1}{2}\{L_j^\dagger L_j, O\}\Big). \tag{14.3}$$

The Heisenberg picture $O(t) = \mathrm{e}^{\mathcal{L}^\dagger t}(O)$ then satisfies $\mathrm{Tr}[\rho(t)O] = \mathrm{Tr}[\rho(0)O(t)]$, exactly as for closed systems.

Example 14.3 (Classical Markov chains as Lindblad dynamics)**.** Take $H = 0$ and, for a rate matrix $q_{ij} \geq 0$ $(i \neq j)$, jump operators $L_{ij} = \sqrt{q_{ij}}\;|i\rangle\langle j|$. For a diagonal (classical) state $\rho = \sum_j p_j\,|j\rangle\langle j|$ one has $L_{ij}\rho L_{ij}^\dagger = q_{ij}p_j\,|i\rangle\langle i|$ and $L_{ij}^\dagger L_{ij} = q_{ij}\,|j\rangle\langle j|$, so (14.1) keeps ρ diagonal and the populations obey $\dot p_i = \sum_{j\neq i} q_{ij}p_j - \big(\sum_{k\neq i} q_{ki}\big)p_i$, i.e. $\dot p = Qp$ with the classical generator $Q_{ij} = q_{ij}$ $(i \neq j)$, $Q_{jj} = -\sum_{i\neq j} q_{ij}$. Restricted to diagonal states the Lindblad semigroup $\mathrm{e}^{\mathcal{L}t}$ is exactly the column-stochastic Markov semigroup e^{Qt}: Lindblad dynamics is the quantum analogue of a continuous-time Markov chain.

Example 14.4 (Photon loss in a cavity)**.** A single cavity mode with annihilation operator a, Hamiltonian $H = \omega a^\dagger a$, and a single jump operator $L = \sqrt{\kappa}\,a$ models photon leakage at rate κ. With $n = a^\dagger a$ the adjoint (14.3) gives $\mathcal{L}^\dagger(n) = \kappa(a^\dagger a^\dagger a a - n^2) = \kappa(n(n-1) - n^2) = -\kappa n$, so $\mathrm{d}\langle n\rangle/\mathrm{d}t = -\kappa\langle n\rangle$ and $\langle n\rangle(t) = \langle n\rangle(0)\mathrm{e}^{-\kappa t}$: the mean photon number decays exponentially and the vacuum $|0\rangle\langle 0|$ is the steady state.

14.2 Dilation to a larger unitary system

The dilated Hamiltonian. The obstruction to simulating (14.1) directly is that Φ_t is not unitary. Stinespring's theorem (Theorem 2.12) says every channel is a unitary on a larger space followed by discarding an ancilla; we now realize the *short-time* Lindblad channel this way, using an ancilla of $a = \lceil \log_2(J+1)\rceil$ qubits with basis $\{|0\rangle, |1\rangle, \dots, |J\rangle\}$. Define the *dilated Hamiltonian*

$$\widetilde{H} = |0\rangle\langle 0| \otimes \frac{H}{\sqrt{\Delta t}} + \sum_{j=1}^{J}\big(|j\rangle\langle 0| \otimes L_j + |0\rangle\langle j| \otimes L_j^\dagger\big) = \begin{pmatrix} H/\sqrt{\Delta t} & L_1^\dagger & \cdots & L_J^\dagger \\ L_1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ L_J & 0 & \cdots & 0 \end{pmatrix}, \tag{14.4}$$

and set the short-time unitary $U = \mathrm{e}^{-\mathrm{i}\widetilde{H}\sqrt{\Delta t}}$. The corresponding channel on the system, obtained by initializing the ancilla in $|0\rangle$ and tracing it out, is

$$\Phi(\rho) = \mathrm{Tr}_a\Big(U\big(|0\rangle\langle 0| \otimes \rho\big)U^\dagger\Big). \tag{14.5}$$

The factors of $\sqrt{\Delta t}$ are chosen precisely so that the dissipative terms enter at the right order. Writing $\widetilde{H} = |0\rangle\langle 0| \otimes H/\sqrt{\Delta t} + R$ with $R = |j\rangle\langle 0| \otimes L_j + \mathrm{h.c.}$, one has $R^2 = \mathrm{diag}(\sum_j L_j^\dagger L_j, \dots)$, so $\big\|\widetilde{H}\big\|^2 = \mathcal{O}(\|H\|^2\,\Delta t + \sum_j \|L_j\|^2)$ stays bounded once $\Delta t = \mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}}^{-1})$, where

$$\|\mathcal{L}\|_{\mathrm{be}} := 1 + \|H\| + \sum_{j=1}^{J}\|L_j\|^2\,. \tag{14.6}$$

Theorem 14.5 (Dilated-Hamiltonian simulation of Lindblad dynamics)**.** *Let* Φ *be the channel* (14.5)*,* $\|\mathcal{L}\|_{\mathrm{be}}$ *as in* (14.6)*, and* $\Delta t = \mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}}^{-1})$*. Then the one-step error is* $\left\|\mathrm{e}^{\mathcal{L}\Delta t} - \Phi\right\|_\diamond = \mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}}^2\,\Delta t^2)$*, and hence with* $t = N_t\Delta t$*,*

$$\left\|\mathrm{e}^{\mathcal{L}t} - \Phi^{N_t}\right\|_\diamond \le N_t\,\left\|\mathrm{e}^{\mathcal{L}\Delta t} - \Phi\right\|_\diamond = \mathcal{O}(t\,\|\mathcal{L}\|_{\mathrm{be}}^2\,\Delta t). \tag{14.7}$$

Consequently, achieving diamond-norm error ϵ *at time* t *requires* $N_t = \mathcal{O}(t^2\,\|\mathcal{L}\|_{\mathrm{be}}^2/\epsilon)$ *applications of the Hamiltonian simulation* $U = \mathrm{e}^{-\mathrm{i}\widetilde{H}\sqrt{\Delta t}}$*, each followed by resetting the ancilla to* $|0\rangle$*.*

Proof sketch. Expand $U = \mathrm{e}^{-\mathrm{i}\widetilde{H}\sqrt{\Delta t}}$ to third order in $\sqrt{\Delta t}$ as $W = \mathbb{I} - \mathrm{i}\widetilde{H}\sqrt{\Delta t} - \frac{1}{2}\widetilde{H}^2\Delta t + \frac{\mathrm{i}}{6}\widetilde{H}^3\Delta t^{3/2}$, with $\|U - W\| = \mathcal{O}(\left\|\widetilde{H}\right\|^4\,\Delta t^2) = \mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}}^2\,\Delta t^2)$ by the bound $\left\|\widetilde{H}\right\|^2 = \mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}})$ above. Because $\|\cdot\|_\diamond$ is defined over ancillary extensions of trace-norm-one inputs and the partial trace is a contraction in the trace norm, replacing U by W in (14.5) changes the channel by $\mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}}^2\,\Delta t^2)$ in diamond norm. Tracing out the ancilla kills all odd powers of $\sqrt{\Delta t}$, and a direct expansion of $\mathrm{Tr}_a(W(|0\rangle\langle 0|\otimes\rho)W^\dagger)$ yields $\rho + \Delta t\,\mathcal{L}(\rho) + \mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}}^2\,\Delta t^2)$, which matches $\mathrm{e}^{\mathcal{L}\Delta t}(\rho) = \rho + \Delta t\,\mathcal{L}(\rho) + \mathcal{O}(\|\mathcal{L}\|_{\mathrm{be}}^2\,\Delta t^2)$. Subtracting gives the one-step bound; the long-time estimate is the telescoping (linear error growth) argument of Theorem 14.1. □

This construction, illustrated schematically as a chain of *repeated interactions*—couple to a fresh ancilla, evolve for $\sqrt{\Delta t}$, discard—turns any Hamiltonian simulation method of Chapters 11 and 12 into a Lindblad simulator.

Example 14.6 (Ising evolution with dephasing)**.** For the transverse-field Ising model $H = -\sum_{j=1}^{n-1} Z_jZ_{j+1} - g\sum_{j=1}^n X_j$ with site-local dephasing $L_j = \sqrt{\gamma}\,Z_j$, the Lindbladian is $\dot\rho = -\mathrm{i}[H,\rho] + \gamma\sum_j(Z_j\rho Z_j - \rho)$. The dilated Hamiltonian (14.4) uses $\lceil\log_2(n+1)\rceil$ ancilla qubits with the $L_j = \sqrt{\gamma}Z_j$ in its first block-column. With $g,\gamma = \mathcal{O}(1)$ one has $\|\mathcal{L}\|_{\mathrm{be}} = \mathcal{O}(n)$, so error ϵ at time T needs $\Delta t = \mathcal{O}(\epsilon/(T\,\|\mathcal{L}\|_{\mathrm{be}}^2))$ and $N_t = \mathcal{O}(T^2n^2/\epsilon)$ dilated-Hamiltonian simulation steps.

Oblivious amplitude amplification of channels. Each dilated step succeeds only when the ancilla postselects back to $|0\rangle$; naively this success probability compounds and decays, exactly the pathology of Section 13.3. The cure generalizes oblivious amplitude amplification (Chapter 12) from unitaries to *channels*. Given a channel $\mathcal{Q}(\rho) = \sum_m K_m\rho K_m^\dagger$ with $M = 2^a$ Kraus operators and a SELECT oracle $U_{\mathrm{SEL}} = \sum_m |m\rangle\langle m|\otimes U_{K_m}$, where $U_{K_m} \in \mathrm{BE}_\alpha(K_m)$, the operator $W = U_{\mathrm{SEL}}(\mathsf{H}^{\otimes a}\otimes\mathbb{I})$ is a block encoding of the Stinespring isometry $V = \sum_m |m\rangle\otimes K_m$ (Theorem 2.12), with $(\langle 0^b|\otimes\mathbb{I})W(|0^a\rangle|\psi\rangle|0^b\rangle) = (\alpha\sqrt{M})^{-1}V|\psi\rangle$. Since V is an *isometry*, all its nonzero singular values equal 1, so amplitude amplification on the singular value $(\alpha\sqrt{M})^{-1}$ is exact: choosing $(\alpha\sqrt{M})^{-1} = \sin\frac{\pi}{2}(2k+1)$ maps it to 1 and eliminates the postselection, yielding a unitary that realizes $\mathcal{Q}$ deterministically on the traced-out ancilla. This removes the exponential success-probability loss from the repeated-interaction chain.

14.3 Dissipative state preparation

Fixed points as targets. A state σ is a *fixed point* (stationary state) of the dynamics if $\mathcal{L}(\sigma) = 0$. For closed systems any σ commuting with H is fixed—far too many to be useful. The dissipative part breaks this degeneracy: a well-chosen $\mathcal{L}$ can have a *unique* fixed point σ, in which case the semigroup is *ergodic* and $\mathrm{e}^{\mathcal{L}t}(\rho_0) \to \sigma$ for every initial ρ_0. This is the engine of *dissipative state preparation*: design H and the L_j so that the target (a thermal state $\mathrm{e}^{-\beta H_0}/Z$ or a ground state)

is that unique fixed point, then run the dynamics until it mixes. It is the open-system counterpart of the adiabatic and Zeno solvers of Chapter 10: instead of slowly deforming a Hamiltonian whose ground state is tracked, one relaxes irreversibly toward a designed steady state.

Building spectrally selective jump operators. The construction never diagonalizes H. From simple coupling operators A_a (e.g. Paulis) one forms *filtered* jump operators by convolving the Heisenberg-evolved coupling against a filter f_ω,

$$L_a(\omega) = \int_{-\infty}^{\infty} f_\omega(s)\, \mathrm{e}^{\mathrm{i}Hs} A_a\, \mathrm{e}^{-\mathrm{i}Hs}\, \mathrm{d}s, \tag{14.8}$$

which, in the eigenbasis $H|\psi_i\rangle = \lambda_i |\psi_i\rangle$, acts as $L_a(\omega) = \sum_{i,k} \hat{f}_\omega(\lambda_i - \lambda_k)\, \langle\psi_i|A_a|\psi_k\rangle\, |\psi_i\rangle\langle\psi_k|$, with $\hat{f}_\omega$ the Fourier transform of f_ω. Thus the filter $\hat{f}_\omega$ selects which energy *transitions* the jump operator drives. Because (14.8) is an integral of Hamiltonian evolutions, a block encoding of $L_a(\omega)$ is assembled by truncating, discretizing, and applying LCU (Chapter 7) to the Hamiltonian simulations of Chapters 11 and 12.

Proposition 14.7 (Ground state as a dark state)**.** *Let H have a nondegenerate ground state $|\psi_0\rangle$ with gap $\Delta = \lambda_1 - \lambda_0 > 0$, and take a single coupling A and filter with $\hat{f}(\nu) = 1$ for $\nu \le -\Delta$ and $\hat{f}(\nu) = 0$ for $\nu \ge 0$. Then the jump operator $L = \sum_{i,k} \hat{f}(\lambda_i - \lambda_k)\, \langle\psi_i|A|\psi_k\rangle\, |\psi_i\rangle\langle\psi_k|$ only lowers energy, so $L|\psi_0\rangle = 0$, and $|\psi_0\rangle\langle\psi_0|$ is a fixed point of the Lindbladian (14.1) with H as its coherent term.*

Proof. The support condition on $\hat{f}$ makes every surviving term $|\psi_i\rangle\langle\psi_k|$ carry $\lambda_i \le \lambda_k - \Delta < \lambda_k$, so L maps each eigenstate strictly downward in energy and annihilates the lowest, $L|\psi_0\rangle = 0$; hence $L^\dagger L|\psi_0\rangle = 0$ and $L|\psi_0\rangle\langle\psi_0|L^\dagger = 0$. Since also $[H, |\psi_0\rangle\langle\psi_0|] = 0$ ($|\psi_0\rangle$ is an eigenstate), every term of (14.1) annihilates $|\psi_0\rangle\langle\psi_0|$, so $\mathcal{L}(|\psi_0\rangle\langle\psi_0|) = 0$. Thus $|\psi_0\rangle$ is a *dark state*: population flows down the spectrum and pools in the ground state, which cannot decay further. □

Uniqueness of the dark state (and hence convergence) requires that the coupling connect every excited state to some lower state—an irreducibility condition on A—and quantitative convergence is measured by the *mixing time*

$$\tau_{\mathrm{mix}}(\eta) = \inf\Big\{ t : \sup_{\rho_0} \big\| \mathrm{e}^{\mathcal{L}t}(\rho_0) - \sigma \big\|_1 \le \eta \Big\}. \tag{14.9}$$

On n sites one speaks of *fast mixing* when $\tau_{\mathrm{mix}} = \mathcal{O}(\mathrm{poly}(n))$ and *rapid mixing* when $\tau_{\mathrm{mix}} = \mathcal{O}(\mathrm{polylog}(n))$; proving such bounds (the open-system analogue of the classical Markov-chain spectral-gap estimates) is problem-specific and an active research area. The total simulation cost is then τ_{mix} multiplied by the per-time cost of one of the simulators of Sections 14.2 and 14.4.

Unraveling. For intuition and for classical numerics it helps to represent the same semigroup as an average over stochastic pure-state trajectories, the *quantum-jump* (Monte Carlo wavefunction) unraveling: the Itô stochastic Schrödinger equation $\mathrm{d}|\psi_t\rangle = (-\mathrm{i}H - \frac{1}{2}\sum_j L_j^\dagger L_j)|\psi_t\rangle\, \mathrm{d}t + \sum_j L_j |\psi_t\rangle\, \mathrm{d}W_t^j$ satisfies $\mathbb{E}[|\psi_t\rangle\langle\psi_t|] = \rho(t)$ whenever $\mathbb{E}[|\psi_0\rangle\langle\psi_0|] = \rho(0)$. Between jumps the state evolves under the non-Hermitian effective Hamiltonian $H_{\mathrm{eff}} = H - \frac{\mathrm{i}}{2}\sum_j L_j^\dagger L_j$; this is exactly the dissipative non-unitary dynamics of Chapter 13, and it motivates the truncated Dyson method below.

Operator splitting. An alternative to dilation splits the generator $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2$ into two groups (typically coherent versus dissipative) and applies the first-order product $\Phi(\Delta t) = \mathrm{e}^{\mathcal{L}_1 \Delta t}\mathrm{e}^{\mathcal{L}_2 \Delta t}$. The error is governed by the same superoperator-commutator bound as the Trotter formula (Section 11.1):

$$\left\| \mathrm{e}^{\mathcal{L}\Delta t} - \mathrm{e}^{\mathcal{L}_1 \Delta t}\mathrm{e}^{\mathcal{L}_2 \Delta t} \right\|_\diamond \le \tfrac{1}{2}\Delta t^2 \, \|[\mathcal{L}_1, \mathcal{L}_2]\|_\diamond , \tag{14.10}$$

so if $[\mathcal{L}_1, \mathcal{L}_2] = 0$ the splitting is exact. With the coarse bounds $\|\mathcal{L}_H\|_\diamond \le 2\,\|H\|$ and $\|\mathcal{D}_j\|_\diamond \le 2\,\|L_j\|^2$ for the single-jump generator $\mathcal{D}_j(\rho) = L_j \rho L_j^\dagger - \frac{1}{2}\{L_j^\dagger L_j, \rho\}$, the long-time error is $\mathcal{O}(t\,\|\mathcal{L}\|_{\mathrm{be}}^2\,\Delta t)$ as in (14.7). Crucially, unlike the Hamiltonian case, *higher-order* product formulas cannot be used naïvely: they require negative time steps, but $\mathrm{e}^{-\mathcal{L}t}$ need not be a channel, so one may not run dissipation backwards. Splitting is attractive when the pieces are exactly solvable: for site-local dephasing $\mathcal{D}_j$ commutes with $\mathcal{D}_k$ and $\mathrm{e}^{\mathcal{D}_j \Delta t}(\rho) = \frac{1+\mathrm{e}^{-2\gamma\Delta t}}{2}\rho + \frac{1-\mathrm{e}^{-2\gamma\Delta t}}{2} Z_j \rho Z_j$ is an elementary single-qubit dephasing channel.

14.4 Nonlinear PDEs via Carleman linearization

From nonlinear to infinite-linear. Quantum evolution is linear, so a nonlinear differential equation must be recast as a linear one before a quantum solver applies. Consider the quadratic autonomous ODE—the generic form after semi-discretizing a quadratic PDE such as Navier–Stokes or a reaction–diffusion equation—

$$\frac{\mathrm{d}z}{\mathrm{d}t} = F_1 z + F_2\,(z \otimes z), \qquad z(0) = z_0 \in \mathbb{C}^d, \tag{14.11}$$

with $F_1 \in \mathbb{C}^{d\times d}$ and $F_2 \in \mathbb{C}^{d\times d^2}$. *Carleman linearization* lifts this to a linear system on the tower of tensor powers $z^{\otimes k}$. Introduce the variables $y_k := z^{\otimes k} \in \mathbb{C}^{d^k}$ (so $y_1 = z$) and differentiate using the Leibniz rule and (14.11):

$$\frac{\mathrm{d}y_k}{\mathrm{d}t} = \sum_{\ell=1}^{k} z^{\otimes(\ell-1)} \otimes \dot z \otimes z^{\otimes(k-\ell)} = A_k^k\, y_k + A_k^{k+1}\, y_{k+1}, \tag{14.12}$$

where A_k^k collects the k copies of the *linear* part F_1 (each acting on one tensor slot, the others carried by identities) and A_k^{k+1} collects the k copies of the *quadratic* part F_2, which raises the level from $k+1$ to k. Explicitly,

$$A_k^k = \sum_{\ell=1}^{k} \mathbb{I}^{\otimes(\ell-1)} \otimes F_1 \otimes \mathbb{I}^{\otimes(k-\ell)}, \qquad A_k^{k+1} = \sum_{\ell=1}^{k} \mathbb{I}^{\otimes(\ell-1)} \otimes F_2 \otimes \mathbb{I}^{\otimes(k-\ell)}. \tag{14.13}$$

Equation (14.12) is an *exact* but infinite linear system: the k-th block couples only to itself and the next level $k+1$, an upper-bidiagonal (in block index) structure. Truncating at level N—setting $y_{N+1} \equiv 0$—gives a finite linear ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat y = \mathcal{A}\,\hat y, \qquad \hat y = (y_1; y_2; \ldots; y_N) \in \mathbb{C}^{d+d^2+\cdots+d^N}, \tag{14.14}$$

whose block-bidiagonal generator $\mathcal{A}$ carries A_k^k on the diagonal and A_k^{k+1} on the super-diagonal. This is precisely a linear ODE of the form treated in Chapter 13: solve it with the linear-system, time-marching, LCHS, or Schrödingerization method, then read off $z(T)$ from the first block y_1.

Convergence and the dissipativity condition. The truncation is useful only if the neglected levels contribute negligibly. This is controlled by a dimensionless ratio comparing the nonlinear forcing to the linear dissipation. Assume F_1 is diagonalizable with eigenvalues of negative real part, order them $\operatorname{Re}\lambda_d \le \cdots \le \operatorname{Re}\lambda_1 < 0$, and set

$$R := \frac{\|z_0\|\,\|F_2\|}{|\operatorname{Re}\lambda_1|}. \tag{14.15}$$

R is the ratio of the quadratic strength $\|F_2\|\,\|z_0\|$ to the linear decay rate $|\operatorname{Re}\lambda_1|$; it plays the role of a Reynolds number.

Theorem 14.8 (Carleman truncation error)**.** *Suppose F_1 is diagonalizable with $\operatorname{Re}\lambda_1 < 0$ and $R < 1$ as in (14.15). Let $z(t)$ solve (14.11) and let $\hat{y}(t)$ solve the level-N truncation (14.14) with the exact initial data $y_k(0) = z_0^{\otimes k}$. Then the error in the recovered first block obeys*

$$\max_{0\le t\le T} \|y_1(t) - z(t)\| \le \|z_0\|\, R^N \left(1 - \mathrm{e}^{-|\operatorname{Re}\lambda_1| t}\right)^N \le \|z_0\|\, R^N, \tag{14.16}$$

so the truncation error decays geometrically *in the level N; achieving accuracy ϵ needs only $N = \mathcal{O}\big(\log(\|z_0\|/\epsilon)/\log(1/R)\big)$.*

Proof sketch. Diagonalizing F_1 and passing to the eigenbasis, the linear part contracts each mode at rate at least $|\operatorname{Re}\lambda_1|$, while the quadratic coupling A_k^{k+1} feeds level $k+1$ into level k with strength $\le k\,\|F_2\|$. Writing the error $\eta_k(t) = y_k(t) - z(t)^{\otimes k}$ (with η_k zero above the truncation contributing the sole inhomogeneity at level N through the dropped $A_N^{N+1} y_{N+1}$ term), Duhamel's principle (Theorem 13.1) bounds $\|\eta_k\|$ by a convolution of the decaying propagator against the neglected term. The neglected term at the top level is $\left\|A_N^{N+1}\right\|\,\|y_{N+1}\| \le N\,\|F_2\|\,\|z(t)\|^{N+1}$, and $\|z(t)\| \le \|z_0\|\,\mathrm{e}^{-|\operatorname{Re}\lambda_1| t}$ under $R < 1$ (the quadratic term cannot overturn the linear decay). Solving the resulting recursion $\|\eta_1\| \lesssim \left(\|F_2\|\,\|z_0\|\,/\,|\operatorname{Re}\lambda_1|\right)^N \|z_0\| = R^N\,\|z_0\|$, with the extra $(1-\mathrm{e}^{-|\operatorname{Re}\lambda_1| t})^N$ factor from the finite-time integrals, gives (14.16). The condition $R < 1$ is what makes the geometric series converge; when $R \ge 1$ the nonlinear cascade can dominate and the truncation is not controlled. □

Remark 14.9 (Cost and caveats). The truncated system (14.14) has dimension $\sum_{k=1}^{N} d^k = \mathcal{O}(d^N)$, so it uses $\mathcal{O}(N \log d)$ qubits—exponentially smaller than the classical $\mathcal{O}(d^N)$ state, provided the block-encoding oracles for $\mathcal{A}$ are efficient (they inherit the sparsity of F_1, F_2). Combined with the linear-ODE solvers of Chapter 13, the end-to-end cost is $\operatorname{poly}(N, d, \alpha_{\mathcal{A}} T, \log(1/\epsilon))$ and, because $N = \mathcal{O}(\log(1/\epsilon)/\log(1/R))$, polylogarithmic in $1/\epsilon$. The essential restriction is $R < 1$: strongly nonlinear (high-Reynolds) regimes fall outside the guarantee. An alternative for scalar conservation laws and Hamilton–Jacobi equations is the *level-set* approach, which lifts the nonlinear equation to a *linear* Liouville (transport) equation in phase space at the cost of extra dimensions; we do not develop it here.

14.5 Example: Burgers' equation via Cole–Hopf

An exact linearization. Some nonlinear PDEs linearize *exactly*, with no truncation. The archetype is the viscous Burgers equation, the simplest model combining nonlinear advection with diffusion,

$$\partial_t u + u\,\partial_x u = \nu\,\partial_{xx} u, \qquad \nu > 0, \tag{14.17}$$

where ν is the viscosity. The *Cole–Hopf transformation* substitutes

$$u = -2\nu\,\frac{\partial_x \phi}{\phi} = -2\nu\,\partial_x(\ln \phi), \tag{14.18}$$

for a strictly positive potential $\phi(x,t) > 0$, and turns (14.17) into the linear heat equation.

Theorem 14.10 (Cole–Hopf)**.** *If $\phi(x,t) > 0$ solves the heat equation $\partial_t \phi = \nu\,\partial_{xx}\phi$, then $u = -2\nu\,\partial_x\phi/\phi$ solves the viscous Burgers equation (14.17).*

Proof. Write $u = -2\nu(\ln\phi)_x$. We compute the three terms of (14.17) directly. For the spatial derivatives,

$$u_x = -2\nu(\ln\phi)_{xx} = -2\nu\Big(\frac{\phi_{xx}}{\phi} - \frac{\phi_x^2}{\phi^2}\Big), \qquad u_{xx} = -2\nu(\ln\phi)_{xxx} = -2\nu\Big(\frac{\phi_{xxx}}{\phi} - 3\frac{\phi_x\phi_{xx}}{\phi^2} + 2\frac{\phi_x^3}{\phi^3}\Big).$$

The nonlinear term is

$$u\,u_x = \Big(-2\nu\frac{\phi_x}{\phi}\Big)(-2\nu)\Big(\frac{\phi_{xx}}{\phi} - \frac{\phi_x^2}{\phi^2}\Big) = 4\nu^2\Big(\frac{\phi_x\phi_{xx}}{\phi^2} - \frac{\phi_x^3}{\phi^3}\Big).$$

For the time derivative, $u_t = -2\nu(\ln\phi)_{xt} = -2\nu(\phi_{xt}/\phi - \phi_x\phi_t/\phi^2)$. Now use the hypothesis $\phi_t = \nu\phi_{xx}$, hence $\phi_{xt} = \nu\phi_{xxx}$, to get

$$u_t = -2\nu^2\Big(\frac{\phi_{xxx}}{\phi} - \frac{\phi_x\phi_{xx}}{\phi^2}\Big).$$

Adding the advection term,

$$u_t + u\,u_x = -2\nu^2\frac{\phi_{xxx}}{\phi} + 2\nu^2\frac{\phi_x\phi_{xx}}{\phi^2} + 4\nu^2\frac{\phi_x\phi_{xx}}{\phi^2} - 4\nu^2\frac{\phi_x^3}{\phi^3} = -2\nu^2\frac{\phi_{xxx}}{\phi} + 6\nu^2\frac{\phi_x\phi_{xx}}{\phi^2} - 4\nu^2\frac{\phi_x^3}{\phi^3}.$$

On the other hand,

$$\nu\,u_{xx} = -2\nu^2\Big(\frac{\phi_{xxx}}{\phi} - 3\frac{\phi_x\phi_{xx}}{\phi^2} + 2\frac{\phi_x^3}{\phi^3}\Big) = -2\nu^2\frac{\phi_{xxx}}{\phi} + 6\nu^2\frac{\phi_x\phi_{xx}}{\phi^2} - 4\nu^2\frac{\phi_x^3}{\phi^3}.$$

The two expressions coincide, so $u_t + uu_x = \nu u_{xx}$, as claimed. □

Remark 14.11 (Why the substitution is guessed). The choice (14.18) is natural in two steps. First set $u = \partial_x\psi$ (assuming u has a potential); Burgers becomes $\partial_x(\psi_t + \frac{1}{2}\psi_x^2 - \nu\psi_{xx}) = 0$, so up to an x-independent function $\psi_t + \frac{1}{2}\psi_x^2 = \nu\psi_{xx}$, a Hamilton–Jacobi equation. Second, the exponential (Hopf–Cole) substitution $\psi = -2\nu\ln\phi$ converts the quadratic $\frac{1}{2}\psi_x^2$ into a linear term: $\psi_t = -2\nu\phi_t/\phi$, $\psi_x = -2\nu\phi_x/\phi$, $\psi_{xx} = -2\nu(\phi_{xx}/\phi - \phi_x^2/\phi^2)$, and substituting collapses everything to $\phi_t = \nu\phi_{xx}$. The nonlinearity is absorbed because the same ϕ_x^2/ϕ^2 term appears in $\frac{1}{2}\psi_x^2$ and in $\nu\psi_{xx}$ with matching coefficients.

The quantum algorithm. Cole–Hopf reduces solving Burgers to three classical/quantum steps. (i) Transform the initial data: given $u(x,0) = u_0(x)$, set $\phi(x,0) = \exp(-\frac{1}{2\nu}\int_0^x u_0(x')\,dx') > 0$, which is exactly the inversion of (14.18) at $t = 0$. (ii) Solve the linear heat equation $\partial_t\phi = \nu\partial_{xx}\phi$ to time T by any method of Chapter 13 (linear-system, LCHS, or Schrödingerization); after spatial discretization this is the tridiagonal Laplacian system of Section 13.6 with $\alpha_A = \mathcal{O}(\nu N^2)$. (iii) Recover $u(x,T) = -2\nu\,\partial_x\phi/\phi$, a division and a derivative on the grid. Because $\phi > 0$ stays strictly positive (the heat semigroup preserves positivity, and a positive initial ϕ stays positive), the recovery is numerically well posed. The nonlinear solve thus inherits the near-optimal $\widetilde{\mathcal{O}}(q\,\nu N^2 T)$ scaling of the linear heat-equation solver, with the norm-growth factor q now referring to the potential ϕ.

Remark 14.12 (Scope). Cole–Hopf is special to Burgers (and a handful of related equations); general quadratic PDEs have no exact linearization and require the truncated Carleman route of Section 14.4, whose guarantee is conditional on $R < 1$. Together the two examples bracket the state of the art: *exact* linearization where a lucky change of variables exists, and *approximate* linearization with controlled error in the weakly nonlinear regime otherwise.

Exercises

Exercise 14.1. *Trace preservation and the adjoint.* (a) Show directly from (14.1) that $\mathrm{Tr}[\mathcal{L}(\rho)] = 0$ for every ρ, so the Lindblad flow preserves the trace. (b) Verify that $\mathcal{L}^\dagger$ in (14.3) is the Hilbert–Schmidt adjoint of $\mathcal{L}$, i.e. $\mathrm{Tr}[\mathcal{L}(X)^\dagger Y] = \mathrm{Tr}[X^\dagger \mathcal{L}^\dagger(Y)]$ for all X, Y, and confirm that $\mathcal{L}^\dagger(\mathbb{I}) = 0$ (unitality of the Heisenberg dynamics).

Exercise 14.2. *Classical Markov chains.* Complete the computation of Theorem 14.3: with $H = 0$ and $L_{ij} = \sqrt{q_{ij}}\ |i\rangle\langle j|$, substitute into (14.1) and show that a diagonal $\rho = \sum_j p_j\, |j\rangle\langle j|$ stays diagonal with $\dot p = Qp$, identifying the entries of Q. Verify that each column of Q sums to zero, so e^{Qt} is column-stochastic.

Exercise 14.3. *The dilated Hamiltonian reproduces $\mathcal{L}$.* For a single jump operator L_1, take $\widetilde{H}$ from (14.4) and the channel (14.5). Expand $U = \mathrm{e}^{-\mathrm{i}\widetilde{H}\sqrt{\Delta t}}$ to second order, trace out the ancilla, and show $\mathrm{Tr}_a(U(|0\rangle\langle 0| \otimes \rho)U^\dagger) = \rho + \Delta t\, \mathcal{L}(\rho) + \mathcal{O}(\Delta t^2)$. Explain why the odd powers of $\sqrt{\Delta t}$ vanish under the partial trace, and why the factor $H/\sqrt{\Delta t}$ in (14.4) is needed for the coherent term to appear at order Δt.

Exercise 14.4. *Photon loss.* For the cavity model of Theorem 14.4 with $H = \omega a^\dagger a$ and $L = \sqrt{\kappa}\, a$, use the adjoint Lindbladian to compute $\mathcal{L}^\dagger(n)$ for $n = a^\dagger a$ and deduce $\langle n\rangle(t) = \langle n\rangle(0)\mathrm{e}^{-\kappa t}$. Then compute $\mathcal{L}^\dagger(a)$ and find the decay of the coherence $\langle a\rangle(t)$.

Exercise 14.5. *Dark state and fixed point.* Prove Theorem 14.7: given the energy-lowering filter, show $L\,|\psi_0\rangle = 0$ and that every term of the Lindbladian annihilates $|\psi_0\rangle\langle\psi_0|$, so $\mathcal{L}(|\psi_0\rangle\langle\psi_0|) = 0$. Explain what additional structural condition on the coupling A is needed for $|\psi_0\rangle\langle\psi_0|$ to be the *unique* fixed point, and relate the convergence rate to the mixing time (14.9).

Exercise 14.6. *Carleman linearization.* Starting from the quadratic ODE (14.11), derive the block recursion (14.12)–(14.13) for $y_k = z^{\otimes k}$ by differentiating $z^{\otimes k}$ with the product rule and substituting $\dot z = F_1 z + F_2(z \otimes z)$. Explain why the resulting infinite system is block upper-bidiagonal in the level index k, and why truncating at level N (setting $y_{N+1} = 0$) leaves a finite linear ODE solvable by the methods of Chapter 13. State the role of the ratio R of (14.15) in controlling the truncation error.

Exercise 14.7. *Cole–Hopf details.* (a) Re-derive Theorem 14.10 through the potential substitution of Theorem 14.11: set $u = \psi_x$, obtain the Hamilton–Jacobi equation $\psi_t + \frac{1}{2}\psi_x^2 = \nu\psi_{xx}$, then substitute $\psi = -2\nu \ln \phi$ and show it collapses to $\phi_t = \nu\phi_{xx}$. (b) Verify that the initial-data transform $\phi(x,0) = \exp(-\frac{1}{2\nu}\int_0^x u_0)$ inverts (14.18) at $t = 0$, and argue that $\phi > 0$ is preserved by the heat flow so the recovery $u = -2\nu\phi_x/\phi$ is well defined for all t.

Further reading

- Lin Lin, *live notes*, Ch. 21 (open quantum systems): the Lindblad equation and the proof that $\mathrm{e}^{\mathcal{L}t}$ is CPTP, the GKSL characterization, dissipative thermal and ground-state preparation, the

dilated-Hamiltonian (Stinespring) simulation, oblivious amplitude amplification of channels, the operator-splitting method, and the truncated Dyson method.

- H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems*, Oxford University Press, 2002, for physical background on Lindblad dynamics, quantum-jump unravelings, and Markovian master equations.

- G. Lindblad, *On the generators of quantum dynamical semigroups*, Commun. Math. Phys. **48**, 119 (1976); and V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, *Completely positive dynamical semigroups of N-level systems*, J. Math. Phys. **17**, 821 (1976), for the original GKSL theorem of Theorem 14.2.

- J.-P. Liu, H. Ø. Kolden, H. K. Krovi, N. F. Loureiro, K. Trivisa, and A. M. Childs, *Efficient quantum algorithm for dissipative nonlinear differential equations*, Proc. Natl. Acad. Sci. USA **118**, e2026805118 (2021) (arXiv:2011.03185), for the Carleman-linearization algorithm and the $R < 1$ convergence condition of Section 14.4.

- J. D. Cole, *On a quasi-linear parabolic equation occurring in aerodynamics*, Quart. Appl. Math. **9**, 225 (1951); and E. Hopf, *The partial differential equation $u_t + uu_x = \mu u_{xx}$*, Comm. Pure Appl. Math. **3**, 201 (1950), for the exact linearization of Burgers' equation (Section 14.5); see also L. C. Evans, *Partial Differential Equations*, 2nd ed., AMS, 2010, §4.4.

- S. Jin and N. Liu, *Quantum algorithms for nonlinear partial differential equations*, Bull. Sci. Math. **194**, 103457 (2024) (arXiv:2212.00681), for the level-set / Liouville (Hamilton–Jacobi) approach mentioned in Theorem 14.9.

Part IV

Tensor Networks

Chapter 15

Tensor-Network States

Learning objectives

- Read and manipulate tensor-network diagrams.
- Derive matrix product states (MPS) from sequential Schmidt decompositions.
- Put an MPS into canonical form and bound its contraction cost.
- Relate bond dimension to entanglement and the area law.

The state of n qubits lives in a space of dimension 2^n, so a generic state needs exponentially many amplitudes to write down—a wall that blocks any naive classical simulation past a few dozen qubits. Yet the states that occur in physics are far from generic: the ground states of local Hamiltonians, and the states reached by shallow circuits, are only *lightly entangled*, and a lightly entangled state can be compressed. *Tensor networks* are the data structure that performs this compression. This lecture develops the most important one-dimensional example, the *matrix product state* (MPS). We start with a graphical calculus that turns index bookkeeping into pictures, use the singular value decomposition (SVD) to read off the entanglement across any cut of a bipartite state (the Schmidt decomposition first met in Chapter 1), and then chain $n-1$ such decompositions together to factor an arbitrary state into a line of small tensors. The single number controlling the whole construction is the *bond dimension* χ: it measures entanglement, it sets the storage cost $n\,d\,\chi^2$ and the contraction cost, and—through the *area law* for gapped one-dimensional systems—it stays bounded exactly when the state is physically relevant. The MPS machinery here is the foundation for the DMRG ground-state algorithm of Chapter 16 and the time-evolution methods of Chapter 17.

15.1 Tensors and diagrammatic notation

Definition 15.1 (Tensor)**.** A *tensor* of *rank* (or *order*) r is a multi-index array of complex numbers

$$T_{i_1 i_2 \cdots i_r} \in \mathbb{C}, \qquad i_k \in \{1, \dots, d_k\},$$

with one index per *leg*. A rank-0 tensor is a scalar, a rank-1 tensor is a vector, and a rank-2 tensor is a matrix. The number of independent entries is $\prod_{k=1}^{r} d_k$, which grows exponentially in the rank when the leg dimensions are comparable.

An n-qubit state $|\psi\rangle = \sum_{s_1 \cdots s_n} \psi_{s_1 \cdots s_n} |s_1 \cdots s_n\rangle$ is exactly a rank-n tensor $\psi_{s_1 \cdots s_n}$ with each $s_k \in \{0, 1\}$ (leg dimension $d = 2$). The exponential cost of Theorem 15.1 is the cost of storing a generic state; the rest of the lecture is about avoiding it.

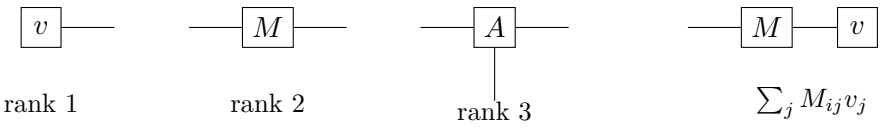


Figure 15.1: Penrose notation. Boxes are tensors, legs are indices, and a leg joining two boxes is a summed (contracted) index. The rightmost picture is the matrix–vector product $(Mv)_i = \sum_j M_{ij} v_j$.

15.1.1 Diagrammatic (Penrose) notation

Tensor-network diagrams replace indices by pictures. The rule is simple:

- a tensor is a *box* (or node);
- each *leg* (line) sticking out of the box is one index;
- a leg joining two boxes is a *contracted* (summed) index, shared by the two tensors.

Thus a vector v_i has one leg, a matrix M_{ij} has two, and a rank-3 tensor A_{ijk} has three. Contraction—summation over a shared index—is drawn by connecting the corresponding legs.

For example, matrix multiplication $C_{ik} = \sum_j A_{ij} B_{jk}$ is two boxes A, B joined by the leg j, leaving external legs i and k; the trace $\mathrm{Tr}[A] = \sum_i A_{ii}$ is a single box with its two legs joined into a loop, and the inner product $\langle u|v\rangle = \sum_i \overline{u_i}\, v_i$ is two vectors joined along their only leg. A *tensor network* is a collection of tensors with some legs contracted (internal) and some free (external); contracting all internal legs produces one tensor whose legs are the external ones.

15.1.2 The cost of contracting a network

The order in which we contract legs does not change the answer, but it changes the cost dramatically. The cost of a single pairwise contraction is easy to count.

Proposition 15.2 (Cost of a pairwise contraction)**.** *Let A and B share a set of legs whose dimensions multiply to D_{c} (the contracted dimension), and let their free legs have dimensions multiplying to D_A and D_B respectively. Forming*

$$C_{\alpha\beta} = \sum_{\gamma} A_{\alpha\gamma} B_{\gamma\beta}$$

costs $\mathcal{O}(D_A D_B D_{\mathrm{c}})$ scalar multiply–adds, since each of the $D_A D_B$ entries of C is a sum over D_{c} products.

Proof. Group the free legs of A into a single index α of size D_A, those of B into β of size D_B, and the shared legs into γ of size D_{c}. Then A is a $D_A \times D_{\mathrm{c}}$ matrix, B is $D_{\mathrm{c}} \times D_B$, and $C = AB$ is an ordinary matrix product, which costs $D_A D_{\mathrm{c}} D_B$ multiply–adds. □

Example 15.3 (Contraction order matters)**.** Consider three matrices of size $d \times d$ contracted along a chain, $T_{il} = \sum_{j,k} A_{ij} B_{jk} C_{kl}$. Contracting (AB) first, then $\cdot C$, costs $d^3 + d^3 = 2d^3$; the same holds for $A\,(BC)$. But for the network $\sum_{jk} A_{ij} B_{jk} C_{ik}$ (a triangle with external legs on each node) the contraction cost depends on which pair is contracted first, and choosing the pair that leaves the smallest intermediate tensor is essential. Finding the optimal contraction order of a general network is NP-hard, but for the chain-like networks of an MPS the left-to-right order is optimal and gives the linear scaling we derive in Section 15.5.

The moral of Theorem 15.3: a tensor network is efficient to contract only when it has favorable structure. Matrix product states are the structure that makes one-dimensional quantum states tractable.

15.2 Schmidt decomposition and the SVD

The engine behind every MPS is the singular value decomposition applied to a bipartition of the state. We first recall the SVD, then translate it into the language of quantum states.

Theorem 15.4 (Singular value decomposition)**.** *Every matrix $M \in \mathbb{C}^{m\times n}$ factors as*

$$M = U\Sigma V^\dagger,$$

where $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ are unitary and $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal with nonnegative entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ (the singular values*). The number of nonzero singular values equals* $\operatorname{rank} M$*.*

We take the SVD as known from numerical linear algebra. Its use here is the following restatement of the decomposition of a bipartite pure state, already previewed for the Bell pair in Chapter 1.

Theorem 15.5 (Schmidt decomposition)**.** *Let $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ be a pure state, with $\dim \mathcal{H}_A = d_A$, $\dim \mathcal{H}_B = d_B$. There exist orthonormal sets $\{|a_k\rangle\} \subset \mathcal{H}_A$ and $\{|b_k\rangle\} \subset \mathcal{H}_B$ and real numbers $\lambda_k > 0$ such that*

$$|\psi\rangle = \sum_{k=1}^{\chi} \lambda_k \, |a_k\rangle \otimes |b_k\rangle \,, \qquad \sum_{k=1}^{\chi} \lambda_k^2 = 1. \tag{15.1}$$

The number $\chi \leq \min(d_A, d_B)$ of terms is the Schmidt rank*, and the λ_k are the* Schmidt coefficients*. The decomposition is unique up to the ordering of terms and phases within any degenerate block.*

Proof. Expand $|\psi\rangle = \sum_{i,j} \psi_{ij} \, |i\rangle_A \, |j\rangle_B$ in product bases and read the coefficients as a $d_A \times d_B$ matrix $\psi = (\psi_{ij})$. By Theorem 15.4 write $\psi = U\Sigma V^\dagger$, i.e. $\psi_{ij} = \sum_k U_{ik}\sigma_k \overline{V_{jk}}$. Define $|a_k\rangle = \sum_i U_{ik} \, |i\rangle_A$ and $|b_k\rangle = \sum_j \overline{V_{jk}} \, |j\rangle_B$. Because U and V are unitary, $\{|a_k\rangle\}$ and $\{|b_k\rangle\}$ are orthonormal: $\langle a_k | a_{k'}\rangle = \sum_i \overline{U_{ik}} U_{ik'} = (U^\dagger U)_{kk'} = \delta_{kk'}$, and likewise for the $|b_k\rangle$. Substituting,

$$|\psi\rangle = \sum_{i,j} \Big(\sum_k U_{ik}\sigma_k \overline{V_{jk}} \Big) \, |i\rangle_A \, |j\rangle_B = \sum_k \sigma_k \, |a_k\rangle \otimes |b_k\rangle \,.$$

Keep only the $\chi = \operatorname{rank}\psi$ terms with $\sigma_k > 0$ and set $\lambda_k = \sigma_k$. Normalization of $|\psi\rangle$ gives $1 = \langle\psi|\psi\rangle = \sum_k \lambda_k^2$ by orthonormality. Since $\operatorname{rank}\psi \leq \min(d_A, d_B)$ we have $\chi \leq \min(d_A, d_B)$. □

The Schmidt decomposition is a bipartition diagonalization: it simultaneously diagonalizes *both* reduced density operators.

Corollary 15.6 (Reduced states and entanglement entropy)**.** *With the notation of Theorem 15.5, the reduced density operators are*

$$\rho_A = \operatorname{Tr}_B |\psi\rangle\langle\psi| = \sum_k \lambda_k^2 \, |a_k\rangle\langle a_k| \,, \qquad \rho_B = \operatorname{Tr}_A |\psi\rangle\langle\psi| = \sum_k \lambda_k^2 \, |b_k\rangle\langle b_k| \,.$$

They share the spectrum $\{\lambda_k^2\}$, so the entanglement entropy *of the cut,*

$$S(\rho_A) = -\operatorname{Tr}[\rho_A \log_2 \rho_A] = -\sum_{k=1}^{\chi} \lambda_k^2 \log_2 \lambda_k^2, \tag{15.2}$$

equals $S(\rho_B)$ and satisfies $0 \leq S \leq \log_2 \chi$. In particular $S = 0$ iff $\chi = 1$ (a product state), and $S = \log_2 \chi$ iff all $\lambda_k^2 = 1/\chi$ (a maximally entangled cut).

Proof. Compute $\mathrm{Tr}_B\,|\psi\rangle\langle\psi|$ using (15.1) and $\mathrm{Tr}_B[|b_k\rangle\langle b_{k'}|] = \langle b_{k'}|b_k\rangle = \delta_{kk'}$:

$$\rho_A = \sum_{k,k'} \lambda_k \lambda_{k'}\, |a_k\rangle\langle a_{k'}|\, \langle b_{k'}|b_k\rangle = \sum_k \lambda_k^2\, |a_k\rangle\langle a_k|\,,$$

and symmetrically for ρ_B; the two operators have the same nonzero eigenvalues λ_k^2. Formula (15.2) is the von Neumann entropy evaluated on this spectrum. The probability vector $p_k = \lambda_k^2$ has Shannon entropy in $[0, \log_2 \chi]$ by concavity, with the stated equality cases. □

Remark 15.7 (Schmidt rank certifies entanglement). Theorem 15.6 recovers the criterion of Chapter 1: a pure state is a product state across a cut iff its Schmidt rank there is $\chi = 1$, and it is entangled iff $\chi \geq 2$. The entanglement entropy (15.2) quantifies *how* entangled the cut is. Because ρ_A and ρ_B share a spectrum, the entanglement of a pure state is a property of the *cut*, not of either side; this symmetry is why a single number χ will suffice to label each bond of an MPS.

15.2.1 Truncation error

The point of the SVD for compression is that discarding the smallest singular values is the *optimal* low-rank approximation, and the error it incurs is exactly the tail of the squared Schmidt coefficients.

Theorem 15.8 (Optimal truncation). *Let $|\psi\rangle$ have Schmidt decomposition (15.1) with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\chi > 0$. For a target rank $\chi' < \chi$, the truncated (unnormalized) state $|\tilde\psi\rangle = \sum_{k=1}^{\chi'} \lambda_k\, |a_k\rangle \otimes |b_k\rangle$ minimizes $\||\psi\rangle - |\phi\rangle\|$ over all states $|\phi\rangle$ of Schmidt rank $\leq \chi'$, and the minimal error is*

$$\left\||\psi\rangle - |\tilde\psi\rangle\right\|^2 = \sum_{k=\chi'+1}^{\chi} \lambda_k^2 \;=\; \varepsilon_{\chi'} \qquad \text{(the discarded weight)}. \tag{15.3}$$

Proof. By orthonormality of the Schmidt vectors, $\left\||\psi\rangle - |\tilde\psi\rangle\right\|^2 = \sum_{k>\chi'} \lambda_k^2$, giving the right-hand side. Optimality is the Eckart–Young theorem: for the coefficient matrix ψ the best rank-χ' approximation in Frobenius norm is obtained by keeping the χ' largest singular values, and the Frobenius norm of the coefficient matrix equals the vector norm $\|\cdot\|$ of the state. Any Schmidt rank-$\leq \chi'$ state $|\phi\rangle$ corresponds to a rank-$\leq \chi'$ coefficient matrix, so no such $|\phi\rangle$ beats the truncated SVD. □

Equation (15.3) is the workhorse of the whole subject: to compress a state we throw away the smallest Schmidt coefficients, and the squared error is simply their summed squares. If the coefficients decay quickly—as they do for lightly entangled states—then a modest χ' already captures the state to high fidelity.

15.3 Matrix product states

We now factor an arbitrary n-body state into a chain of small tensors by applying the Schmidt decomposition, equivalently the SVD, across each of the $n-1$ bonds in turn.

Theorem 15.9 (Existence of the MPS form). *Every state $|\psi\rangle = \sum_{s_1\cdots s_n} \psi_{s_1\cdots s_n}\, |s_1 \cdots s_n\rangle$ of n subsystems of local dimension d can be written as a* matrix product state

$$\psi_{s_1 s_2 \cdots s_n} = \sum_{\alpha_1,\ldots,\alpha_{n-1}} A^{[1]s_1}_{\alpha_1}\, A^{[2]s_2}_{\alpha_1\alpha_2} \cdots A^{[n]s_n}_{\alpha_{n-1}}, \tag{15.4}$$

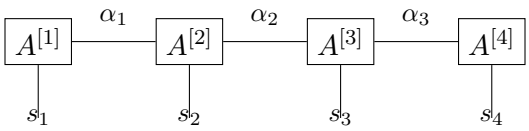


Figure 15.2: A matrix product state on $n = 4$ sites. Downward legs s_k are physical indices; horizontal legs α_k are virtual bonds of dimension χ_k. Contracting the network yields the amplitude $\psi_{s_1 s_2 s_3 s_4}$.

i.e. each amplitude is a product of matrices $A^{[k]s_k}$ (one $\chi_{k-1} \times \chi_k$ matrix for each value of the physical index s_k), with $\chi_0 = \chi_n = 1$. The bond dimension χ_k *across the cut between sites k and $k+1$ can be taken equal to the Schmidt rank there, and $\chi = \max_k \chi_k$.*

Proof. Reshape ψ into a matrix $\Psi^{(1)}$ with row index s_1 (size d) and column index $(s_2 \cdots s_n)$ (size d^{n-1}). Its SVD gives $\Psi^{(1)} = U^{(1)}\Sigma^{(1)}(V^{(1)})^\dagger$ with inner dimension $\chi_1 = \operatorname{rank}\Psi^{(1)}$. Absorb $U^{(1)}$ into the first tensor, $A^{[1]s_1}_{\alpha_1} = U^{(1)}_{s_1\alpha_1}$, and carry the rest forward as $M^{(1)}_{\alpha_1,(s_2\cdots s_n)} = \sum(\Sigma^{(1)}(V^{(1)})^\dagger)_{\alpha_1,(s_2\cdots s_n)}$, so that

$$\psi_{s_1\cdots s_n} = \sum_{\alpha_1} A^{[1]s_1}_{\alpha_1}\, M^{(1)}_{\alpha_1,(s_2\cdots s_n)}.$$

Now regroup $M^{(1)}$ into a matrix with rows $(\alpha_1 s_2)$ (size $\chi_1 d$) and columns $(s_3 \cdots s_n)$, take its SVD with inner dimension χ_2, set $A^{[2]s_2}_{\alpha_1\alpha_2} = U^{(2)}_{(\alpha_1 s_2),\alpha_2}$, and carry the remainder $M^{(2)}$ forward. Iterating this SVD "sweep" across all $n-1$ bonds peels off one tensor $A^{[k]s_k}$ at a time and terminates with a final vector $A^{[n]s_n}_{\alpha_{n-1}}$. Multiplying the pieces back together reproduces (15.4). Because the k-th SVD is exactly the Schmidt decomposition across the cut between the first k and last $n-k$ sites, its inner dimension is the Schmidt rank χ_k there. □

Diagrammatically an MPS is a line of rank-3 tensors (rank-2 at the ends), one per site: each has a physical leg s_k pointing out and *virtual* (bond) legs α_{k-1}, α_k joining it to its neighbors.

15.3.1 Exponential-to-linear compression

Proposition 15.10 (Storage cost)**.** *An MPS on n sites of local dimension d and bond dimension bounded by χ stores at most $n\,d\,\chi^2$ complex numbers.*

Proof. Each site tensor $A^{[k]}$ holds d matrices of size at most $\chi \times \chi$, hence at most $d\chi^2$ entries; the two boundary tensors hold only $d\chi$ each. Summing over n sites gives the bound $n\,d\,\chi^2$. □

This is the central win: the d^n amplitudes of a generic state are replaced by $\mathcal{O}(n\,d\,\chi^2)$ parameters—linear in n whenever χ is bounded. The representation is *exact* when every Schmidt rank χ_k is at most χ; it is a faithful *approximation*, with error controlled by (15.3) summed over the truncated bonds, when the discarded Schmidt weight is small. As we make precise in Section 15.5, this is exactly the regime of low (area-law) entanglement.

Example 15.11 (MPS of a product state has $\chi = 1$)**.** A product state $|\psi\rangle = \bigotimes_{k=1}^{n} |\phi_k\rangle$ with $|\phi_k\rangle = \sum_s c^{[k]}_s |s\rangle$ has amplitude $\psi_{s_1\cdots s_n} = \prod_k c^{[k]}_{s_k}$, which is already an MPS (15.4) with all bond dimensions $\chi_k = 1$: each tensor $A^{[k]s_k} = c^{[k]}_{s_k}$ is a 1×1 "matrix." Every Schmidt rank is 1, consistent with the state having no entanglement across any cut, and the storage is nd numbers.

Example 15.12 (A three-qubit MPS by explicit SVD)**.** Consider the (unnormalized before we fix it) three-qubit state

$$|\psi\rangle = \tfrac{1}{2}(|000\rangle + |011\rangle + |100\rangle + |111\rangle),$$

which is normalized ($4 \cdot \frac{1}{4} = 1$). *First bond.* Reshape the amplitudes into the 2×4 matrix $\Psi^{(1)}$ with row s_1 and columns $(s_2 s_3)$ ordered $00, 01, 10, 11$:

$$\Psi^{(1)} = \tfrac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Both rows are equal, so $\operatorname{rank}\Psi^{(1)} = 1$ and $\chi_1 = 1$. Its SVD has one singular value $\sigma_1 = \left\|\Psi^{(1)}\right\|_F = \frac{1}{2}\sqrt{1+1+1+1} = 1$, with $U^{(1)} = \frac{1}{\sqrt{2}}\binom{1}{1}$ and row vector $(V^{(1)})^\dagger = \frac{1}{\sqrt{2}}(1,0,0,1)$. Thus $A^{[1]s_1} = \frac{1}{\sqrt{2}}$ for $s_1 \in \{0,1\}$ (a 1×1 block), i.e. $|a_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$, and the state factors as $|\psi\rangle = |+\rangle_1 \otimes |\phi\rangle_{23}$ with $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. So qubit 1 is unentangled from $\{2,3\}$: $S = 0$, $\chi_1 = 1$.

Second bond. The carried remainder is the Bell state $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{23}$ (see Theorem 15.24), whose coefficient matrix across $2 \mid 3$ is $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ with two equal singular values $\frac{1}{\sqrt{2}}$; hence $\chi_2 = 2$ and $S = 1$. Reading off the site tensors, $A^{[1]0} = A^{[1]1} = \frac{1}{\sqrt{2}}$ (scalars),

$$A^{[2]0} = \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 \end{pmatrix}, \quad A^{[2]1} = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} \end{pmatrix}, \qquad A^{[3]0} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A^{[3]1} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

One checks the product reproduces the amplitudes: e.g. for 000, $A^{[1]0}A^{[2]0}A^{[3]0} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cdot 1 = \frac{1}{2}$, and for 011, $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cdot 1 = \frac{1}{2}$, while forbidden strings such as 001 give $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cdot 0 = 0$. The maximum bond dimension is $\chi = 2$.

15.4 Canonical forms and gauge freedom

The MPS representation (15.4) is not unique: the bond indices carry a *gauge freedom* that we now describe and then fix by choosing a canonical form.

Proposition 15.13 (Bond gauge freedom)**.** *Let G be any invertible $\chi_k \times \chi_k$ matrix acting on bond k. The replacement*

$$A^{[k]s_k} \longmapsto A^{[k]s_k}G, \qquad A^{[k+1]s_{k+1}} \longmapsto G^{-1}A^{[k+1]s_{k+1}} \tag{15.5}$$

leaves the physical state (15.4) *unchanged.*

Proof. In the matrix product (15.4) the two altered factors appear adjacently as $\cdots A^{[k]s_k}A^{[k+1]s_{k+1}}\cdots$. After the substitution this becomes $\cdots (A^{[k]s_k}G)(G^{-1}A^{[k+1]s_{k+1}})\cdots = \cdots A^{[k]s_k}(GG^{-1})A^{[k+1]s_{k+1}}\cdots$, and $GG^{-1} = \mathbb{I}$ restores the original product. All other amplitudes are untouched. □

This freedom—one invertible matrix per bond—is exactly enough to impose orthonormality conditions on the tensors.

Definition 15.14 (Left- and right-canonical tensors)**.** A site tensor is *left-canonical* if, viewing $A^{[k]s_k}_{\alpha\beta}$ as a matrix A with combined row index (α, s_k) and column index β, it is left-isometric:

$$\sum_{s_k} (A^{[k]s_k})^\dagger A^{[k]s_k} = \mathbb{I}_{\chi_k}. \tag{15.6}$$

It is *right-canonical* if, with row index α and column index (s_k, β), it is right-isometric:

$$\sum_{s_k} A^{[k]s_k} \big(A^{[k]s_k}\big)^\dagger = \mathbb{I}_{\chi_{k-1}}. \tag{15.7}$$

An MPS is in *left- (right-) canonical form* if every tensor is left- (right-) canonical.

The left-canonical tensors are precisely the U factors produced by the left-to-right SVD sweep in the proof of Theorem 15.9: an SVD factor U is an isometry, $U^\dagger U = \mathbb{I}$, which is exactly (15.6). Sweeping from the right with the $V^\dagger$ factors gives the right-canonical form.

Proposition 15.15 (Diagrammatic meaning of canonical form)**.** *If sites $1, \dots, k$ are left-canonical, then contracting the tensor network of $\langle\psi|\psi\rangle$ from the left collapses those sites to the identity on bond k: the left environment is $\mathbb{I}_{\chi_k}$. Symmetrically, right-canonical sites $k+1, \dots, n$ collapse from the right.*

Proof. Contract site 1: $\sum_{s_1} (A^{[1]s_1})^\dagger A^{[1]s_1} = \mathbb{I}_{\chi_1}$ by (15.6). Feed this identity into site 2: $\sum_{s_2} (A^{[2]s_2})^\dagger \mathbb{I}\, A^{[2]s_2} = \mathbb{I}_{\chi_2}$, again by (15.6). Induction on k propagates the identity rightward to bond k; the right-canonical case is the mirror image. □

15.4.1 Mixed-canonical form and the orthogonality center

Combining the two one-sided forms gives the representation used by essentially every MPS algorithm.

Definition 15.16 (Mixed-canonical form)**.** An MPS is in *mixed-canonical form with orthogonality center at site c* if sites $1, \dots, c-1$ are left-canonical, sites $c+1, \dots, n$ are right-canonical, and the center tensor at site c carries the state's norm. Equivalently one writes

$$\psi_{s_1 \cdots s_n} = A^{[1]s_1} \cdots A^{[c-1]s_{c-1}}\, M^{[c]s_c}\, B^{[c+1]s_{c+1}} \cdots B^{[n]s_n}, \tag{15.8}$$

with A-tensors left-canonical and B-tensors right-canonical.

Theorem 15.17 (Reduced state from the center)**.** *For a normalized MPS in mixed-canonical form (15.8) with center at site c, the Schmidt coefficients across the bond $(c \mid c+1)$ are the singular values of the center tensor reshaped across that bond, and*

$$\langle\psi|\psi\rangle = \sum_{s_c} \mathrm{Tr}\big[(M^{[c]s_c})^\dagger M^{[c]s_c}\big] = 1.$$

Consequently expectation values of an operator supported near site c are computed by contracting only the center tensor against $\mathbb{I}$ environments on both sides.

Proof. By Theorem 15.15 the left $(1, \dots, c-1)$ and right $(c+1, \dots, n)$ blocks each contract to the identity in $\langle\psi|\psi\rangle$, leaving only $\sum_{s_c} \mathrm{Tr}[(M^{[c]s_c})^\dagger M^{[c]s_c}]$, which normalization sets to 1. Grouping the center as a matrix across the bond and taking its SVD exhibits the Schmidt decomposition of the cut $(c \mid c+1)$ directly, since the left block is an isometry into $\mathcal{H}_{1\cdots c}$ and the right block an isometry into $\mathcal{H}_{c+1\cdots n}$; the singular values are the λ_k of Theorem 15.5. An operator supported at site c then contracts against the identity environments on both sides. □

Remark 15.18 (Why canonical forms matter). The gauge freedom of Theorem 15.13 is spent to make the environments trivial. This has three payoffs used throughout Chapters 16 and 17: (i) expectation values and norms are read off *locally* at the center, without contracting the whole chain;

(ii) the Schmidt spectrum at any bond is available directly as singular values, so *truncation* to bond dimension χ by Theorem 15.8 is a local SVD; and (iii) numerical stability is improved because the isometric tensors have unit-norm columns. Moving the orthogonality center from site c to $c+1$ is one local SVD (or QR), so the center can be swept back and forth cheaply—the basic move of the DMRG algorithm.

15.5 Contraction cost, bond dimension, and the area law

We close by quantifying when MPS are efficient and explaining, via the area law, why the physically important states fall in that regime.

Proposition 15.19 (Cost of MPS operations)**.** *For an MPS on n sites with local dimension d and bond dimension $\leq \chi$:*

(a) *the norm $\langle\psi|\psi\rangle$ and any local expectation value cost $\mathcal{O}(n\,d\,\chi^3)$ to evaluate;*

(b) *the overlap $\langle\phi|\psi\rangle$ of two such MPS costs $\mathcal{O}(n\,d\,\chi^3)$;*

(c) *bringing the state to canonical form, or truncating each bond to dimension χ, costs $\mathcal{O}(n\,d\,\chi^3)$ dominated by the per-site SVDs.*

Proof. (a),(b) Contract left to right, carrying a $\chi\times\chi$ "transfer" matrix. At each site one contracts this matrix with the bra tensor (cost $\mathcal{O}(\chi^2\cdot\chi\cdot d)=\mathcal{O}(d\chi^3)$, by Theorem 15.2 with contracted dimension χ and free dimensions $\chi, d\chi$) and then with the ket tensor (another $\mathcal{O}(d\chi^3)$). Repeating for n sites gives $\mathcal{O}(n\,d\,\chi^3)$; a local operator inserts one extra $\mathcal{O}(d^2\chi^2)$ factor at its site, which is subleading. (c) Each bond requires the SVD of a $d\chi\times\chi$ (or $\chi\times d\chi$) matrix, costing $\mathcal{O}(d\chi^3)$; over n bonds this is $\mathcal{O}(n\,d\,\chi^3)$. The left-to-right sweep is the optimal contraction order for this chain topology, so no reordering improves the exponent. □

Thus *every* basic MPS manipulation is linear in the system size n and polynomial (χ^3) in the bond dimension. The entire cost of simulating a one-dimensional quantum system with tensor networks therefore hinges on a single question: *how large must χ be?* By Theorem 15.6 and Theorem 15.8, a bond can be truncated to dimension χ with small error precisely when the entanglement entropy across that bond is small—roughly, $\chi \gtrsim 2^S$. So the efficiency of an MPS is governed by how entanglement entropy scales with system size.

15.5.1 The area law

Definition 15.20 (Area law vs. volume law)**.** Consider a family of states on regions of growing size and the entanglement entropy $S(\ell)$ of a contiguous block of ℓ sites (with the rest of the system). The family obeys an *area law* in one dimension if $S(\ell)=\mathcal{O}(1)$ (bounded, independent of ℓ), and a *volume law* if $S(\ell)$ grows proportionally to ℓ. (In D dimensions "area law" means S scales with the size of the boundary of the block, hence the name.)

Proposition 15.21 (Area law implies an efficient MPS)**.** *If a one-dimensional state obeys an area law $S(\ell)\leq S_{\max}$ across every cut with, in addition, a Schmidt spectrum whose truncation error (15.3) decays as $\varepsilon_{\chi'}\leq C\,\eta^{\chi'}$ for some $\eta<1$ (a rapidly decaying spectrum, typical of gapped systems), then a bond dimension*

$$\chi=\mathcal{O}(\mathrm{polylog}(n/\epsilon))$$

suffices to represent the state to global error ϵ; the total MPS cost is then $\mathrm{poly}(n)$.

Proof sketch. Truncating bond k to dimension χ' incurs squared error $\varepsilon_{\chi'}^{(k)} \le C\eta^{\chi'}$ by hypothesis. Truncating all $n-1$ bonds and using the subadditivity of MPS truncation errors (the global squared error is bounded by the sum of per-bond discarded weights), the total error is at most $(n-1)C\eta^{\chi'}$. Setting this $\le \epsilon^2$ gives $\chi' = \mathcal{O}(\log(n/\epsilon)/\log(1/\eta)) = \mathcal{O}(\mathrm{polylog}(n/\epsilon))$. Substituting into Theorem 15.19 makes every operation $\mathrm{poly}(n)$. □

The physics that guarantees the hypothesis of Theorem 15.21 is the following landmark theorem, which we quote (a full proof uses Lieb–Robinson bounds and is beyond our scope).

Theorem 15.22 (Area law for gapped 1D ground states, Hastings)**.** *Let $H = \sum_j h_j$ be a local Hamiltonian on a one-dimensional chain with a unique ground state and a spectral gap $\Delta > 0$ above it. Then the entanglement entropy of any contiguous block in the ground state is bounded,*

$$S(\ell) \le S_{\max} = \mathcal{O}(1),$$

independent of the block size ℓ and of n.

Remark 15.23 (Why gapped ground states are efficiently representable). Theorem 15.22 says the ground state of a gapped local 1D Hamiltonian has $\mathcal{O}(1)$ entanglement across every cut, so by Theorem 15.21 it admits an MPS of bounded bond dimension—this is the theoretical justification for the spectacular success of DMRG (Chapter 16). Intuitively, a nonzero gap gives a finite correlation length $\xi \sim 1/\Delta$, so correlations (and hence the entanglement that can be built across a cut) decay exponentially and saturate at a length-independent value. *Volume-law* states—generic states, or the highly excited states reached by long-time chaotic evolution—have $S(\ell) \propto \ell$, forcing $\chi \sim 2^{\ell}$ and destroying the efficiency; these are exactly the states an MPS *cannot* compress, and are the frontier that quantum computers aim to reach. In two and higher dimensions, even gapped ground states obey only a boundary-area law $S \sim \ell^{D-1}$, which is why MPS give way to higher-dimensional tensor networks (PEPS) there.

Example 15.24 (Bell state: Schmidt data and entropy)**.** For $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ the coefficient matrix across the single cut is $\psi = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, already diagonal, so the Schmidt coefficients are $\lambda_1 = \lambda_2 = \frac{1}{\sqrt{2}}$ with $\lambda_1^2 = \lambda_2^2 = \frac{1}{2}$ and Schmidt rank $\chi = 2$. The entanglement entropy is

$$S = -\tfrac{1}{2}\log_2 \tfrac{1}{2} - \tfrac{1}{2}\log_2 \tfrac{1}{2} = \log_2 2 = 1 \text{ bit},$$

the maximum for a single qubit—consistent with the maximally mixed reduced state $\rho_A = \frac{1}{2}\mathbb{I}$ found in Chapter 1.

Example 15.25 (GHZ and W states: $\chi = 2$ everywhere)**.** Both the n-qubit *GHZ* state $|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n})$ and the *W* state $|W\rangle = \frac{1}{\sqrt{n}}(|10\cdots0\rangle + |01\cdots0\rangle + \cdots + |0\cdots01\rangle)$ have Schmidt rank 2 across *every* bipartition, so each has an exact MPS with $\chi = 2$—linear storage $\mathcal{O}(n)$ despite being globally entangled.

For GHZ, cutting after any site k gives $|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|0\rangle^{\otimes k}|0\rangle^{\otimes(n-k)} + |1\rangle^{\otimes k}|1\rangle^{\otimes(n-k)})$, already in Schmidt form with $\lambda_1^2 = \lambda_2^2 = \frac{1}{2}$, so $S = 1$ bit at every cut.

For the W state, cut the first site from the rest: the reduced state on the remaining $n-1$ qubits has the single "excitation" either on qubit 1 (weight $1/n$) or among qubits $2,\dots,n$ (weight $(n-1)/n$), giving Schmidt squares $\{1/n, (n-1)/n\}$ and

$$S = -\tfrac{1}{n}\log_2 \tfrac{1}{n} - \tfrac{n-1}{n}\log_2 \tfrac{n-1}{n} = \log_2 n - \tfrac{n-1}{n}\log_2(n-1).$$

For $n = 2$ this is 1 bit (the W state on two qubits is a Bell state); as $n \to \infty$ it decays as $S \approx \frac{\log_2 n}{n} \to 0$. In both cases the *rank* is 2, so a bond dimension $\chi = 2$ suffices exactly, independent of n.

Exercises

Exercise 15.1. Using Penrose notation, draw the diagrams for (a) the trace $\mathrm{Tr}[AB]$ of a product of two matrices, and (b) the contraction $C_{i\ell} = \sum_{j,k} A_{ijk} B_{jk\ell}$ of two rank-3 tensors. In each case give the arithmetic cost in terms of the leg dimensions (take all legs of dimension d), and identify the contracted and free legs.

Exercise 15.2. Let $|\psi\rangle = \frac{1}{\sqrt{3}}|00\rangle + \frac{1}{\sqrt{3}}|01\rangle + \frac{1}{\sqrt{3}}|11\rangle$ be a two-qubit state. (a) Write its 2×2 coefficient matrix and compute its singular values. (b) Give the Schmidt rank, the Schmidt coefficients, and the entanglement entropy S. (c) Is the state entangled?

Exercise 15.3. Prove that a bipartite pure state $|\psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is a product state if and only if its reduced density operator ρ_A is a pure state (rank-one projector). *Hint:* use Theorem 15.6.

Exercise 15.4. For the three-qubit state $|\psi\rangle$ of Theorem 15.12, verify by direct matrix multiplication that the MPS tensors $A^{[1]}, A^{[2]}, A^{[3]}$ given there reproduce all eight amplitudes $\psi_{s_1 s_2 s_3}$ (four nonzero, four zero). Then check that $A^{[2]}$ is left-canonical, i.e. satisfies (15.6).

Exercise 15.5. Consider the GHZ state on n qubits. (a) Write an explicit MPS with bond dimension $\chi = 2$: give the boundary vectors and the two bulk matrices $A^{[k]0}, A^{[k]1}$ (independent of k in the bulk). (b) Verify that your MPS reproduces exactly the two nonzero amplitudes $\psi_{0\cdots 0} = \psi_{1\cdots 1} = \frac{1}{\sqrt{2}}$ and vanishes on all other strings.

Exercise 15.6. Let $|\psi\rangle$ have Schmidt coefficients (across some cut) $\lambda_1^2 = 1 - 3\delta$, $\lambda_2^2 = \lambda_3^2 = \lambda_4^2 = \delta$ for a small $\delta > 0$. (a) Compute the truncation error (15.3) if we keep only $\chi' = 1$ Schmidt value, and if we keep $\chi' = 2$. (b) For a target squared error $\epsilon^2 = 10^{-3}$ and $\delta = 10^{-4}$, what is the smallest χ' that suffices? (c) Compute the exact entanglement entropy S as a function of δ and its limit as $\delta \to 0$.

Exercise 15.7. Explain, in terms of the area law, why the ground state of a gapped local one-dimensional Hamiltonian admits an efficient MPS while a Haar-random state of n qubits does not. Specifically: (a) quote the entanglement scaling $S(\ell)$ in each case (for the random state, use the Page value $S \approx \frac{n}{2}$ across a half-chain cut); (b) deduce the bond dimension χ required in each case; and (c) state the resulting storage cost and say which is efficient.

Further reading

- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, Ann. Phys. **326**, 96 (2011); arXiv:1008.3477. The canonical reference for MPS, canonical forms, and DMRG.
- R. Orús, *A practical introduction to tensor networks: matrix product states and projected entangled pair states*, Ann. Phys. **349**, 117 (2014); arXiv:1306.2164. Diagrammatic notation, contraction, and higher-dimensional networks.
- J. Eisert, M. Cramer, and M. B. Plenio, *Colloquium: Area laws for the entanglement entropy*, Rev. Mod. Phys. **82**, 277 (2010); arXiv:0808.3773. The area law and its consequences for simulability.

Chapter 16

DMRG and Ground States

Learning objectives

- Represent operators as matrix product operators (MPO).
- Formulate ground-state search as variational optimization over MPS.
- Run DMRG sweeps and diagnose convergence.

[TODO: Write the introductory paragraph orienting the lecture.]

16.1 Matrix product operators

[TODO: Draft this section.]

16.2 Variational optimization over MPS

[TODO: Draft this section.]

16.3 The DMRG sweep algorithm

[TODO: Draft this section.]

16.4 Convergence and truncation error

[TODO: Draft this section.]

Exercises

Exercise 16.1. [TODO: Add exercise.]

Exercise 16.2. [TODO: Add exercise.]

Exercise 16.3. [TODO: Add exercise.]

Further reading

- U. Schollw
öck, Ann. Phys. 326 (2011), DMRG sections.
- S. R. White, *Density matrix formulation for quantum renormalization groups*, PRL 69 (1992).

Chapter 17

Time Evolution with Tensor Networks

Learning objectives

- Evolve an MPS in real and imaginary time with TEBD.
- Contrast TEBD with the time-dependent variational principle (TDVP).
- Use tensor networks to classically simulate quantum circuits.

[TODO: Write the introductory paragraph orienting the lecture.]

17.1 Time-evolving block decimation (TEBD)

[TODO: Draft this section.]

17.2 Time-dependent variational principle (TDVP)

[TODO: Draft this section.]

17.3 Simulating quantum circuits with tensor networks

[TODO: Draft this section.]

17.4 Beyond 1D: a look at PEPS

[TODO: Draft this section.]

Exercises

Exercise 17.1. [TODO: Add exercise.]

Exercise 17.2. [TODO: Add exercise.]

Exercise 17.3. [TODO: Add exercise.]

Further reading

- R. Or
  'us, Ann. Phys. 349 (2014).
- **[TODO: add TEBD (Vidal) and tensor-network circuit-simulation references]**.

Part V

Optimization and Quantum Machine Learning

Chapter 18

Variational Quantum Algorithms and QAOA

Learning objectives

- State the variational principle behind VQE.
- Construct the QAOA ansatz for combinatorial optimization.
- Relate QAOA to adiabatic evolution.
- Discuss trainability and barren-plateau concerns.

[TODO: Write the introductory paragraph orienting the lecture.]

18.1 The variational quantum eigensolver (VQE)

[TODO: Draft this section.]

18.2 The quantum approximate optimization algorithm (QAOA)

[TODO: Draft this section.]

18.3 The adiabatic connection

[TODO: Draft this section.]

18.4 Trainability and expressibility

[TODO: Draft this section.]

Exercises

Exercise 18.1. [TODO: Add exercise.]

Exercise 18.2. [TODO: Add exercise.]

Exercise 18.3. [TODO: Add exercise.]

Further reading

- E. Farhi, J. Goldstone, S. Gutmann, *A Quantum Approximate Optimization Algorithm*, arXiv:1411.4028.
- M. Cerezo et al., *Variational Quantum Algorithms*, Nat. Rev. Phys. 3 (2021).

Chapter 19

Quantum Algorithms for LP and SDP

Learning objectives

- Recall the LP/SDP problems and their classical solvers.
- Explain the multiplicative-weights framework for SDP.
- Describe quantum SDP solvers based on Gibbs-state sampling.
- Outline quantum interior-point methods and their assumptions.

[TODO: Write the introductory paragraph orienting the lecture.]

19.1 Linear and semidefinite programming

[TODO: Draft this section.]

19.2 Multiplicative weights and the Arora–Kale framework

[TODO: Draft this section.]

19.3 Quantum SDP solvers via Gibbs sampling

[TODO: Draft this section.]

19.4 Quantum interior-point methods

[TODO: Draft this section.]

19.5 Speedups, caveats, and input/output costs

[TODO: Draft this section.]

Exercises

Exercise 19.1. [TODO: Add exercise.]

Exercise 19.2. **[TODO: Add exercise.]**

Exercise 19.3. **[TODO: Add exercise.]**

Further reading

- F. Brand
  ao, K. Svore, *Quantum Speed-ups for SDPs*, arXiv:1609.05537.
- J. van Apeldoorn et al., *Quantum SDP-Solvers*, arXiv:1705.01843.
- I. Kerenidis, A. Prakash, *A Quantum Interior Point Method for LPs and SDPs*, arXiv:1808.09266.

Chapter 20

Quantum Machine Learning

Learning objectives

- Load classical data into quantum states and feature maps.
- Define and use quantum kernels.
- Train variational classifiers / quantum neural networks.
- Assess barren plateaus and the practical caveats of QML.

[TODO: Write the introductory paragraph orienting the lecture.]

20.1 Data loading and feature maps

[TODO: Draft this section.]

20.2 Quantum kernels

[TODO: Draft this section.]

20.3 Variational classifiers and QNNs

[TODO: Draft this section.]

20.4 Linear-algebra-based QML (HHL, PCA)

[TODO: Draft this section.]

20.5 Barren plateaus and caveats

[TODO: Draft this section.]

Exercises

Exercise 20.1. [TODO: Add exercise.]

Exercise 20.2. **[TODO: Add exercise.]**

Exercise 20.3. **[TODO: Add exercise.]**

Further reading

- M. Schuld, N. Killoran, *Quantum Machine Learning in Feature Hilbert Spaces*, PRL 122 (2019).
- J. Biamonte et al., *Quantum Machine Learning*, Nature 549 (2017).